

Diagnostic Manual

Air-Cooled Product with Power Zone® 200 Control

Air-Cooled Product Generator Set



SAVE THIS MANUAL FOR FUTURE REFERENCE

 WARNING

Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information, go to:
www.p65warnings.ca.gov

(P000394)

Table of Contents

Section 1: Safety

Safety	5
Read This Manual Thoroughly	5
Replacement Parts	5
General Hazards	5
Exhaust Hazards	7
Electrical Hazards	7
Fire Hazards	7
Explosion Hazards	8
Battery Hazards	8
General Rules	9

Section 2: General Troubleshooting Guidelines

Introduction	11
Recommended Tools	11
Troubleshooting Reminders and Tips	11
GenService	11
Contacting Technical Support	11
Troubleshooting Basic	12
Back Probes	12
Measuring Voltage	12
Measuring Resistance/continuity	12
Measuring Frequency	13
Diode Check	13
Mesuring Amperage	13
Checking Connections	13
Checking For Shorted Circuits	13

Section 3: Field Pro App

Field Pro App	15
---------------------	----

Section 4: Generator Specifications

Specification	17
Component Locator	21

Section 5: E-Codes

Controller E-Codes	25
--------------------------	----

Section 6: Operational Analysis

Introduction	33
Field Boost	33
Operation	33
Utility Source Voltage Available	35
Initial Dropout of Utility Source Voltage	37
Utility Voltage Failure and Engine Cranking	39
Engine Startup and Running	41
Transfer to Standby	43
Utility Voltage Restored and Re-transfer to Utility	45
Engine Shutdown	47

Section 7: Power Zone 200 Controller

Introduction	53
Customer Connection	53
Power Zone 200 Controller	53
Control Panel Interface	54
Controller Mode Indicator	54
Model Sync	54
External LED Indicators	55
Controller Battery Charger	55
AUTO-OFF-MANUAL	56
Fuse	56
Common Alarm Relay	56
Circuit Pin Description	56
Power Zone 200 Controller	57

Section 8: AC Troubleshooting

Introduction	61
Safety	61
AC Troubleshooting	61
AC Excitation Winding Test	61
AC Output Voltage Test	62
AVR PWM Output Test	63
Brushes And Slip Ring Test	63
Current Calibration Procedure	64
Field Boost Voltage Test	65

Internal Diode Test	65	Fuel Pressure Sensor Test.....	99
Overload Condition Test.....	66	Fuel Shutoff 2 (SOV2) Control Test.....	101
Power Cycle and Sync Test.....	66	Fuel Supply and Pressure Test	102
Preliminary Output Voltage Test.....	66	Ignition Coil Test	103
Rotor Assembly Test	67	Intake Manifold Pressure Sensor Test	104
Sensing Circuits Test.....	67	Intake Manifold Temperature Sensor Test	106
Stator Test	69	Master Power Relay Test	107
Utility Voltage Test and Calibration.....	70	Mixer Valve Power Test	108
Voltage And Frequency Under Load Test	70	Oil Level Sensor Test	109
Voltage Calibration	71	Oil Pressure Switch Test	111
Radio and Field Pro Test	71	Oxygen Sensor Test.....	112
Field Pro Status	72	Spark Plug Condition Test.....	113
Radio Test.....	73	Stator Battery Charger (Voltage Regulator) Test	114
Section 9: Engine Control Module		Starter Motor Test.....	115
Engine Control Components	75	Starter Contactor Test	115
Engine Control Location	76	Starter Control Relay Test	116
Theory Of Operation – ECM ON/OFF	82	Throttle Position Sensor Test	117
Engine Control Module.....	84	Transfer Control Wires Test.....	118
ETMC Connector Harness	85	Wire 14 Test.....	120
Section 10: Engine Control Module Troubleshooting		Intermittent Alarm Test	121
Introduction.....	87	Power Zone 200 Fuse Testing	121
Safety	87	ECM Harness Fuse Testing	123
Engine/DC Troubleshooting	87	Engine Cranks But Does Not Run, Lacks Power, Runs Rough, Hunts or Backfires.....	123
AC Battery Charger Test	87	Section 11: Electrical Data	
AC Input Voltage Test.....	88	Introduction.....	125
Battery and Cables Test	89	A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459	126
Battery Potential/Power Output 1 Test	91	A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997	132
CAN bus Test.....	92	A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997	138
Cylinder Head Temperature Sensor Test.....	93	Section 12: Major Disassembly	
Cylinder Leak Down Test.....	94	Major Disassembly	145
Electronic Throttle Control Power Test.....	95	Electrical Formulas.....	149
Engine Compression Test	96		
Engine Speed Test	97		
Engine Timing Sensor Test	97		
Fuel Mixer Valve Test	98		

Section 1: Safety

Safety

Throughout this publication and on tags and decals affixed to the generator, DANGER, WARNING, and CAUTION blocks are used to alert personnel to special instructions about a particular operation that may be hazardous if performed incorrectly or carelessly. Observe them carefully. Their definitions are as follows:



Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

(D000001)



Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

(000002)



Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

(C000003)

NOTE: Notes provide additional information important to a procedure or component.

NOTE: These safety alerts cannot eliminate the hazards they indicate. Observing safety precautions and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

Read This Manual Thoroughly

Read This Manual Thoroughly

This diagnostic manual has been written and published by Generac to aid qualified Generac dealer technicians and company service personnel when servicing the products described herein. This manual should not be distributed outside of Generac's dealer network.

It is assumed that these personnel are familiar with the servicing procedures for these products, or like or similar products manufactured and marketed by Generac, and that they have been trained in the recommended servicing procedures for these products, including the use of common hand tools and any special Generac tools or tools from other suppliers.

Generac could not possibly know of and advise the service trade of all conceivable procedures by which a

service might be performed and of the possible hazards and/or results of each method. We have not undertaken any such wide evaluation. Therefore, anyone who uses a procedure or tool not recommended by Generac must first satisfy themselves that neither their own nor the product's safety will be endangered by the service procedure selected.

All information, illustrations and specifications in this manual are based on the latest product information available at the time of publication.

When working on these products, remember that the electrical system and engine ignition system are capable of violent and damaging short circuits or severe electrical shocks. If you intend to perform work where electrical terminals could be grounded or touched, the battery cables should be disconnected at the battery.

Any time the intake or exhaust openings of the engine are exposed during service, they should be covered to prevent accidental entry of foreign material. Entry of such materials will result in extensive damage when the engine is started.

During any maintenance procedure, replacement fasteners must have the same measurements and strength as the fasteners that were removed. Metric bolts and nuts have numbers that indicate their strength. Customary bolts use radial lines to indicate strength while most customary nuts do not have strength markings. Mismatched or incorrect fasteners can cause damage, malfunction and possible injury.

Replacement Parts

Replacement Parts

When servicing this equipment, it is extremely important that all components be properly installed and tightened. If improperly installed and tightened, sparks could ignite fuel vapors from fuel system leaks.

General Hazards



Loss of life. Property damage. Installation must always comply with applicable codes, standards, laws and regulations. Failure to do so will result in death or serious injury.

(000190)

⚠ DANGER

Automatic start-up. Disconnect utility power and render unit inoperable before working on unit. Failure to do so will result in death or serious injury. (000191)

⚠ WARNING



Electrocution. Potentially lethal voltages are generated by this equipment. Render the equipment safe before attempting repairs or maintenance. Failure to do so could result in death or serious injury. (000187)

⚠ WARNING



Loss of life. This product is not intended to be used in a critical life support application. Failure to adhere to this warning could result in death or serious injury. (000209b)

⚠ WARNING

Accidental Start-up. Disconnect the negative battery cable, then the positive battery cable when working on unit. Failure to do so could result in death or serious injury. (000130)

⚠ WARNING

Equipment damage. Only qualified service personnel may install, operate, and maintain this equipment. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage. (000182a)

⚠ WARNING

Equipment damage. This unit is not intended for use as a prime power source. It is intended for use as an intermediate power supply in the event of temporary power outage only. Doing so could result in death, serious injury, and equipment damage. (000247a)

⚠ WARNING

Electric Shock. Only a trained and licensed electrician should perform wiring and connections to unit. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage. (000155a)

⚠ WARNING



Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury. (000115)

⚠ WARNING



Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury. (W000111)

⚠ WARNING



Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire. (000108)

⚠ WARNING

Equipment and property damage. Do not alter construction of, installation, or block ventilation for generator. Failure to do so could result in unsafe operation or damage to the generator. (W000146)

⚠ WARNING

Risk of injury. Do not operate or service this machine if not fully alert. Fatigue can impair the ability to service this equipment and could result in death or serious injury. (000215a)

⚠ WARNING

Environmental Hazard. Always recycle batteries at an official recycling center in accordance with all local laws and regulations. Failure to do so could result in environmental damage, death or serious injury. (000228)

⚠ WARNING

Injury and equipment damage. Do not use generator as a step. Doing so could result in falling, damaged parts, unsafe equipment operation, and could result in death or serious injury. (000216)

Inspect the generator regularly, and contact the nearest IASD for parts needing repair or replacement.

Exhaust Hazards



⚠ DANGER

Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.
(000103)



⚠ DANGER

Asphyxiation. Carbon monoxide can kill in minutes. Operate this unit outdoors only. Failure to do so will result in death or serious injury.
(000525)

⚠ WARNING

Equipment and property damage. Do not alter construction of, installation, or block ventilation for generator. Failure to do so could result in unsafe operation or damage to the generator.
(000146)



⚠ WARNING

Asphyxiation. Always use a battery operated carbon monoxide alarm indoors and installed according to the manufacturer's instructions. Failure to do so could result in death or serious injury.
(W000178)

Electrical Hazards



⚠ DANGER

Electrocution. Contact with bare wires, terminals, and connections while generator is running will result in death or serious injury.
(000144)



⚠ DANGER

Electrocution. Never connect this unit to the electrical system of any building unless a licensed electrician has installed an approved transfer switch. Failure to do so will result in death or serious injury.
(000150)



⚠ DANGER

Electrical backfeed. Use only approved switchgear to isolate generator from the normal power source. Failure to do so will result in death, serious injury, and equipment damage.

(000237)



⚠ DANGER

Electrocution. Verify electrical system is properly grounded before applying power. Failure to do so will result in death or serious injury.
(000152)



⚠ DANGER

Electrocution. Do not wear jewelry while working on this equipment. Doing so will result in death or serious injury.
(000188)



⚠ DANGER

Electrocution. Water contact with a power source, if not avoided, will result in death or serious injury.
(000104)



⚠ DANGER

Electrocution. In the event of electrical accident, immediately shut power OFF. Use non-conductive implements to free victim from live conductor. Apply first aid and get medical help. Failure to do so will result in death or serious injury.
(D000145)

Fire Hazards



⚠ WARNING

Fire hazard. Do not obstruct cooling and ventilating airflow around the generator. Inadequate ventilation could result in fire hazard, possible equipment damage, death or serious injury.
(000217)



⚠ WARNING

Fire and explosion. Installation must comply with all local, state, and national electrical building codes. Noncompliance could result in unsafe operation, equipment damage, death or serious injury.
(000218)



⚠ WARNING

Fire hazard. Use only fully-charged fire extinguishers rated "ABC" by the NFPA. Discharged or improperly rated fire extinguishers will not extinguish electrical fires in automatic standby generators.

(000219)



WARNING

Electrocution. Refer to local codes and standards for safety equipment required when working with a live electrical system. Failure to use required safety equipment could result in death or serious injury.

(000257)



WARNING

Risk of Fire. Unit must be positioned in a manner that prevents combustible material accumulation underneath. Failure to do so could result in death or serious injury.

(000147)

Comply with regulations the local agency for workplace health and safety has established. Also, verify that the generator is installed in accordance with the manufacturer's instructions and recommendations. Following proper installation, do nothing that might alter a safe installation and render the unit in noncompliance with the aforementioned codes, standards, laws, and regulations.

Explosion Hazards



DANGER

Explosion and fire. Fuel and vapors are extremely flammable and explosive. No leakage of fuel is permitted. Keep fire and spark away. Failure to do so will result in death or serious injury.

(000192)

DANGER

Explosion and fire. Connection of fuel source must be done by a qualified professional technician or contractor. Incorrect installation of this unit will result in death, serious injury, and property and equipment damage.

(000151)



DANGER

Risk of fire. Allow fuel spills to completely dry before starting engine. Failure to do so will result in death or serious injury.

(D000174)



WARNING

Risk of Fire. Hot surfaces could ignite combustibles, resulting in fire. Fire could result in death or serious injury.

(W000110)

Battery Hazards



DANGER

Electrocution. Do not wear jewelry while working on this equipment. Doing so will result in death or serious injury.

(000188)



WARNING

Explosion. Do not dispose of batteries in a fire. Batteries are explosive. Electrolyte solution can cause burns and blindness. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(000162)



WARNING

Explosion. Batteries emit explosive gases while charging. Keep fire and spark away. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000137a)



WARNING

Electrical shock. Disconnect battery ground terminal before working on battery or battery wires. Failure to do so could result in death or serious injury.

(000164)



WARNING

Risk of burns. Batteries contain sulfuric acid and can cause severe chemical burns. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000138a)



WARNING

Risk of burn. Do not open or mutilate batteries. Released electrolyte solution has been known to be harmful to the skin and eyes and to be toxic. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(000163b)

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit Call2Recycle website at: <http://Call2Recycle.org/locator>.

- Incorrect installation can result in personal injury and damage to the unit. It may also result in the warranty being suspended or voided. All instructions listed below must be followed including location clearances and pipe sizes.

General Rules

DANGER

Loss of life. Property damage. Installation must always comply with applicable codes, standards, laws and regulations. Failure to do so will result in death or serious injury.

(000190)



DANGER

Electrical backfeed. Use only approved switchgear to isolate generator from the normal power source. Failure to do so will result in death, serious injury, and equipment damage.

(000237)

WARNING

Equipment damage. Only qualified service personnel may install, operate, and maintain this equipment. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage.

(000182a)



WARNING

Electrocution. Refer to local codes and standards for safety equipment required when working with a live electrical system. Failure to use required safety equipment could result in death or serious injury.

(000257)



WARNING

Consult Manual. Read and understand manual completely before using product. Failure to completely understand manual and product could result in death or serious injury.

(000100)

- Follow all safety precautions in the owner's manual, installation guidelines manual, and other documents included with the equipment.
- Never energize a new system without opening all disconnects and breakers.
- Always consult local code for additional requirements for where unit is being installed.

This page intentionally left blank.

Section 2: General Troubleshooting Guidelines

Introduction

This section familiarizes the service technician with recommended procedures for testing and evaluating various problems that can occur on air-cooled home standby products. Become familiar with these guidelines before attempting to troubleshoot any of the three main generator components: AC generator, air-cooled engine, transfer switch. The test procedures in each section require a basic knowledge of electricity and electrical safety, hand tool skills, and use of multimeters. Testing and troubleshooting methods covered in this manual are not exhaustive. No attempt has been made to discuss, evaluate, or advise the home standby service trade of all conceivable ways in which service and trouble diagnosis must be performed. Any test method not recommended herein must be deemed safe for personnel and equipment.

Recommended Tools

In addition to standard hand tools, some test procedures require the use of specialized test equipment as follows:

- General mechanics toolbox
- A 1/4" and 3/8" metric and SAE socket set
- Allen wrenches (metric and SAE)
- Wrenches for flywheel nuts
- T30 star bit for enclosure disassembly
- Generator key
- A meter capable of measuring frequency (Hz), AC and DC volts, DC amps, and Ohms
- J7 breakout harness (P/N A0008081735)
- Needle test probes
- Back test probes
- A clamp-on ammeter
- Manometer
- Spark tester
- Compression gauge
- Oil pressure gauge
- Leak down tester with an air compressor

Troubleshooting Reminders and Tips

The most important step in troubleshooting is to correctly identify the problem. Use Field Pro app to access the run log of the Power Zone 200 controller to help identify what the controller is seeing. Use the alarm log to view the faults that caused the warning or alarm shutdown. The

Date/Time stamp provides the date and time (to the second) that the alarm event occurred. If there are several alarms that all have the same date-time stamps, go to the alarm priority list, above the Ecode table. Some failures can cause a cascading series of faults to occur, one right after the other. Compare the alarm log and the run log to each other to see the operational sequence of events.

GenService

Any and all relevant bulletins must be reviewed before making repairs to ensure these generators are up to manufacturing standards..

Contacting Technical Support

Any repair using parts from the Level 1 or Level 2 required stock part kit does not require a control number, a call to technical support, or any kind of preauthorization. Troubleshoot unit, perform repair, and file standard claim on GenService.

Repairs exceeding \$3,000, replacement of major component, or parts that needed to be ordered and a second trip made can have a control number assigned for the job.

Technical support will not send out parts other than engines, rotors, stators, or other large/heavy major components. Warranty parts should be stocked or ordered from GenService.

Be on-site and be sure to check the warranty status of the unit before calling. Some generators may be under full warranty, others may be in parts-only coverage that will require billing the end user for the work..

Before calling in, have the following information ready:

- Dealer account number
- Technician ID number
- Unit serial number
- Test results from completed tests.

Technical Support is working from the basic test results that should already be performed. If this information is not available, you may be asked to call back after testing is complete.

Don't overlook the basics - weak batteries, fuel pressure, installation issues, loose electrical connectors or terminals, maintenance issues.

Following these guidelines will provide the best service and shortest wait times to all technicians working in the field.

NOTE: Do not replace more than one component (ECM, Controller, Radio, etc...) at a time without explicit direction from technical support. Failure to follow this policy may result in additional failures and misdiagnosed troubleshooting results. Component replacements deemed in neglect of this policy are not warrantable and will result in warranty charge backs.

Troubleshooting Basic

Proper diagnosis requires an understanding of how to perform basic checks with a digital Multimeter (DMM). This tool (used in conjunction with the appropriate [Back Probes](#)) will allow the technician to diagnose any wiring circuit within the generator. A high quality DMM and test leads are necessary to avoid false diagnoses due to inadequate/substandard equipment. Diagnostic steps were designed using a Fluke 117.

NOTE: Prior to troubleshooting, perform a thorough inspection of the wiring on the generator. Look for chafed wires, exposed connectors, or pinched wires. Verify all Molex pins and spade terminals are making a positive connection. Verify status of the battery voltage.

Back Probes

When testing a wiring harness, the harness connector will need to be back probed to perform the testing. There are two types of back probes used for testing. The first is the flexible test lead used to back probe Molex connectors.



Figure 2-1. Flexible Test Leads and Probes

The second type is the needle probe used to test ECM and weather tight connectors. These are designed to slide between the seal and wire as shown.

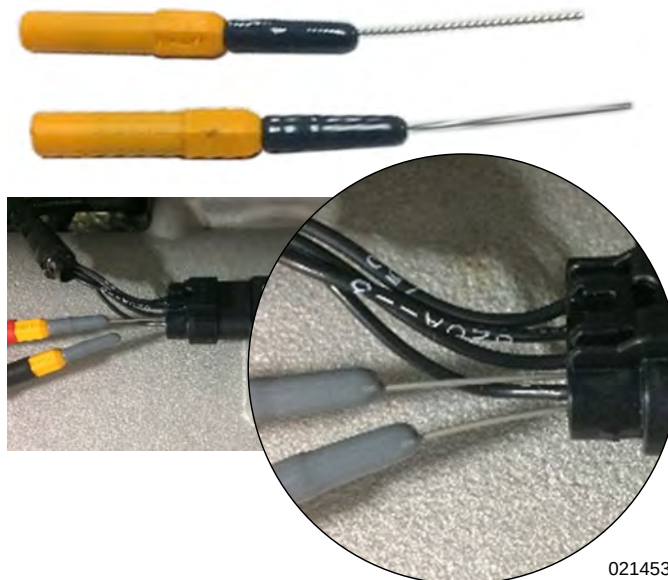


Figure 2-2. Needle Test Probes

NOTE: A needle type probe must be used for any connector with a weather tight seal (to fit between the wire and seal). If a flexible test lead is used, it can damage the seal, which can lead to premature failure of the connector and harness.

Measuring Voltage

A DMM can be used to make AC and DC voltage measurements. On a DMM, the $\overline{V}$ icon symbolizes the DC voltage selection. The $\tilde{V}$ icon symbolizes the AC voltage selection. To check voltage, there must be a power differential between the meter leads in order to provide a reading. In simpler terms, there must be a power source and a ground source to take measurements from.

Measuring Resistance/continuity

Resistance and continuity can be used to check the integrity of a circuit.

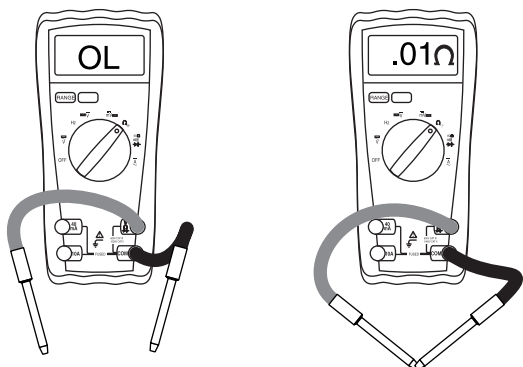
- Resistance will provide a measurement that can be compared to a specification chart to verify if the circuit/component has the right amount of resistance.
- Continuity will sound an audible alarm if the circuit is complete from end to end.

There are two important rules to remember when checking a circuit for resistance or continuity:

1. The circuit must not be energized while measuring.
2. The circuit being tested must be isolated from the rest of the generator.

On a DMM, the Ω icon symbolizes the resistance selection (measured in Ohms). The |||| icon symbolizes the Continuity selection.

See [Figure 2-3](#). When checking resistance, the measurement will be displayed as a number. If OL (Over Limit) is shown, this does not symbolize zero resistance; rather this means that the circuit is broken or open.



002397

Figure 2-3. INFINITY (Left) and CONTINUITY (Right) Meter Readings

Measuring Frequency

The selection for checking frequency can vary depending on the DMM. Most will have the frequency selection as part of the AC voltage selection $\tilde{V}$. Typically, AC voltage would be selected and then a button pressed on the meter to switch between voltage and frequency.

Diode Check

A diode is an electrical “check valve” that only allows voltage to pass through it in one direction. This is done in order to protect a circuit from a voltage backfeed that may damage it. The meter will supply a small amount of voltage through the positive meter lead to push through the diode and make the measurement (typically 0.5 VDC). If the meter leads are switched, a functioning diode will provide an OL measurement on the meter. This would indicate a functioning diode.

On the DMM, the $\rightarrow|$ picture symbolizes the diode check function.

Mesuring Amperage

As with voltage, a DMM can measure both AC and DC amperage. Most meters will require the positive meter lead to be moved to the “A” port. Also note that most meters are only capable of measuring 10 amps of current. Anything greater will damage the meter fuse.

On the DMM, the $\overline{A}$ picture symbolizes the DC amperage selection. The $\tilde{A}$ picture symbolizes the AC amperage selection.

Checking Connections

It is important to verify all connections are clean and tight to make positive connections. A loose or corroded electrical connection can create numerous issues such

as increased resistance, voltage drops, excess heat, and loss of output.

A few examples of these failed connection can be:

- loose screws (like the ones that tighten down a battery terminal).
- Molex pins that have been damaged from incorrect testing procedures.
- corroded wiring due to improper wire testing

Checking For Shorted Circuits

To check for a short-to-ground, select the Continuity selection on the DMM. Prepare the circuit for testing continuity by removing power and isolating the circuit. Use the red meter test lead to connect to the suspect circuit. Place the black meter test lead on an appropriate ground. If the meter reads continuity, a short-to-ground is present and the circuit will need to be repaired.

To check for a short-to-power, select the Continuity selection on the DMM. Prepare the circuit for testing continuity by removing power and isolating the circuit. Use the red meter test lead to connect to the suspect circuit. Place the black meter test lead on the suspected power source (battery positive terminal wire or utility voltage source wire). If the meter registers continuity, a short-to-power is present and the circuit will need to be repaired.

This page intentionally left blank.

Section 3: Field Pro App

Field Pro App

The generator details page is the dashboard view of the unit and is the landing page for users upon connecting to a previously commissioned generator.

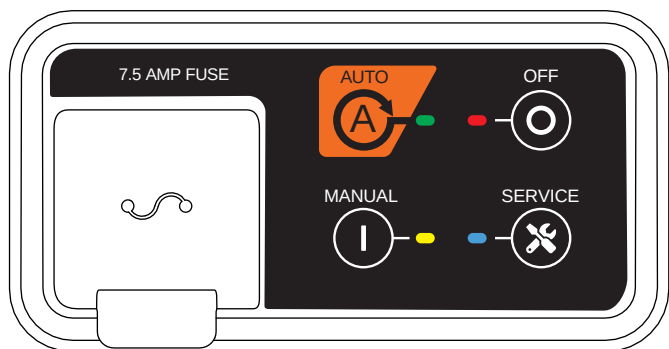


Figure 3-1. Controller Mode Buttons

NOTE: Connection between generator and Field Pro app can only be accomplished when the SERVICE button on the control panel interface is selected and a working GGCAC is connected and powered.

Users can view the generator's state/mode, status, telemetry data which dynamically updates when running, diagnostics info such as active alarms, warnings, log data

(e-codes) and is also the location to find the generator's controller settings (e.g. exercise schedule)

The Status card at the top displays the mode and status of the generator.

Controller settings are available from the Generator Details page.

Available settings will be based on user permissions. The telemetry data displayed on the generator details dashboard changes based on the state of the equipment.

For Advanced / Certified Users Only

Allows a user to calibrate the Generator Voltage, Current, and Utility Voltage (recognized by the generator), and enable 2-wire start.

GGCAC Device

The Generac Generator Connectivity Accessory – Cellular otherwise known as the GGCAC or the radio connects to both Wi-Fi and/or Cellular to connect with Generac servers for remote monitoring, model sync, remote firmware updates, and other remote commands. The radio also provides Bluetooth® connection for use with the Field Pro app. It has a green light that may blink corresponding to different modes. Refer to [Radio and Field Pro Test](#) for more information.

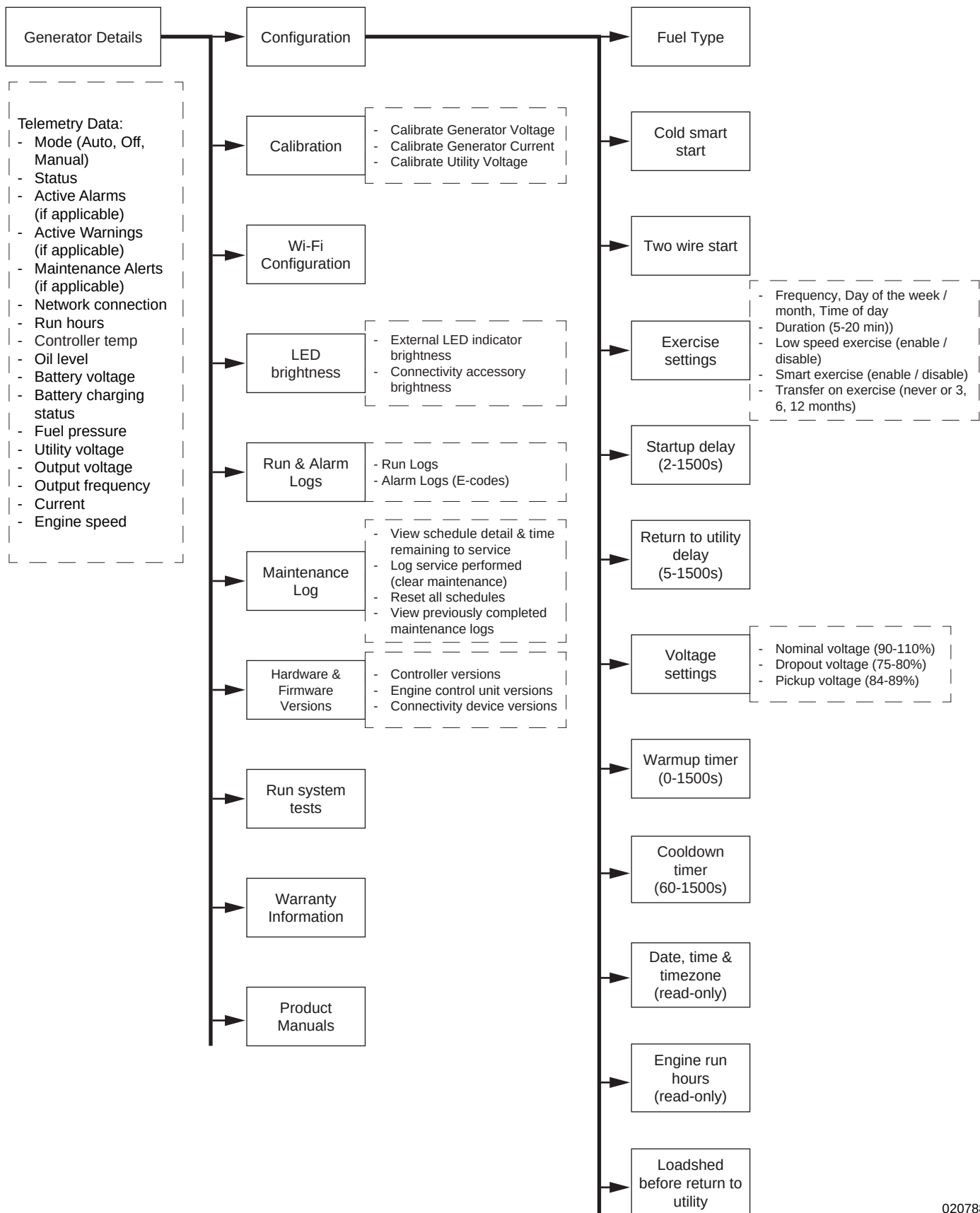


Figure 3-2.

020788

Section 4: Generator Specifications

Specification

For rated power capacity, rated voltages, maximum load, harmonic distortion, main line circuit breaker sizes, number of phases, number of rotor poles, rated frequency, power factor, battery requirement, unit weight, dimensions, sound output, exercise duration, engine type, number of cylinders, displacement, cylinder block design, valve arrangement, ignition system, compression ratio, starter system, oil capacity, operating rpm, and fuel consumption, please refer to the specific unit's spec sheet located at www.generac.com.

NOTE: All unit specifications are subject to change.

Table 3-1. Generator

Model	10 kW	14 kW	18 kW	20 KW	22 kW	24 kW	25 kW	26 kW	28 kW
Rated voltage	240			208	240				
Rated maximum load current (amps) at rated voltage with LP*	42	58	75	69	92	100	104	108	116
Main line circuit breaker (generator disconnect)	45 amp	60 amp	80 amp	80 amp	100 amp		110 amp		125 amp
Phase	1			3	1				
Rated AC frequency	60 Hz								
Battery requirement (field supplied)	A0010137451 / A0010137452 ‡ 12 volts, AGM Powersport Battery Group BTX30L 400CCA minimum								
Enclosure	Aluminum								
Weight (lb / kg) (without battery)	344 / 156	391 / 177	-0 Models: 426 / 193 -10 Models: 425 / 193	472/ 214	-0 Models: 451 / 205 -10 Models: 463 / 210	461 / 209	461 / 209	524 / 238	524 / 238
Normal operating range	This unit is tested to perform within an operating temperature of -20 °F (-29 °C) to 122 °F (50 °C). For areas where temperatures fall below 32 °F (0 °C), a cold weather kit is recommended. When operated above 77 °F (25 °C), there may be a decrease in engine power.								
These generators are rated in accordance with UL 2200, Safety Standard for Stationary Engine Generator Assemblies, and CSA-C22.2 No. 100-04 Standard for Motors and Generators. * NG ratings will depend on specific fuel joules/BTU content. Typical derates are between 10-20% off the LP gas rating. ‡ Kit G00735230 has a bracket to be used when upgrading from 20L to 30L batteries.									

Table 3-2. Engine

Model	10 kW	14/18 kW	20–28 kW
Engine type	G-Force™ 400 Series	G-Force™ 800 Series	G-Force™ 1000 Series
Number of cylinders	1	2	2
Displacement	459 cc	817 cc	997 cc
Cylinder block	Aluminum with cast iron sleeve		

Generator Specifications

Model	10 kW	14/18 kW	20–28 kW
Nominal Cylinder Compression ± 15%	125 psi	161 psi	187 psi
Recommended spark plug	A0003637864		
Spark plug gap	0.020 in (0.508 mm)		
Starter	12 VDC		
Oil capacity including filter	Approx. 1.4 qt (1.3 L)	Approx. 2.5 qt (2.4 L)	Approx. 2.2 qt (2.1 L)
Recommended oil filter	070185ES		
Recommended air filter	0E9371AS	0J8478S	
Engine power is subject to and limited by such factors as fuel BTU/joules, ambient temperature, and altitude. Engine power decreases approximately 3.5% for each 1,000 ft (304.8 m) above sea level, and also will decrease approximately 1% for each 10 ° F (6 °C) above 60 °F (15 °C) ambient temperature.			
NOTE: Break in oil to be used until 25 hours. After please convert to full synthetic.			

Table 3-3. Stator Winding Resistance Values / Rotor Resistance*

	Power Windings Across 11–22	Power Windings Across 33–44	Power Windings Across 11–44 at Stator Terminal Block	Excitation Windings Across 2–6	Rotor Resistance
ALL	0.00–0.30	0.00–0.30	0.00–1.0	0.8–1.5	4–15

* Resistance (ohms) values shown are based on new windings at 68 ° F (20 °C) with neutrals connected. Actual readings may vary based on type of meter used and any other components or connections included in the circuit being tested.

Table 3-4. Fuel Consumption*

Unit	Natural Gas**		Propane***	
	1/2 Load	Full Load	1/2 Load	Full Load
10 kW	2.77 / 98	3.74 / 132	0.94 / 3.55 / 35	1.60 / 6.07 / 58
14 kW	5.35 / 189	6.89 / 243	1.76 / 6.66 / 63	2.92 / 11.03 / 106
18 kW	4.65 / 164	6.64 / 235	1.65 / 6.26 / 60	2.87 / 10.87 / 105
20 kW	5.35 / 189	8.21 / 290	1.99 / 7.53 / 72.4	3.4 / 12.86 / 123.6
22 kW	6.27 / 221	8.80 / 311	2.45 / 9.28 / 89	3.71 / 14.03 / 135
24 kW	5.58 / 197	8.23 / 291	2.45 / 9.28 / 89	3.71 / 14.03 / 135
25 kW	5.95 / 210	9.26 / 327	2.24 / 8.49 / 82	3.75 / 14.19 / 136
26 kW	5.16 / 182	8.96 / 316	2.05 / 7.74 / 74	3.95 / 14.94 / 144
28 kW	5.15 / 182	8.41 / 297	2.23 / 8.45 / 81	3.97 / 15.02 / 144

*Values given are approximate

** Natural gas is in m3/h / ft3/h

***Propane is in US gal/h / L/h (LP) / ft3/h (LPV)

Table 3-5. Torque Specifications

Spark plug	14.75-22.1 ft-lbs	20–30 Nm
M6.3 Hi-lo into plastic base	30 in-lbs	3.4 Nm
M6.3 Hi-lo into controller	20 in-lbs	2.2 Nm

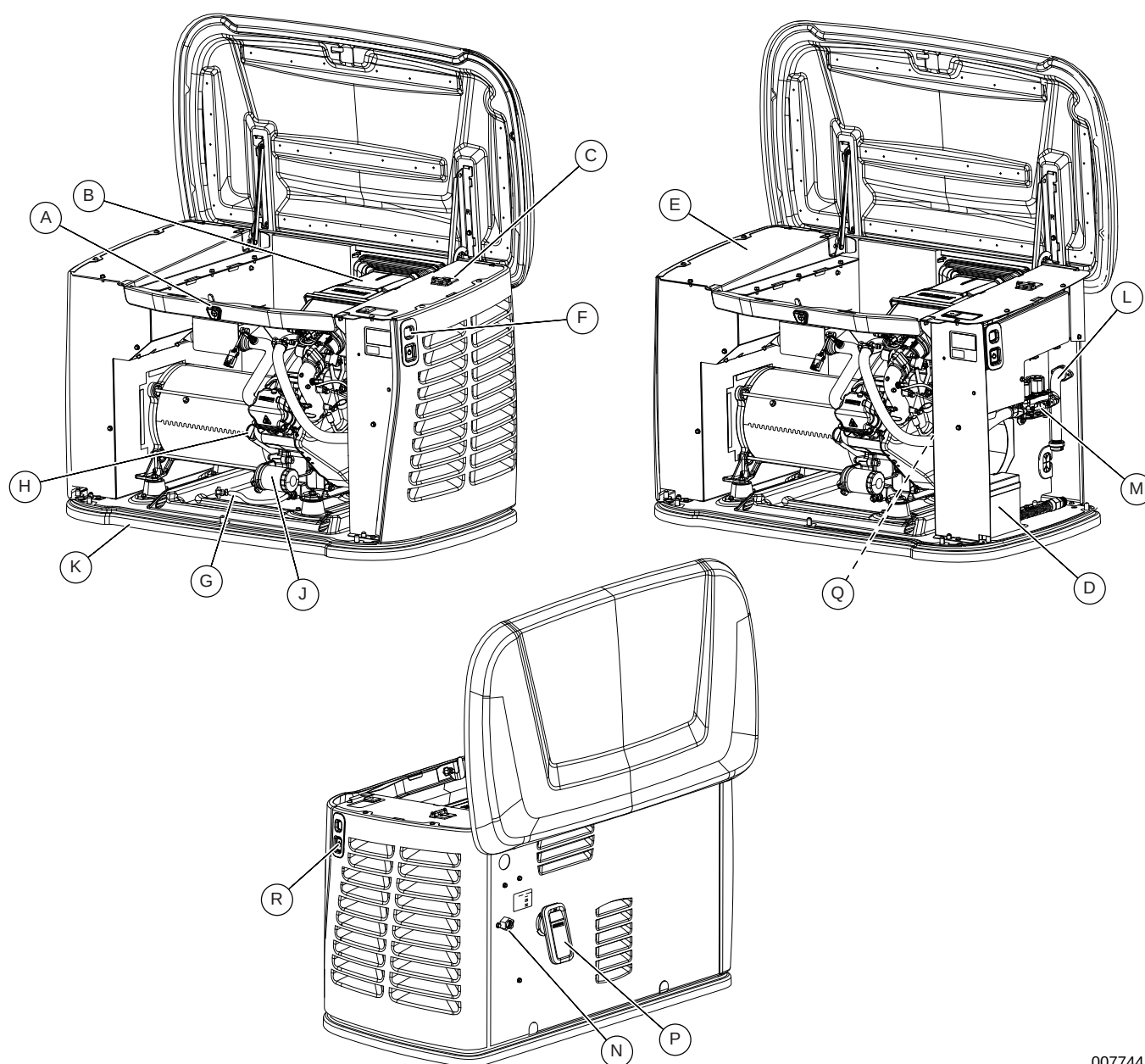
Table 3-5. Torque Specifications

M8 - 1.25 Nylok nut onto vibration isolator stud	100 in-lbs	11.3 Nm
M8 - 1.25 Stator bolts	78 in-lbs	8.8 Nm
3/8 - 24 Rotor bolt	30 ft-lbs	40 Nm
M6 - 1.0 Nut starter motor battery cable - 459cc	40 in-lbs	4.5 Nm
M8 - 1.25 Nut starter motor battery cable - 817cc–997cc	84 in-lbs	9.5 Nm
M8 - 1.25 Bolt exhaust manifold	18 ft-lbs	24 Nm
M12 - 1.25 O2 sensor	13 ft-lbs	17.6 Nm
60 Hz Breaker terminal screw - #10 wire	25 in-lbs	2.8 Nm
60 Hz Breaker terminal screw - #8, #6, #4, #3, & #2/0 wire	45 in-lbs	5 Nm
50 Hz Breaker terminal screw - #8 & #10 wire	25 in-lbs	2.8 Nm
50 Hz Breaker terminal screw - #6 wire	45 in-lbs	5 Nm
1/4 - 28 Set screw neutral bus bar	35 in-lbs	3.9 Nm
M5 - 0.8 Nut for neutral bar	30 in-lbs	3.3 Nm
7/16 - 20 Set screw neutral & ground lug	10 ft-lbs	13.5 Nm
M6 - 1.0 Grade 8.8 bolt into base threaded inserts	60 in-lbs	6.8 Nm
M6 - 1.0 Grade 8.8 bolt on sediment trap and fuel SOV	84 in-lbs	9.5 Nm
M6 - 1.0 Exhaust flex pipe band clamp	10 ft-lbs	13.5 Nm
M6 - 1.0 Bolt for air-box thru mixer to intake	96 in-lbs	10.8 Nm
M6 - 1.0 Bolt for air-box to valve cover	78 in-lbs	8.8 Nm
M5 - 0.8 Taptite screw into aluminum	40 in-lbs	4.5 Nm
M5 - 0.8 Taptite screw into extruded hole	35 in-lbs	3.9 Nm
M6 - 1.0 Taptite screw into extruded hole	55 in-lbs	6.2 Nm
M8 - 1.25 Taptite screw into extruded hole	14 ft-lbs	18.9 Nm
M5 - 0.8 Nylok nut	40 in-lbs	4.5 Nm
#10 Hi-lo into plastic	20 in-lbs	2.2 Nm
M5 - 0.8 Bolt stator terminal block cover	15 in-lbs	1.7 Nm
M6 - 1.0 Taptite stator terminal block mounting	35 in-lbs	3.9 Nm
M6 - 1.0 Taptite stator duct mounting	35 in-lbs	3.9 Nm
M6 - 1.0 Taptite screw into intake side roof hinge	84 in-lbs	9.5 Nm
M5 - 0.8 Nylok nut onto stator block	45 in-lbs	5 Nm
1/4 - 28 T-bolt clamp fuel hose	78 in-lbs	8.8 Nm
3/4 - 24 Jam nut lock body on crossbar	60 in-lbs	6.8 Nm
1/4 - 20 Screw for lock pawl to lock body	40 in-lbs	4.5 Nm
M6 - 1.0 Grade 8.8 Bolt tailpipe extension clamp	14 ft-lbs	18.9 Nm
M6 - 1.0 Taptite screw into weld nut	55 in-lbs	6.2 Nm

Table 3-5. Torque Specifications		
M6-1.0 459 & 817cc Engine Rocker Arm Studs	5.5-6.7 ft-lbs	7.5-9.1 Nm
M6-1.0 x25 Rocker Cover bolts	57.5 - 66.4 in-lbs	6.5-7.5 Nm
M8-1.0 - 997cc Engine Rocker Arm Studs	11.1-12.4 ft-lbs	15.1-16.8 Nm

NOTE: Torques values are dynamic, with $\pm 10\%$ tolerance unless otherwise noted.

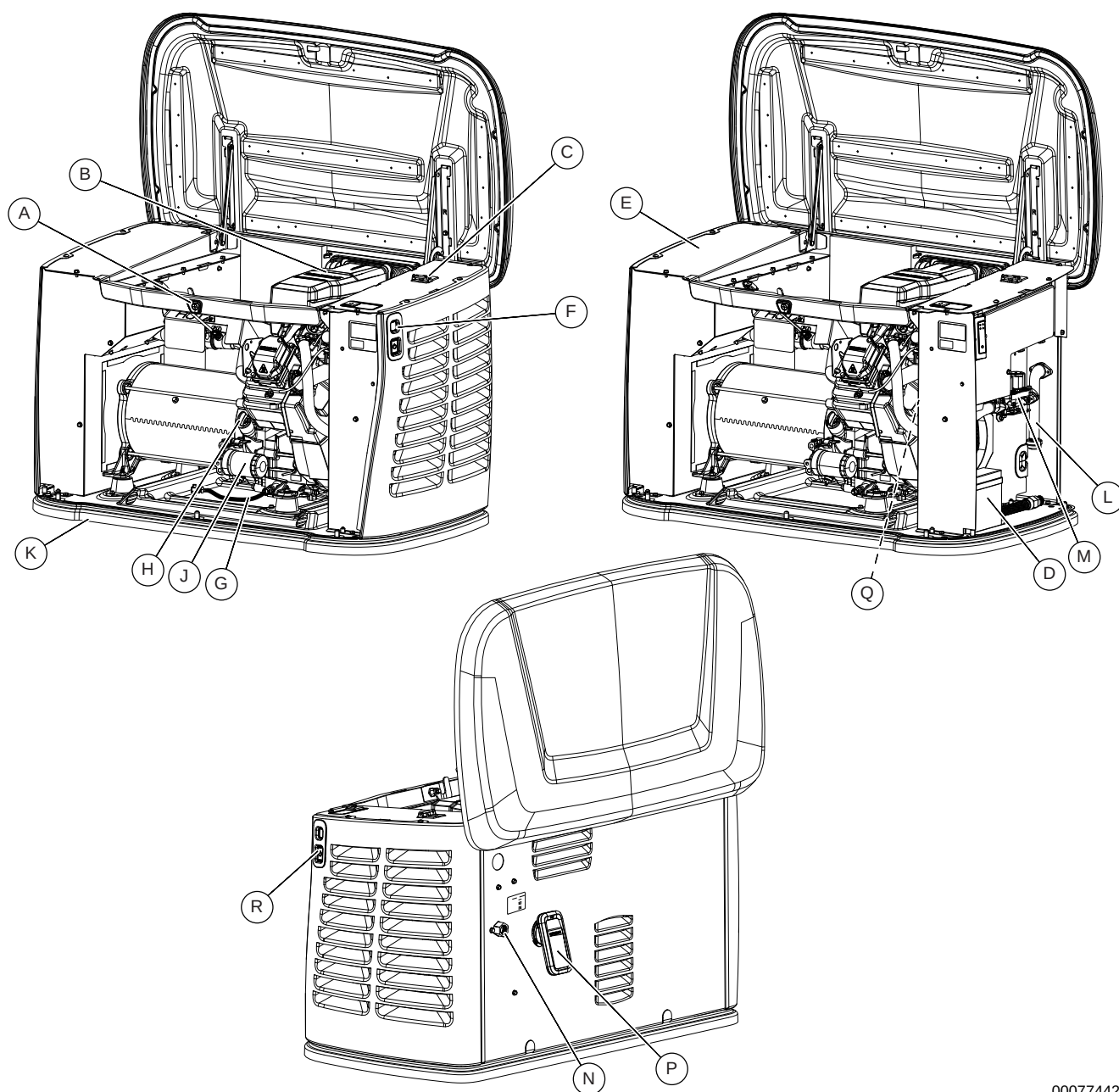
Component Locator



0077441

Figure 4-1. 10 kW—Components and Control Locations

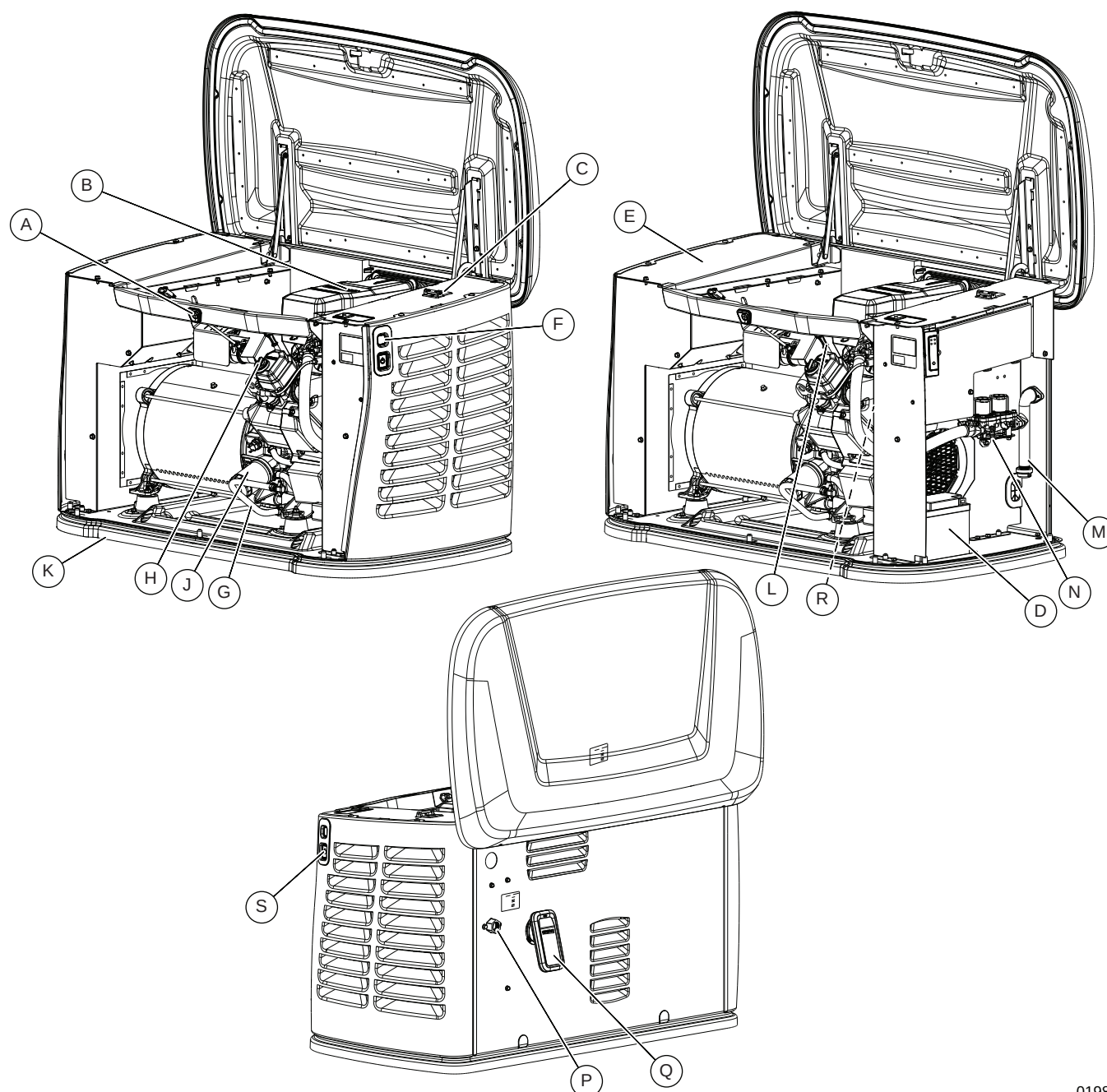
A Lock with cover	G Oil drain hose	M Dual shutoff valve
B Airbox with air cleaner	H Oil fill cap/dipstick	N Fuel inlet
C Main line circuit breaker (generator disconnect)	J Oil filter	P Radio
D Battery compartment (battery not supplied)	K Composite base	Q Data decal location
E Exhaust enclosure	L Sediment trap	R Generator emergency shutdown button
F Status LED indicators		



00077442

Figure 4-2. 14 kW–18 kW—Components and Control Locations

A Lock with cover	G Oil drain hose	M Dual shutoff valve
B Airbox with air cleaner	H Oil fill cap/dipstick	N Fuel inlet
C Main line circuit breaker (generator disconnect)	J Oil filter	P Radio
D Battery compartment (battery not supplied)	K Composite base	Q Data decal location
E Exhaust enclosure	L Sediment trap	R Generator emergency shutdown button
F Status LED indicators		



019847

Figure 4-3. 22 kW–28 kW—Components and Control Locations

A Lock with cover	G Oil drain hose	N Dual shutoff valve
B Airbox with air cleaner	H Oil fill cap	P Fuel inlet
C Main line circuit breaker (generator disconnect)	J Oil filter	Q Radio
D Battery compartment (battery not supplied)	K Composite base	R Data decal location
E Exhaust enclosure	L Oil dipstick	S Generator emergency shutdown button
F Status LED indicators	M Sediment trap	

This page intentionally left blank.

Section 5: E-Codes

Controller E-Codes

NOTE: For any Displayed Alarm in the Field Pro app, check GenService for any relevant bulletins before troubleshooting the unit. Any clear-able alarms triggered by connecting or disconnecting the battery or removing or inserting fuses or after firmware updates are to be ignored. First, attempt to clear the alarm and run the unit before navigating to the action step. Perform the [Intermittent Alarm Test](#) for help to recreate if uncertain what triggered the alarm. Once the alarm has been recreated refer to the corresponding Action Step.

Remote Reset of Alarms

Alarms considered critical to generator or user safety require a physical reset at the controller. When the generator has the Generac Generator Connectivity Accessory - Cellular (GGCAC) installed and communicating, an alarm can be remotely reset from the FLEET platform without visual confirmation of unit state if it is deemed safe for both generator and user to be reset remotely.

Warnings are self-clearing and are not eligible for remote reset.

NOTE: Alarms that are not remotely reset-able will need to be resolved before a firmware update will be accepted by the controller.

Alarm Priority

When receiving multiple E-Codes, any codes that refer to battery issues, charging issues, or CANBus, will be higher priority.

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
Missing Config - Parameter Group	Alarm	502	Generator Model not selected, finish setup	Power Cycle and Sync Test	No
Missing Config - Fuel Type	Alarm	504	Generator fuel not selected, finish setup.	Power Cycle and Sync Test	No
Invalid Serial Number	Alarm	505	Invalid serial number in controller.	Power Cycle and Sync Test	No
Controller Fault	Alarm	1000	Controller memory failure.	Power Cycle and Sync Test	No
Controller Fault	Alarm	1005	Controller timer failure.	Power Cycle and Sync Test	No
Controller Fault	Alarm	1006	Internal Controller alarm.	Power Cycle and Sync Test	No
Overcrank	Alarm	1100	Engine cranks but will not start.	Fuel Supply and Pressure Test	Yes
Overspeed	Alarm	1200	Prolonged, Over 72 Hz for 3 seconds	Engine Speed Test	No
Overspeed	Alarm	1205	Prolonged, Over 75 Hz for 0.1 seconds (100 milliseconds) using reported ECM speed.	Engine Speed Test	No
Overspeed	Alarm	1208	System overspeed from DPE sensing wires, over 75 Hz for 0.5 seconds, using DPE zero crossing.	Verify Power Zone 200 FW is at V1.4.1 or higher. Verify ECM FW is at 14304 or higher. Check for other E-codes first; if others present, diagnose those first before attempting to fix this one. Power Cycle. Call Technical support.	No

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
RPM Sensor	Alarm	1522	Crankshaft Position Sensor RPM and DPE Frequency are both measured at 0 after crank command.	<u>Battery and Cables Test</u>	Yes
Underspeed	Alarm	1600	Unit could not run at operating speed.	<u>Engine Speed Test</u>	Yes
Overvoltage	Alarm	1800	Prolonged Overvoltage	<u>AC Output Voltage Test</u>	Yes
Overvoltage	Alarm	1801	Instantaneous Overvoltage	<u>AC Output Voltage Test</u>	Yes
Undervoltage	Alarm	1900	Prolonged Undervoltage, Undervoltage due to loss of voltage for some time (10+ seconds).	<u>Preliminary Output Voltage Test</u>	Yes
Undervoltage	Alarm	1901	Instantaneous Undervoltage, Sudden loss of voltage (Less than 15V after 2 seconds past start-up).	<u>Preliminary Output Voltage Test</u>	Yes
Undervoltage	Alarm	1902	Both Zero Crosses Missing. No voltage for over 1.5 seconds.	<u>Preliminary Output Voltage Test</u>	Yes
Undervoltage	Alarm	1906	Single Zero Cross Missing. Zero cross missing for more 1.5 seconds.	<u>Preliminary Output Voltage Test</u>	Yes
Undervoltage	Alarm	1907	After attempting to shed load, the generator remains overloaded.	<u>Overload Condition Test</u>	Yes
Keypad Missing	Alarm	2070	Controller does not recognize keypad presence.	<u>Power Cycle and Sync Test</u>	No
Quadclops Missing	Warning	2075	Controller does not recognize external LED indicators. Applicable in firmware version 1.4.1 and below.	<u>Power Cycle and Sync Test</u>	No
Quadclops Missing	Alarm	2076	Controller does not recognize external LED indicators. Applicable in firmware version 1.5.0 and higher	<u>Power Cycle and Sync Test</u>	No
Model ID Sync Failed	Warning	2090	Communication failure between controller/radio/ECM.	<u>Power Cycle and Sync Test</u>	No
Wiring Error	Alarm	2094	DC Wires 395A/B sees AC power.	Verify customer wiring to ATS/Generator.	No
Fuel Type Sync Failed	Warning	2095	Communication failure between controller/radio/ECM for fuel type.	<u>Power Cycle and Sync Test</u>	No
Wiring Error	Alarm	2098	Wiring error, low voltage on 194, look for short or overload condition.	<u>Transfer Control Wires Test</u>	No
Wiring Error	Alarm	2099	DC Wires 194/23 sees AC voltage.	<u>Transfer Control Wires Test</u>	No
Overload Remove Load	Alarm	2100	Controller is detecting an overload condition. Hold off time is dependent on how far field current is above normal operation.	<u>Overload Condition Test</u>	No
AVR PWM Overload	Alarm	2115	AVR sensed overload.	<u>Overload Condition Test</u>	No

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
Gen kW Overload	Alarm	2116	Generator three phase total power (kW) is greater than 102% for 20 seconds. Three phase models only.	Overload Condition Test	No
Phase kW Overload	Alarm	2117	Generator three phase power on one of the three phases (A, B or C) is greater than 102% for 20 seconds. Three phase models only.	Overload Condition Test	No
Serial Num Sync Failed	Warning	2120	Communication failure between controller/radio/ECM for Serial number.	Power Cycle and Sync Test	No
Run Hours Sync Failed	Warning	2125	Communication failure between controller/radio/ECM for run hours.	Power Cycle and Sync Test	No
Low Oil Level	Warning	2126	Engine oil level is low.	Oil Level Sensor Test	No
CANBus Error	Alarm	2670	Communication lost between ECM and Controller.	CAN bus Test	Yes
No Transfer Detected	Warning	2738	The command to transfer has been issued by the controller, but no transfer has been detected.	Transfer Control Wires Test	No
Low Battery	Warning	2750	Battery is less than 12.1 Volts for 60 seconds.	Battery and Cables Test	No
Very Low Battery	Alarm	2751	Battery voltage level less than 9.0 Volts for 60 seconds.	Battery and Cables Test	Yes
Battery Problem	Warning	2760	More than 16 Volts of battery voltage or 600 milliamperes of charge current at the end of an 18 hour charge cycle.	AC Battery Charger Test	No
Charger Warning	Warning	2770	Battery voltage is less than 12.5 Volts after 18 hour charge cycle.	AC Battery Charger Test	No
Charger Missing AC	Warning	2780	During charging, AC input to battery charger missing for greater than 3 minutes.	AC Battery Charger Test	No
SEEPROM Abuse	Warning	2790	Internal Controller alarm	Power Cycle and Sync Test	No
Emergency Shutdown Pressed	Alarm	2800	Emergency Shutdown active.	With unit off, press and hold OFF button on the controller for 3 seconds to reset. If error reoccurs replace controller.	No
Auxiliary Shutdown	Alarm	2801	Remote Emergency Shutdown active.	Verify remote E-Stop switch is not activated and wiring is correct. If no remote E-Stop kit has been installed, verify jumper is present at TB1 as shown in wiring diagram. Once remote E-Stop is reset, clear E-Stop on Generator. With unit off, press and hold OFF button on the controller for 3 seconds to reset. If error reoccurs contact Technical support.	No
ECM Faulted	Alarm	3000	Alarm SPN/FMI from ECM not supported by controller.	Verify Power Zone 200 FW is at V1.4.1 or higher. Verify ECM FW is at 14605 or higher. Power cycle	No

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
				unit, if error reoccurs call Technical support.	
ECM Faulted	Alarm	3001	The engine is not running at commanded speed.	Verify Power Zone 200 FW is at V1.4.1 or higher. Verify ECM FW is at 14605 or higher. Power cycle unit, if error reoccurs call Technical support.	No
ECM Faulted	Alarm	3002	The ECM has communicated it is set to the wrong fuel type.	Verify Power Zone 200 FW is at V1.4.1 or higher. Verify ECM FW is at 14605 or higher. Power cycle unit, if error reoccurs call Technical support.	No
Engine Throttle Valve 1 Position 1	Alarm	3100	The engine throttle valve data received is valid but above normal operating range; high severe level.	<u>Throttle Position Sensor Test</u>	No
Engine Throttle Valve 1 Position 1	Alarm	3101	The engine throttle valve data received is valid but below normal operating range; high severe level.	<u>Throttle Position Sensor Test</u>	No
Engine Throttle Valve 1 Position 1	Alarm	3103	The engine throttle valve data received is valid but above normal operating range. Throttle Position signal voltage read above 4900 mV.	<u>Throttle Position Sensor Test</u>	No
Engine Throttle Valve 1 Position 1	Alarm	3104	The engine throttle valve data received is valid but below normal operating range. Throttle Position signal voltage read below 250 mV	<u>Throttle Position Sensor Test</u>	No
Engine Throttle Valve 1 Position 1	Alarm	3107	The engine throttle valve mechanical system does not respond as commanded.	<u>Electronic Throttle Control Power Test</u>	No
Engine Throttle Valve 1 Position 1	Warning	3115	The engine throttle valve is receiving voltage but the voltage is above range.	<u>Throttle Position Sensor Test</u>	No
Engine Throttle Valve 1 Position 1	Warning	3117	The engine throttle valve is receiving voltage but the voltage is below range.	<u>Throttle Position Sensor Test</u>	No
Engine Fuel Delivery Pressure	Alarm	3200	The fuel pressure is reading above normal range - high severe level.	<u>Fuel Pressure Sensor Test</u>	No
Engine Fuel Delivery Pressure	Alarm	3201	The fuel pressure is reading below normal range - high severe level.	<u>Fuel Pressure Sensor Test</u>	No
Engine Fuel Delivery Pressure	Alarm	3203	The fuel pressure signal is reading above normal range. Signal is reading above 4900 mV.	<u>Fuel Pressure Sensor Test</u>	No
Engine Fuel Delivery Pressure	Alarm	3204	The fuel pressure signal is reading below normal range. Signal is reading below 120 mV	<u>Fuel Pressure Sensor Test</u>	No
Engine Fuel Delivery Pressure	Warning	3215	The fuel pressure is reading above normal range - least severe level.	<u>Fuel Pressure Sensor Test</u>	No

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
Engine Fuel Delivery Pressure	Warning	3217	The fuel pressure is reading below normal range - least severe level.	Fuel Pressure Sensor Test	No
Engine Oil Pressure	Alarm	3301	The engine oil pressure signal is below normal range - high severe level.	Oil Pressure Switch Test	No
Engine Oil Pressure	Alarm	3304	The engine oil pressure signal is reading above normal range.	Oil Pressure Switch Test	No
Engine Intake Manifold #1 Pressure	Alarm	3400	The engine intake manifold pressure signal is above normal range - high severe level.	Intake Manifold Pressure Sensor Test	No
Engine Intake Manifold #1 Pressure	Alarm	3401	The engine intake manifold pressure signal is below normal range - high severe level.	Intake Manifold Pressure Sensor Test	No
Engine Intake Manifold #1 Pressure	Alarm	3403	The engine intake manifold pressure signal is reading above normal range. Signal voltage is above 4900 mV.	Intake Manifold Pressure Sensor Test	No
Engine Intake Manifold #1 Pressure	Alarm	3404	The engine intake manifold pressure signal is reading below normal range. Signal voltage is below 250 mV.	Intake Manifold Pressure Sensor Test	No
Engine Intake Manifold #1 Pressure	Warning	3415	The engine intake manifold pressure signal is reading above normal range - least severe level.	Intake Manifold Pressure Sensor Test	No
Engine Intake Manifold #1 Pressure	Warning	3417	The engine intake manifold pressure signal is reading below normal range - least severe level.	Intake Manifold Pressure Sensor Test	No
Engine Intake Manifold 1 Temperature	Alarm	3500	The engine intake manifold temperature reading is valid but above normal operating range; high severe level.	Intake Manifold Temperature Sensor Test	Yes
Engine Intake Manifold 1 Temperature	Alarm	3501	The engine intake manifold temperature reading is valid but below normal operating range; high severe level.	Intake Manifold Temperature Sensor Test	Yes
Engine Intake Manifold 1 Temperature	Alarm	3503	The engine intake manifold temperature is reading below normal range. Resistance on signal is reading below 100 ohms.	Intake Manifold Temperature Sensor Test	Yes
Engine Intake Manifold 1 Temperature	Alarm	3504	The engine intake manifold temperature is reading above normal range. Resistance on signal is reading OL or above 65 kohms.	Intake Manifold Temperature Sensor Test	Yes
Engine Intake Manifold 1 Temperature	Warning	3515	The engine intake manifold temperature reading is valid but above normal operating range; least severe level.	Intake Manifold Temperature Sensor Test	No
Engine Intake Manifold 1 Temperature	Warning	3517	The engine intake manifold temperature reading is valid but below normal operating range; least severe level.	Intake Manifold Temperature Sensor Test	No

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
Battery Potential / Power Input 1	Alarm	3803	The battery voltage received is above the normal range, high severe level. Battery is reading voltage above 16 volts.	Battery Potential/Power Output 1 Test	No
Battery Potential / Power Input 1	Alarm	3804	The battery voltage received is below the normal range, high severe level. Battery is reading voltage below 8.5 volts.	Battery Potential/Power Output 1 Test	No
Battery Potential / Power Input 1	Warning	3815	The battery voltage received is above the normal range, least severe level. Battery is reading voltage above 15 volts.	Battery Potential/Power Output 1 Test	No
Battery Potential / Power Input 1	Warning	3817	The battery voltage received is below the normal range, least severe level. Battery is reading voltage below 10 volts.	Battery Potential/Power Output 1 Test	No
Engine Position Sensor	Alarm	4101	The crankshaft sensor signal is reading below the normal range, high severe level.	Engine Timing Sensor Test	No
Secondary Sync	Alarm	4201	The engine timing sensor signal is reading below the normal range, high severe level.	Engine Timing Sensor Test	No
O2 Sensor	Alarm	4300	The O2 sensor data received is valid but above normal operating range; high severe level.	Oxygen Sensor Test	No
O2 Sensor	Alarm	4301	The O2 sensor data received is valid but below normal operating range; high severe level.	Oxygen Sensor Test	No
O2 Sensor	Alarm	4303	The O2 sensor is reading voltage above normal range, may be shorted to high voltage source.	Oxygen Sensor Test	No
O2 Sensor	Alarm	4304	The O2 sensor is reading voltage below normal range.	Oxygen Sensor Test	No
O2 Sensor	Alarm	4307	Engine has run for 5 minutes without being able to run at the required air/fuel ratio.	Engine Cranks But Does Not Run, Lacks Power, Runs Rough, Hunts or Backfires	No
O2 Sensor	Warning	4315	The O2 sensor data received is valid but above normal operating range; least severe level.	Oxygen Sensor Test	No
O2 Sensor	Warning	4317	The O2 sensor data received is valid but below normal operating range; least severe level.	Oxygen Sensor Test	No
Engine Ignition Coil #1	Alarm	4503	The engine ignition coil 1 voltage data received is valid but above normal range.	Ignition Coil Test	No
Engine Ignition Coil #1	Alarm	4504	The engine ignition coil 1 voltage data received is valid but below normal range.	Ignition Coil Test	No

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
Engine Ignition Coil #1	Alarm	4507	The engine ignition coil 1 mechanical system is not responding correctly.	<u>Ignition Coil Test</u>	No
Engine Ignition Coil #2	Alarm	4603	The engine ignition coil 2 voltage data received is valid but above normal range.	<u>Ignition Coil Test</u>	No
Engine Ignition Coil #2	Alarm	4604	The engine ignition coil 2 voltage data received is valid but below normal range.	<u>Ignition Coil Test</u>	No
Engine Ignition Coil #2	Alarm	4607	The engine ignition coil 2 mechanical system is not responding correctly.	<u>Ignition Coil Test</u>	No
Engine Fuel Valve 1 Position	Alarm	4900	The fuel mixer position data received is valid but above normal operating range; high severe level.	<u>Fuel Mixer Valve Test</u>	No
Engine Fuel Valve 1 Position	Alarm	4901	The fuel mixer position data received is valid but below normal operating range; high severe level.	<u>Fuel Mixer Valve Test</u>	No
Engine Fuel Valve 1 Position	Alarm	4903	The fuel mixer position is reading voltage above normal range. Fuel mixer position signal is reading above 4900 mV.	<u>Fuel Mixer Valve Test</u>	No
Engine Fuel Valve 1 Position	Alarm	4904	The fuel mixer position is reading voltage below normal range. Fuel mixer position signal is reading below 120 mV.	<u>Fuel Mixer Valve Test</u>	No
Engine Fuel Valve 1 Position	Alarm	4907	The fuel mixer mechanical system is not responding as commanded.	<u>Mixer Valve Power Test</u>	No
Engine Fuel Valve 1 Position	Warning	4915	The fuel mixer position data received is valid but above normal operating range; least severe level.	<u>Fuel Mixer Valve Test</u>	No
Engine Fuel Valve 1 Position	Warning	4917	The fuel mixer position data received is valid but below normal operating range; least severe level.	<u>Fuel Mixer Valve Test</u>	No
Engine Speed	Alarm	5100	The engine speed data received is valid but above normal operating range; high severe level.	<u>Engine Speed Test</u>	No
Engine Speed	Alarm	5101	The engine speed data received is valid but below normal operating range; high severe level.	<u>Engine Speed Test</u>	No
Engine Speed	Warning	5115	The engine speed data received is valid but above normal operating range; least severe level.	<u>Engine Speed Test</u>	No
Engine Speed	Warning	5117	The engine speed data received is valid but below normal operating range; least severe level.	<u>Engine Speed Test</u>	No
Cylinder Head Temperature 1	Alarm	5200	The cylinder head temperature reading is valid but above normal operating range; high severe level.	<u>Cylinder Head Temperature Sensor Test</u>	No

Displayed Alarm	Alarm/ Warning	E-Code	Condition Description	Action Step	Remote Reset
Cylinder Head Temperature 1	Alarm	5201	The cylinder head temperature reading is valid but below normal operating range; high severe level.	<u>Cylinder Head Temperature Sensor Test</u>	No
Cylinder Head Temperature 1	Alarm	5203	The cylinder head temperature is reading resistance above normal range. Resistance on signal is reading above 1.8 M ohms.	<u>Cylinder Head Temperature Sensor Test</u>	No
Cylinder Head Temperature 1	Alarm	5204	The cylinder head temperature is reading resistance below normal range. Resistance on signal is reading below 15 ohms.	<u>Cylinder Head Temperature Sensor Test</u>	No
Cylinder Head Temperature 1	Warning	5215	The cylinder head temperature reading is valid but above normal operating range; least severe level.	<u>Cylinder Head Temperature Sensor Test</u>	No
Cylinder Head Temperature 1	Warning	5217	The cylinder head temperature reading is valid but below normal operating range; least severe level.	<u>Cylinder Head Temperature Sensor Test</u>	No
Engine Fuel Shutoff 2 Control	Alarm	5303	The voltage for fuel shutoff signal is above nominal.	<u>Fuel Shutoff 2 (SOV2) Control Test</u>	No
Engine Fuel Shutoff 2 Control	Alarm	5304	The fuel shutoff signal is below nominal voltage. Short reading is detected for over 10 seconds or in 6 occurrences.	<u>Fuel Shutoff 2 (SOV2) Control Test</u>	No
Engine Oil Level Sensor	Warning	5415	Oil level sensor signal is out of range high.	<u>Oil Level Sensor Test</u>	No
Engine Oil Level Sensor	Warning	5417	Oil level sensor signal is out of range low.	<u>Oil Level Sensor Test</u>	No
Overload Cooldown	Warning	7000	Controller has detected 5 overload shutdowns in succession. It will run for 5 minutes with the AVR turned off allowing the engine to cool. After the 5 minutes expire the warning will turn off and the AVR will enable thus powering the load once again.	<u>Overload Condition Test</u>	No

across the stationary stator windings, the greater the voltage induced into the stator.

Initially, the AC power windings output voltage sensed by the AVR is low. The AVR reacts by increasing the excitation voltage (and hence current flow) to the rotor until AC output voltage increases to a preset level. The AVR then maintains the voltage at this level. For example, if voltage exceeds the desired level, the AVR will decrease excitation. Conversely, if voltage drops below the desired level, the AVR responds by increasing excitation.

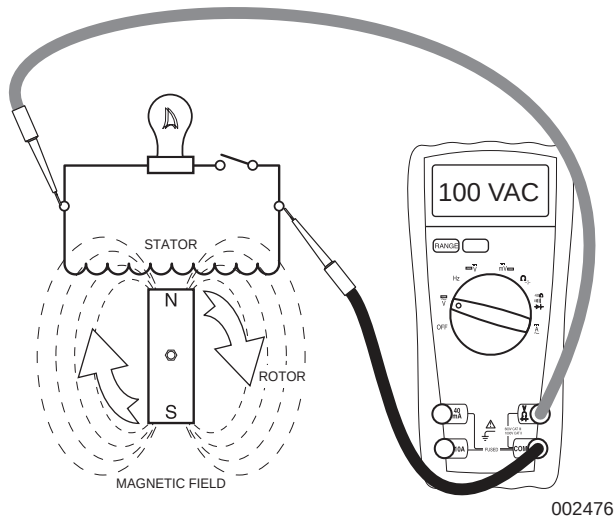


Figure 6-2. Low Field Excitation Voltage = Low Magnetic Lines of Flux = Low AC Output

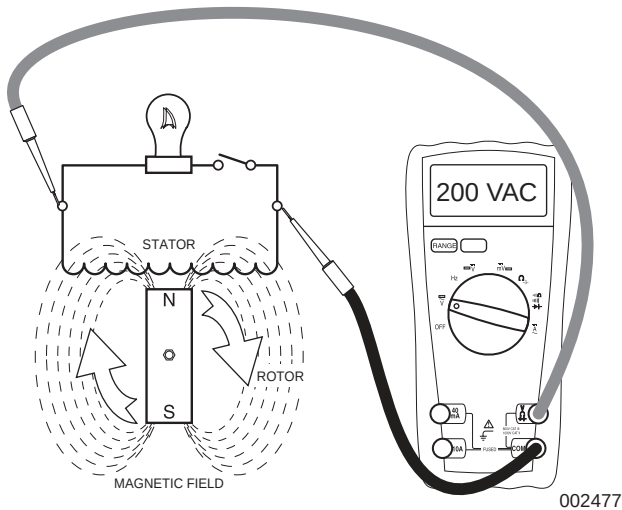


Figure 6-3. Increased Field Excitation Voltage = Increased Magnetic Lines of Flux = Increased AC Output Voltage

AC Power Winding Output

When electrical loads are connected across the AC power windings to complete the circuit, current flows through the circuit powering the loads.

Load changes will result in a corresponding change in voltage. As load demand increases the voltage will decrease. As load demand decreases the voltage will increase. The AVR changes excitation to provide a constant output voltage with minimal increase or decrease during load changes. Frequency is also affected during load changes. However, frequency is a function of rotor speed which is directly coupled to the engine. The ECM controls multiple engine functions to compensate for any engine speed changes to maintain a stable, isochronous, frequency output based on the specifications of the unit.

The automatic voltage regulator and the ECM work together to provide output voltage regulation of $\pm 1\%$ voltage regulation and $+ 0.25\%$ steady state, isochronous, frequency (speed) regulation.

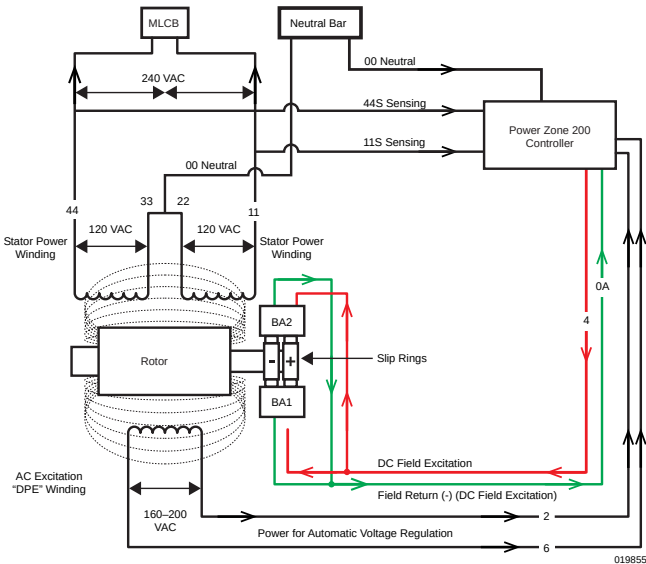
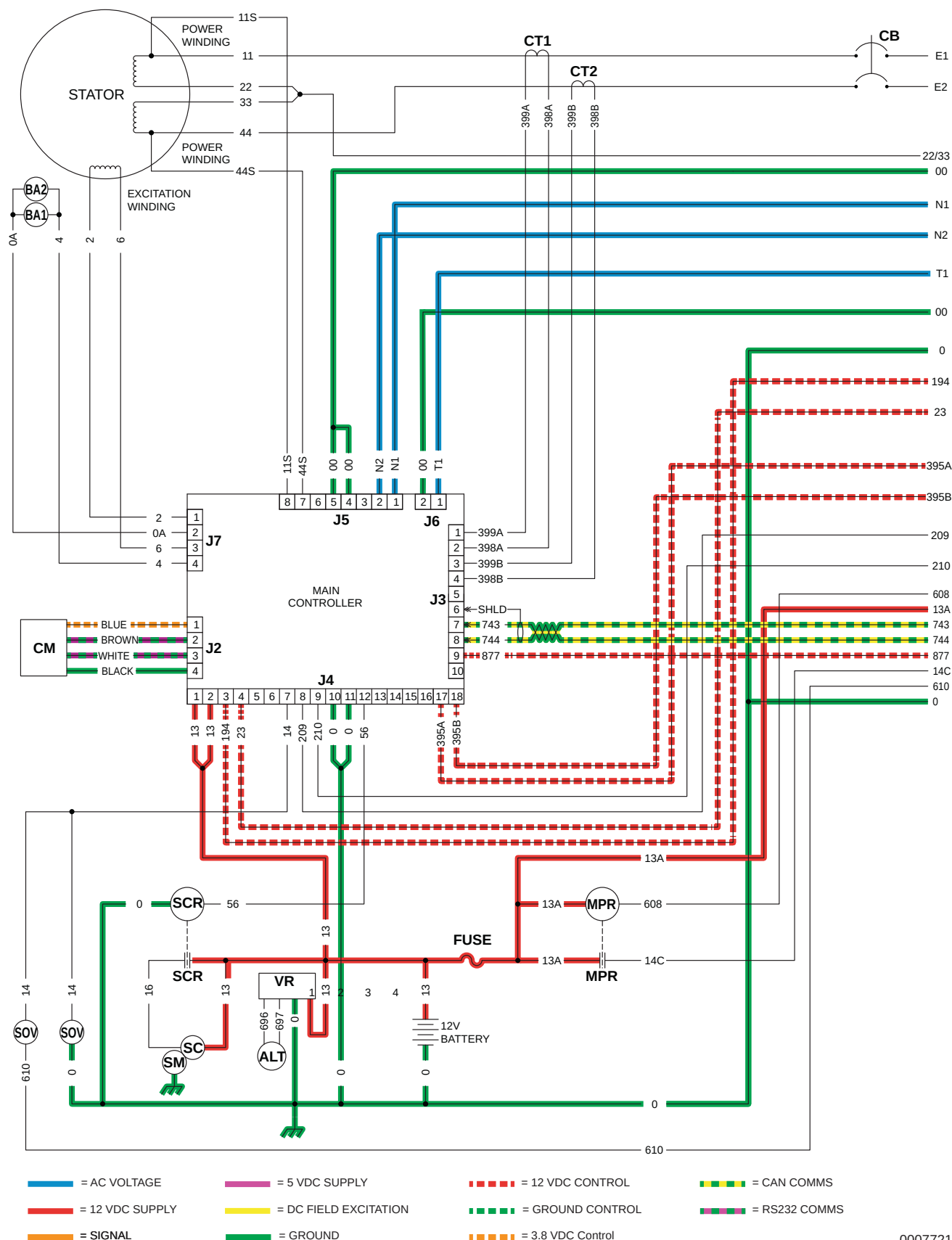


Figure 6-4. Voltage Regulator Schematic

AVR = Automatic Voltage Regulator
DPE = Displaced Phase Excitation
VAC = Volts Alternating Current
MLCB = Main Line Circuit Breaker

Utility Source Voltage Available



00077213

Figure 6-5. Utility Source Voltage Available

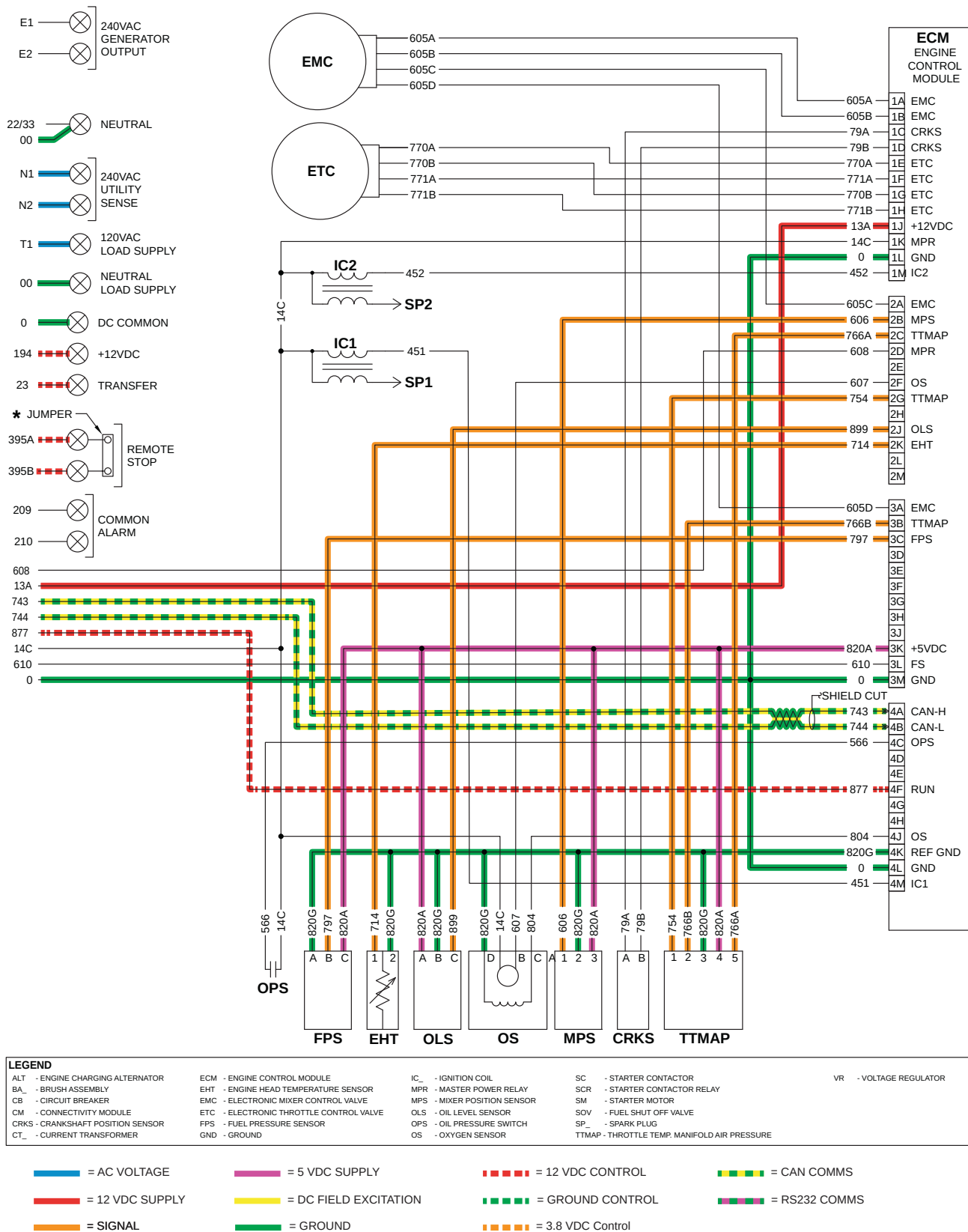
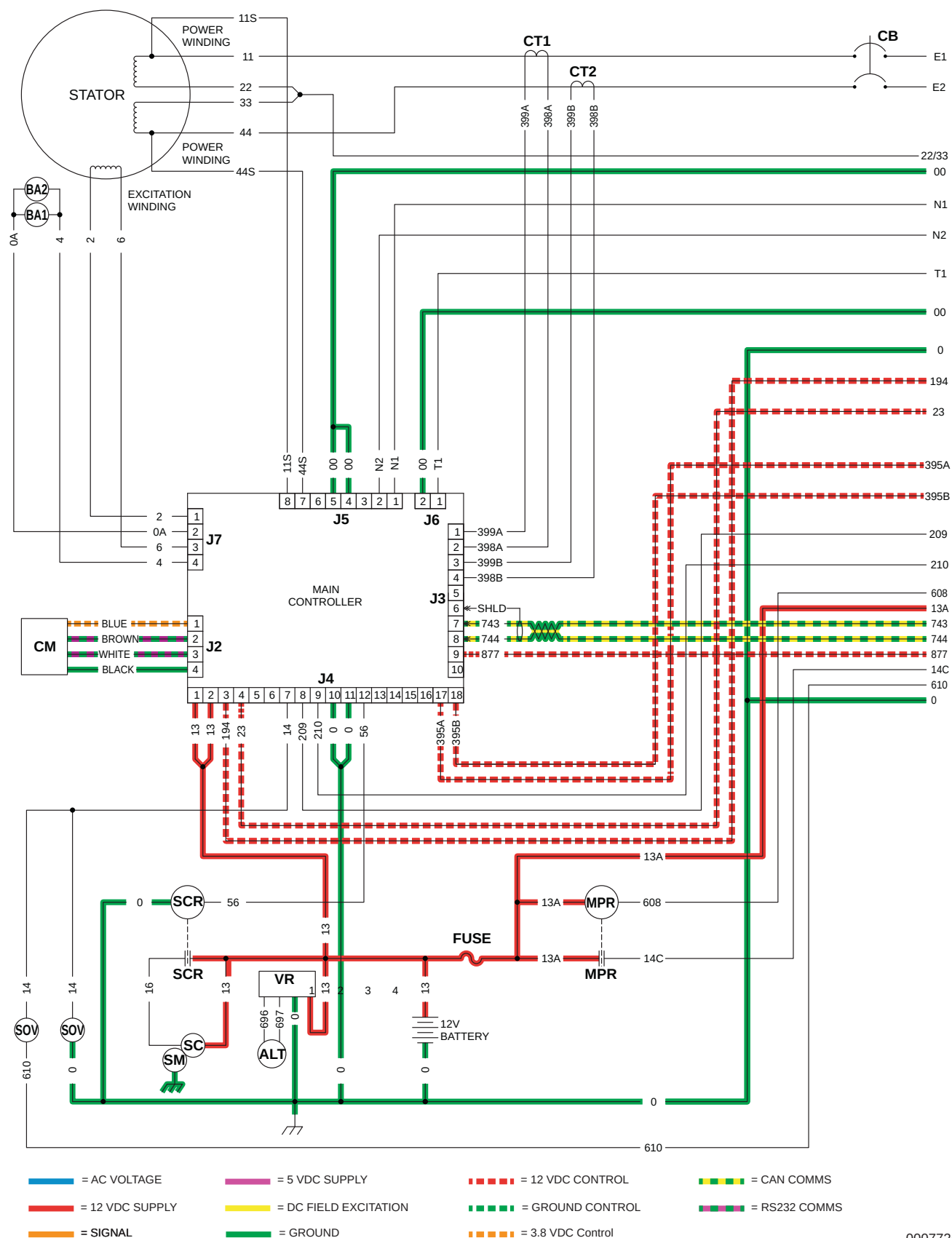


Figure 6-6. Utility Source Voltage Available

00077214

Initial Dropout of Utility Source Voltage



00077253

Figure 6-7. Initial Dropout of Utility Source Voltage

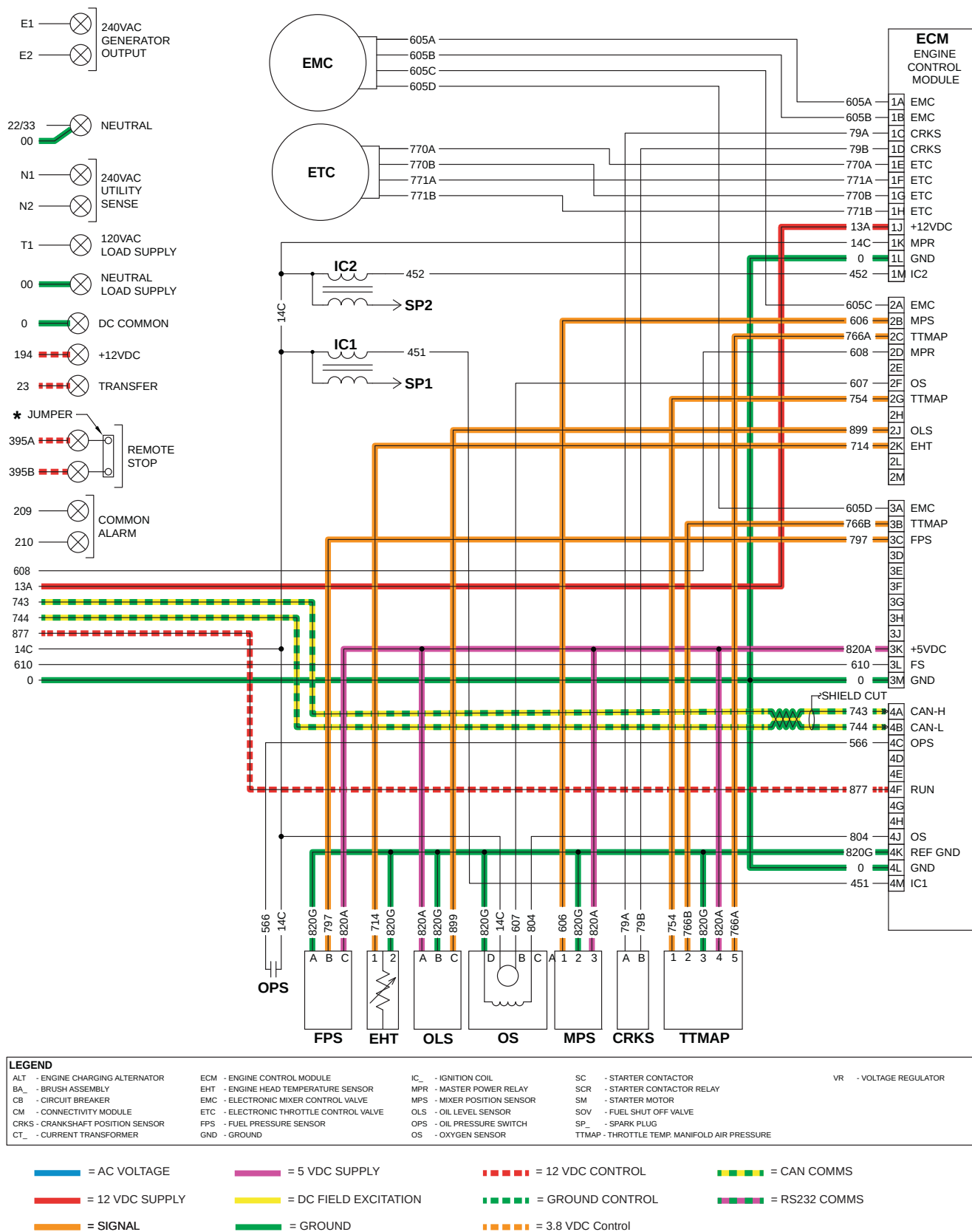
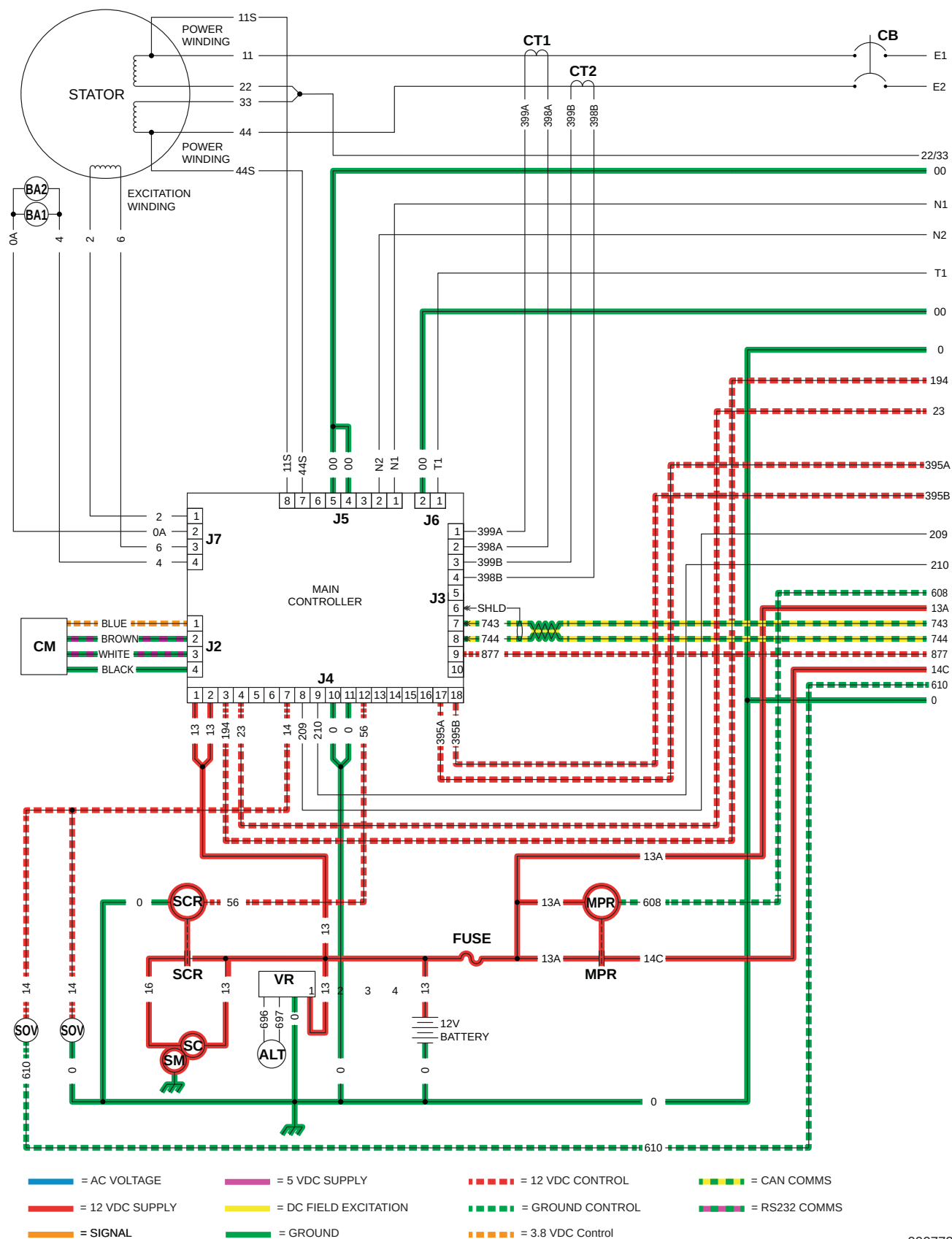


Figure 6-8. Initial Dropout of Utility Source Voltage

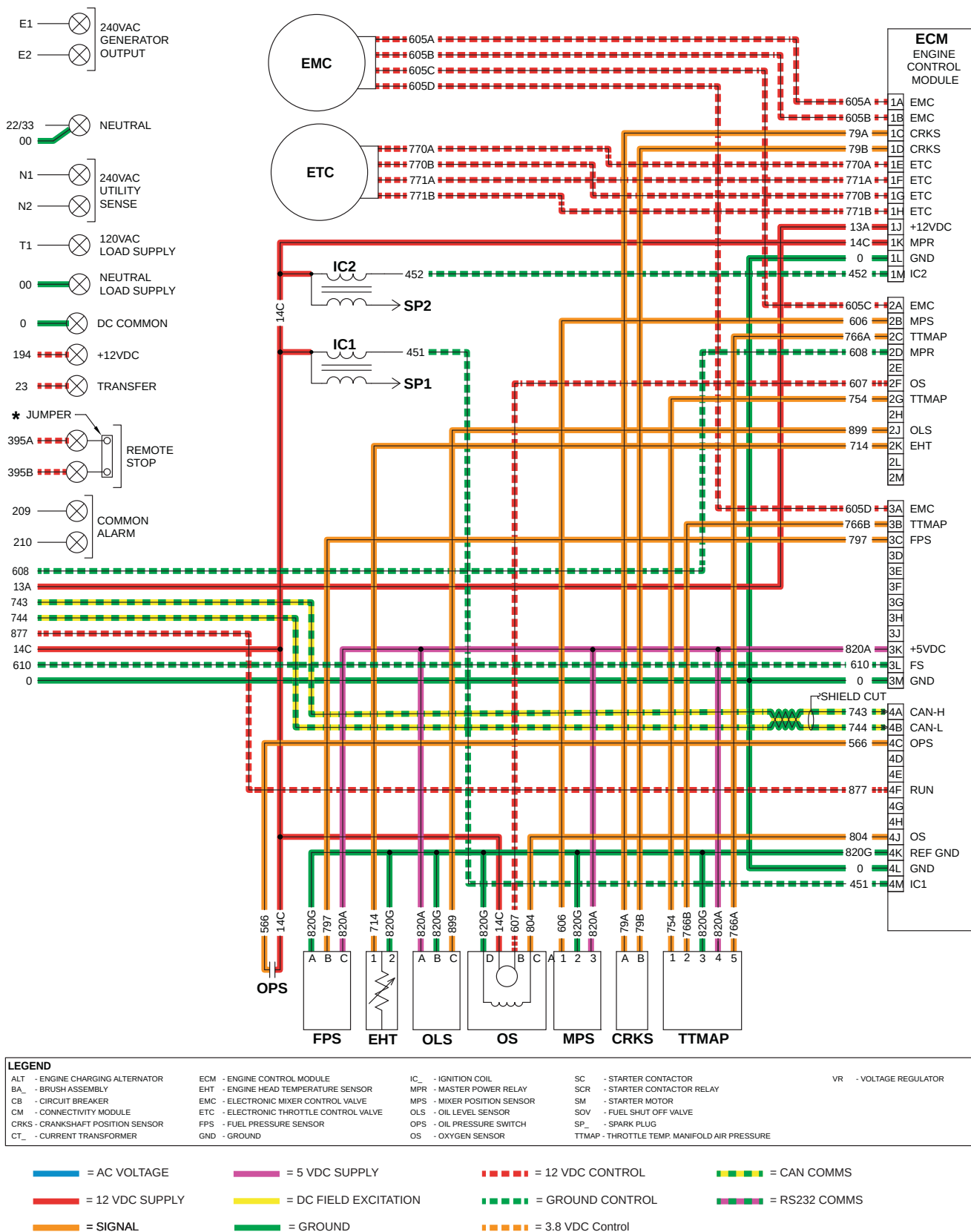
00077252

Utility Voltage Failure and Engine Cranking



00077251

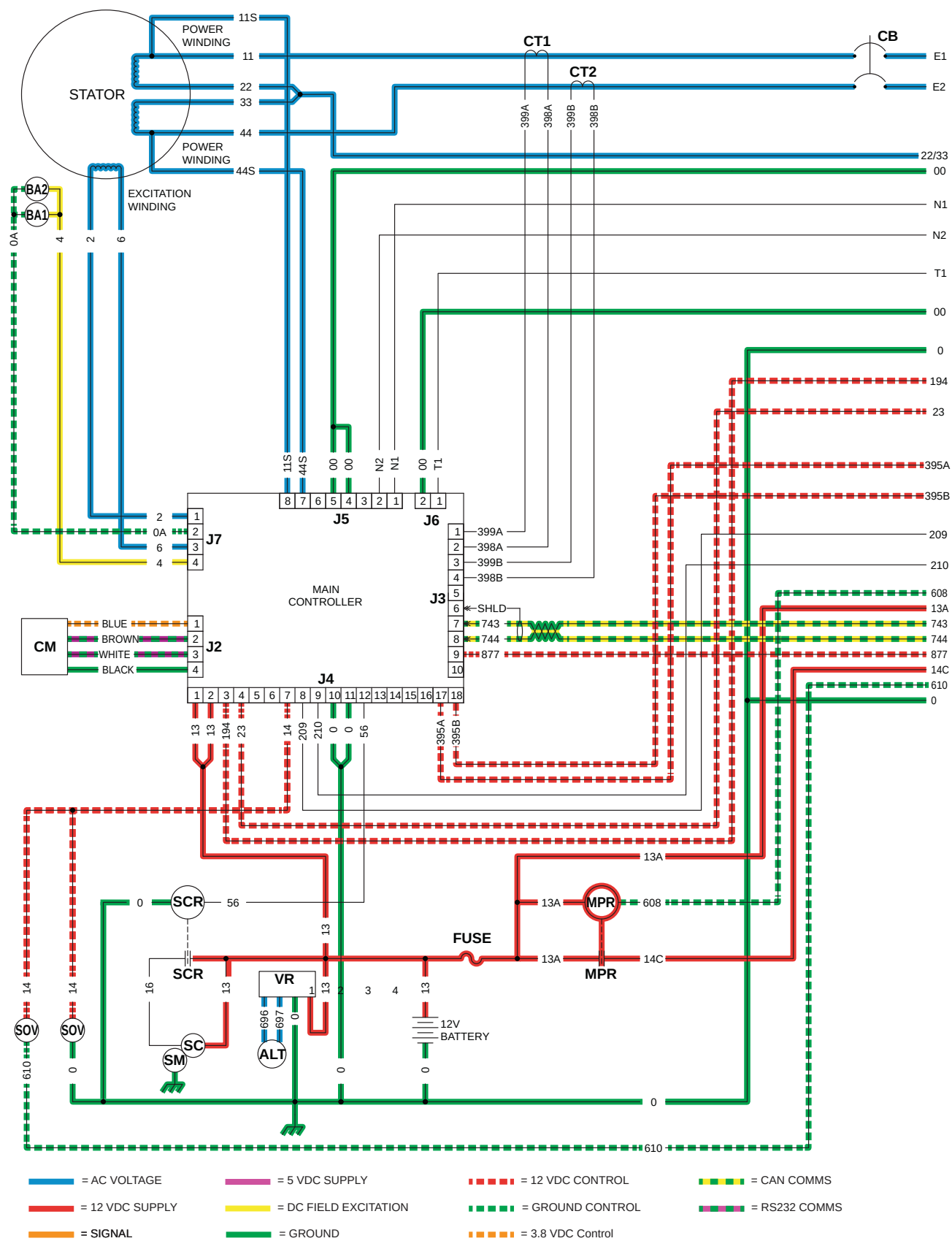
Figure 6-9. Utility Voltage Failure and Engine Cranking



00077250

Figure 6-10. Utility Voltage Failure and Engine Cranking

Engine Startup and Running



00077217

Figure 6-11. Engine Startup and Running

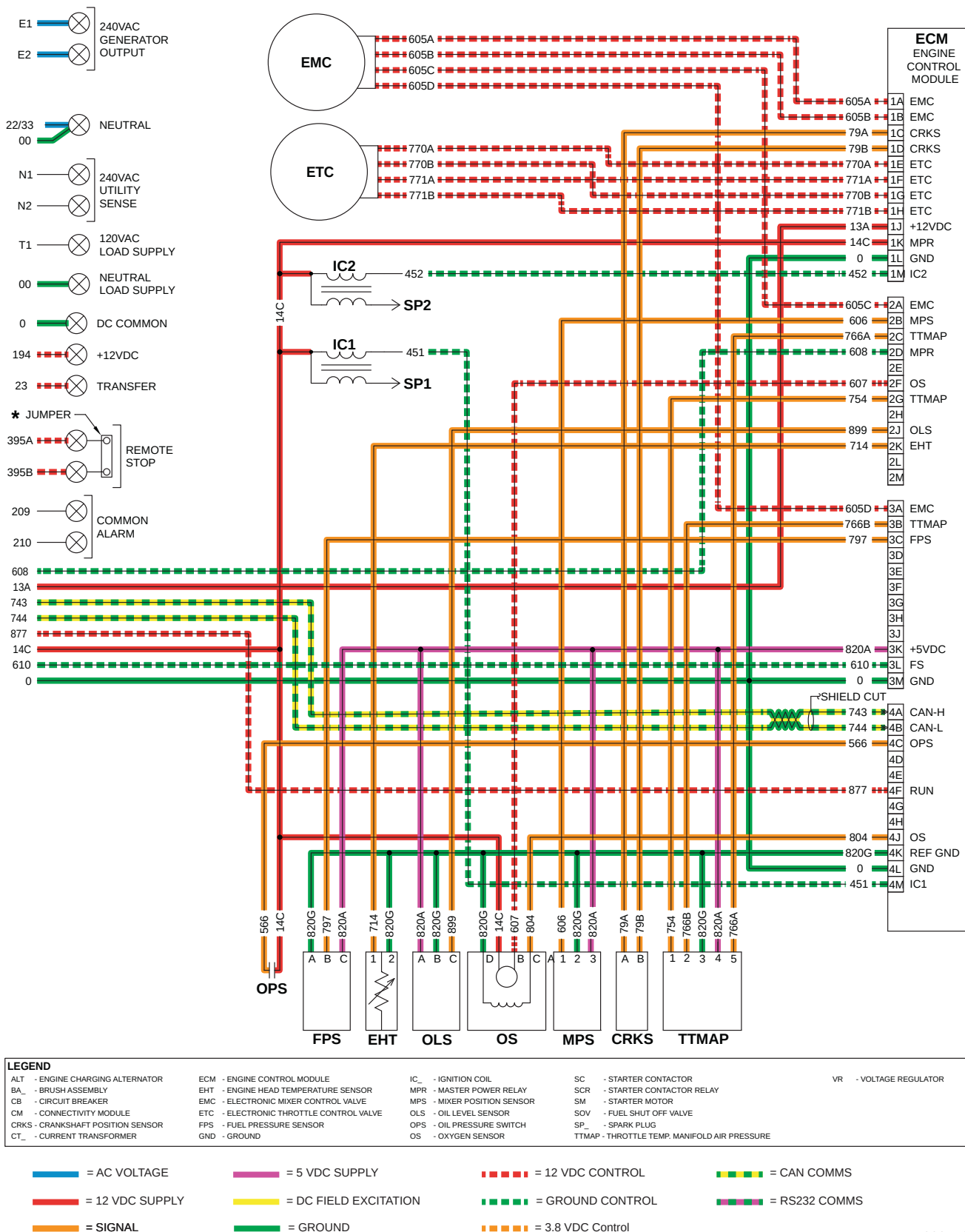


Figure 6-12. Engine Startup and Running

00077249

Transfer to Standby

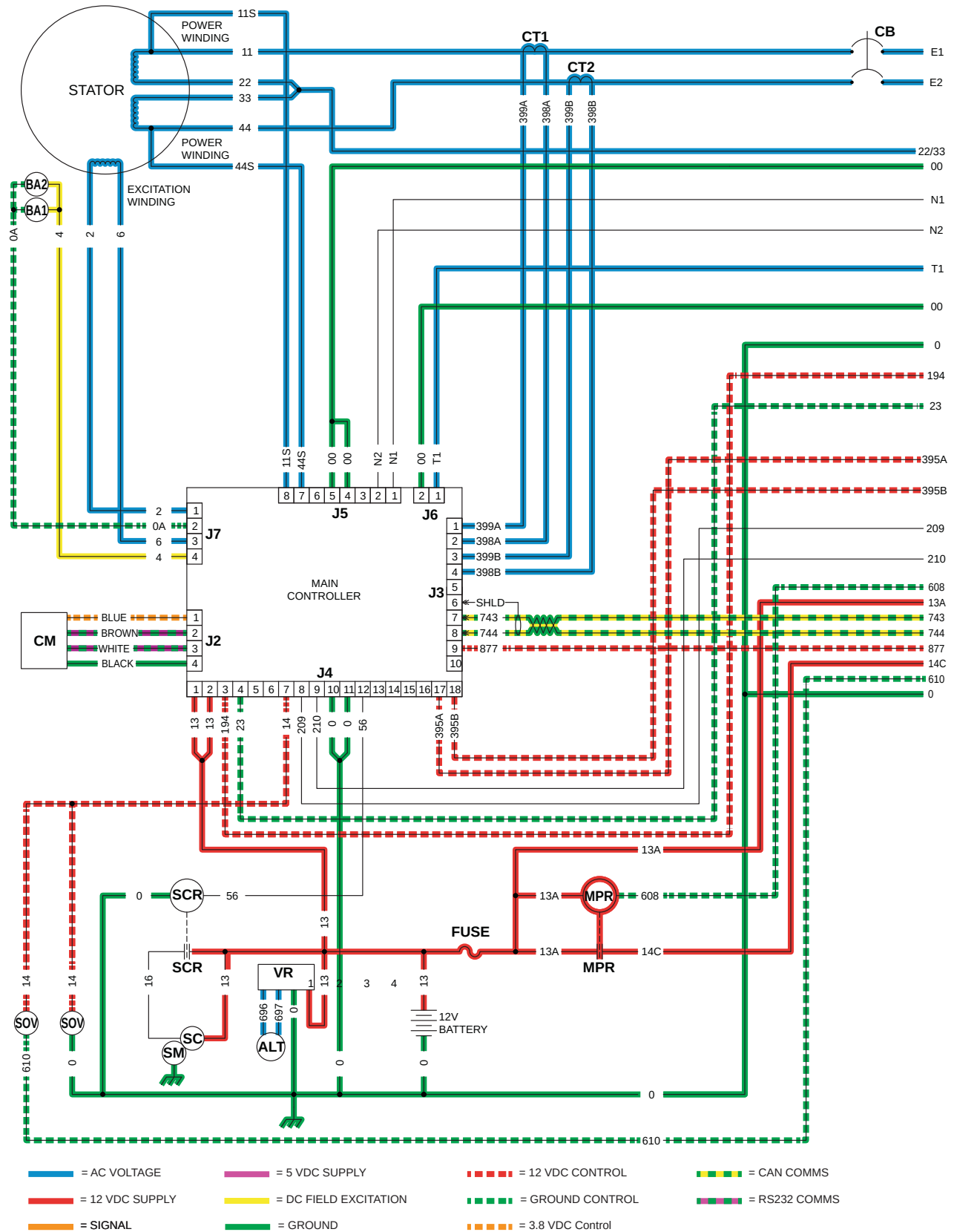


Figure 6-13. Transfer to Standby

00077248

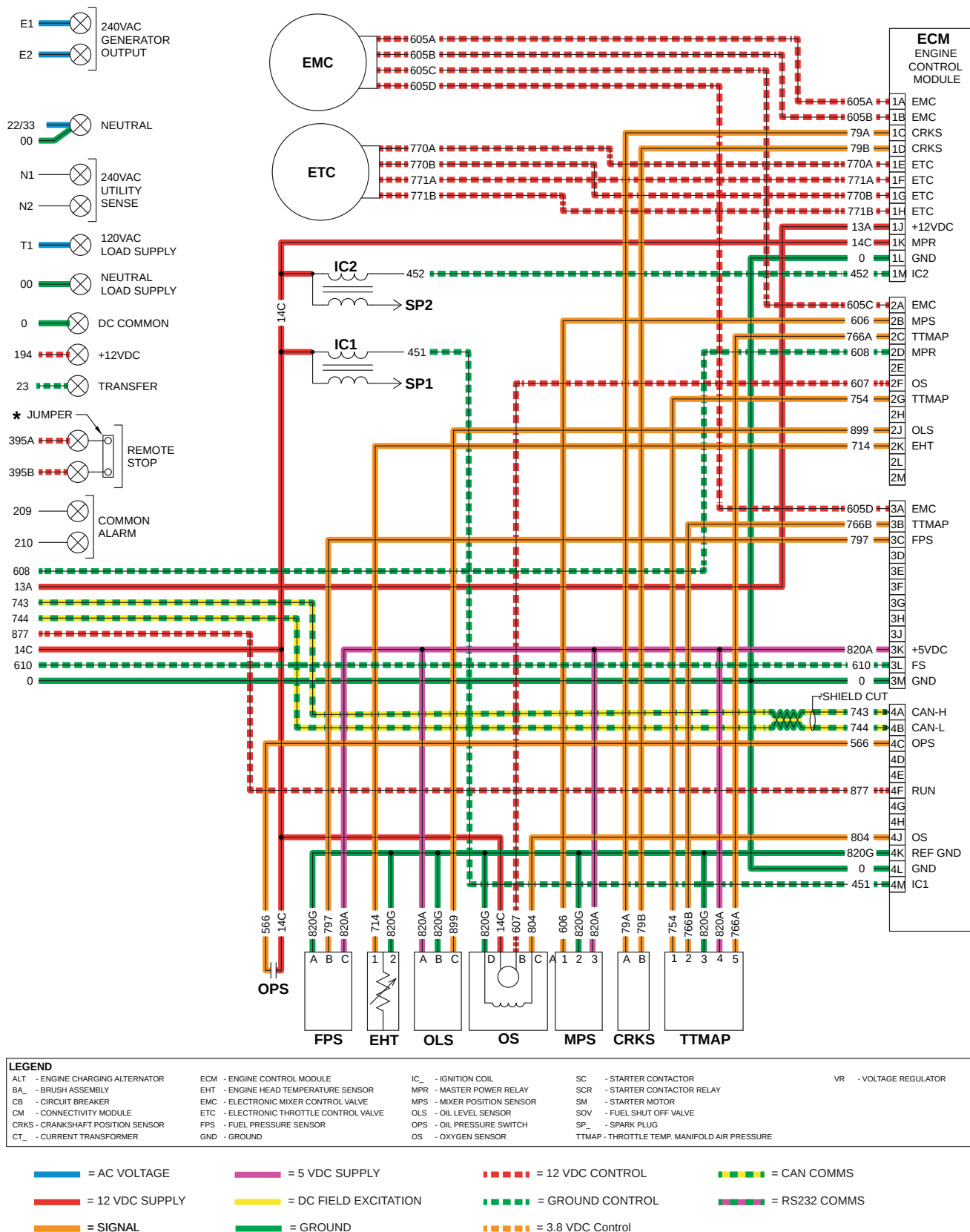
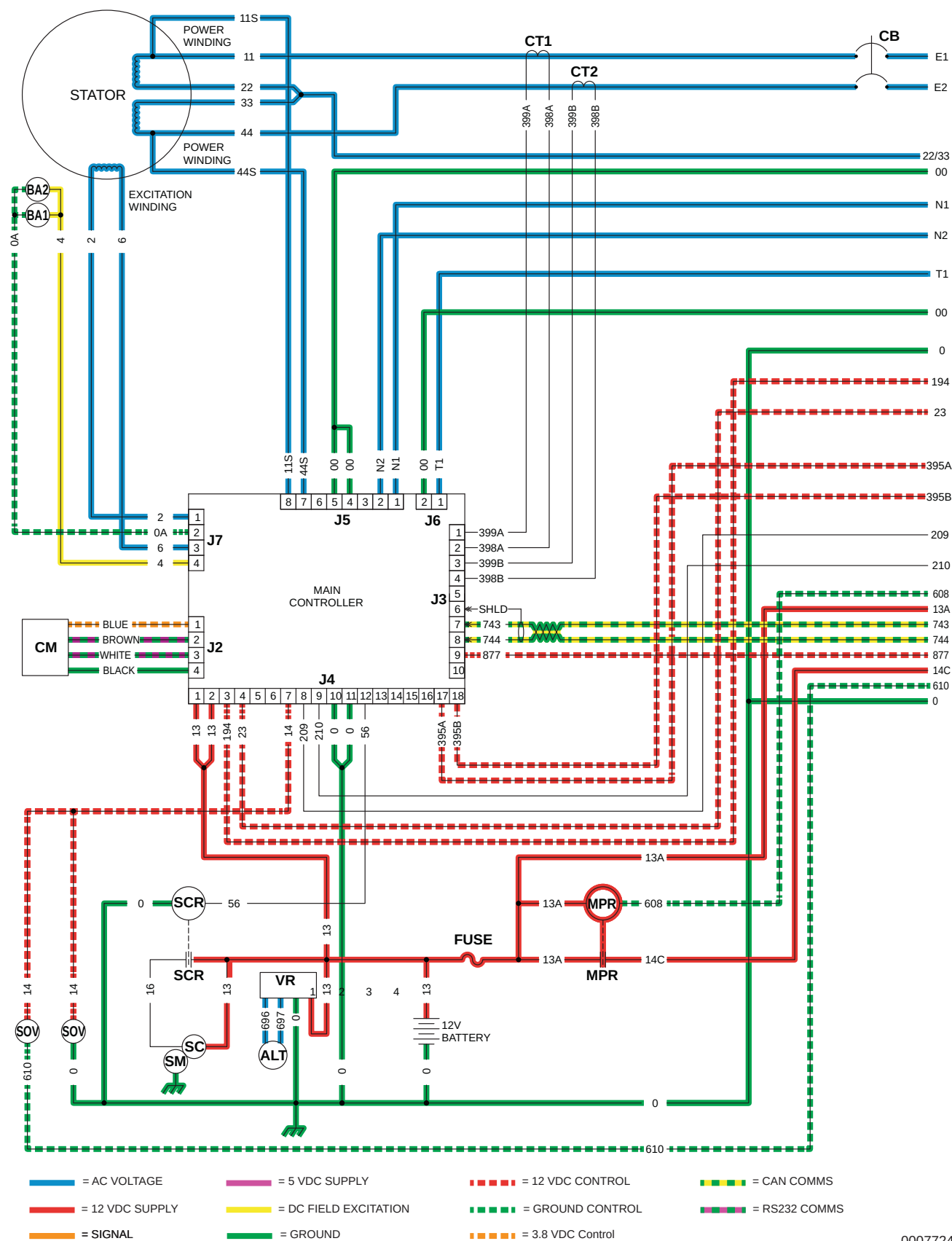


Figure 6-14. Transfer to Standby

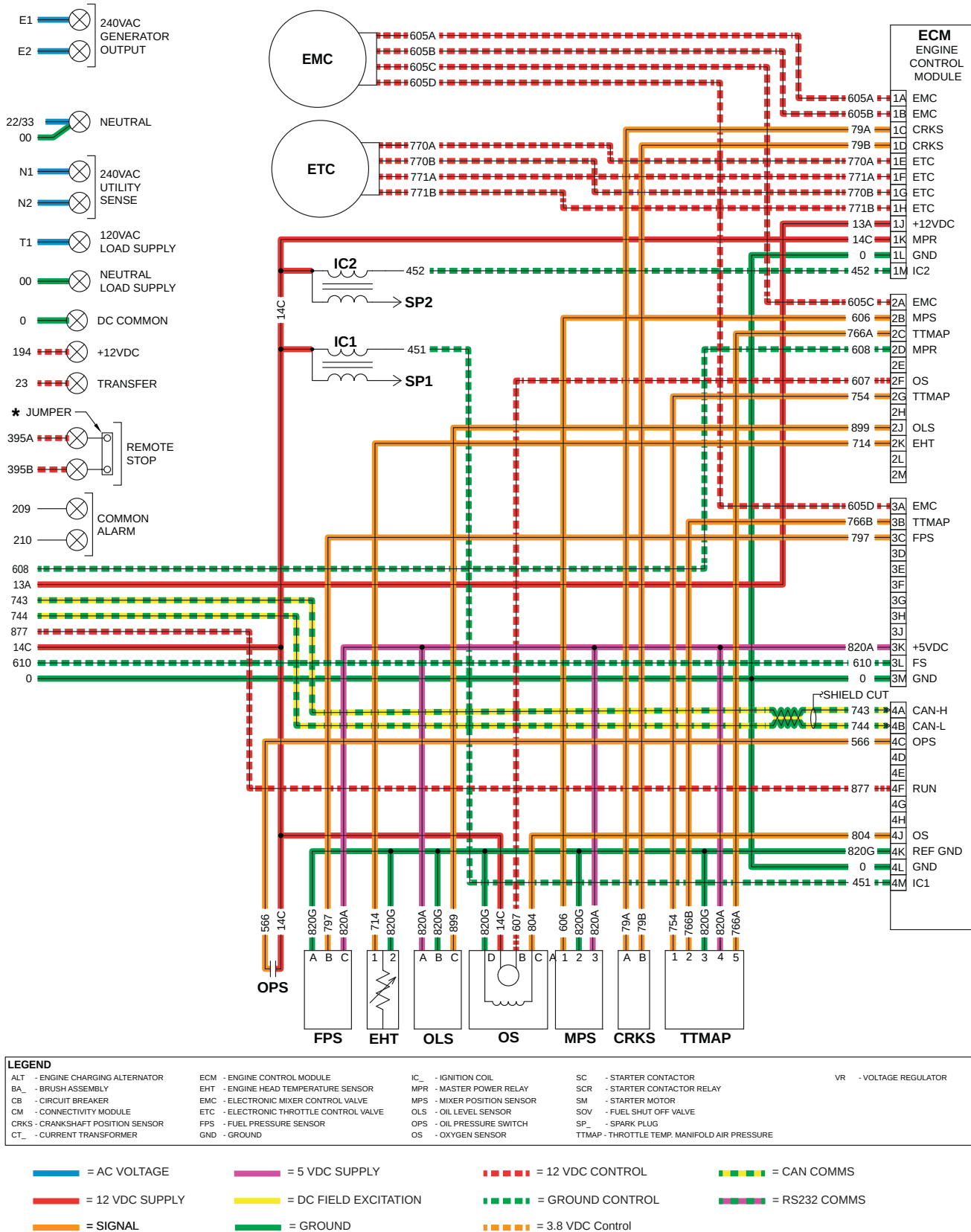
00077247

Utility Voltage Restored and Re-transfer to Utility



00077246

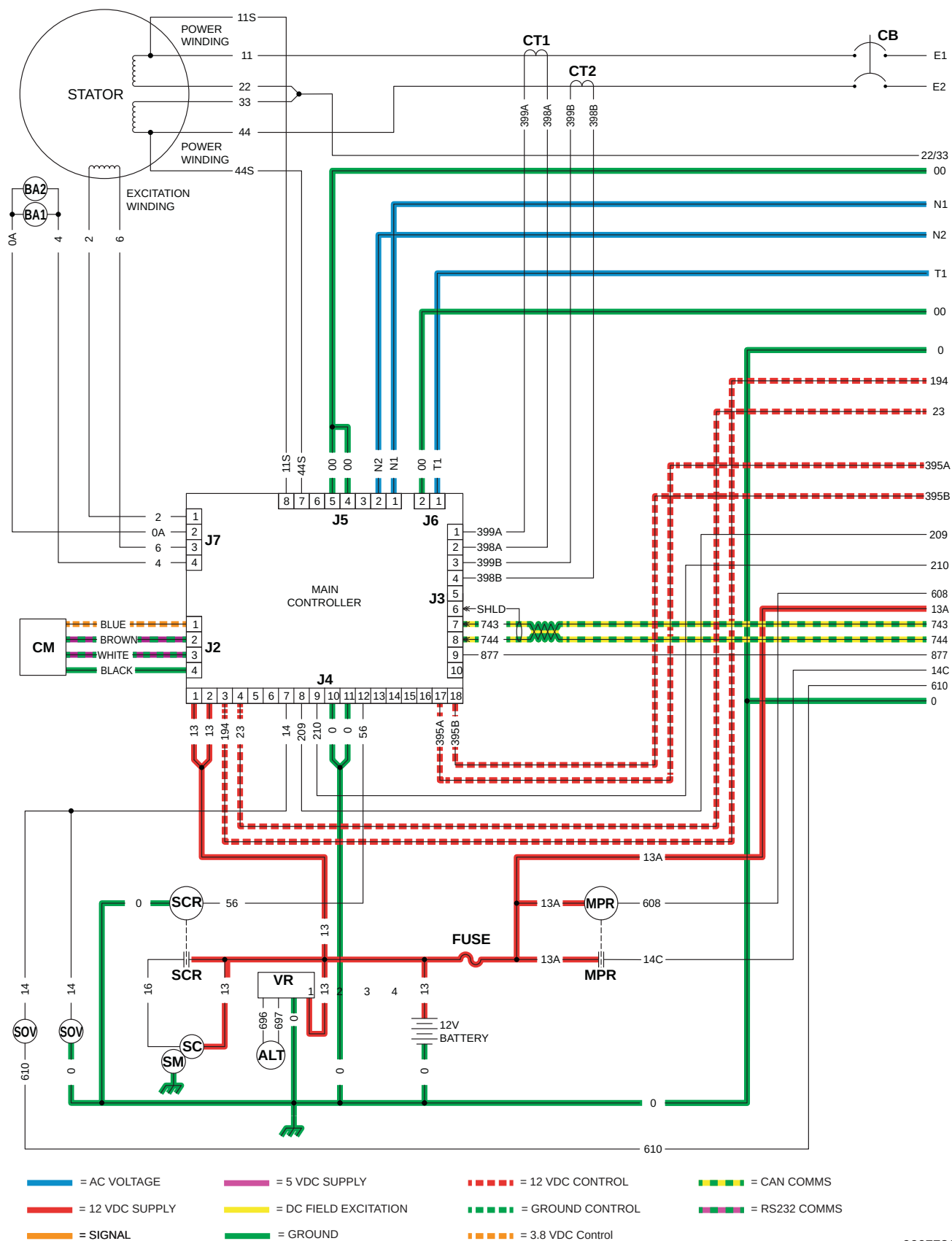
Figure 6-15. Utility Voltage Restored and Re-transfer to Utility



00077245

Figure 6-16. Utility Voltage Restored and Re-transfer to Utility

Engine Shutdown



00077215

Figure 6-17. Engine Shutdown

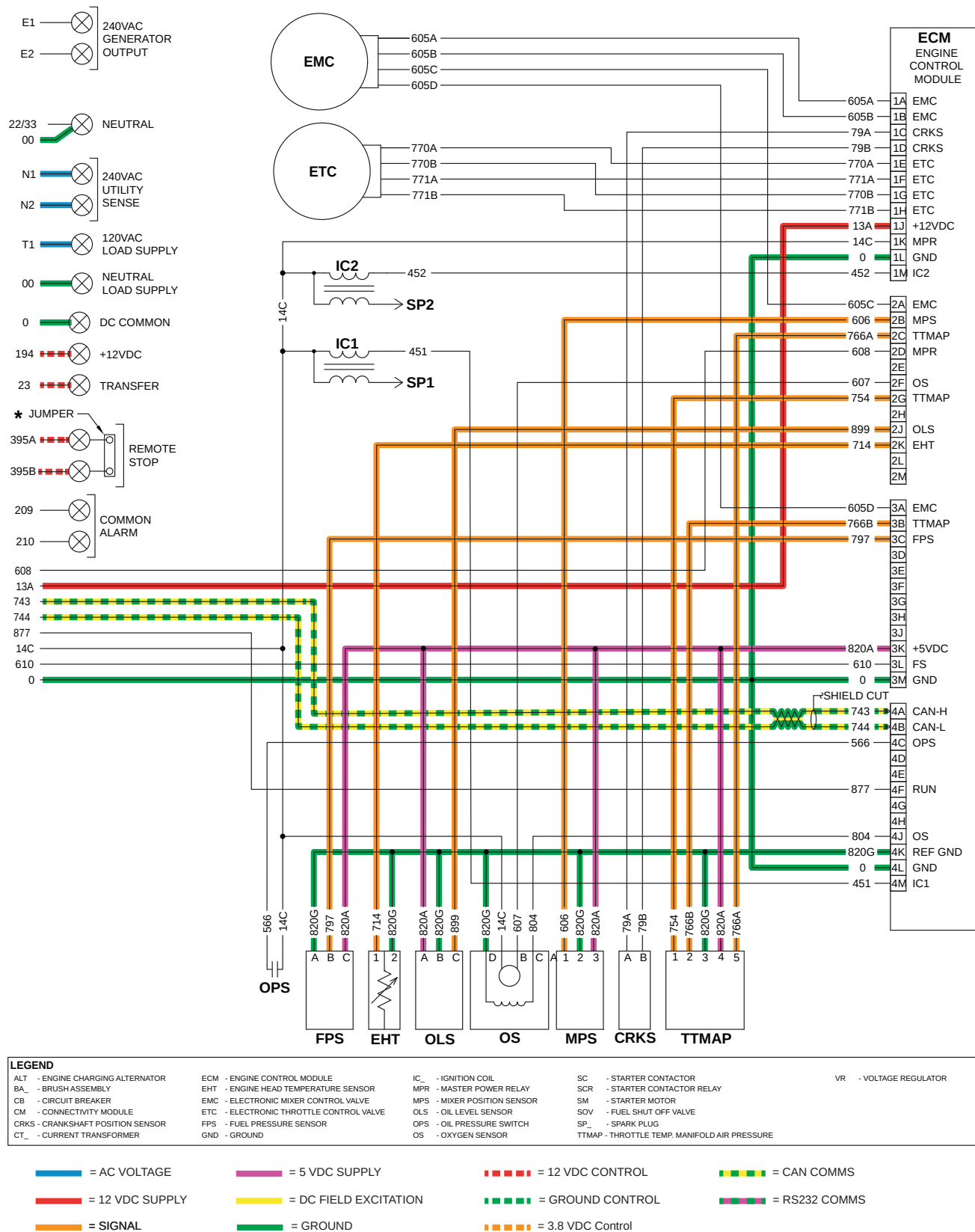


Figure 6-18. Engine Shutdown

00077216

Utility Source Voltage Available

See [Figure 6-5](#) and [Figure 6-6](#). The circuit condition with the controller set to AUTO and with Utility source power available can be briefly described as follows:

- Utility source voltage is available to the transfer switch Terminal Lugs N1, N2, and Neutral. The CONTACTOR is in the "Utility" position.
- Utility voltage is available to the controller via Wire N1, N2 and neutral (00).
- Load voltage (120 VAC) is available to the controller via Wire T1 and 00 (Neutral) for Battery Charger. (220-240 VAC on T1 and T2 50 Hz only)
- The controller is shown in the AUTO mode. Battery voltage is available to the circuit board via Wire 13, the 7.5 amp fuse (F1). Wire 194 provides 12 VDC to the transfer relay in the transfer switch.
- The quad light annunciator is part of the controller and is visible on the side of the controller.
- The ECM is monitoring the Battery V, FPS, EHT, OLS, MPS, TTMAP, for fault conditions.

Initial Dropout of Utility Source Voltage

See [Figure 6-7](#) and [Figure 6-8](#). Should a Utility power failure occur, circuit condition may be briefly described as follows:

- The controller continually monitors for acceptable Utility voltage via N1, N2 and neutral (00). Should Utility voltage drop below approximately 77% (adjustable, 75–80% see chart) of the nominal source voltage, a programmable timer on the controller will turn on.
- The 5-second timer (factory default, adjustable from 2-1500 seconds) is still timing and engine cranking has not yet begun.

Utility Dropout	
Factory Default	Util Volts Low Value
60 Hz = 186 VAC	60 Hz = 180-192 VAC

NOTE: Dropout / pickup thresholds and associated timers can be configured in the Field Pro app during the guided install flow or the "Configurations" page available from the Generator Details dashboard.

Utility Voltage Failure and Engine Cranking

See [Figure 6-9](#) and [Figure 6-10](#).

- After the controller's adjustable programmed timer has timed out, if Utility voltage is still below the programmed utility drop out level, the controller's logic will energize the internal crank relay followed by the internal run relay.
- When the internal crank relay energizes, 12 VDC is delivered to the starter contactor relay (SCR) via Wire 56. When the SCR energizes, its contacts close and battery voltage is delivered to a starter contactor (SC). When the SC energizes, its contacts close and battery voltage is delivered to the starter motor (SM); the engine is now cranking.
- With the engine cranking, oil pressure will begin to build in the engine, activating the contact in the oil pressure switch (OPS).

10 kW with 459cc Engine	14-28 kW with 817 / 997cc Engine
5 psi Normally Closed (NC) contact will open	10 psi Normally Open (NO) contact will close

- The ECM will open the throttle and mixer valves.
- With the engine cranking, a pulsing AC speed reference signal is generated by the CRKS and is delivered to the ECM. If a valid signal is received the ECM will begin firing the ignition coils. The ECM and controller will open their respective fuel valve to allow fuel to flow to the engine.

NOTE: If the controller does not see a RPM signal, it will not energize Wire 14.

- With ignition and fuel flow available the engine RPM will begin to increase.

- The ECM is monitoring the Battery V, OPS FPS, EHT, OLS, OS, MPS, TTMAP, IC1, IC2 (if V-twin), and SOV2 for fault conditions and adjusting engine parameters for optimal engine performance.
- The ECM is grounding the MPR via Wire 610 which provides power for the ignition coils and the EMC and ETC.

Engine Startup and Running

See [Figure 6-11](#) and [Figure 6-12](#). With the fuel solenoid open and ignition occurring, the engine starts. Engine startup and running may be briefly described as follows:

- The CRKS delivers a speed signal to the ECM. The ECM sends the RPM data to the controller over the CAN communication signal. Once the controller determines that the engine is running, the controller:
 - terminates cranking by de-energizing Wire 56 at approximately 2160 RPM.
 - energizes a field flash relay in the controller at 3400 RPM which delivers 12 VDC internally to the controller to the Wire 4 output. The field boost will continue for a pre-determined time, or until field boost parameters are achieved, whichever occurs first.
- NOTE:** Dropout / pickup thresholds and associated timers can be configured in the Field Pro app during the guided install flow or the "Configurations" page available from the Generator Details dashboard.
- also at 3400 RPM the hold off timers activate and the 5 second "warm-up timer" goes active.
 - The "engine warm-up timer" will run for 5 seconds (adjustable). When this timer finishes timing, the controller's logic will initiate a transfer to the "Standby" position. As shown in the next series, the timer is still running and transfer has not yet occurred.
 - Generator AC output is available to the transfer switch Terminal Lugs E1 and E2 and to the normally open contacts of the transfer relay. However, the transfer relay is de-energized and its contacts are open.
 - The ECM is monitoring the Battery V, OPS, FPS, EHT, OLS, OS, MPS, TTMAP, IC1, IC2 (if V-twin), and SOV2 for fault conditions and adjusting engine parameters for optimal engine performance.
 - The ECM is grounding the MPR via Wire 610 which provides power for the ignition coils and the EMC and ETC.

Transfer to Standby

In [Figure 6-13](#) and [Figure 6-14](#) the Generator is running, the controller's "engine warm-up" timer has expired and generator AC output is available to the transfer switch Terminal Lugs E1 and E2 and to the open contacts on the transfer relay. Transfer to Standby may be briefly described as follows:

- 12 VDC is delivered to the transfer relay coil via Wire 194. The 12 VDC circuit is completed back to the controller via Wire 23. However, controller logic holds Wire 23 open from ground and the transfer relay is de-energized.
- When the "engine warm-up timer" expires, the controller will take Wire 23 to ground. The transfer relay energizes and its normally open contacts close (standby position).
- When the transfer switch closes to standby, the Charger AC voltage input detect is used to detect that transfer switch has made contact. If the switch does not transfer, there will be a warning with E-code 2738.
- While running, the RPM data from the ECM over CAN will be used for the following functions:
 - govern speed to maintain frequency through different loads
 - overspeed
 - underspeed
- With loads applied, CT1 - Wires 398A/399A and CT2 - Wires 398B/399B deliver approximately 0- 1.5 VAC based on percentage of Amps (load).

Approximate Values (when back-probed at connector):

25 Amps = 0.260 VAC

50 Amps = 0.518 VAC

75 Amps = 0.766 VAC

150 Amps = 1.552 VAC

- The ECM is monitoring the Battery V, OPS, FPS, EHT, OLS, OS, MPS, TTMAP, IC1, IC2 (if V-twin), and SOV2 for fault conditions and adjusting engine parameters for optimal engine performance.
- The ECM is grounding the MPR via Wire 610 which provides power for the ignition coils and the EMC and ETC.


⚠ DANGER

Electrocution. Lethal voltage may be present at current transformers (CTs). Do not touch CTs while primary current is applied. Doing so will result in death or serious injury.

(D000310)

Utility Voltage Restored and Re-transfer to Utility

See [Figure 6-15](#) and [Figure 6-16](#). The Load is powered by Generator voltage. On restoration of Utility voltage, the following events will occur:

- On restoration of Utility voltage above approximately 87% (adjustable, 84-89%, see chart) of the nominal rated voltage, a "re-transfer time delay" on the controller starts timing. The timer will run for 15 seconds (factory default, adjustable from 15-1500 seconds).
- At the end of the adjustable programmed timer, the "re-transfer time delay" will stop timing. The controller will open the Wire 23 circuit from ground and the transfer relay will de-energize.
- The generator continues to run for 60 seconds in its cooling down mode (factory default, adjustable 60-1500 seconds).

NOTE: If utility fails during the cool-down timer cycle, the controller will transfer back to standby.

NOTE: If programmed for 2 Wire Start mode, the transfer switch will operate as it does in Utility Loss mode.

- The ECM is monitoring the Battery V, OPS, FPS, EHT, OLS, OS, MPS, TTMAP, IC1, IC2 (if V-twin), and SOV2 for fault conditions and adjusting engine parameters for optimal engine performance.
- The ECM is grounding the MPR via Wire 610 which provides power for the ignition coils and the EMC and ETC.

Utility Pickup	
Factory Default	Util Recovery Volts
60 Hz = 208 VAC	60 Hz = 202-214 VAC

Engine Shutdown

See [Figure 6-17](#) and [Figure 6-18](#). Following re-transfer back to the Utility source an "engine cool-down timer" on the controller starts timing. When the timer has expired (adjustable, default is 60 seconds), the controller will de-energize the internal run relay removing fuel from the engine. The following events will occur:

- The ECM is monitoring the Battery V, OPS, FPS, EHT, OLS, OS, MPS, TTMAP, IC1, IC2 (if V-twin), and SOV2 for fault conditions and adjusting engine parameters for optimal engine performance.
- The ECM is grounding the MPR via Wire 610 which provides power for the ignition coils and the EMC and ETC.
- Wire 14 (run circuit) will de-energize and the fuel shutoff valves (SOV) will close to terminate the fuel supply to the engine.
- After a short fuel burn off, when engine speed reduces to 350 rpm, the ECM will stop firing the ignition coils.
- Without fuel flow and without ignition the engine will shut down.
- Once engine is shutdown the generator controller will remove power to the ECM by taking power away from the ECM enable, Wire 877. The MPR loses ground on Wire 608 from the ECM which turns off power to the ECM, IC1 (and IC2), the OS heater, and the OPS. The ECM enable is then turned back on after 2.5 seconds.
- The ECM then takes 608 to ground, re-powering Wire 14C through the MPR.
- The mixer and throttle motors will sweep and automatically set for the next engine run cycle. Once the sweep is complete the ECM takes ground away from 608, deactivating the MPR.

This page intentionally left blank.

Section 7: Power Zone 200 Controller

Introduction

This section will familiarize the reader with the various components that make up the Engine and DC Control systems.

Topics covered in this section are:

- Customer Connections
- Power Zone 200 Controller
- LED Display
- Battery Charger
- AUTO-OFF-MANUAL
- 7.5 Amp Fuse
- Starter Contactor Relay
- Connector Pin Descriptions

Customer Connection

See [Figure 7-1](#). The terminals of this terminal strip connect to identical terminals in the transfer switch. The terminal block provides the electrical connection for the controller at the following connection points:

- UTILITY N1 (Utility Sensing)
- UTILITY N2 (Utility Sensing)
- LOAD T1 (Internal Battery Charger) - 60 Hz Unit
- LOAD 00/T2 (Internal Battery Charger) - 60/50 Hz Unit
- Wire 194 (Transfer Relay)
- Wire 23 (Transfer Relay)
- Wire 0 (DC Common)
- Wire 395A (E-Stop)
- Wire 395B (E-Stop)

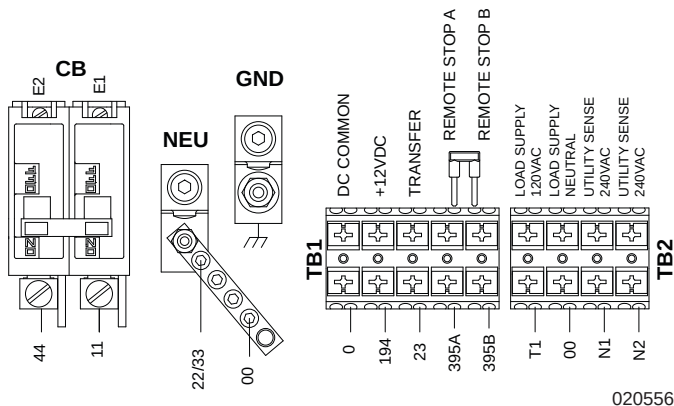


Figure 7-1. Customer Connections

Power Zone 200 Controller

See [Figure 7-2](#). The Power Zone 200 controller is responsible for all standby electric system operations including (a) engine startup, (b) engine running, (c) automatic transfer, (d) automatic re-transfer, and (e) engine shutdown. In addition, the controller performs the following functions:

- Automatic voltage regulation.
- Automatic engine shutdown in the event of High voltage, low voltage, loss of CAN communication, over speed, no RPM sense, over crank, low battery, or other alarms.
- Maintains proper battery charge.

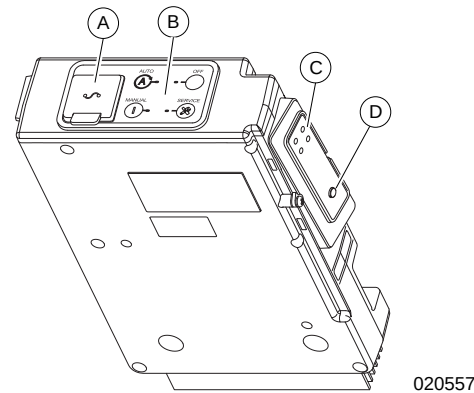


Figure 7-2. Power Zone 200 Controller

A	7.5A Fuse (under cover)
B	Controller Mode Indicators
C	External LED Indicators
D	Generator Emergency Shutdown Button

Controller Harness Connectors

The controller harness connectors are used to interconnect the controller with the various circuits of the DC and AC systems. Connector pin locations, numbers, associated wires and circuit functions are listed in the [Power Zone 200 Controller](#) diagram in this section, as well as in Section 12 [Engine Control Module \(ECM\) and Controller Pin-outs](#).

IMPORTANT NOTE: The generator engine will crank and start when exercise settings are configured. The unit will also crank and start every programmed interval thereafter, on the programmed day and time.

IMPORTANT NOTE: The exercise interval can also be set to transfer on exercise at 3, 6, or 12 months. The generator will run this at normal speed even if low speed is selected for exercise.

Control Panel Interface

AUTO/OFF/MANUAL/SERVICE Buttons

Button	Description of Operation
AUTO	Activates fully automatic system operation. Automatic operation allows unit to automatically start and exercise generator according to exercise timer settings (see Setting the Exercise Timer).
OFF	Shuts down engine and prevents automatic operation and exercise of unit.
MANUAL	Cranks and starts generator. Transfer to standby power will not occur unless there is a utility failure.
SERVICE	Suspends Mobile Link® and Fleet notifications and locks out the ability to receive remote start commands, such as exercise now. Also blocks the ability to receive remote firmware updates. Allows unit to be placed in any other mode (AUTO, OFF, MANUAL) at the same time as SERVICE mode for diagnostics or servicing purposes.

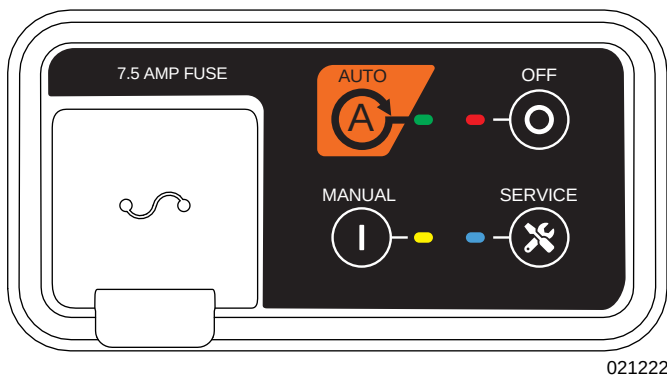


Figure 7-3. Controller Mode Indicators

Controller Mode Indicator

See [Control Panel Interface](#). Located on the generator controller, the LED Indicators provide a visual indication of the generator status.

- Green LED - Illuminates when the controller is set to AUTO mode.

- Red LED - Illuminates when the controller is set to OFF mode. Red LED will flash when the Emergency Shutdown is activated.
- Yellow LED - Illuminates when the controller is set to MANUAL mode.
- Blue LED - Illuminates when the controller is set to SERVICE mode. This LED will be lit in addition to the other Mode selection (Auto, Off or Manual). Blue LED will flash when controller is executing bootloader.
- Power Up Sequence - When the controller begins to power up, the green red and blue LEDs will flash once, then the blue LED will flash for a few seconds, then the red off LED and the red external LED will turn on solid for a few more seconds. The system will be ready to operate approximately 5 seconds after the mixer learn sweep completes.

Model Sync

The Generac Air-cooled Standby generator with the Power Zone 200 control is equipped with Model Sync. Model Sync is designed to ensure that the control components within the system share all critical operational information, such as serial number, model ID, run hours, and fuel type, between the following three components:

- Power Zone 200 controller
- Engine Control Module (ECM)
- Generac Generator Connectivity Accessory - Cellular (GGCAC)

The three components listed utilize a "best two of three" to share the information and attempt to automatically update a newly introduced component. This requires that replacement of any of the above listed components occurs one at a time, to ensure the model sync function occurs. If making a replacement, disconnect J6 and the battery before swapping a device.

Replacement of two or more of the listed components at once will interrupt the model sync and may cause device conflicts or other inability to update the required operational information listed above.

External LED Indicators

OFF		solid red
OFF + ALARM		flashing red
OFF + WARNING/MAINT		solid red / flashing yellow
AUTO		solid green
AUTO + ALARM		solid green / flashing red
AUTO + WARNING/MAINT		solid green / flashing yellow
running in AUTO		solid green
AUTO + Utility Loss Delay Countdown		flashing green (2 sec)
running in AUTO + Transfer		flashing green (1 sec)
MANUAL + ALARM		flashing red / solid yellow
running in MANUAL + WARNING/MAINT		flashing yellow
running in MANUAL		solid yellow
running in MANUAL + Transfer		flashing green (1 sec) / solid yellow
Emergency Shutdown		flashing red
SERVICE (service button pressed)		solid blue + other LED color (green, yellow, or red)
Controller Firmware Update In Progress		all lights / flashing counterclockwise (red-green-yellow-blue)
ECM Firmware Update In Progress		all lights / flashing clockwise (red-blue-yellow-green)

020555a

Figure 7-4.

Controller Battery Charger

NOTE: Battery Charging status can be viewed in the Field Pro app on the Generator Details dashboard.

NOTE: 50 Hz units use fuses F3 and F4 and a 220 VAC supply to the battery charger.

NOTE: The battery charger is not designed to charge a dead or discharged battery. Any battery below 7V must be charged separately until fully charged before installing into the generator.

The charger operates between 12.2 VDC and 16.0 VDC at a maximum current of 5 amps. It will adjust the charging voltage depending on ambient temperature. It will charge at the higher voltage level when cold and the lower voltage level when hot. The battery charger is powered from a 120 VAC load side connection through a fuse (F3) in the transfer switch. It will also require a neutral wire to the charger power connection. This 120 VAC source must be connected to the generator in order to operate the charger.

NOTE: During a utility failure, the charger will momentarily be turned off until the generator is connected to the load. During normal operation, the battery charger supplies power to the controller but in case of failure, the generator battery is used to supply power.

The battery charger will begin its charge cycle when voltage drops below 12.7 VDC. The charger provides current directly to the starting battery at a voltage determined by the ambient temperature for the 18 hour period. At the end of the 18 hour period battery charge current is measured when the generator is off. If battery charge current at the end of the 18 hour charge time is greater than a pre-set level, a “Battery problem” is raised. If the battery voltage at the end of the 18 hour charge time is less than approximately 12.7 VDC open circuit, a “Charger warning” is raised. If the battery voltage is above 16.5 VDC, charger warning is raised and charging is disabled until the condition is corrected. There is also a warning for Charger AC missing which means the T1 or T2/00 wire is not powered and charger will not operate. If the battery voltage is above 18.5 VDC at any time, the charger output is disconnected from the battery and an alarm is raised. Any charger warnings will self clear if the battery voltage corrects to a normal level. Any Low

Battery alarms will not self-clear and will require manual alarm reset or remote-command Clear Alarm to resolve Low Battery alarms. The next 18 hour charge cycle will also need to be successful.

The battery has a similar role as that found in an automotive application. It sits doing nothing (but can still supply current to the controller if the charger AC voltage is removed) until it either self-discharges below 12.7 VDC or an engine crank occurs (i.e. such as occurs during the weekly exercise). If either condition occurs the battery charger will begin its 18 hour charge cycle.

AUTO-OFF-MANUAL

This feature permits the operator to (a) select fully automatic operation, (b) start the generator manually, or (c) stop the engine and prevent the automatic startup. The controller has OFF-MANUAL-AUTO Mode membrane push buttons. See [Figure 7-4Figure 1](#) for the location of the push buttons.

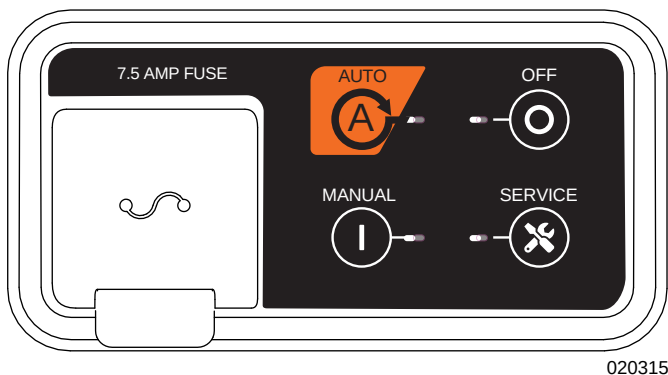


Figure 7-5. Off-Manual-Auto Buttons

Fuse

See [Figure 7-5](#). The fuse protects the controller against excessive current. If the fuse has blown, engine cranking

and operation will not be possible. The controller will remain powered as long as charger AC power is still present from Utility. Should fuse replacement become necessary, use only an equivalent 7.5 amp replacement fuse. Refer to controller fuse testing for troubleshooting.

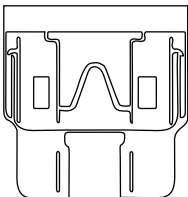


Figure 7-6. Typical 7.5 Amp Fuse

Common Alarm Relay

The common alarm relay provides a set of normally closed (N. C.) contacts to drive a customer provided external alarm indication. When the control is powered up, if there are no Alarms, the relay contacts will be OPEN. Any ALARM (not warning) will trigger the common alarm relay to operate, closing the contacts.

The connections are made to the controller via Wires 209 and 210 by connecting to the dedicated wire connections.

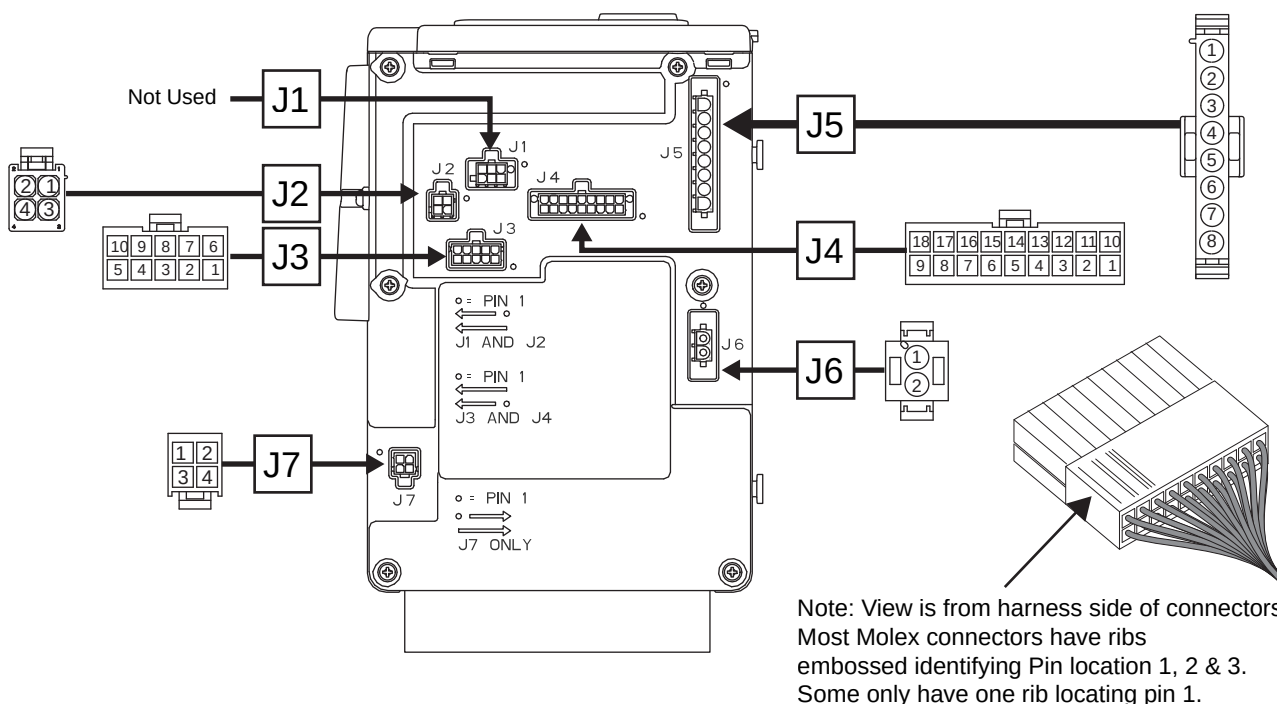
Contact Rating:	200 mA at 12 VDC
-----------------	------------------

NOTE: Contact rating is for resistive load only

Circuit Pin Description

The [Power Zone 200 Controller](#) diagram on the following page, as well as in Section 12 [Engine Control Module \(ECM\) and Controller Pin-outs](#), provides the physical wire identification and circuit functions.

Power Zone 200 Controller



020785

Figure 7-7. Air-cooled Panel and Connectors (Harness End)

J2 Connector		
Pin	Wire	Circuit Function
J2-1	Blue	Radio 3.8V DC Power
J2-2	Brown	Radio RS232RX Coms
J2-3	White	Radio RS232TX Coms
J2-4	Black	Radio DC Ground

J3 Connector		
Pin	Wire	Circuit Function
J3-1	399A	Generator Phase A+ Current Sense
J3-2	398A	Generator Phase A - Current Sense
J3-3	399B	Generator Phase B + Current Sense
J3-4	398B	Generator Phase B - Current Sense
J3-5	399C	Generator Phase C +Current Sense
J3-6	SHLD	Shielding Wire for CAN bus Wiring
J3-7	743	CAN bus Coms +
J3-8	744	CAN bus Coms -
J3-9	877	ECM Enable
J3-10	398C	Generator Phase C - Current Sense

J4 Connector

Pin	Wire	Circuit Function
J4-1	13	12 VDC Unfused for the Controller
J4-2	13	12 VDC Unfused for the Controller
J4-3	194	12 VDC for the Transfer Relay
J4-4	23	Switched to Ground (Internally) to Energize the Transfer Relay
J4-5	—	Optional 2-Wire Start Kit (Advance access menu only)
J4-6	—	Optional 2-Wire Start Kit (Advance access menu only)
J4-7	14	12 VDC for Fuel Shut Off Valves 1 & 2 (powered when Running)
J4-8	209	Common Alarm Relay Input
J4-9	210	Common Alarm Relay Output
J4-10	0	Common Ground (DC)
J4-11	0	Common Ground (DC)
J4-12	56	12 VDC Output to Starter Control Relay
J4-13	—	Unused Digital Output
J4-14	—	Unused Ground for Digital Output
J4-15	—	Unused Digital Input 1
J4-16	—	Unused Digital Input 2
J4-17	395A	Remote Stop A
J4-18	395B	Remote Stop B

J5 Connector

Pin	Wire	Circuit Function
J5-1	N1	Utility Phase A Sensing Voltage
J5-2	N2	Utility Phase B Sensing Voltage
J5-3	N3	Utility Phase C Sensing Voltage
J5-4	00	Neutral Connection for Controller
J5-5	00	Neutral Connection for Controller
J5-6	S3S	Phase C
J5-7	44S/S2S	Phase B
J5-8	11S/S1S	Phase A

J6 Connector		
Pin	Wire	Circuit Function
J6-1	T1	120 VAC Power for Internal Battery Charger
J6-2	00	Neutral Connection for Internal Battery Charger

J7 Connector		
Pin	Wire	Circuit Function
J7-1	2	DPE Winding (AC Excitation Power)
J7-2	0A	Field Return (-) (DC Field Excitation)
J7-3	6	DPE Winding (AC Excitation Power)
J7-4	4	Field Flash/DC Excitation (+)

This page intentionally left blank.

Section 8: AC Troubleshooting

Introduction

This section provides acceptable procedures for the testing and evaluation of various problems that can occur on standby generators with air-cooled engines. Some test procedures in this section require the use of specialized test equipment, meters or tools. Most tests can be performed with a Digital Multimeter (DMM). An AC frequency meter is required where frequency readings must be taken. To measure AC Current it is acceptable to use a clamp-on ammeter.

Testing and troubleshooting methods covered in this section are not exhaustive. No attempt has been made to discuss, evaluate and advise the home standby service trade of all conceivable ways in which service and trouble diagnosis must be performed. Accordingly, anyone who uses a test method not recommended herein must first satisfy themselves that the procedure or method they have selected will jeopardize neither their own nor the product's safety.

Safety

Service personnel who work on this equipment should be aware of the dangers of such equipment. Extremely high and dangerous voltages are present that can kill or cause serious injury. Gaseous fuels are highly explosive and can ignite by the slightest spark. Engine exhaust gases contain deadly carbon monoxide gas that can cause unconsciousness or even death. Contact with moving parts can cause serious injury.

When working on this equipment, use common sense and remain alert at all times. Never work on this equipment while physically or mentally fatigued. If you do not understand a component, device or system, do not work on it.

While servicing the generator, verify Service mode is active by pressing the service mode button and verifying the blue LED on the right side of the generator is illuminated. If a requires a power cycle of the generator, be sure to turn service mode back on once the controller reboots. If a test requires turning service mode off, beware—the generator may start and run at any moment.

AC Troubleshooting

Continue to ask questions during the troubleshooting process. Asking some of these questions may help identify the problem more quickly.

- What is the generator supposed to do?
- What fault (Alarm) is shutting the generator down?
- Is the fault a symptom of another problem?

- Once the alarm is cleared and the unit is placed back into auto or manual does the alarm reoccur?
- Does the same fault happen consistently?
- When does the fault occur?
- After the fault occurred what was displayed in the Field Pro app?
- Why would this happen?
- How would this happen?
- What type of test will either prove or disprove the cause of the fault?

AC Excitation Winding Test

General Theory

The controller's internal voltage regulator requires unregulated voltage from the stator to supply excitation power to the AVR. The regulator supplies DC field excitation current to the rotor. The alternator may be producing this voltage, but if the voltage is not being supplied to the regulator, it will fault out for under-voltage. This test will verify the integrity of the excitation (DPE) winding inside the stator and connections to the voltage regulator.

NOTE: Measure the resistance of the meter leads and subtract that reading from the actual reading to ensure accurate results.

Procedure A — using J7 Breakout Harness, P/N A0008081735

1. Set the DMM to measure resistance.
2. Disconnect the generator harness J7 connector.
3. Connect the female end of the J7 breakout harness to the unit harness.

IMPORTANT NOTE: DO NOT CONNECT the male end of the J7 breakout harness to the controller.
4. Insert the meter test leads into the AC excitation test points of the breakout harness (2 and 6).
5. Record the reading.
 - a. If the meter reading shows a resistance value consistent with the values found in Section 3 [Generator Specifications](#), stop testing and refer back to the [Preliminary Output Voltage Test](#).
 - b. If the meter reading shows an OPEN, proceed to **Procedure B**.

Procedure B

1. See [Figure 8-1](#). Disconnect the STR connector located in the wire harness and inspect the pins inside both sides of the connector.

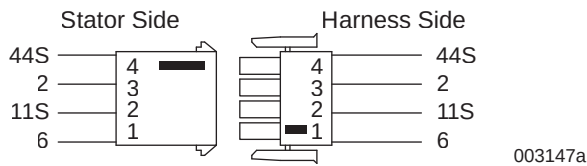


Figure 8-1. Stator Connector (STR) Pin Locations

- a. If the pins are found to be damaged, replace wiring harness and retest.
2. Set the DMM to measure resistance.
 3. On the STR connector that leads to the stator, connect one meter test lead to Wire 2 and the other meter test lead to Wire 6. Record the reading.
 - a. If the meter reading shows a resistance value consistent with the values found in Section 3 [Generator Specifications](#), replace the wiring harness and retest.
 - b. If the meter indicated an OPEN or a value inconsistent with the values found in Section 3 [Generator Specifications](#), proceed to **Procedure C**

Procedure C

1. See [Figure 8-2](#) and [Figure 8-3](#). Remove the stator terminal block cover (A) and identify the studs containing Wires 2 and 6.

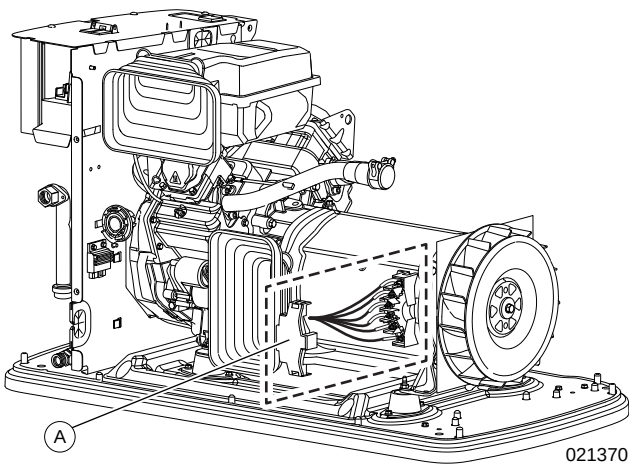
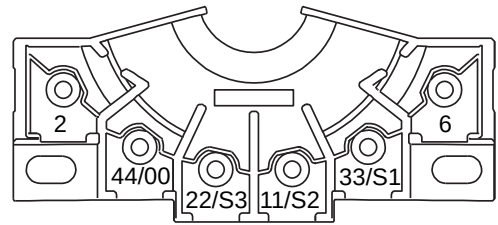


Figure 8-2. Stator Terminal Block Cover

WITH 254 ALTERNATOR



WITH 315 ALTERNATOR

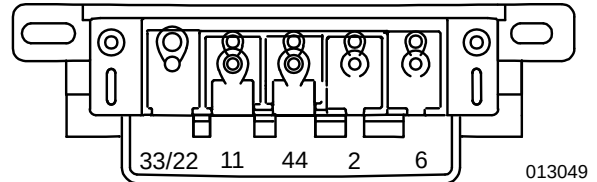


Figure 8-3. Stator Terminal Block Wires

2. Check to see that harness connections to the stator terminal block are tight and secure.
 - a. If the connections are faulty, repair the connections and retest.
3. Set the DMM to measure resistance.
4. Connect one meter test lead to the stud for Wire 2 and the other meter test lead to the stud for Wire 6. Record the reading.
 - a. If the meter reading shows a resistance value consistent with the values found in Section 3 [Generator Specifications](#), proceed to Step 5.
 - b. If the meter indicated an OPEN or a value inconsistent with the values found in Section 3 [Generator Specifications](#), replace the stator and retest.
5. Measure Wire 2 between the STR (stator) connector and the STB (stator terminal block).
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and proceed to Step 6.
6. Measure Wire 6 between the STR (stator) connector and the STB (stator terminal block).
 - a. If resistance measures less than 5 ohms, reinstall the stator block cover and proceed to the [AVR PWM Output Test](#).
 - b. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

AC Output Voltage Test

General Theory

Use a DMM to check the generator's output voltage. Test output voltages at the unit's main circuit breaker (MLCB)

terminals. Refer to the unit's Data Plate for rated line-to-line and line-to-neutral voltages.



⚠ DANGER

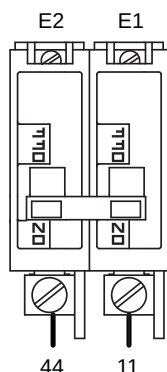
Electrocution. High voltage is present at test terminals. Contact with live terminals will result in death or serious injury.

(D000311)

IMPORTANT NOTE: During this test the generator will be running. Prior to starting the generator connect meter test clamps to the high voltage terminals. Stay clear of power terminals during the test. Verify the meter clamps are securely attached and will not shake loose.

Procedure

1. Gain access to the MLCB on the generator by removing the controller fascia.
2. Set the DMM to measure AC voltage.
3. See [Figure 8-4](#). With the engine shut down, connect the meter test leads across the load terminals of the generator MLCB. This will measure line-to-line voltage.



021124

Figure 8-4. MLCB Test Points

4. Set the MLCB to the OFF position. Verify that all electrical loads are disconnected from the generator.
5. Set the controller to MANUAL.
6. Set the MLCB to the ON position. Record the voltage.
 - a. If the DMM shows approximately 238–244 VAC, the output voltage is good.
 - b. If the DMM shows a voltage outside of the above range, proceed to the [AVR PWM Output Test](#).

AVR PWM Output Test

General Theory

The AVR can change the amount of current to the rotor at any time. The AVR PWM runs at a much higher frequency, and the faster response allows the generator to handle large load changes much better than previous designs. A low duty cycle (on-time) will give a very low amount of current to the rotor and the output will reduce the voltage at the breaker. A high duty cycle (on-time) will give a large amount of current to the rotor and the output will increase the voltage at the breaker. The controller will sense the output voltage and change the PWM duty cycle to lower or raise the current and control the output voltage to the target as quickly as possible while matching the engine speed for the best response. This test will verify that PWM is working as designed.

NOTE: See [Field Boost](#) in Section 5 [Operational Analysis](#) for analysis of field boost parameters.

Procedure

1. Set the DMM to measure hertz (HZ).
2. Disconnect the J7 connector from the controller.
3. Connect the male end of the J7 breakout harness to the controller.
4. Connect the female end of the J7 breakout harness to the generator harness.
5. Place the red meter lead into the WAGO block for Wire 4.
6. Place the black meter lead into the WAGO block for Wire 0A.
7. Clear all faults and place generator in MANUAL.

Record the Hz reading.

- a. If approximately 1500-2000 Hz (1.5kHz-2 kHz) is measured, AVR output regulation passes test. Proceed to the [Brushes and Slip Rings Test](#).
- b. If 0 Hz is measured, replace the Power Zone 200 controller and retest.

Brushes And Slip Ring Test

General Theory

The brushes and slip rings function to provide an electrical connection for excitation current from the stationary components to the rotating rotor. Made of a special long-lasting material, brushes seldom wear out or fail. However, slip rings can develop a tarnish or film that can impede the flow of current. This film usually develops during non-operational periods. Broken or disconnected wiring can also cause loss of field excitation current to the rotor.

Procedure

1. Disassemble the generator until the brushes and slip rings are exposed. Refer to Section 11 [Major Disassembly](#).

2. Inspect the brush wires and verify they are secured and properly connected.

NOTE: Verify straight brush connectors for Wire 4 are connected to the brush pin closest to the bearing for both brush assemblies. Verify the right angle connector for Wire 0A is behind the straight connector.

3. Inspect the brush assemblies for excessive wear, or damage.
4. Inspect the rotor slip rings. If their appearance is dull or tarnished, polish with a fine grade abrasive material.

IMPORTANT NOTE: Do not use metallic grit (such as emery cloth) to polish slip rings. This may cause irreversible damage to the rotor.

5. Disconnect the J7 harness connector from the controller containing Wires 4 and 0A.
6. Set the DMM to measure resistance.
7. Place the red meter lead on one of the disconnected wire ends for Wire 0A at the brushes. Using the appropriate back probe, insert the other lead to Wire 0A at the J7 connector and record the reading.

- a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

8. On the other brush assembly, place the red meter lead on the disconnected wire end for Wire 0A at the brushes. Using the appropriate back probe, insert the other lead to Wire 0A at the J7 connector and record the reading.

- a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

9. Place the red meter lead on one of the disconnected wire ends for Wire 4 at the brushes. Using the appropriate back probe, insert the other lead to Wire 4 at the J7 connector and record the reading.

- a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

10. On the other brush assembly, place the red meter lead on the disconnected wire end for Wire 4 at the brushes. Using the appropriate back probe, insert the other lead to Wire 4 at the J7 connector and record the reading.

- a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
11. With Wire 4 still disconnected at the brush assemblies, place the red meter lead on one of the disconnected wire ends for Wire 4 at the brushes and place the black lead on a frame ground. Record the reading. Move the red lead to the other Wire 4 connector at the brushes and record the reading.
 - a. If either wire shows a resistance value, replace wiring harness and retest.
 12. Place the meter leads across the terminals of each brush holder. Record the readings.
 - a. If the resistance value is consistent with the values found in Section 3 [Generator Specifications](#), proceed to the [Rotor Assembly Test](#).
 13. Remove the affected brush assembly.
 14. Set the DMM to measure continuity.
 15. Place the red meter lead on the Wire 4 terminal and the black meter lead on the slip ring side of the brushes. Record the reading. Move the red meter lead to the Wire 0 terminal and record the reading.
 - a. If OL is measured, replace the brush assembly and retest.
 - b. If continuity is measured, proceed to the [Rotor Assembly Test](#).

Current Calibration Procedure

General Theory

When a Power Zone 200 controller is replaced, the calibration of both voltage and amperage will need to be verified so the controller is aligned with the generator's output. Calibration of current (amperage) is done with the use of a load bank that is set to at least 20% of the generator rated capacity.

NOTE: Calibration between the generator and controller is preset from the factory, and does not need adjustment. Only when replacing a controller will this need to be checked.

Procedure

NOTE: In the Field Pro app, CT1 correlates to "AN" (Wire 11)/ S1 and CT2 correlates to "BN" (Wire 44/S2). CT3 correlates to CN (wire S3)



⚠ DANGER

Electrocution. Lethal voltage may be present at current transformers (CTs). Do not touch CTs while primary current is applied. Doing so will result in death or serious injury.

(D000310)

1. Gain access to the MLCB on the generator by removing the controller fascia.
2. Connect a load bank to the output circuit of the generator. This should be done at the generator MLCB.
3. Set the Amp Clamp to measure AC Amperage.
4. Using the Field Pro app, navigate to the "Calibration" screen.
5. Place the Amp Clamp over the circuit being calibrated.
6. Start the generator and allow it to warm up for 10 seconds.
7. Place a load on the generator that is at least 20% of the generator rated capacity.
8. On the Field Pro app under the "Current Calibration" box, enter the recorded amperage for the circuit being measured in the "Measured Current" box. Click the save icon.
9. Repeat Steps 5–8 on the other circuit.
10. Once completed, turn the generator OFF and remove test equipment.

Field Boost Voltage Test

General Theory

Wire 4 is used for both carrying the field flash voltage to the rotor to excite it, and also to send regulated voltage from the AVR to the rotor. This test will verify the integrity of Wire 4.

NOTE: [Field Boost](#) in Section 5 [Operational Analysis](#) for analysis of field boost parameters

Procedure

1. Set DMM to measure resistance
2. Disconnect the J7 connector from the controller.
3. Connect the male end of the J7 breakout harness to the controller and the female end of the J7 breakout harness to the harness.
4. Disconnect the battery and the J6 battery charger connector.
5. Insert the red meter lead into the WAGO block for Wire 4.
6. Insert the black meter lead into the WAGO block for Wire 0A. Record the reading.
 - a. If 4-15 ohms measured, proceed to Step 6.
 - b. If any other resistance is, proceed to the [Brushes and Slip Rings Test](#).
7. Set the DMM to measure DC voltage.

8. Leave the red meter lead in WAGO block terminal for Wire 4. Place the black meter lead in Wago block terminal for wire 0A.
9. Clear any faults and start the generator in MANUAL. Record the voltage.
 - a. If the voltage measures approximately 14 VDC, proceed to [AVR PWM Output Test](#).
 - b. If the voltage measures approx. 3–4 VDC, check the connection and pins for the J7 connector. If the pins are found to be damaged, replace the wiring harness and/or controller as necessary and retest. If the harness and controller pins pass inspection, proceed to the [AC Excitation Winding Test](#).
 - c. If the voltage measures 0 VDC, check the connection and pins for the J7 connector. If the pins are found to be damaged, replace the wiring harness and/or controller as necessary and retest. If all passes, proceed to the [Internal Diode Test](#).

Internal Diode Test

General Theory

A diode acts like an electronic one-way check valve that allows current to pass in only one direction to protect a circuit from an electrical "back feed". When testing the internal diode, a DMM is used in the "Diode" function to apply a small amount of voltage across the circuit. This is used to verify that voltage will pass through in one direction (as the voltage is applied by the positive lead of the DMM) and not the other. Typically, when a diode is tested and working properly, one way will show approximately 0.5 VDC and the other will show OL.

Procedure

1. Disconnect all connectors from the controller.
2. Set the DMM to the Diode function.
3. Locate the J7 connector on the back of the controller and connect the J7 breakout harness to the controller.

IMPORTANT NOTE: DO NOT CONNECT the female end of the J7 breakout harness to the generator harness.
4. Insert the red meter lead into the WAGO block for Wire 0A and the black lead into the WAGO block for Wire 4. Record the reading.
 - a. If approx. 0.5 VDC is measured, proceed to Step 5.
 - b. If OL or 0.00V is measured, replace the controller and retest.

5. Insert the red meter lead into the WAGO block for Wire 2 and the black lead into the WAGO block for Wire 4. Record the reading.
 - a. If approx. 0.5 VDC is measured, proceed to Step 6.
 - b. If OL or 0.00V is measured, replace the controller and retest.
6. Insert the red meter lead into the WAGO block for Wire 6 and the black lead into the WAGO block for Wire 4. Record the reading.
 - a. If approx. 0.5 VDC is measured, proceed to Step 7.
 - b. If OL or 0.00V is measured, replace the controller and retest.
7. Set the DMM to measure resistance.
8. Insert the red meter lead into the WAGO block for Wire 2 and the black lead into the WAGO block for Wire 6. Record the reading.
 - a. If resistance is greater than 1 Megohm, controller passes test. Proceed to the [Brushes and Slip Rings Test](#).
 - b. If less than 100 ohms is measured, replace the controller and retest.

Overload Condition Test

General Theory

An "overload" condition exists when the generator rated amperage capacity has been exceeded. To test for an overload condition on an installed unit, the recommended method is to use an ammeter.

NOTE: Verify the Current Calibration is set correctly before proceeding. See the Current Calibration Procedure for information.

Procedure

1. Set the ammeter to measure AC amperage.
2. Connect the clamp-on ammeter over Wires 11 and 44 on the generator side of the MLCB.
 - a. If 3-phase, perform Step 2 for all 3 phases, S1, S2, and S3. No individual leg can exceed the load maximum.
3. Transfer all of customer's typical electrical loads to the generator. Record the amperage.
 - a. If the ammeter displays amperage readings that are higher than the rated capacity of the generator, reduce the loads to within the rated capacity and advise the customer accordingly.
 - b. If the ammeter displays amperage readings that are below the generator's rated capacity, but the engine RPM and/or frequency dropped

excessively, proceed to the [Voltage and Frequency Under Load Test](#).

Power Cycle and Sync Test

General Theory

Some circuits inside the controller have a latching condition where the circuit must be powered down entirely to reset. By removing power from the controller, the circuits return to default conditions. The Radio, ECM and PZ200 communicate between each other, when they are all disconnected from power and repowered, their communication is also reset to default conditions where they can reestablish connections. Model Sync occurs when 1 new device is connected to the system, if more than 1 new device is connected to the system, sync cannot occur. Sync ensures fuel type, hours, model type, and serial number are synced together within the system when parts are replaced.

Procedure

1. Disconnect the negative battery terminal.
2. Disconnect J6 from the back of the controller.
3. Wait 60 seconds.
4. Reconnect the negative battery terminal.
5. Reconnect J6.
6. Wait 5 minutes for the controller to reboot.
7. Activate service mode and navigate to the Field Pro App.
8. Rerun the system test.
 - a. Verify selected fuel type in the configuration.
9. If errors reoccur were multiple devices replaced? (PZ200 and ECM and Radio)
 - a. Yes, reinstall 2 of the original devices and retest.
 - b. No, continue to Step 10.
10. Check data tag and confirm the SN in Field Pro.
11. Click the Service Mode On, wait 5 seconds, then click the Service Mode Off.
12. Wait 15 minutes for sync to occur.
13. If there are still errors contact Technical Support.

Preliminary Output Voltage Test

General Theory

When an alarm of the 1900 group (Undervoltage) is active, certain tests need to be performed to determine the actual fault. When Cold Smart Start is enabled within Field Pro and the generator is not producing enough voltage, the controller will allow up to four minutes to

measure output voltage before shutting down on “undervoltage” during which the Field Pro app will read “Warming Up”. Measuring output voltage as described in this test will help determine the next step in troubleshooting.

When measuring output voltage, it is important to look at the output voltage displayed on the Field Pro app as well as measuring actual output voltage of the generator.

NOTE: If the generator shuts down before this test is completed, clear the alarm and restart the test as needed to obtain the correct measurements.

Procedure

1. Gain access to the main line circuit breaker (MLCB) on the generator by removing the controller fascia.
2. Set the MLCB to the OFF position.
3. Using the Field Pro app, navigate to view the “Output Voltage” input box.
4. Set the DMM to measure AC voltage.
5. Connect one meter test lead to Wire 11 and the other meter test lead to Wire 44 at the generator side of the MLCB.

NOTE: For 3-phase units its important to test S1–S2, S2–S3, and S1–S3.

6. Set the controller to MANUAL. Allow generator to start and stabilize.
 - a. If the generator does not start and run at 3600 rpm proceed to [Engine Cranks But Does Not Run, Lacks Power, Runs Rough, Hunts or Backfires](#).
7. Record the voltage shown on the meter.
8. On the Field Pro app, add the AN and BN (and CN for 3-phase) voltage together in the Output Voltage box. Record the total output voltage.
9. Turn OFF the generator.
10. Use [Table 8-1](#) to determine which test to perform.

Table 8-1. Preliminary Output Voltage Test Results

Field Pro AC Voltage	AC Voltage measured at MLCB	Perform Test(s)
0	0 (approx.)	Brushes and Slip Rings Test
0	300 (approx.)	Sensing Circuits Test
40–90 (approx.)	40–90 (approx.)	AC Excitation Winding Test

Field Pro AC Voltage	AC Voltage measured at MLCB	Perform Test(s)
80–160 (approx.)	200–300 (approx.)	Sensing Circuits Test
240 (approx.)	240 (approx.)	Voltage and Frequency Under Load Test

Rotor Assembly Test

General Theory

A rotor having open windings will cause loss of excitation current flow and as a result generator AC output voltage will drop to “residual” voltage. A “shorted” rotor winding can result in a low voltage condition.

Procedure

1. Disassemble the generator until the brushes and slip rings are exposed. Refer to Section 11 [Major Disassembly](#).
2. Remove the brush assemblies from the slip rings to prevent interaction.
3. Set the DMM to measure resistance.
4. Connect the red meter lead to the positive slip ring (nearest the rotor bearing) and the black lead to the negative slip ring. Record the reading. Compare the reading with Section 3 [Generator Specifications](#).
 - a. If the resistance is out of specifications, replace the rotor and proceed to the [Stator Test](#) to determine if it also needs to be replaced.
5. Set the DMM to measure continuity.
6. Connect the red meter lead to the positive slip ring and the black meter lead to a chassis ground. Record the reading.
 - a. If no continuity is measured, proceed to the [Stator Test](#).
 - b. If continuity is measured, replace the rotor and proceed to the [Stator Test](#) to determine if it also needs to be replaced.

Sensing Circuits Test

General Theory

The automatic voltage regulator (AVR) built into the Power Zone 200 controller needs a way to monitor the voltage the generator is providing to the customer's load. This is needed to regulate the voltage provided to the

brushes to maintain 240 VAC under all load conditions. This test will verify the integrity of the sensing circuit.

- If one sensing wire is damaged, 230–300 VAC will be measured at the MLCB. However, the Field Pro app will display some various voltage on AN or BN (depending on which leg is damaged).
- If both sensing wires are damaged, up to 300 VAC may be measured on the MLCB. However, the Field Pro app will display 0 VAC on both AN and BN.

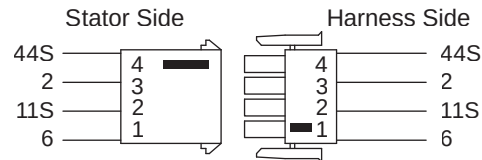
NOTE: Stator winding resistance values are very low. Some meters will not read such a low resistance and thus will show different ranges of resistance. Resistance of the meter leads themselves will need to be taken into account.

Procedure A

1. Open the Field Pro App to the Generator Details screen.
2. Start the generator in MANUAL. Once the generator stabilizes, look at the Output Voltage input box. The readings will be voltage to neutral on Line A and Line B. Each line should display 120 VAC $\pm$ 5%.
 - a. If both lines display 120 VAC $\pm$ 5%, stop testing as the sensing circuit is working correctly.
3. Set the DMM to measure resistance.
4. Remove the 7.5 Amp fuse from the controller.
5. Locate and unplug the J5 connector on the back of the controller.
6. Locate the pins for Wires 11S and 44S.
7. Using appropriate back probes, record the resistance between Wires 11S and 44S.
 - a. If the meter displays a resistance value consistent with the values found in Section 3 [Generator Specifications](#), check the connection and pins for the J5 connector. If the pins are found to be damaged, replace the wiring harness and/or controller as necessary and retest. If the harness and controller pins pass inspection, replace the controller and retest.
 - b. If meter displays OL, proceed to Procedure B.

Procedure B

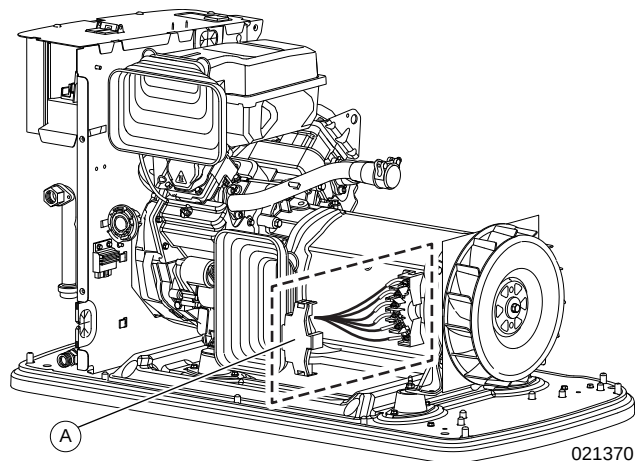
1. Set DMM to measure resistance.
2. See [Figure 8-5](#). Locate and disconnect STR connector in wire harness.
 - a. If the pins pass inspection, proceed to Step 3.
 - b. If the pins are found to be damaged, replace the wiring harness and retest.



003147a

Figure 8-5. Stator Connector (STR) Pin Locations

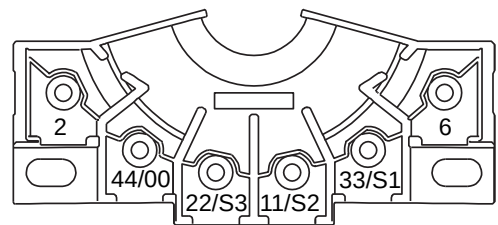
3. Connect meter test leads the appropriate harness pins containing Wire 11S and Wire 44S in the STR connector leading to the Stator. Record the resistance.
 - a. If the meter displays a resistance value consistent with the values found in Section 3 [Generator Specifications](#), replace the wiring harness and retest.
 - b. If the meter displays OL, proceed to Step 4.
4. See [Figure 8-6](#) and [Figure 8-7](#). Remove the stator terminal block cover and identify the studs containing Wires 11S and 44S.



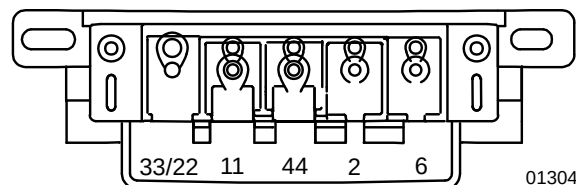
021370

Figure 8-6. Stator Terminal Block Cover

WITH 254 ALTERNATOR



WITH 315 ALTERNATOR



013049

Figure 8-7. Stator Terminal Block Wires

5. Verify harness connections to the stator terminal block are tight and secure. If they are loose, tighten and retest.
6. Place the meter leads on the terminals for Wires 11 and 44. Record the reading.

- a. If the meter displays a resistance value of less than 1 ohm in Step 6, but displayed OL in Step 3, replace the wiring harness and retest.
 - b. If the meter displays OL or a value inconsistent with the values found in Section 3 [Generator Specifications](#), replace stator and retest.
 - c. If values are correct but a sensing issue is still present, go to **Procedure C**.
9. Remove Wire 33 jumper (if equipped) from the STB.
 10. Measure resistance across the disconnected Wire 33.
 - a. If the meter displays less than 5 ohms, return to Procedure A and test again.
 - b. If the meter displays OL, replace the wiring harness and retest.

Procedure C

1. Set DMM to measure resistance.
2. Disconnect the J5 connector from the back of the controller. Visually inspect the controller and the harness pins for damage.
 - a. If the pins are found to be damaged, replace the wiring harness and/or controller as necessary and retest.
3. Using appropriate back probes, connect one meter test lead to J5-4. Connect the other meter test lead to J5-5. Record the resistance.
 - a. If the meter displays a resistance value of up to 10 ohms, return to Procedure A and test again.
 - b. If the meter displays OL, proceed to Step 4.
4. Disconnect Wire 00 (18 gauge white) from the neutral bar.
5. Connect one meter lead to the disconnected Wire 00 from the neutral bar and other meter lead to J5-4. Record the resistance.
 - a. If the meter displays less than 5 ohms go to Step 6.
 - b. If the meter displays OL, replace the wiring harness and retest.
6. Connect one meter lead to the disconnected Wire 00 from the neutral bar and other meter lead to J5-5. Record the resistance.
 - a. If the meter displays less than 5 ohms go to Step 7.
 - b. If the meter displays OL, replace the wiring harness and retest.
7. Disconnect Wire 22/33 from the neutral bar in the customer connection compartment and Wire 22/33 from the stator terminal block (STB).
8. Connect one meter lead to the disconnected Wire 22/33 at the neutral bar the other meter lead to the disconnected Wire 22/33 from the stator terminal block. Record the resistance.
 - a. If the meter displays less than 5 ohms go to Step 9.
 - b. If the meter displays OL, replace the wiring harness and retest.

Stator Test

General Theory

This test will use a digital multimeter (DMM) to test the stator windings for the following faults:

- An OPEN circuit condition
- A “short-to-ground” condition
- A short circuit between windings

[Table 8-2](#) has been provided to record the results. These results may be required when requesting factory support.

Procedure

1. Disconnect Wires 11 and 44 from the main line circuit breaker (MLCB).
2. Disconnect Wires 22 and 33 from the NEUTRAL connection and separate the leads.
3. Remove the stator terminal block cover to expose all wires/studs (33/22, 11, 44, 2, and 6).
4. Verify harness connections to the stator terminal block are tight and secure. If any are found loose, tighten and retest.
5. Set the DMM to measure resistance.
6. Measure and record the resistance values for each set of windings between the A and B test points as shown in [Table 8-2](#) and follow the instructions within the table for repair procedures.

Table 8-2. Stator Results (w/STB terminal block)

Test Point A	Test Point B	Results
Resistance Test with wires removed from terminal Block		
Stator Stud Wire 11	Stator Stud Wire 44	
Stator Stud Wire 2	Stator Stud Wire 6	
1. If results are OL or out of specification, replace the stator. 2. If results are within specification, continue to next chart.		

Test Point A	Test Point B	Results
Test Windings for a Short-to-Ground		
Stator Stud Wire 11	Ground	
Stator Stud Wire 2		
1. Any results other than OL indicates a short-to-ground condition; replace the stator. 2. If results are OL, continue to next chart.		
Test For A Short Circuit Between Windings		
Stator Stud Wire 2 or Wire 6	Stator Stud Wire 11	
	Stator Stud Wire 22/33	
	Stator Lead Wire 44	
1. If resistance is measured between any test points, replace the stator. 2. If results are OL, restart test again.		

Utility Voltage Test and Calibration

General Theory

When a Power Zone 200 controller is replaced, the calibration of utility voltage will need to be verified so the controller is aligned with the incoming utility source voltage. Calibration of utility voltage is done with the generator not running and the utility breaker set to ON.

Procedure A

NOTE: In the Field Pro app, “AN” correlates to Wire N1 and “BN” correlates to Wire N2.

- Gain access to the Customer Connections on the generator by removing the controller fascia.
- Set the DMM to measure AC voltage.
- Using the Field Pro app, navigate to the “Calibration” screen.
- Place the red meter lead in the WAGO block terminal for Wire N1 and the black meter lead to the neutral block. Record the reading.
 - If no or improper voltage is measured proceed to Procedure B.
- On the Field Pro app under the “Voltage Calibration” box, enter the recorded voltage under the “Measured Voltage” column for A to Neutral. Click the save icon.
- Place the red meter lead on one of the disconnected wire ends for Wire 0A at the brushes. Using the appropriate back probe, insert the other

lead to Wire 0A at the J7 connector and record the reading.

- On the Field Pro app under the “Voltage Calibration” box, enter the recorded voltage under the “Measured Voltage” column for B to Neutral. Click the save icon.
- Repeat Steps 6–7 for C to Neutral if 3-phase.
- Once completed, remove test equipment.
 - If utility voltage is still not sensed proceed to Procedure B.

Procedure B

- Set the DMM to measure AC voltage. Insert one meter lead into N1 at the WAGO block, and the other meter lead to N2 at the WAGO block. Record the reading. (If 3-phase repeat these steps for N1-N3, and N2-N3)
 - If 110-130 VAC is measured skip to Step 3.
- At the transfer switch board check for voltage at N1 to N2 with each meter lead. Record the reading. (If 3-phase repeat these steps for N1 to N3, and N2 to N3)
 - If 110– 130 VAC is measured repair the connections from the transfer switch to the generator.
 - If 110– 130 VAC is not measured check the fuses within the transfer switch or repair the transfer switch wiring.
- Using the appropriate back probes, connect a meter lead to J5-1 at the back of the controller, and the other meter lead to J5-2. Record the reading. (If 3 phase repeat these steps for J5-1 to J5-3 and J5-2 to J5-3)
 - If 110– 130 VAC is measured and the Power Zone 200 controller still does not detect utility voltage replace the controller and retest.
 - If 110– 130 VAC is not measured at J6 but is still measured at the WAGO block, replace the wiring and retest.

Voltage And Frequency Under Load Test

General Theory

It is possible for generator AC output frequency and voltage to be good at no-load, but they may drop excessively when electrical loads are applied. This condition can be caused by (a) overloading the generator, (b) loss of engine power or performance, or (c) a shorted condition in the stator windings or in one or more connected loads.

Procedure

1. Set the DMM to measure AC voltage.
2. Connect an AC frequency meter and the DMM meter leads across the stator AC power winding leads.
3. Start the engine. Let it stabilize and warm-up.
4. Using a clamp-on ammeter to verify the load current, apply electrical loads to the generator equal to the rated capacity of the unit. Record the frequency and the voltage.
 - a. If the DMM displays 60 Hz and approximately 240 VAC during full load, test passes.
 - b. If the DMM displays a frequency and voltage that dropped while under full load with an engine that is affected, refer to the Field Pro app for the applicable engine E-code.
 - c. If the DMM displays a frequency and voltage that dropped while under full load but rpm recovers to near 3600 rpm, proceed to [AVR PWM Output Test](#).
 - d. If the RPM dropped while under load and cannot recover, proceed to [Engine Cranks But Does Not Run, Lacks Power, Runs Rough, Hunts or Backfires](#).

Voltage Calibration

General Theory

When a Power Zone 200 controller is replaced, the calibration of both voltage and amperage will need to be verified so the controller is aligned with the generator's output. Calibration of voltage is done **without** the use of a load bank.

NOTE: Calibration between the generator and controller is preset from the factory, and does not need adjustment. Only when replacing a controller will this need to be checked.

Procedure

NOTE: In the app, "AN" correlates to Wire 11 and "BN" correlates to Wire 44.

1. Gain access to the main line circuit breaker (MLCB) on the generator by removing the controller fascia.
2. Set the DMM to measure AC voltage.
3. Set the MLCB to the "OFF" position.
4. Using the Field Pro app, navigate to the "Calibration" screen.
5. Place the red meter lead on the generator side of the MLCB terminal for Wire 11 and the black meter lead to the neutral block.

6. Start the generator in MANUAL and record the meter reading.
7. On the Field Pro app under the "Voltage Calibration" box, enter the recorded voltage under the "Measured Voltage" column for A to Neutral. Click the save icon.
8. Move the red meter lead to the generator side of the MLCB terminal for Wire 44 while leaving the black meter lead in place. Record the meter reading.
9. On the Field Pro app under the "Voltage Calibration" box, enter the recorded voltage under the "Measured Voltage" column for B to Neutral. Click the save icon.
10. Repeat Steps 8–9 for C to Neutral if 3-phase.
11. Once completed, turn the generator OFF and remove test equipment.

Radio and Field Pro Test

General Theory

The Generac Generator Connectivity Accessory - Cellular (GGCAC or *Radio*), connects to both Wi-Fi and Cellular to connect with Generac servers for remote monitoring, remote firmware updates, and other remote commands. The radio takes 3.8 VDC power from the controller via J2-1 and J2-4. It also uses RS232 communication with the controller via pins J2-2 and J2-3. The radio also provides Bluetooth® connection for use with the Field Pro app. Refer to for more information.

Field Pro Test

Refer to the steps below for troubleshooting various problems with Field Pro.

1. Verify Field Pro app is up to date. Retest.
2. Connect the device to Wi-Fi (if customer allows). Open Field Pro again.
3. Verify unit SN within Field Pro matches the unit decal.
 - a. If unit SN does not match, locate the SN printed on the radio and perform a device swap within Fleet with this radio SN and the SN printed on the unit sticker. Contact Technical Support for assistance. Retest.
4. Check grounds and fuses. Retest.
5. Disconnect J2 connector from the back of the controller. Wait 10 seconds before reconnecting J2 connector. Retest.
6. Perform [Power Cycle and Sync Test](#). Retest.
7. Try the "prepare for a future visit" feature within Field Pro. Retest.

8. Force the app on the device to close or quit, then reopen it. Retest.
9. Sign out of the account on the app and relog in. Retest.
10. Uninstall Field Pro and reinstall Field Pro. Retest.
11. Perform [Radio Test](#).

Field Pro Status

Field Pro Status Description	Meaning	Troubleshooting
Generator is ready to run	Unit is in Auto and will start during a utility outage	N/A
Utility Loss Delay	Unit is in Auto, it sensed the utility has lost voltage. It will start once the delay timer has expired.	Consult the Start up delay timer within the configuration menu within Field Pro.
Engine is Cranking	Unit is cranking the engine over attempting to start.	N/A
Generator is running	The unit is running	N/A
Generator is running in cooldown	The unit has detected utility and transferred back to utility. It is running without load to cool the engine down.	Consult the Cooldown timer within the configuration menu within Field Pro.
Generator is running with a warning	The controller has detected a warning E-Code but is still running.	Consult the E-Code table.
Generator is stopped with an alarm	The controller has detected a shutdown alarm E-Code and is stopped.	Consult the E-Code table.
Generator is stopped – start inhibit active	The generator is stopped and the controller will not allow the generator to start in any state.	- The engine is not stopped. - The oil pressure sensor is still detecting oil pressure. Proceed to Oil Pressure Switch Test. - The controller is performing an update wait 5 minutes if still in start inhibit active then proceed to Power Cycle and Sync Test.
Engine cranking paused.	Engine is in between crank attempts.	Wait 15 seconds. If the engine does not crank proceed to Power Cycle and Sync Test .
Generator is off	Generator is switched to off	N/A
ECM Firmware update in progress	The generator's ECM is updating	Wait 5 minutes. If message is still present, proceed to Power Cycle and Sync Test .
ECM Data request in progress	The PZ200 has requested data from the ECM.	N/A
Generator is running and has transferred.	The generator is powering the load.	Proceed to Utility Voltage Test and Calibration .

Field Pro Status Description	Meaning	Troubleshooting
Generator is exercising and has transferred	The generator is powering the load in exercise mode.	Consult the Transfer on exercise setting within the Exercise settings within the configuration menu within Field Pro.
Generator is running in warm up	The generator has started and is waiting for it to warm up.	Consult the Warmup timer within the configuration menu within Field Pro.

Table 8-3. Radio LED Light Pattern Description

Description	Pattern	Description	Troubleshooting
Solid	–	The device is fully functional.	None.
Pulsing	Slow On and Off once every 3.5 seconds	The device is booting and attempting to establish a cellular connection.	This is expected behavior.
LED off	–	The device is not powered.	Verify battery is connected.
			Verify cable harness is connected to device (threaded connection is fully seated), and is connected to the generator controller.
			Perform Radio Test
Blinking	1 Blink	There is an issue communicating with the generator controller.	Perform Radio Test
	2 Blinks	There is a cellular communication error*.	Perform Radio Test
			Inspect environment for any metal objects or other mechanical interference which may block cellular performance.
			Clear any debris which may be contributing to signal degradation, if possible.
			Communicate location of generator installation to Generac for further investigation of the area if problem persists.
	Fast On and Off 5 times per second	There is a Bluetooth device connected to this radio	If a known device is connected this is expected behavior. If no device is connected, proceed to Field Pro Test .

* It may take up to 15 minutes to establish a cellular connection.

Radio Test

Procedure

- Unplug J6 connector and remove fuse from Power Zone 200 controller. Disconnect J2 connector. Set DMM to measure resistance. Use one probe to touch pin 2 (J2 top left pin) and the other probe to touch pin 4 (J2 bottom left). Record the resistance.
 - If less than 4.5 kohms is measured replace controller and retest.
- With the DMM measuring resistance use one probe to touch pin 3 (J2 bottom right pin) and the other probe to touch pin 4 (J2 bottom left). Record the resistance.
 - If less than 750 kohms is measured replace the controller and retest.
- Reinsert the fuse in the Power Zone 200 controller and reconnect J6 connector. Using the appropriate back probes connect the red lead to J2-1 (blue wire) and connect the black lead to J2-4 (black wire). Record the results.
 - If 3.5–4V is measured skip to Step 5.
- Unplug J2 connector. Connect the red lead on the back of the controller to pin J2-1 (top right) and

connect the black lead to J2-4 (black wire) Record the results.

- a. If 3.5– 4V is not measured, replace the controller and retest.
5. With both J2 and the radio connector disconnected set the DMM to measure resistance/continuity. Verify the wire color matches the position on the connector according to [Power Zone 200 Controller](#). Measure the resistance of each of the 4 wires between J2 and the radio connector. Verify the wire corresponds to the correct position on the radio connector according to the wiring diagram (they are not 1-1, 2-2, etc). The pin to the right of the notch corresponds to pin 1, pin 2, 3, 4 is counterclockwise from pin 1. Back probe the connector at J2 and gently front probe the pins in the radio connector. Measure each result.
 - a. If more than 5 ohms was measured on any of the 4 wires or if the wires are out of place, replace the wiring harness and retest.
6. Plug all connectors back into their respective locations. Set the DMM to measure DC voltage. Using the appropriate back probes, connect the red DMM meter lead to J2-2 (brown wire) and connect the black meter lead to J2-3 (white wire). Record the results.
 - a. If 5.2–5.6V is measured, replace the radio and retest.
7. Unplug J2 connector. On the controller connect the red meter lead to the pin for J2-2 and the black meter lead to the pin for J2-3. Record the results.
 - a. If 5.2– 5.6V is not measured, replace the controller and retest.
8. If issues are still present, contact Technical Support.

Section 9: Engine Control Module

Engine Control Components

Engine Control Module

The engine is controlled by an engine control module (ECM). The ECM controls fuel flow, air flow and ignition timing to maintain optimal engine performance. It does this by reading values from multiple sensors located on various engine components, interpreting the data and adjusting engine fuel flow, timing and intake air flow to maintain proper speed, performance and emissions characteristics

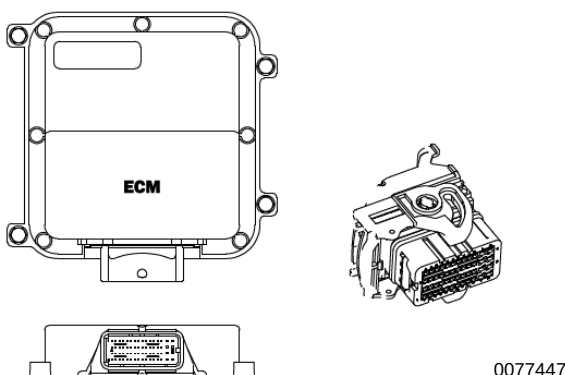


Figure 9-1. Engine Control Module

Sensors are used to monitor engine performance:

- **TTMAP** – Throttle Position/Manifold Absolute Pressure/Intake Air Temperature
- **CRKS** – Crankshaft Position
- **OS** – Oxygen Sensor
- **OLS** – Oil Level Sensor
- **OPS** – Oil Pressure Switch

- **MPS** – Mixer Position Sensor
- **FPS** – Fuel Pressure Sensor
- **EHT** – Engine Head Temperature Sensor

Engine performance and speed are controlled by:

- **ETC** – Electronic Throttle Control Valve
- **EMC** – Electronic Mixer Control Valve
- **IC** – Ignition Coil

Other control devices:

- **MPR** – Master Power Relay
- **SOV** – Fuel Shut Off Valve

The ECM is located behind the Power Zone 200 controller in the customer connection area. A large single connector connects the ECM to the engine harness and also to the generator main control panel. Communication to the main control panel is achieved via a digital communications link CANBUS 2.0 J1939.

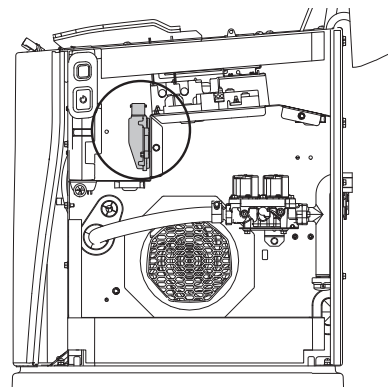
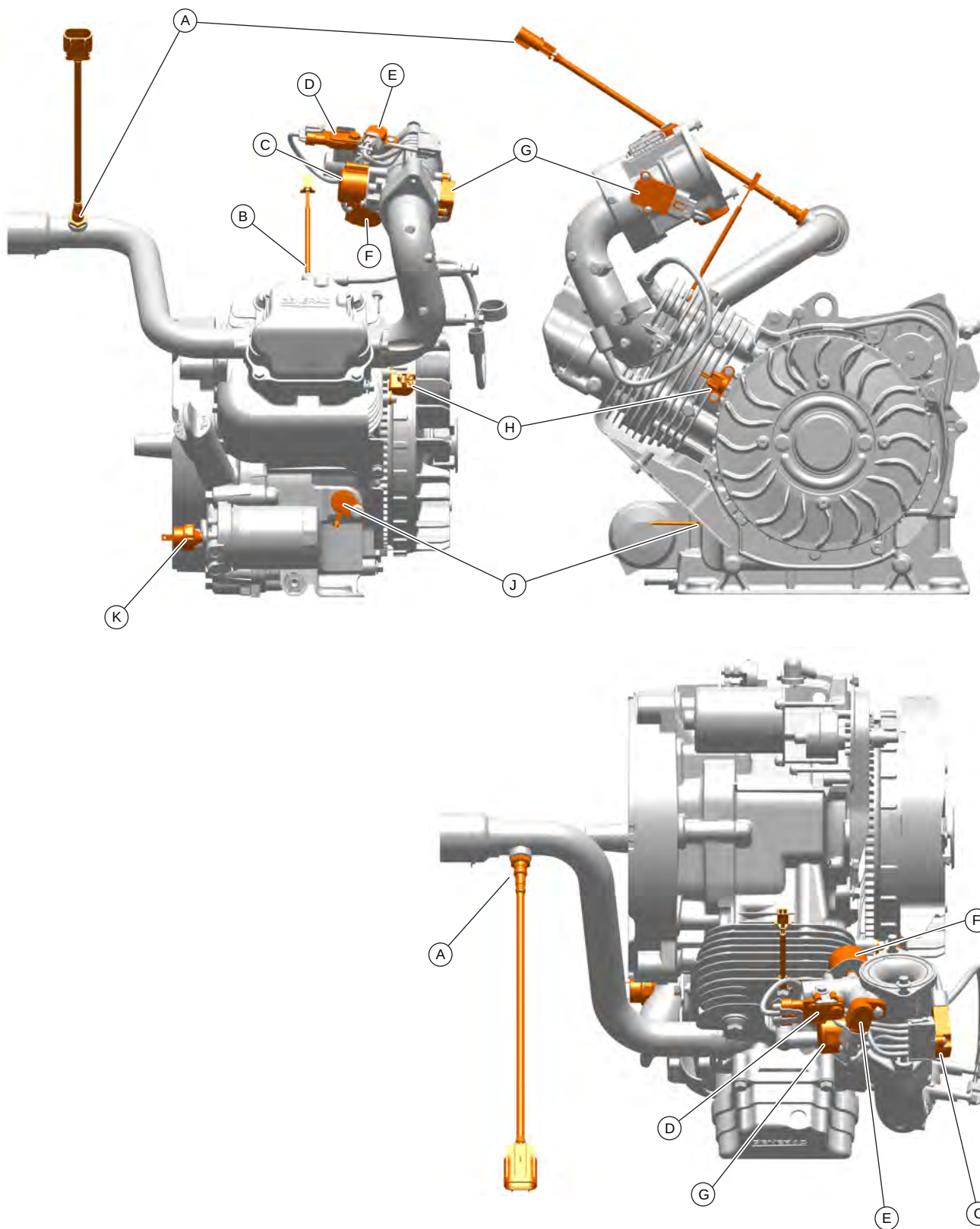


Figure 9-2. ECM Location

Engine Control Location

459 cc Single Cylinder Engine

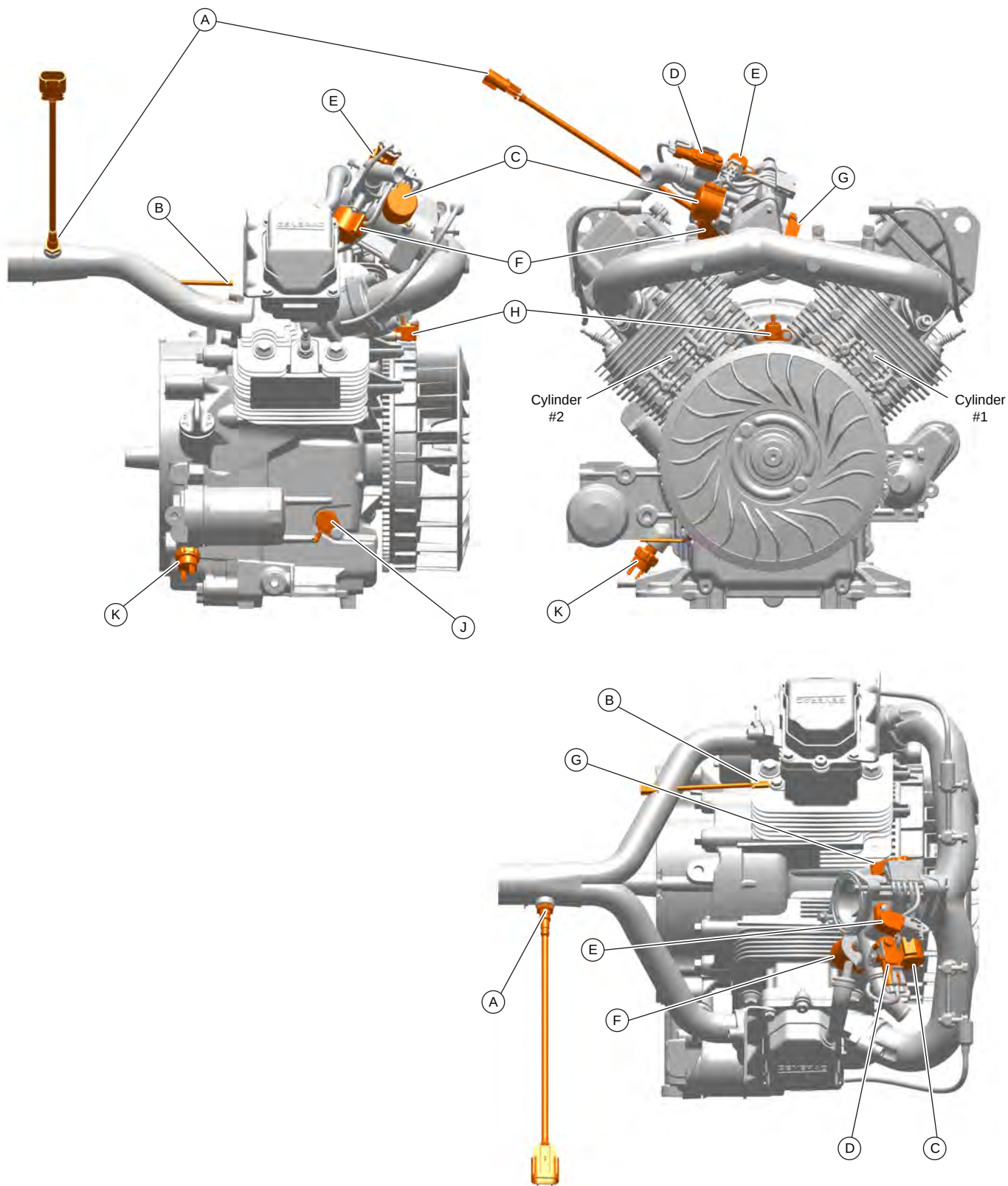


021191

Figure 9-3. Engine Control Locations

A	OS – Oxygen sensor
B	EHT – Engine Head Temperature Sensor
C	ETC – Electronic Throttle Control Valve
D	FPS – Fuel Pressure Sensor
E	MPS – Mixer Position Sensor
F	EMC – Electronic Mixer Control Valve
G	TTMAP – Throttle Position, Temperature and Manifold Absolute Pressure Sensor
H	CRKS – Crankshaft Position
J	OLS – Oil Level Sensor
K	OPS – Oil Pressure Switch

817 cc V-Twin Engine

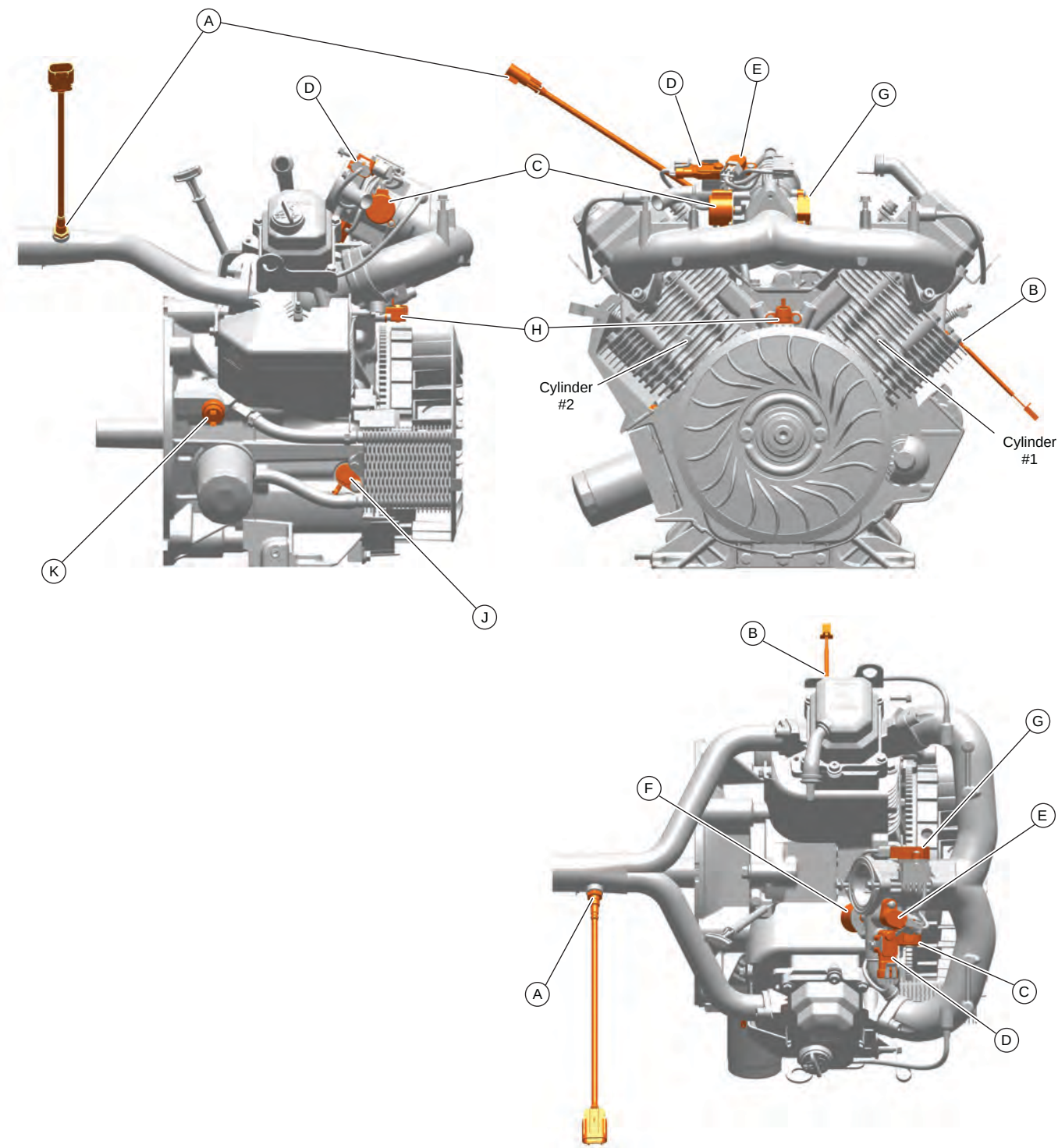


021192

Figure 9-4. Engine Control Locations

A	OS – Oxygen sensor
B	EHT – Engine Head Temperature Sensor
C	ETC – Electronic Throttle Control Valve
D	FPS – Fuel Pressure Sensor
E	MPS – Mixer Position Sensor
F	EMC – Electronic Mixer Control Valve
G	TTMAP – Throttle Position, Temperature and Manifold Absolute Pressure Sensor
H	CRKS – Crankshaft Position
J	OLS – Oil Level Sensor
K	OPS – Oil Pressure Switch

997 cc V-Twin Engine



021193

Figure 9-5. Engine Control Locations

A	OS – Oxygen sensor
B	EHT – Engine Head Temperature Sensor
C	ETC – Electronic Throttle Control Valve
D	FPS – Fuel Pressure Sensor
E	MPS – Mixer Position Sensor
F	EMC – Electronic Mixer Control Valve
G	TTMAP – Throttle Position, Temperature and Manifold Absolute Pressure Sensor
H	CRKS – Crankshaft Position
J	OLS – Oil Level Sensor
K	OPS – Oil Pressure Switch

Theory Of Operation – ECM ON/OFF

The ECM receives constant power via Wire 877 from the Controller. The ECM then provides ground to the Master Power Relay (MPR). The “always on” design allows for engine sensor testing to be performed even when the Generator is not running.

NOTE: At engine shutdown, the ECM will be powered off for approximately 2.5 seconds, then powered back on. This allows for the throttle/mixer assembly to perform a sweep and set itself for the next engine start cycle.

The only time the engine will start is when the ECM sees the command from the Controller to start, which comes through the CAN bus network and if Wire 877 has power. The Controller and ECM will continue to communicate via the CAN bus network during the engine run cycle. If the ECM senses a problem or malfunction with the engine, a fault code will be triggered (depending on the severity, the engine will shut down). The ECM will communicate the fault to the Controller where the code can be retrieved using the Field Pro app.

When the Controller commands the engine to stop, it will send a command to the ECM for the engine to shut down via the CAN bus network. The ECM will then react accordingly. Engine shutdown will also occur if power to Wire 877 is turned off.

Wiring Harness

The wiring harness connects the ECM directly to the control panel and to the various engine sensors and controllers.

Master Power Relay

See [Figure 9-6](#). The master power relay (MPR), located under the control panel, supplies 12 VDC to:

- Ignition Coils
- Oxygen Sensor
- ECM Drivers
- Oil Pressure Switch

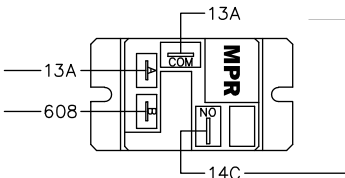


Figure 9-6. Master Power Relay

Starter Contactor Relay

See [Figure 9-7](#). The starter contactor relay (SCR) provides a safe and controlled method of energizing the solenoid located on the starter. The controller is responsible for energizing the relay when the start command is given.

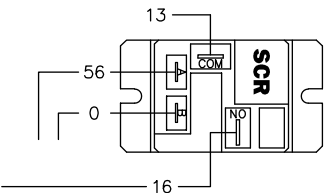


Figure 9-7. Starter Control Relay

Throttle Body / Mixer and Electronic Throttle / Mixer Control

See [Figure 9-8](#). The electronic throttle control valve (ETC) (Item C) and electronic mixer control valve (EMC) (Item D) are integrated with the throttle body / mixer assembly. The ETC (which controls air flow into the engine) and EMC (which controls fuel flow into the engine) receive signals from the engine control module (ECM) to open and adjust the valves in the throttle body and allow air and fuel to flow into the combustion chamber.

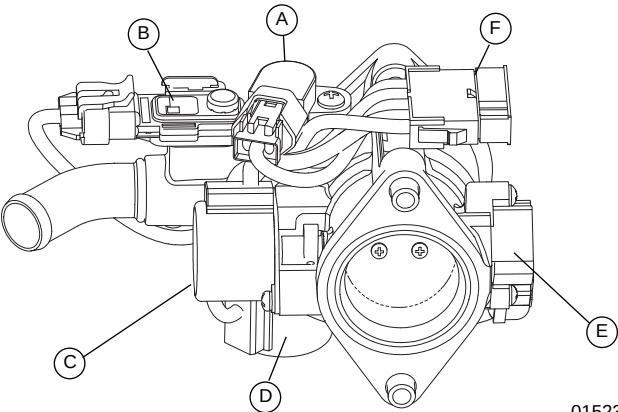


Figure 9-8. Throttle / Electronic Control Valve

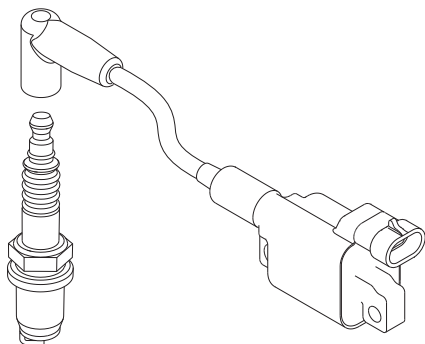
A	Mixer Position Sensor (MPS)
B	Fuel Pressure Sensor (FPS)
C	Electronic Throttle Control Valve (ETC)
D	Electronic Mixer Control Valve (EMC)
E	Throttle Position, Temperature and Manifold Absolute Pressure Sensor (TTMAP)
F	Engine Throttle/Mixer Connector (ETMC) Connector

Throttle Position, Temperature and Manifold Absolute Pressure Sensor (TTMAP)

See . The throttle position, temperature and manifold absolute pressure sensor (TTMAP) (Item E) monitors the throttle position, the temperature of the air entering the engine, and the intake manifold absolute pressure. These parameters are sent to the ECM to assist in maintaining optimal spark and fuel delivery while the unit is running.

Ignition Coils

See [Figure 9-9](#). The ignition coils (IC1 and IC2) receive signal from the ECM to generate and deliver a high voltage signal to the spark plugs.

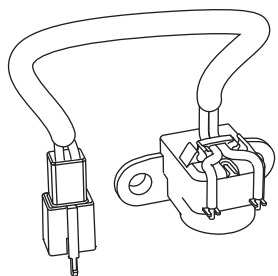


014997

Figure 9-9. Ignition Coil

Crankshaft Position Sensor (CRKS)

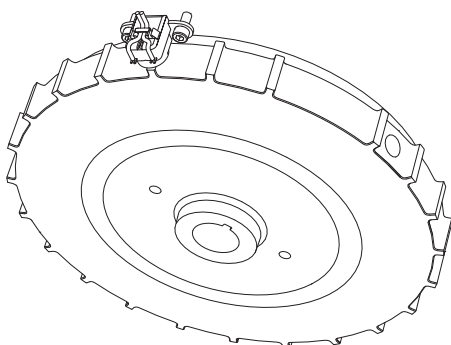
See [Figure 9-10](#). The crankshaft position sensor (CRKS) senses the timing teeth on the flywheel as it rotates.



014999

Figure 9-10. Crankshaft Position Sensor

See [Figure 9-11](#). The flywheel has a 24 minus one tooth design where one position has no tooth present. The CRKS senses this gap during rotation and sends that signal to the ECM to determine the position of the crankshaft and camshaft.

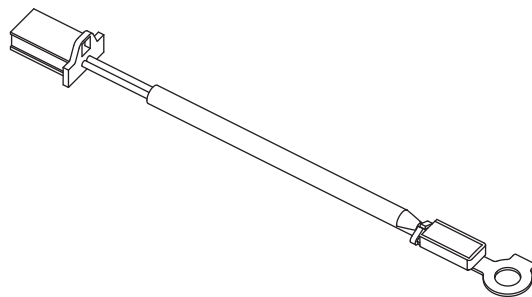


015000

Figure 9-11. Crankshaft Position Sensor and Flywheel

Engine Head Temperature Sensor (EHT)

See [Figure 9-12](#). The engine head temperature sensor (EHT) is located on cylinder #1 and secured in place by a bolt. The EHT monitors engine head temperature. This data is sent to the ECM to assist in maintaining optimal spark and fuel delivery while the unit is running.

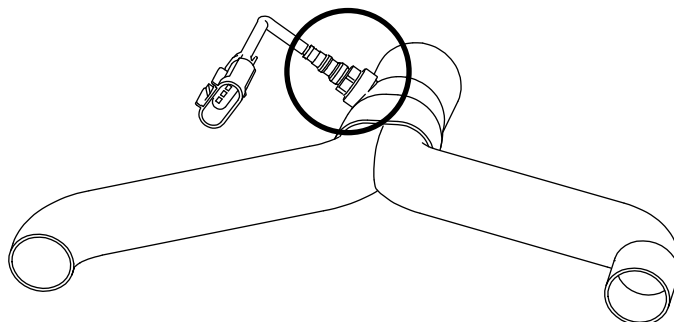


015001

Figure 9-12. Engine Head Temperature Sensor

Oxygen Sensor (OS)

See [Figure 9-13](#). The oxygen sensor (OS), located on the exhaust manifold, sends data to the ECM to assist in maintaining optimal fuel delivery while the unit is running. The OS is heated by voltage from the ECM for optimum performance. The sensor sends a voltage signal indicating the oxygen level in the exhaust. The ECM uses that data to determine how much fuel to add for optimal emissions and power.



015237

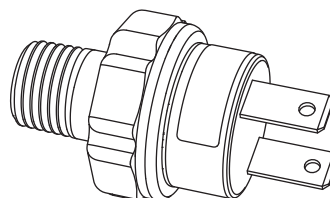
Figure 9-13. Oxygen Sensor

Oil Pressure Switch (OPS)

See [Figure 9-14](#). The engine is equipped with a low oil pressure switch (OPS). The sensor works differently depending on the engine.

- 459cc 10 kW - The switch is a normally closed (NC) type that closes when the oil pressure drops below the set threshold.
- 817/997cc 14–28 kW - The switch is a normally open (NO) type that opens when the oil pressure drops below the set threshold.

Both types are connected to the ECM which will shutdown the engine and activate an alarm on a low pressure signal.



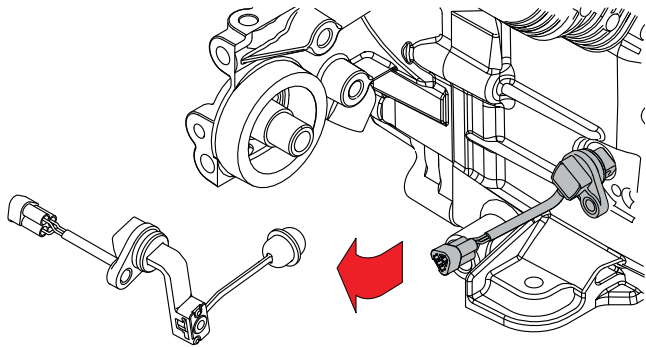
015003

Figure 9-14. Oil Pressure Switch

Generator	PSI	Type
459cc 10 kW	5 PSI	Normally Closed (NC)
817 / 997cc 14–28 kW	10 PSI	Normally Open (NO)

Oil Level Sensor (OLS)

See [Figure 9-15](#). The engine is equipped with a float style oil level sensor (OLS) to provide oil level reference to the ECM. The sensor will display an oil level within the field pro app. It will provide warnings if the ECM detects out of range sensor measurements. If the oil is low the controller will give a low oil warning.



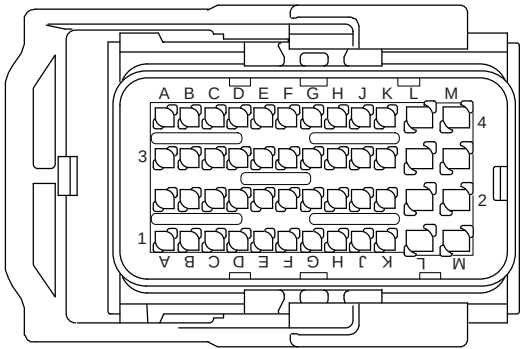
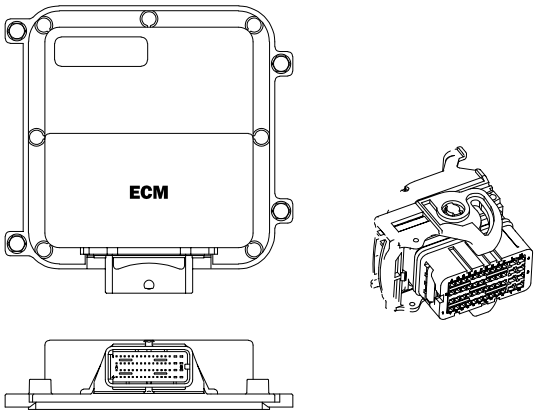
00080933

Figure 9-15. Oil Level Sensor

Fuse

The fuse protects the ECM against excessive current. If the fuse has blown, engine cranking and operation will not be possible. The ECM will not be powered. Should fuse replacement become necessary, use only an equivalent 15 amp replacement fuse. Refer to [ECM Harness Fuse Testing](#) for diagnostic help.

Engine Control Module



CONNECTOR END VIEW

020974

Figure 9-16. Generac Engine Control Module (ECM) and Harness Connector

Pin	Wire	To	Pin	Wire	To	Pin	Wire	To	Pin	Wire	To
1A	605A	ETMC-13	2A	605C	ETMC-15	3A	605D	ETMC-16	4A	743	J3-7
1B	605B	ETMC-14	2B	606	ETMC-7	3B	766B	ETMC-4	4B	744	J3-8
1C	79A	CRKS-A	2C	766A	ETMC-6	3C	797	ETMC-1	4C	566	OPS
1D	79B	CRKS-B	2D	608	MPR-B	3D	—	—	4D	—	—
1E	770A	ETMC-9	2E	—	—	3E	—	—	4E	—	—
1F	771A	ETMC-11	2F	607	OS-B	3F	—	—	4F	877	J3-9
1G	770B	ETMC-10	2G	754	ETMC-3	3G	—	—	4G	—	—
1H	771B	ETMC-12	2H	—	—	3H	—	—	4H	—	—
1J	13A	SPLICE 2	2J	899	OLS-C	3J	—	—	4J	804	OS-C
1K	14C	SPLICE 6	2K	714	EHT-1	3K	820A	SPLICE 7	4K	820G	SPLICE 8

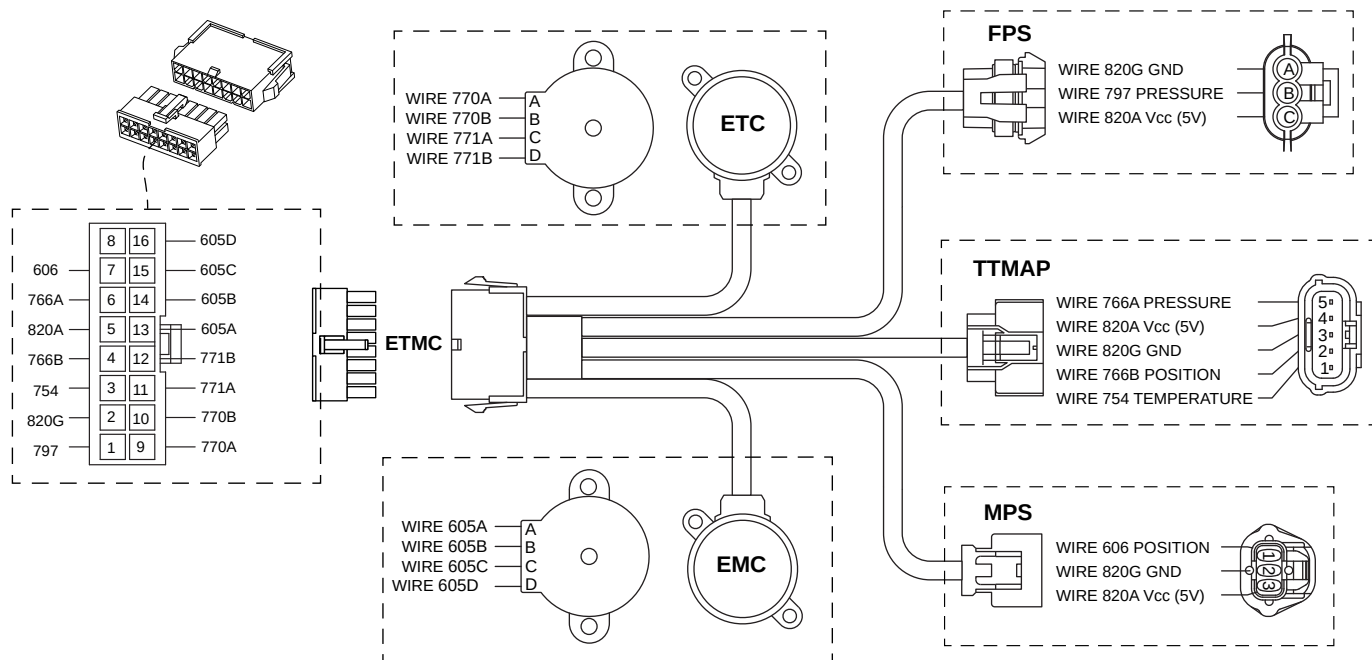
Pin	Wire	To
1L	0	GND
1M	452	IC2-C

Pin	Wire	To
2L	—	—
2M	—	—

Pin	Wire	To
3L	610	SOV2
3M	0	GND

Pin	Wire	To
4L	0	GND
4M	451	IC1-C

ETMC Connector Harness



020950

Figure 9-17. ETMC Connector Harness

This page intentionally left blank.

Section 10: Engine Control Module Troubleshooting

Introduction

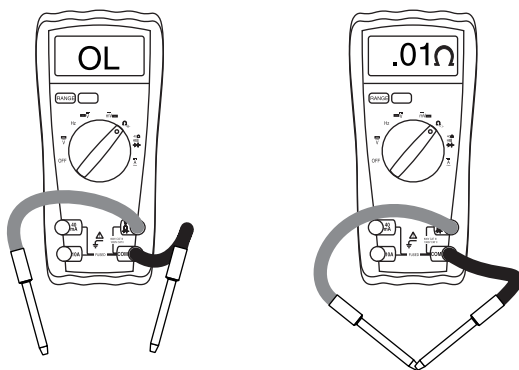
This section familiarizes the service technician with acceptable procedures for the testing and evaluation of various problems that can occur on standby generators with air-cooled engines.

Some test procedures in this section require the use of specialized test equipment, meters, or tools. Most tests can be performed with a digital multimeter (DMM). An AC frequency meter is required where frequency readings must be taken.

Testing and troubleshooting methods covered in this section are not exhaustive. No attempt has been made to discuss, evaluate and advise the home standby service trade of all conceivable ways in which service and trouble diagnosis must be performed. Accordingly, anyone who uses a test method not recommended herein must first satisfy themselves that the procedure or method they have selected will jeopardize neither their own nor the products safety.

Figure 10-1 shows the DMM in two different states. The left DMM indicates an OPEN circuit or INFINITY. The right DMM indicates a dead short or CONTINUITY. Throughout the troubleshooting, refer back to **Figure 10-1** as needed to understand what the meter is indicating about the particular circuit that was tested.

NOTE: CONTINUITY is equal to 0.01 ohms of resistance or a dead short.



002397

Figure 10-1. INFINITY (Left) and CONTINUITY (Right) Meter Readings

Safety

Service personnel who work on this equipment should be aware of the dangers of such equipment. Extremely high and dangerous voltages are present that can kill or cause serious injury. Gaseous fuels are highly explosive and can ignite by the slightest spark. Engine exhaust gases

contain deadly carbon monoxide gas that can cause unconsciousness or even death. Contact with moving parts can cause serious injury. The list of hazards is seemingly endless.

When working on this equipment, use common sense and remain alert at all times. Never work on this equipment while you are physically or mentally fatigued. If you do not understand a component, device or system, do not work on it.

Engine/DC Troubleshooting

It is good practice to continue asking questions during the troubleshooting process. Asking some of these questions during evaluation may help identify a problem more quickly.

- What is the generator doing?
- What is the fault that the generator is shutting down for?
- After the fault occurred, what was the Field Pro app displaying?
- Is there another Alarm in the log just previous to the shutdown?
- Is the fault causing the shutdown a symptom of another problem?
- Does the generator have the same fault consistently, and when does it occur?
- What was the generator supposed to do?
- Who is controlling it?
- Exactly what is occurring?
- When is it happening?
- Why would this happen?
- How would this happen?
- What type of test will either prove or disprove the cause of the fault?

AC Battery Charger Test

General Theory

The generator depends on the 12 VDC circuit to operate many components in the generator. The 12V battery needs to remain charged to be prepared for a power outage event. When the generator is sitting in utility, the Power Zone 200 controller handles the battery charging through an internal charger. The charging rate will vary

depending on ambient temperature and battery voltage. The battery problem and charger warning alarms only trigger after an 18 hour charge cycle. If the charger still pushes over 600mA of charge current after an 18 hour charge cycle it will trip the 'Battery Problem' warning. This will clear with a power cycle or if the charger current returns to normal after the next 18 hour charge cycle. If the battery voltage stays below 12.5V after an 18 hour charge cycle it will trip the 'Charger Warning' warning. This will clear with a power cycle or, if the battery voltage is 12.6V or higher after the next 18 hour charge cycle.

NOTE: The internal battery charger receives AC voltage from the load side of the transfer switch whether the generator is operating or in standby.

Procedure

1. Perform the [Battery and Cables Test](#) to verify the battery condition.
 - a. If the battery and cables pass inspection, proceed to Step 2.
 - b. If the battery or cables fail the inspection, repair/replace as necessary and retest.
2. Verify the E-Code(s) present.
 - a. If E-code 2760 is present, proceed to **Step 3**.
 - b. If E-code 2760 is NOT present, proceed to Step 5.
3. Set DMM to measure DC voltage.
4. Place the red meter lead on the battery positive terminal and the black meter lead on the battery negative terminal.
 - a. If battery voltage is below 16.4 VDC, proceed to **Step 5**.
 - b. If voltage is above 16.5 VDC, disconnect J6 connector and retest.
 - c. If the battery voltage falls below 14.0 VDC, replace the controller and retest.
 - d. If the voltage does not fall, replace the battery and retest.
5. Set the DMM to measure AC voltage.
6. Using the appropriate back probes, insert the red meter lead to the T1 wire at the J6 connector on the back of the controller and the black meter lead to the 00 Wire at the J6 connector. Record the reading.
 - a. If 120 VAC is not measured, proceed to the [AC Input Voltage Test](#).
7. Set DMM to measure DC voltage.
8. Place the red meter lead on the battery positive terminal and the black meter lead on the battery negative terminal.
 - a. If battery voltage is below 12.6 VDC, proceed to Step 13.
9. Turn OFF the fuel source to the generator.
10. Set Controller to MANUAL.
 - a. Allow generator to go through 5 crank cycles.
 - b. Set generator to OFF.

NOTE: This step may cause an E-code of 1100 for Overcrank. Clear this fault and proceed to Step 11.
11. Using the Field Pro app, verify the Battery Charging status shows "Charging". Check the Battery Voltage display in the app.
 - a. If the battery voltage shows less than 12.9 VDC, replace the controller and retest.
12. Turn ON the fuel source to the generator.
13. Locate connector J4 on the controller. Leaving the connector connected, use the appropriate back probes and connect the red meter lead to one of the Wire 13 pins and the black lead to one of the Wire 0 pins.
 - a. If 12.8 VDC or less is measured, check the connections and pins in the J4 connector for damage. If the pins are found to be damaged, replace the wiring harness and/or controller as necessary and retest. Check all harness ground and the main battery ground. If the harness and controller pins pass inspection, replace the controller and retest.
14. Place the red meter lead on Wire 13 at the Starter Contactor (SC) and the black meter lead to the negative battery terminal.
 - a. If 12.8 VDC or less is measured check all harness ground and the main battery ground. Replace the wiring harness and retest.
15. Place the red meter lead on the battery positive terminal and the black meter lead on the battery negative terminal.
 - a. 12.9 VDC or greater is measured, the battery charger is working and test is complete.
 - b. If 12.8 VDC or less is measured check all harness ground and the main battery ground. Replace the wiring harness and retest.

AC Input Voltage Test

General Theory

AC power needs to be provided to the controller to supply the internal battery charger with voltage to charge. This power comes from the T1 circuit that is in the transfer switch. If this is missing, the battery will not charge while in Standby.

NOTE: The transfer switch breaker needs to be switched to ON and power available to the breaker in order to power the T1 circuit.

Procedure

1. Set the DMM to measure AC voltage.
2. Using appropriate back probes, insert the red meter lead into the T1 Wire at the J6 connector at the controller. Insert the black meter lead into the 00 Wire at the J6 connector at the controller. Record the reading.
 - a. If 120 VAC is measured, voltage is available for battery charging/transfer signal. Test is completed.
3. See [Figure 10-2](#). At the customer connection block, locate the T1 and 00 terminal connections at the WAGO block.

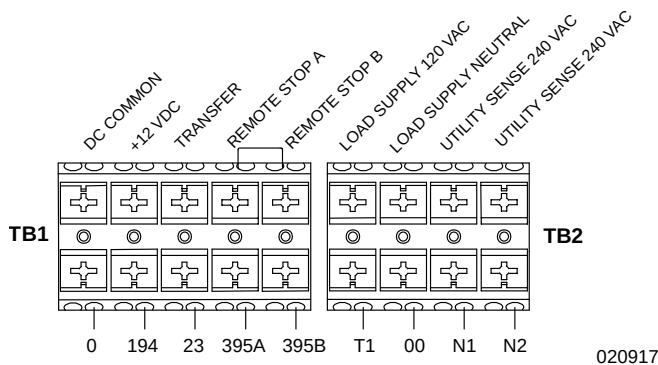


Figure 10-2. Customer Connection Block

4. Insert the red meter lead into the T1 Wire and the black meter lead into the 00 Wire. Record the reading.
 - a. If 120 VAC is present, proceed to Step 5.
 - b. If 120 VAC is not measured, proceed to **Step 6**.
5. At the controller locate J6 connector. Unplug J6 connector or keep plugged in as needed. Using the appropriate back probes, probe the blue wire with the red meter lead and the white wire with the black meter lead. Record the reading.
 - a. If 120 VAC is present and E-code 2780 is still present when J6 is plugged back in, replace the controller and retest.
 - b. If 120 VAC is present and E-code 2780 is not present proceed to the [Stator Battery Charger \(Voltage Regulator\) Test](#). If this test passes, replace the controller and retest.
 - c. If 120 VAC is not present, plug everything back in and repeat Step 4. If 120 VAC is present replace the harness and retest.
6. In the transfer switch, locate the terminals for the T1 Wire and the 00 Wire on the SACM/FCM. Connect the red meter lead to the T1 terminal and

the black meter lead to the 00 terminal. Record the reading.

- a. If 120 VAC is present, repair the T1/00 wire between the transfer switch and the generator and retest.
7. Remove the T1 fuse from the SACM/FCM and test integrity of fuse.
 - a. If fuse is blown, replace fuse and retest. If fuse continues to blow, check for short to ground in T1/00 circuit.
8. Verify there is 120 VAC available to the fuse holder.
 - a. If 120 VAC is being measured at the fuse holder, replace the SACM/FCM and retest.
 - b. If 120 VAC is not measured, repair the wire between the SACM/FCM and the transfer switch breaker and retest.

Battery and Cables Test

General Theory

Battery power is used for three key purposes, (a) crank the engine, (b) power the Power Zone 200 controller, and (c) power the ECM. Low or no battery voltage can result in failure of the engine to crank. Additionally, if there is a loose connection or corrosion associated with a wire (positive or negative), battery voltage may be present, but because of the high resistance, will not allow current to flow. Electrical voltage drop varies according to current flow. Unless the circuit is operated so current flows through it, voltage drop cannot be measured. To properly measure voltage drop, a crank attempt will need to be performed. This test will determine whether the battery, battery cables, or both are at fault.

NOTE: The integral battery charger in the Power Zone 200 controller is not designed to recharge a dead battery.

Procedure A

1. Inspect battery cables at both battery posts, the back engine foot, and the starter terminals.
 - a. If cable clamps or terminals are loose/corroded, tighten/clean away all corrosion and retest.
- NOTE:** If corrosion cannot be cleaned or eliminated, replace the component in question.
2. Verify all cable clamps are tight. The red battery cable from the starter contactor (SC) must be securely attached to the positive (+) battery post. The black cable from the frame ground stud must be tightly attached to the negative (-) battery post.
 - a. If any connections are found to be loose/corroded, tighten/clean away all corrosion and retest.

Procedure B

1. Remove the 7.5A fuse from the controller and inspect fuse.
 - a. If fuse is blown, replace fuse and recharge battery on external battery charger. Once battery is fully charged, restart test.
2. Disconnect the J6 connector to disable the battery charger.
3. Set the DMM to measure DC voltage.
4. Connect the red meter test lead to the positive battery post and connect the black meter test lead to the negative battery post.
5. Measure and record the OCV (Open Circuit Voltage).
 - a. If battery voltage is 12.1 VDC or less, or if engine does not crank (turn over), proceed to [Procedure C](#).
 - b. If battery voltage is 12.2–12.6 VDC or above and the engine does or does not crank, reinstall controller fuse, reconnect J6 connector, then proceed to [Procedure D](#).

Procedure C

1. Perform a Conductance Test with a Conductance Type Battery Tester (follow the conductance battery tester manufacturer's instructions). Test results should not show anything lower than 75% of the battery's rated CCA.
 - a. If battery CCA is 75% or less, replace battery with new and retest.
 - b. If the above test results have been verified good, proceed to **Procedure D**.

Procedure D

1. Reconnect all unplugged connectors and the battery. While feeling the SCR, attempt to run the unit. Verify there are no prior alarms that would prevent a run, such as CANBus, etc.
 - a. If the SCR makes a click, disconnect the battery, unplug Wires 13 and 16 from the SCR and skip to Step 5.
2. Disconnect the battery. Disconnect all wires on the SCR. Set the DMM to measure resistance. Using the appropriate back probes connect a meter lead to Wire 56 at J4-12 and the other meter lead to the connector at the SCR for Wire 56. Record the results.
 - a. If more than 5 ohms is measured, replace the harness and retest.
3. Set the DMM to measure resistance. Connect a meter lead to Wire 0 at the SCR and the other

meter lead to the negative battery cable. Record the results.

- a. If more than 5 ohms is measured, replace the harness and retest.
4. Set the DMM to measure resistance. Measure the resistance across the 2 bare pins on the SCR where Wires 56 and 0 were disconnected. Record the results.
 - a. If less than 130 ohms or greater than 165 ohms is measured, replace the SCR and retest.
5. Set the DMM to measure resistance. Connect a meter lead to Wire 13 at the SCR and the other meter lead to the positive battery cable. Record the results.
 - a. If more than 5 ohms is measured, replace the harness and retest.
6. Set the DMM to measure resistance. Connect a meter lead to Wire 16 at the SCR and the other meter lead to the starter motor solenoid Wire 16. Record the results.
 - a. If more than 5 ohms is measured, replace the harness and retest.
7. Reconnect the battery and all wires. With DMM set to measure DC voltage place the red probe to the starter contactor and the negative to battery negative. Attempt a manual run.
 - a. If 12 VDC is measured at the starter, replace the starter and retest.
 - b. If 12 VDC is not measured, replace the SCR and retest.

Procedure E

NOTE: Engine must crank to properly measure voltage drop. A crank attempt will need to be performed. If the engine does not crank, proceed to the [Starter Control Relay Test](#) before moving to Procedure D.

1. Turn OFF the fuel source to the generator to prevent startup.
2. Set the DMM to read DC voltage.
3. Set the controller to MANUAL. Measure and record the voltage at test points V1, V2, V3, and V4 as depicted in [Figure 10-3](#) and [Figure 10-4](#).

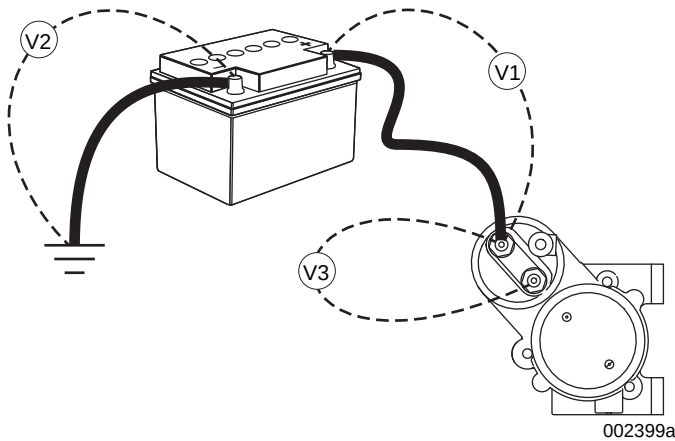


Figure 10-3. Starter Circuit Voltage Drop Test

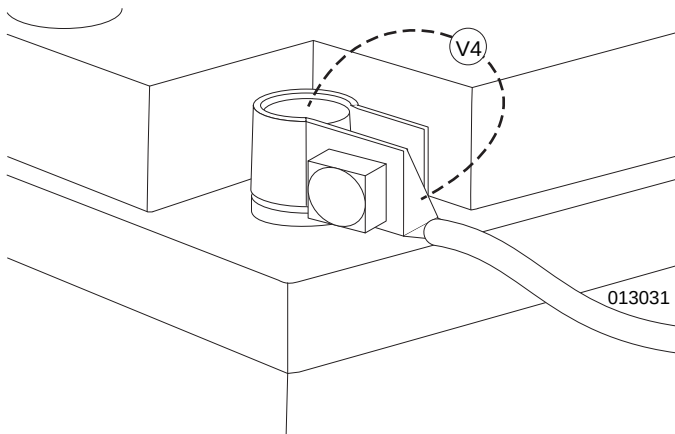


Figure 10-4. Starter Circuit Voltage Drop Test

4. If any of the voltages are greater than shown in [Table 10-1](#), repair or replace the component/harness as needed and retest.

Table 10-1. Maximum Voltage Readings

DC Volts	Connection	Test
0.20-0.30	Across the cable	(V1 & V2)
0.20-0.30	Across the starter contactor	(V3)
0.00-0.10	Battery Post to Battery Terminal	(V4)

Battery Potential/Power Output 1 Test

General Theory

The ECM requires DC voltage to operate. DC voltage is provided by the generator battery. Once the ECM is powered up, it will monitor the battery voltage in the system. It does this to protect itself from damage that can occur from an overvoltage or undervoltage condition. If the voltage is outside of the normal range (10–15 VDC), it will trigger a fault to shut the generator down.

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Check the connections at the battery.
 - a. If the connections are found to be loose or corroded, clean and tighten connections and retest.
3. Perform the [Battery and Cables Test](#).
 - a. If the battery fails the test, replace the battery and retest.
4. Perform the [AC Battery Charger Test](#).
 - a. If the AC battery charger fails the test, repair as necessary and retest.
5. Perform the [Stator Battery Charger \(Voltage Regulator\) Test](#).
 - a. If the stator battery charger fails the test, repair as necessary and retest.
6. Set the DMM to measure DC voltage.

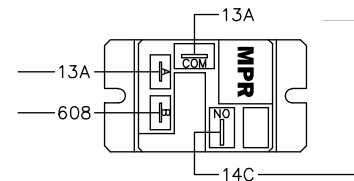


Figure 10-5. Master Power Relay (MPR)

7. See [Figure 10-5](#). Using the appropriate back probe, connect the red meter lead to Wire 13A at the COM terminal. Connect the black meter lead to the battery negative terminal. Record the reading. Move the red meter lead to Wire 13A at the Pin A terminal and record that reading.
 - a. If battery voltage is measured at both wires, proceed to **Step 9**.
 - b. If battery voltage is measured at only 1 wire, replace the wiring harness and retest.
 - c. If battery voltage is not measured at either wire, proceed to **Step 8**.
8. Check the 15A fuse located in line with Wire 13.
 - a. If the fuse is good, replace the wiring harness and retest.
 - b. If the fuse is blown, replace and retest. If the fuse continues to blow, proceed to [ECM Harness Fuse Testing](#).
9. Set the DMM to measure continuity.
10. Disconnect the battery.
11. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.

- a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
12. Disconnect all 4 wires from the MPR. Visually inspect all the terminals and spade connectors for looseness/damage.
 - a. If the terminals/spade connectors are found to be loose/damaged, replace the wiring harness and/or MPR and retest.
13. Place one meter lead on the MPR terminal for Wire 608 and the other meter lead on the terminal for Wire 13A from Terminal A at the MPR. Record the reading.
 - a. if less than 130 ohms or greater than 165 ohms is measured, replace MPR and retest.
14. Connect one meter lead to the disconnected 13A Wire from the COM terminal at the MPR. Using the appropriate back probe, connect the other meter lead to the 13A Wire at the ECM. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace wiring harness and retest.
15. Connect one meter lead to the disconnected 13A Wire from terminal A at the MPR. Using the appropriate back probe, connect the other meter lead to the 13A Wire at the ECM. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace wiring harness and retest.
16. Connect one meter lead to the disconnected 608 Wire at the MPR. Using the appropriate back probe, connect the other meter lead to the 608 Wire at the ECM. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace wiring harness and retest.
17. Connect one meter lead to the disconnected 14C Wire at the MPR. Using the appropriate back probe, connect the other meter lead to the 14C Wire at the ECM. Record the reading.
 - a. If resistance measures less than 5 ohms, replace the ECM and retest.
 - b. If resistance measures greater than 5 ohms or reads OL, replace wiring harness and retest.

CAN bus Test

General Theory

See [Figure 10-6](#). A CAN bus network is a high-speed communication network that allows modules to “speak” with each other. Each module has a set internal resistance that each module recognizes to verify the

network integrity. This test will verify that the network is working correctly.

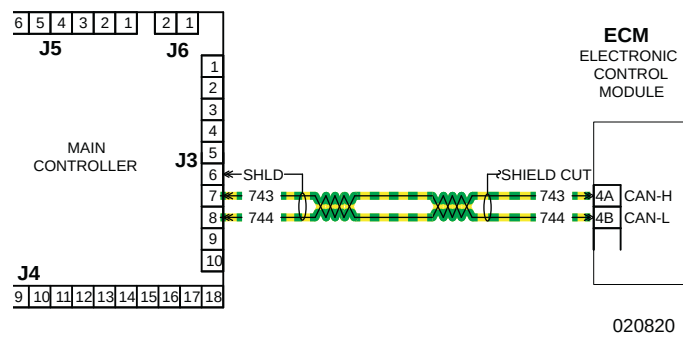


Figure 10-6. CAN bus Diagram

Procedure

NOTE: Note: If an ECM replacement was recently made, it was shipped without any firmware. Clear all alarms and leave the unit in off. Turn service mode on, wait 10 seconds then turn Service mode off. Wait at least 15 minutes for the update to occur.

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Verify battery voltage and all fuses.
3. Disconnect the J6 connector from the back of the Controller.
4. Remove the 7.5A fuse from the Controller.
5. Remove the 15A fuse from the Wire 13 fuse holder.
6. Set the DMM to measure resistance.
7. Using the appropriate back probes, connect one meter lead to Wire 743 at the Controller and the other meter lead to Wire 744 at the Controller. Record the reading.
 - a. If the resistance measures 60– 65 ohms, proceed to **Step 12**.
 - b. If the resistance measures anything other than 60–65 ohms, proceed to **Step 8**.
8. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
9. Disconnect the J3 harness connector at the Controller. Visually inspect the pins for both the Controller and the Controller harness for damage.
 - a. If the pins are found to be damaged, replace the Controller harness and/or Controller as necessary and retest.

10. See [Power Zone 200 Controller](#). With J3 disconnected, connect one meter lead to the Pin at controller J3-7 for Wire 743 and the other to the Pin at controller J3-8 for Wire 744. Record your reading.
 - a. If resistance measures other than 120– 125 ohms, replace the Controller and retest.
11. With J3 disconnected, connect one meter lead to Wire 743 at J3 and the other to Wire 744 at J3. Record the reading.
 - a. If resistance measures other than 120– 125 ohms, replace the ECM and retest.
12. Set the DMM to measure DC voltage.
13. Reconnect the J3 and J6 connectors to the Controller.
14. Reinstall the 7.5A fuse into the Controller.
15. Reinstall the 15A fuse into the Wire 13 fuse holder.
16. See [Power Zone 200 Controller](#). Using the appropriate back probe, insert one meter lead to Wire 877 at the J3 connector at the back the controller and the other to the battery negative terminal. Record the reading.
 - a. If battery voltage is measured, proceed to **Step 18**.
 - b. If no voltage is measured, proceed to Step 17.
17. With J3 disconnected, connect one meter lead to pin J3-9 for Wire 877 at the back of the controller and the other to the battery negative terminal. Record the reading.
 - a. If no voltage is measured, replace the controller.
18. Reconnect all connectors and the battery. Wait for the Power Zone 200 to reboot. Set the DMM to measure AC voltage. Using the appropriate back probes. Connect one meter lead to Wire 743 at the Controller and the other to Wire 744 at the Controller. Record the reading.
 - a. If 0.75V to 0.95V was measured the ECM needs to be programmed. Proceed to Field Pro Testing.
 - b. If 0.4V to 0.5V was measured see if there is an active firmware update. Wait for it to finish and retest.
 - c. If 1.2V-1.35V was measured replace the PZ200 and retest.
 - d. If more than 1.35V was measured replace the ECM and retest.

Cylinder Head Temperature Sensor Test

General Theory

See [Figure 10-7](#). The ECM monitors the engine head temperature to protect the engine from damage due to overheating. The engine head temperature sensor (EHT) converts a temperature reading to a resistance number for the ECM to calculate the engine head temperature. The ECM will trigger a fault if the sensor resistance is below or above the specified programmed range in the ECM.

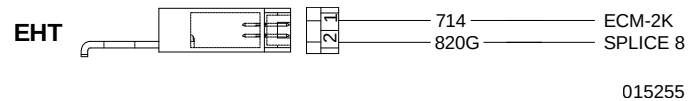


Figure 10-7. Engine Head Temperature Sensor EHT

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Check the connector for the EHT sensor.
 - a. If the connector is found to be loose or improperly connected, repair connection and retest.

NOTE: If diagnosing E-code 5200, 5201, 5215, or 5217 skip to **Step 11**.
3. Set the DMM to measure resistance.
4. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
5. Disconnect the harness connector at the EHT sensor. Visually inspect the EHT sensor and harness pins for damage.
 - a. If the pins are found to be damaged, replace the EHT sensor and/or replace the wiring harness and retest.
6. Using the appropriate back probes, connect one meter lead to Wire 714 at the ECM and the other lead to Wire 714 at the EHT sensor.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
7. Disconnect the ETMC connector.
8. Using appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the EHT sensor.

- a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
9. Reconnect the ETMC connector.
10. Reconnect the ECM harness connector.
11. Allow the engine to cool down so that the cylinder head(s) are at least below 200 °F.
12. Using appropriate back probes, back probe both sides of the sensor side of the EHT sensor connector. Record the resistance reading.
 - a. If the sensor resistance is outside of 4 kohms–36.8 kohms, replace the EHT sensor and retest.
 - b. If the sensor measures correctly, replace the ECM and retest.

Cylinder Leak Down Test

General Theory

The cylinder leak down test checks the combustion chamber of the engine for the ability to properly seal for compression by measuring air leakage. This test is used in conjunction with the compression test to properly identify which part of the engine has failed before beginning disassembly.

NOTE: Refer to the cylinder leak down tester tool instructions for proper operation of tester.

Procedure

1. Turn off the fuel supply to the generator.
2. Remove the 15A fuse from Wire 13.
3. Set the main line circuit breaker (MLCB) to OFF.
4. Disconnect the spark plug wire(s) from each spark plug.
5. Connect a spark tester to the cylinder head(s) and attach the spark plug wire (s) to the tester to properly ground the ignition coil(s).
6. Remove the spark plug(s).
7. Remove the valve cover(s).
8. Visually inspect the rockers, guide plates and check the torque of the rocker studs. (See [Specification](#) at the beginning of the book.)
9. Gain access to the generator fan assembly.
10. Using the fan bolt, rotate the engine crankshaft until the piston reaches top dead center (TDC) of the compression stroke of the cylinder being tested. In this position, both the intake and exhaust valves will be closed.

NOTE: Use a socket extension or similar tool in the spark plug hole and watch it move up and down until it reaches the top of the compression stroke. If the engine is not properly positioned at TDC, the test results will be inaccurate.

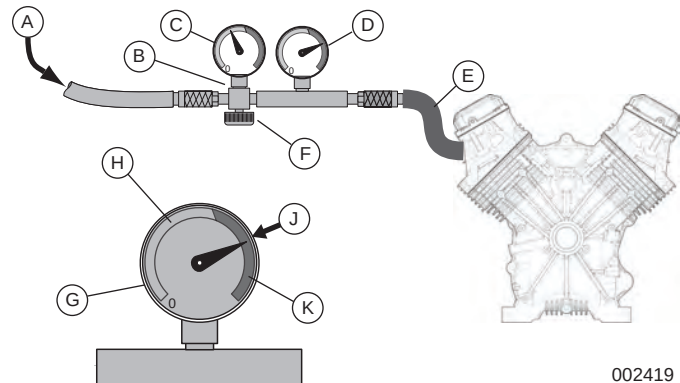


Figure 10-8. Cylinder Leakdown Tester

A	Compressed air in	F	Regulator adjustment knob
B	Air pressure regulator	G	Outlet gauge
C	Inlet gauge pressure set point	H	Red range indicates unacceptable leakage
D	Outlet gauge pressure	J	Needle indicates minimal air leakage
E	To spark plug hole	K	Green range indicates acceptable leakage

11. See [Figure 10-8](#). Attach the cylinder leak down tester adapter to the spark plug hole being tested.

12. Connect an air source to the cylinder leak down tester per the tool manufacturer's requirements.

NOTE: Refer to the tool manufacturer's instructions for proper setting.

13. Connect the tool to the cylinder leak down tester adapter while monitoring the fan for engine rotation.

NOTE: If the fan rotates, repeat Step 9–11 to verify engine is at TDC. Record the leakage percentage.

- a. If testing a single cylinder engine, if leakage is less than 20% and no air leakage is heard, the engine cylinder is good. If testing a V-Twin engine, proceed to Step 13.
- b. If leakage is less than 20% but audible air leakage is heard, or if the leakage is greater than 20%, proceed to Step 14.
14. Repeat Steps 8–11 on the second cylinder.
 - a. If leakage is less than 20% and no air leakage is heard, the engine cylinder is good.
 - b. If leakage is less than 20% but audible air leakage is heard, or if the leakage is greater than 20%, proceed to Step 14.

15. Determine where the air leakage is coming from by listening for leakage as well feeling for leakage at these locations:

- Throttle plate - air escaping past intake valve.
- Muffler/exhaust - air escaping past exhaust valve.
- Breather/dipstick tube - air escaping past piston rings.
- Between cylinder head and block - air escaping between the cylinder head and block.

Record where leakage is coming from and contact Technical support for further instruction.

Electronic Throttle Control Power Test

General Theory

The ECM controls the electronic throttle control valve (ETC). The ECM uses the TTMAP sensor to verify the demand on the engine, and this will move the throttle plate position accordingly to maintain 3600 RPM.

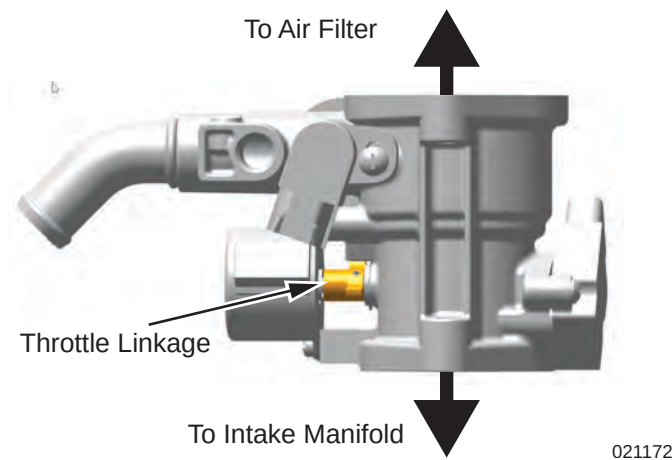


Figure 10-9. External Throttle Valve Linkage

Procedure

NOTE: Note: See [Figure 10-9](#). While visually observing the external throttle valve linkage, remove and reinstall the 15A fuse from fuse holder in Wire 13. The throttle linkage should fully sweep then set itself for engine starting.

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to Step 2.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Check the ETMC connector to verify it is connected correctly.

- a. If connector is found to be loose or improperly connected, repair the connector and retest.
3. Set the DMM to measure resistance.
4. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
5. See [Figure 10-10](#). Disconnect the ETMC connector. Visually inspect the ETMC harness pins for damage.
 - a. If the pins are found to be damaged, replace the wiring harness and/or mixer assembly and retest.

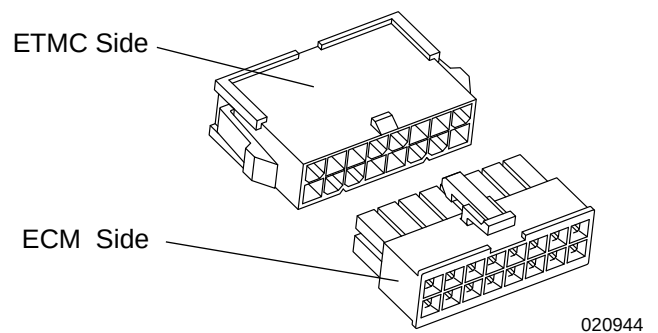


Figure 10-10. ETMC Connector Showing The ECM Connector From The Harness (Wire) Side

6. Using the appropriate back probes, connect one meter lead to Wire 770A at the ECM and the other to Wire 770A at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
7. Using the appropriate back probes, connect one meter lead to Wire 770B at the ECM and the other to Wire 770B at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
8. Using the appropriate back probes, connect one meter lead to Wire 771A at the ECM and the other to Wire 771A at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
9. Using the appropriate back probes, connect one meter lead to Wire 771B at the ECM and the other to Wire 771B at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.

10. Using the appropriate back probes, connect one meter lead to Wire 770A on the mixer side of the ETMC harness and the other lead to Wire 771A on the mixer side of the ETMC harness.
 - a. If the resistance measures between 35– 45 ohms proceed to **Step 11**.
 - b. If the resistance measures outside of the above range, replace the mixer assembly and retest.
11. Using the appropriate back probes, connect one meter lead to Wire 770B on the mixer side of the ETMC harness and the other lead to Wire 771B on the mixer side of the ETMC harness.
 - a. If the resistance measures approximately 35– 45 ohms proceed to **Step 12**.
 - b. If the resistance measures outside of the above range, replace the mixer assembly and retest.
12. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
13. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
14. Set the DMM to measure DC voltage.
15. Reconnect the ETMC harness connector.
16. Reconnect the ECM harness connector.
17. Using the appropriate back probe, connect one meter lead to Wire 766B on the mixer side of the ETMC harness. Connect the other meter lead to the negative battery terminal. With the ECM powered on, the meter should display a voltage range of 0.5–4.5 VDC.
 - a. If the voltage measured is correct, replace the ECM and retest.
 - b. If 5V is measured, disconnect the TMAP sensor and remeasure. If voltage drops, replace mixer assembly.
 - c. If 0V is measured, move the meter lead from the negative battery terminal to the positive terminal. If voltage is measured, disconnect the TMAP sensor. If the voltage drops, replace mixer assembly.

Engine Compression Test

General Theory

A loss or reduction in engine compression can result in a failure of the engine to start, poor engine running, and excessive emissions. Typically, a loss of compression is due to one or more of these factors:

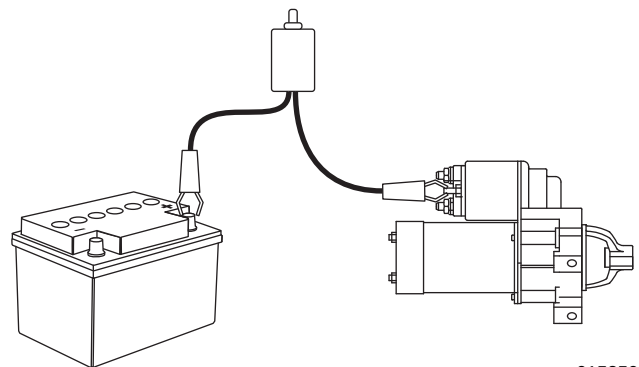
- Leaking or blown cylinder head gasket.
- Improperly seated or sticking valves.
- Worn piston rings or cylinder (this can also result in high oil consumption).

Procedure

NOTE: Battery and starting system need to be properly operational to ensure accurate test results.

NOTE: Remove the spark plugs from both cylinders when testing compression on V-Twin generators to allow for proper engine cranking speed.

1. Turn off the fuel supply to the generator.
2. Remove the 15A fuse from Wire 13.
3. Set the MLCB to OFF.
4. Remove the air cleaner lid.
5. Use a finger to manually open the throttle plate to the wide open position.
6. Disconnect the spark plug wire(s) from each spark plug and ground the wire(s).
7. Connect a spark tester to the cylinder head(s) and attach the spark plug wire (s) to the tester to properly ground the ignition coil(s).
8. Remove the spark plug(s).
9. Disconnect Wire 16 from the starter contactor (SC).
10. See [Figure 10-11](#). Connect one lead from the remote push button starter to the Wire 16 terminal on the SC.



015250

Figure 10-11. Remote Push Button Starter Connected

11. Connect the other lead from the remote push button starter to the positive battery terminal.
12. Install a compression test gauge into a cylinder.

13. Crank the engine until there is no further increase in pressure and record the highest reading obtained.
14. Repeat Steps 12– 13 on the second cylinder if applicable.

NOTE: See Section 4 [Generator Specifications](#) to verify the proper compression reading ranges.

Reading Results

The difference in pressure between the two cylinders should not exceed 25 percent. For example, if the pressure reading of cylinder #1 is 165 PSI and cylinder #2 is 160 PSI, that equates to a pressure difference of 5 PSI. Divide “5” by the highest reading (165 in this case) to obtain the percentages (3 percent in this case).

- a. If the compression is within specifications and each cylinder is within 25 percent of each other, the engine condition is satisfactory.
- b. If the compression is below specifications or the leakage between cylinders is greater than 25 percent, proceed to the [Cylinder Leak Down Test](#).

Engine Speed Test

General Theory

The ECM monitors the engine position using the crankshaft position sensor (CRKS). The engine flywheel has a 24 minus one tooth design. The CRKS uses the gap where the missing tooth is on the flywheel to report the engine position back to the ECM. The ECM uses this information to calculate the position of the crankshaft to control spark timing and monitor engine RPM. If the engine is running at an RPM outside of the pre determined range, an E-code fault will be triggered.

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to Step 2.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Record all codes displayed in the Field Pro app.
 - a. If there are other E-Codes for the throttle/mixer assembly or fuel supply/pressure, proceed to those diagnostic tests before moving to Step 3.
3. Check the generator for an overload condition.
 - a. If the generator is being overloaded, remove overload and retest.
4. Check throttle and mixer assembly for obstruction and/or mechanical damage.

- a. If obstruction is seen, remove and retest. If mechanical damage is found, replace mixer assembly and retest.
5. Verify the ETMC connector is connected correctly.
 - a. If the connector is found to be loose or improperly connected, repair the connector and retest.
6. See [Figure 10-12](#). Clear any fault codes. While visually observing the external throttle valve linkage remove and reinstall the 15A fuse from the fuse holder in Wire 13. The throttle linkage should fully sweep then set itself for engine starting. Once the sweep is done, start generator in MANUAL.

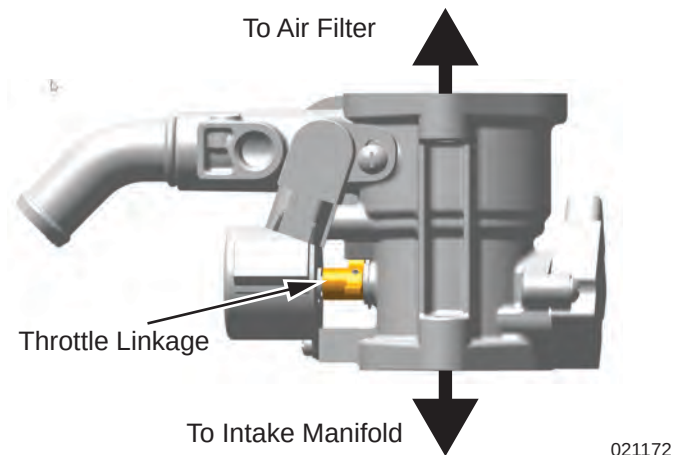


Figure 10-12. External Throttle Valve Linkage

- a. If the throttle assembly performs the sweep and the unit starts with no fault codes returning, discontinue diagnostics.
- b. If the throttle linkage does not move as described, proceed to the [Electronic Throttle Control Power Test](#).
- c. If the generator starts and produces a fault again and the throttle moves as designed, contact Technical Support for further assistance.

Engine Timing Sensor Test

General Theory

The ECM needs to determine the position of the crankshaft and camshaft to command the ignition spark timing correctly for engine running. The ECM uses the crank position sensor (CRKS) and the throttle position, temperature and manifold air pressure sensor (TTMAP) to determine this calculation. When the generator is cranking above 300 RPM, the ECM will look to sync these two signals to calculate the spark timing. If a correlation (or sync) is not achieved, the ECM will trigger an E-code fault.

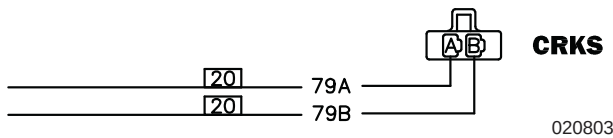


Figure 10-13. Crank Position Sensor (CRKS)

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to Step 2.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Check the CRKS sensor connector to verify it is connected correctly.
 - a. If the connector is found to be loose or improperly connected, repair the connector and retest.
3. Check the ETMC connector to verify it is connected correctly.
 - a. If the connector is found to be loose or improperly connected, repair the connector and retest.
4. Check the TTMAP sensor connector to verify it is connected correctly.
 - a. If the connector is found to be loose or improperly connected, repair the connector and retest.
5. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
6. Disconnect the harness connector at the CRKS. Visually inspect the CRKS harness pins for damage.
 - a. If the pins are found to be damaged, replace the CRKS and/or replace the wiring harness as necessary and retest.
7. Set the DMM to measure resistance.
8. Using the appropriate back probes, connect one meter lead to Wire 79A at the ECM and the other lead to Wire 79A at the harness side of the CRKS sensor.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
9. Using the appropriate back probes, connect one meter lead to Wire 79B at the ECM and the other lead to Wire 79B at the harness side of the CRKS sensor.

- a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

10. Reconnect the ECM harness connector.
11. Using the appropriate back probes, back probe both sides of the sensor side of the CRKS sensor connector. The resistance of the sensor should measure approximately 100–200 ohms. Record the resistance reading.
 - a. If the sensor resistance is outside of this range, replace the CRKS sensor and retest.
12. Proceed to the [Engine Cranks But Does Not Run, Lacks Power, Runs Rough, Hunts or Backfires](#).
 - a. If engine health passes testing replace ECM and retest.

Fuel Mixer Valve Test

General Theory

The fuel valve is controlled by the ECM. The ECM uses the TTMAP sensor to verify the demand on the engine, and this will move the mixer plate position accordingly to maintain 3600 RPM. If the ECM detects a voltage reading outside of the pre-determined range of the MPS, an E-code fault will be triggered.

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. See [Figure 10-14](#). While visually observing the external mixer motor linkage, remove and reinstall the 15A fuse from the fuse holder in Wire 13. The mixer linkage should fully sweep then set itself for engine starting.

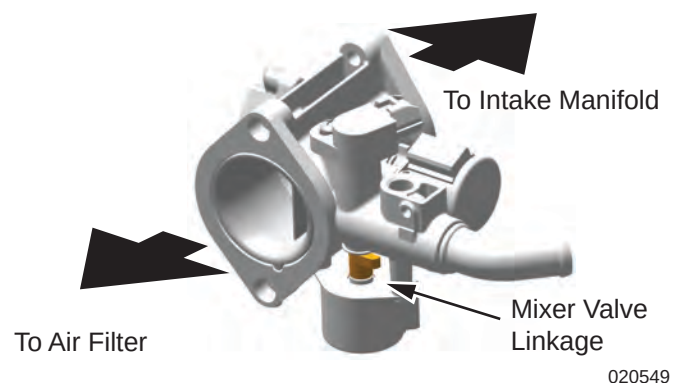


Figure 10-14. External Mixer Valve Linkage

- a. If the mixer plate does not move as described, proceed to the [Mixer Valve Power Test](#).

NOTE: If testing for E-codes 4900, 4901, 4907, 4915, or 4917 skip to **Step 12**.

3. Set the DMM to measure resistance.
4. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
5. Disconnect the harness connector at the MPS. Visually inspect both the MPS and MPS harness pins for damage.
 - a. If the pins are found to be damaged, replace the mixer assembly and retest.
6. Using the appropriate back probes, connect one meter lead to Wire 606 at the ECM and the other lead to Wire 606 at the MPS.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 9**.
7. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the MPS.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 10**.
8. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the MPS.
 - a. If resistance measures less than 5 ohms, proceed to **Step 12**.
 - b. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 11**.
9. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 606 at the ECM and the other lead to Wire 606 at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
10. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
11. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
12. Set the DMM to measure DC voltage.
13. Reconnect the harness connector at the MPS.
14. Reconnect the ECM harness connector.
15. Using the appropriate back probe, connect one meter lead to Wire 820A at the MPS. Connect the other meter lead to battery ground.
 - a. If 5V is not measured, replace ECM and retest.
16. Using the appropriate back probe, connect one meter lead to Wire 820G at the MPS. Connect the other meter lead to battery ground.
 - a. If no voltage is measured, proceed to **Step 18**.
17. Check for a short-to-power between Wire 820G at the ECM and Wire 820G at the mixer assembly.
 - a. If a **short-to-power** is found, replace the wiring harness and/or mixer assembly and retest.
 - b. If no **short-to-power** is found, replace the ECM and retest.
18. Using the appropriate back probe, connect one meter lead to Wire 606 at the MPS. Connect the other meter lead to battery ground. With the ECM powered on, the meter should display a voltage range of 0.5–4.5 VDC.
 - a. If the voltage measured falls within that range, replace ECM and retest.
 - b. If the voltage measured is outside of that range, replace the mixer assembly and retest.

Fuel Pressure Sensor Test

General Theory

The ECM monitors the fuel supply pressure using the Fuel Pressure Sensor (FPS). The sensor converts a pressure reading to a voltage number for the ECM to monitor. This voltage reading is used to monitor the fuel

pressure to confirm to the ECM the pressure range is correct to run the engine without damaging it. If the ECM detects a voltage reading outside of the pre-determined range of the FPS, an E-code fault will be triggered.

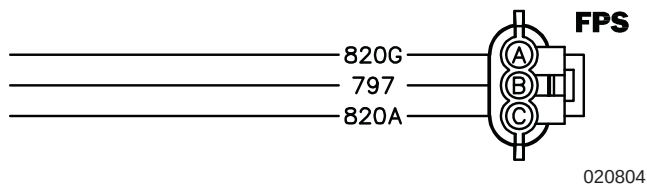


Figure 10-15. Fuel Pressure Sensor

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Perform [Fuel Supply and Pressure Test](#) to verify the generator has the correct fuel pressure.
 - a. If the fuel pressure is out of specification, repair root cause and retest.

NOTE: If testing for E-codes 3200, 3201, 3215, or 3217 skip to **Step 12**.
3. Set the DMM to measure resistance.
4. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
5. Disconnect the harness connector at the FPS. Visually inspect both the FPS and FPS harness pins for damage.
 - a. If the pins are found to be damaged, replace the FPS and/or the mixer assembly as necessary and retest.
6. Using the appropriate back probes, connect one meter lead to Wire 797 at the ECM and the other lead to Wire 797 at the FPS.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 9**.
7. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the FPS.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 10**.
8. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the FPS.
 - a. If resistance measures less than 5 ohms, proceed to **Step 12**.

- b. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 11**.

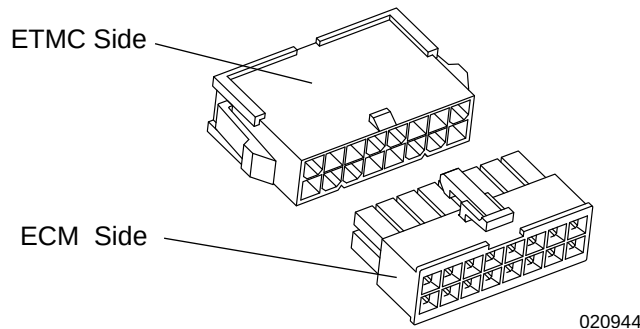


Figure 10-16. ETMC Connector Showing The ECM Connector From The Harness (Wire) Side

9. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 797 at the ECM and the other lead to Wire 797 at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
10. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
11. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace wiring harness and retest.

12. Set the DMM to measure DC voltage.
13. Reconnect the ETMC harness connector.
14. Reconnect the harness connector at the FPS.
15. Reconnect the ECM harness connector.
16. Using the appropriate back probe, connect one meter lead to Wire 820A at the FPS. Connect the other meter lead to battery ground.
 - a. If 5V is not measured, replace ECM and retest.
17. Using the appropriate back probe, connect one meter lead to Wire 820G at the FPS. Connect the other meter lead to battery ground.
 - a. If no voltage is measured, proceed to **Step 19**.
18. Check for a short-to-power between Wire 820G at the ECM and Wire 820G at the mixer assembly.
 - a. If a **short-to-power** is found, replace the wiring harness and/or mixer assembly and retest.
 - b. If no **short-to-power** is found, replace the ECM and retest.
19. Using the appropriate back probe, connect one meter lead to Wire 797 at the FPS. Connect the other meter lead to battery ground. With the ECM powered on, the meter should display a voltage range of 0.5–4.5 VDC.
 - a. If the voltage measured falls within that range, replace ECM and retest.
 - b. If the voltage measured is outside of that range, replace the FPS and retest.

Fuel Shutoff 2 (SOV2) Control Test

General Theory

When the generator is commanded to start, the Power Zone 200 Controller sends battery voltage to fuel shut off valve 1 (SOV1) and fuel shut off valve 2 (SOV2). Once the ECM sees over 75 RPM, it provides the ground (internally) to complete the circuit to power up SOV2. When the generator shuts down, it will remove power from Wire 14 to close the fuel shut off valves, along with the ECM removing the ground from SOV2. The ECM will monitor SOV2 and generate a fault code if the valve does not respond as intended.

Procedure

1. See [Figure 10-17](#). While visually observing the external throttle valve linkage, remove and reinstall the 15A fuse from fuse holder in Wire 13. The throttle linkage should fully sweep then set itself for engine starting.
 - a. If the throttle valve linkage moves as described, proceed to **Step 2**.

- b. If the throttle valve linkage does not move as described, proceed to [Electronic Throttle Control Power Test](#).

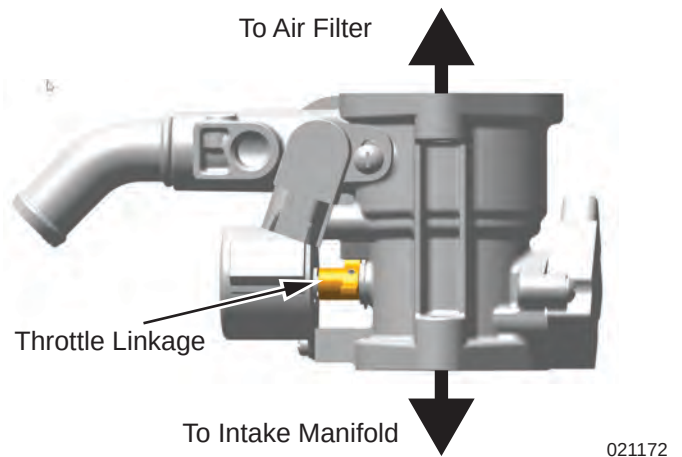


Figure 10-17. External Throttle Valve Linkage

2. Place a hand on both SOVs and crank the generator in MANUAL.
 - a. If the valves clicked, proceed to the [Fuel Supply and Pressure Test](#) to verify the fuel pressure is correct.
 - b. If neither valve clicked, proceed to [Wire 14 Test](#).
 - c. See [Figure 10-18](#). If only the valve CLOSEST to the fuel inlet pipe (A) clicked, proceed to Step 3.

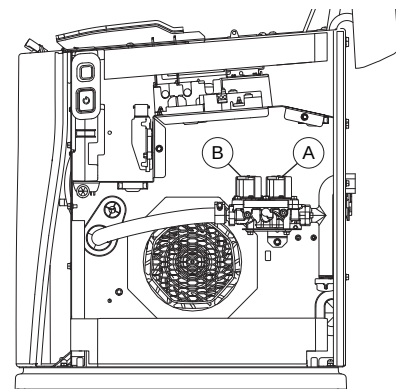


Figure 10-18. Fuel Shut Off Valve 1 (SOV1) and Fuel Shut Off Valve 2 (SOV2)

3. Set the DMM to measure resistance.
4. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins pass inspection, proceed to **Step 5**.
 - b. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
5. Disconnect Wire 610 at SOV2 and visually inspect the SOV2 harness pins for damage.
 - a. If the pins pass inspection, proceed to **Step 6**.

- b. If the pins are found to be damaged, replace SOV2 and/or the wiring harness and retest.
- 6. Using the appropriate back probe, insert one meter lead into the pin for Wire 610 at the ECM harness and connect the other lead to Wire 610 at SOV2. Record the reading.
 - a. If the resistance measures greater than 5 ohms or OL, replace wiring harness and retest.
- 7. Set the DMM to measure DC voltage.
- 8. Reconnect the ECM harness connector.
- 9. Reconnect Wire 610 at SOV2.
- 10. Using the appropriate back probe, insert one meter lead to Wire 610 at SOV2 and connect the other lead to battery ground. Crank the generator in MANUAL. The meter should switch from showing battery voltage to showing nearly no voltage while cranking.
 - a. If the voltage measures less than 0.5V, clear faults and retest.
 - b. If the voltage measures more than 0.5V, proceed to Step 11.
- 11. Check for a short-to-power between Wire 610 at the ECM and Wire 610 at SOV2.
 - a. If a **short-to-power** is found, replace the wiring harness and retest.
 - b. If no **short-to-power** is found, replace the ECM and retest.

Fuel Supply and Pressure Test



⚠ DANGER

Explosion and fire. Fuel and vapors are extremely flammable and explosive. No leakage of fuel is permitted. Keep fire and spark away. Failure to do so will result in death or serious injury.

(D000192)

General Theory

The air-cooled generator is designed to operate on both natural gas (NG) and propane (LP) as a fuel. The following conditions apply for the unit to operate properly:

- An adequate gas supply and sufficient fuel pressure must be available or the engine will not start.
- Minimum recommended gaseous fuel pressure at the generator fuel inlet connection is 3.5 inches water column for natural gas (NG) or 10 inches water column for LP gas.
- Maximum gaseous fuel pressure at the generator fuel inlet connection is 7 inches water column for natural gas or 12 inches water column for LP gas.

- When propane gas is used, only a “vapor withdrawal” system may be used. This system uses the gas that forms above the liquid fuel.
- The gaseous fuel system must be properly tested for leaks following installation and periodically thereafter. **No leakage is permitted.** Leak testing methods must comply strictly with gas codes.

NOTE: Visually inspect the engine air intake and air filter for obstruction prior to performing Fuel Supply and Pressure Test.

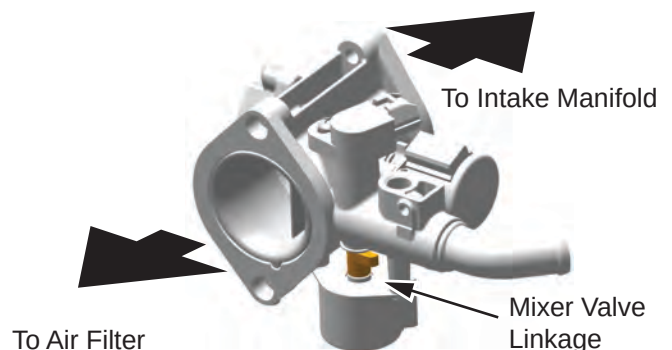
NOTE: Visually inspect the fuel shut off valves for signs of leaks or damage.

NOTE: Refer to installation manual to confirm correct pipe sizing before testing unit. If pipe sizing and/or the regulator is inadequate, correct the pipe sizing/regulator before continuing with diagnostics.

Procedure

A water manometer or a gauge that is calibrated in “inches of water column” should be used to measure the fuel pressure. Fuel pressure at the inlet side of the fuel shut off valves should be between 3.5–7 inches water column for natural gas (NG), or 10–12 inches water column for LP gas.

1. Verify battery voltage and all fuses.
2. Record all codes displayed in the Field Pro app.
 - a. If there are other E-Codes for the throttle/mixer assembly or fuel supply/pressure, proceed to those diagnostic tests before proceeding to Step 2.
3. See [Figure 10-19](#) and [Figure 10-20](#). While visually observing the external mixer/throttle motor linkage, remove and reinstall the 15A fuse from the fuse holder in Wire 13. The mixer/throttle linkage should fully sweep then set itself for engine starting.
 - a. If the mixer plate does not move as described, proceed to the [Fuel Mixer Valve Test](#). If the throttle linkage does not move as described, proceed to the [Electronic Throttle Control Power Test](#).



020549

Figure 10-19. External Mixer Valve Linkage

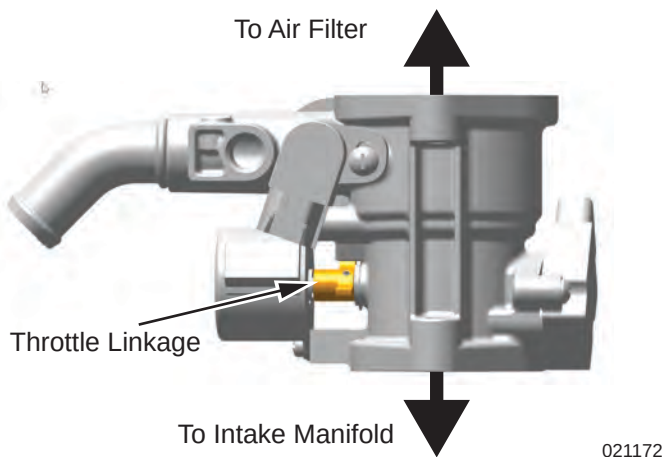


Figure 10-20. External Throttle Valve Linkage

4. See [Figure 10-21](#) for the fuel pressure test point on the fuel shut off valves. Static fuel pressure can be checked at Port 1 (A) and running pressure can be checked at Port 2 (B).

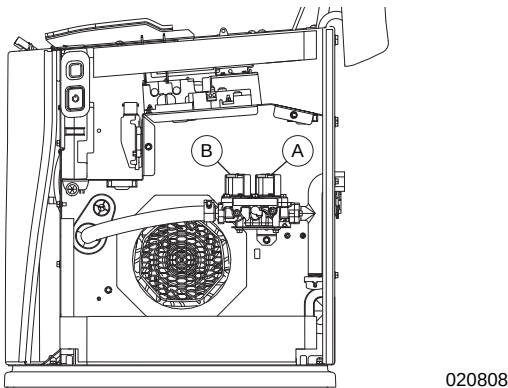


Figure 10-21. Gas Pressure Test Points

NOTE: Where a primary regulator is used to establish fuel inlet pressure, adjustment of that regulator is usually the responsibility of the fuel supplier or the fuel supply system installer.

5. Turn off the fuel supply.
6. Connect a manometer to Port 1. Turn the fuel supply back on and record the fuel pressure reading.
 - a. If the fuel pressure is outside of the range of the fuel type being used, the fuel supply pressure and/or regulator need to be inspected. Once this is corrected, retest fuel pressure.
7. Turn off the fuel supply.
8. Connect a manometer to Port 2. Turn the fuel supply back on. Set the generator to MANUAL and record the fuel pressure reading.
 - a. If the fuel pressure measures below the minimum allowed pressure, but is greater than 0, replace the SOV assembly and retest.
 - b. If no fuel pressure is measured, proceed to the [Wire 14 Test](#).

9. Connect the generator to a load bank capable of loading the generator to its rated capacity. Start the generator in MANUAL and apply the load to the generator. Record the fuel pressure reading.
 - a. If the fuel pressure stays above the minimum pressure range for the fuel type being used, the fuel pressure and supply is good. Refer back to previous test and follow the "Fuel Pressure Within Specifications" path.
 - b. If the fuel pressure drops below the minimum pressure range and/or the pressure changes more than 2 inches of water column erratically, the fuel supply pressure and/or regulator need to be inspected. Once this is corrected, retest fuel pressure.

Ignition Coil Test

General Theory

If the engine cranks but will not start, one cause might be that an ignition system failure has occurred. A special spark tester can be used to check for ignition spark. See [Figure 10-22](#). When using this style spark tester, the adjustment screw must be set to the 10kV mark to accurately test for spark.

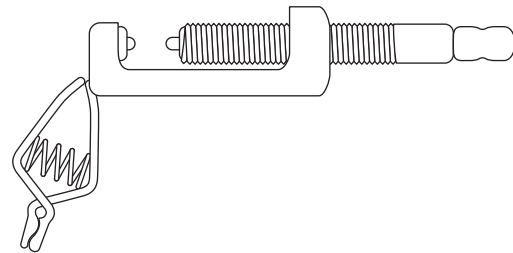


Figure 10-22. Spark Tester

NOTE: On V-Twin engines, the two ignition coil connectors appear identical and can be connected to the wrong coil. IC1 is for cylinder #1 on the back side of the generator (the cylinder on the starter side of the engine) and IC2 is for cylinder #2 on the front side of the generator (the cylinder on the oil filter side of the engine).

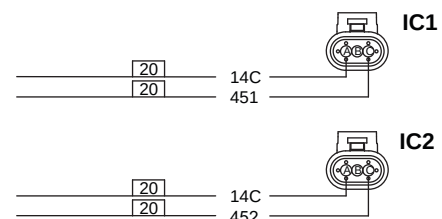


Figure 10-23. Ignition Coils

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to Step 2.

- b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Turn off the fuel supply to the generator.
3. Remove spark plug wire(s) from the spark plug(s).
4. Attach the spark plug wire(s) to a spark tester that is clamped to the cylinder head.
5. Set the controller to MANUAL.
6. While the engine is cranking, observe the spark tester.

a. If the spark jumps the tester gap, the ignition system is providing proper spark for ignition. If testing a single cylinder generator, proceed to the [Spark Plug Condition Test](#). If testing a V Twin generator, proceed to **Step 7**.

b. If the spark fails to jump the tester gap, proceed to **Step 8**.
7. On V-Twin generators, repeat Steps 4 and 5 on the other cylinder of the engine.

a. If the spark jumps the tester gap, the ignition system is providing proper spark for ignition.
8. Remove spark tester and disconnect the ignition coil harness connector. Visually inspect the coil and coil harness pins for damage.

a. If the pins are found to be damaged, replace the ignition coil and/or the wiring harness as necessary and retest.
9. Set the DMM to measure resistance.
10. See **Table 9-2** for test point locations and accepted values.

a. If the values are within specification, replace the ECM and retest.

b. If the values are outside of the specified ranges, replace the ignition coil and retest.

Table 9-2.

Primary resistance – measured between Coil Pin A (for Wire 14C) and Coil Pin C (for Wire 451/452)	1.4 ohms ± 20%
Secondary resistance – measured between Coil Pin C (for Wire 451/452) and the Spark Plug Wire	17–23 kohms ± 20%

Intake Manifold Pressure Sensor Test

General Theory

The ECM monitors the engine intake manifold temperature using the TTMAP sensor. The sensor converts a temperature reading to a resistance number for the ECM to monitor. This resistance reading is used to

monitor the intake manifold air temperature to make the proper fuel trim adjustments for emissions and performance. If the ECM detects a voltage reading outside of the predetermined range of the TTMAP sensor, an E-code fault will be triggered.

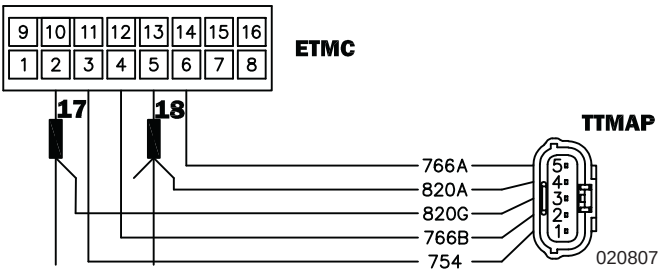


Figure 10-24.

Procedure

1. Clear the alarm and run the unit.

a. If the alarm returns proceed to Step 2.

b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Verify battery voltage and all fuses.
3. Verify the air filter is not clogged and air intake is not obstructed.

a. If obstructions are found, correct condition and retest.

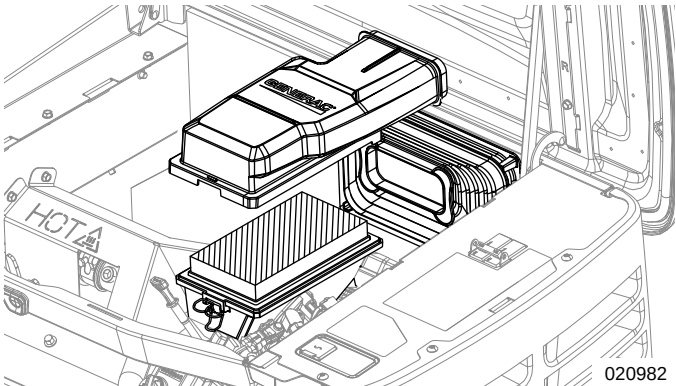


Figure 10-25. Correct Orientation of Air Filter

4. Verify the throttle opening is clear of any obstructions.

a. If obstructions are found, correct condition and retest.
5. Check the TTMAP connector to verify it is connected correctly.

a. If the connector is found to be loose or improperly connected, repair the connector and retest.

NOTE: If testing for E-code 3400, 3401, 3415, or 3417 skip to **Step 15**.

6. Set the DMM to measure resistance.

7. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
8. Disconnect the harness connector at the TTMAP. Visually inspect both the TTMAP and TTMAP harness pins for damage.
 - a. If the pins are found to be damaged, replace the mixer assembly and retest.
9. Using the appropriate back probes, connect one meter lead to Wire 766A at the ECM and the other lead to Wire 766A at the TTMAP.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to Step 12.
10. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the TTMAP.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to Step 13.
11. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the TTMAP.
 - a. If resistance measures less than 5 ohms, proceed to Step 15.
 - b. If resistance measures greater than 5 ohms or reads OL, proceed to Step 14.
- c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
13. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A on the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
14. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and connect the other lead to Wire 820G on the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.

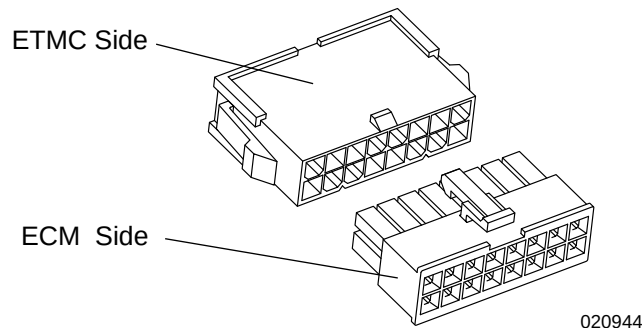


Figure 10-26. ETMC Connector Showing The ECM Connector From The Harness (Wire) Side

12. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 766A at the ECM and the other lead to Wire 766A on the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
15. Set the DMM to measure DC voltage.
16. Reconnect the harness connector at the TTMAP.
17. Reconnect the ECM harness connector.
18. Using the appropriate back probe, connect one meter lead to Wire 820A at the TTMAP. Connect the other meter lead to battery ground.
 - a. If 5V is not measured, replace ECM and retest.
19. Using the appropriate back probe, connect one meter lead to Wire 820G at the TTMAP. Connect the other meter lead to battery ground.
 - a. If no voltage is measured, proceed to Step 21.
20. Check for a **short-to-power** between Wire 820G at the ECM and Wire 820G at the mixer assembly.
 - a. If a **short-to-power** is found, replace the wiring harness and/or mixer assembly and retest.
 - b. If no **short-to-power** is found, replace the ECM and retest.
21. Using the appropriate back probe, connect one meter lead to Wire 766A at the TTMAP. Connect the other meter lead to battery ground. With the ECM powered on, you should see a voltage range of 0.5 - 4.5 VDC.
 - a. If the voltage measured falls within that range, replace ECM and retest

- b. If the voltage measured is outside of that range, replace the mixer assembly and retest.

Intake Manifold Temperature Sensor Test

General Theory

The ECM monitors the engine intake manifold pressure using the TTMAP sensor. This sensor converts a pressure reading to a voltage number for the ECM to monitor. This voltage reading is used to control both the fuel delivery and ignition timing of the engine. If the ECM detects a voltage reading outside of the pre-determined range of the TTMAP sensor, an E-code fault will be triggered.

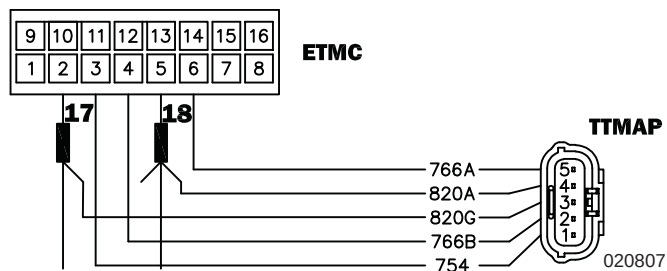


Figure 10-27. Intake Manifold Temperature Sensor

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
 2. Check the connections at both the ETMC and TTMAP.
 - a. If any connection is found to be loose or improperly connected, repair connection and retest.
- NOTE:** If testing for E-code 3500, 3501, 3515, or 3517 skip to **Step 12**.
3. Set the DMM to measure resistance.
 4. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
 5. Disconnect the harness connector at the TTMAP. Visually inspect both the TTMAP and the TTMAP harness pins for damage.
 - a. If the pins are found to be damaged, replace the mixer assembly and retest.

6. Using the appropriate back probes, connect one meter lead to Wire 754 at the ECM and the other lead to Wire 754 at the TTMAP.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 9**.
7. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the TTMAP.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 10**.
8. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the TTMAP.
 - a. If resistance measures less than 5 ohms, proceed to **Step 12**.
 - b. If resistance measures greater than 5 ohms or reads OL, proceed to **Step 11**.

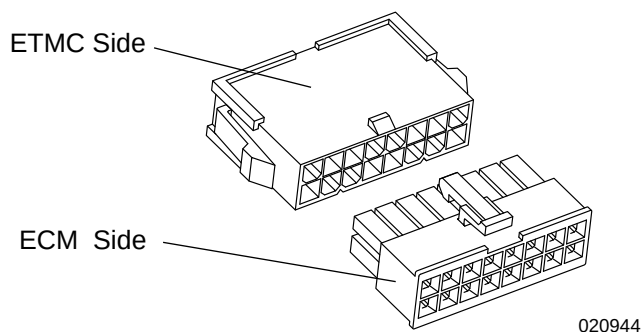


Figure 10-28. ETMC Connector Showing The ECM Connector From The Harness (Wire) Side

9. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 754 at the ECM and the other lead to Wire 754 at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
10. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If resistance measures less than 5 ohms, replace the mixer assembly and retest.

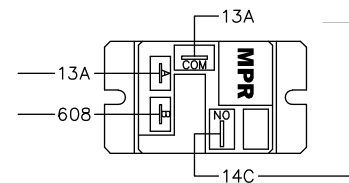
- c. If resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
- 11. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the ECM side of the ETMC connector.
 - a. If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - b. If resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - c. If resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
- 12. Set the DMM to measure DC voltage.
- 13. Reconnect the harness connector at the TTMAP.
- 14. Reconnect the ECM harness connector.
- 15. Using the appropriate back probes, connect one meter lead to Wire 820A at the TTMAP. Connect the other lead to battery ground.
 - a. If 5V is not measured, replace ECM and retest.
- 16. Using the appropriate back probes, connect one meter lead to Wire 820G at the TTMAP. Connect the other lead to battery ground.
 - a. If no voltage is measured, proceed to **Step 18**.
- 17. Check for a **short-to-power** between Wire 820G at the ECM and Wire 820G at the mixer assembly.
 - a. If a **short-to-power** is found, replace the wiring harness and/or mixer assembly and retest.
 - b. If no **short-to-power** is found, replace the ECM and retest.
- 18. Disconnect the ECM connector.
- 19. Set the DMM to measure resistance.
- 20. Using the appropriate back probes, connect one meter lead to Wire 754 at the ECM. Connect the other lead Wire 820G at the ECM. Record the resistance reading.
 - a. If the resistance measured is outside of 570–36.8 kohms replace the mixer assembly and retest.
- 21. Reconnect the ECM connector.
- 22. Using the appropriate back probe, connect one meter lead to Wire 606 on the mixer side of the ETMC harness. Connect the other lead to the negative battery terminal. With the ECM powered on, a voltage range of 0.5–4.5 VDC should be measured.
 - a. If the voltage measured is correct, replace the ECM and retest.

- b. If 5V is measured, disconnect the MPS and remeasure. If voltage drops, replace mixer assembly.
- c. If 0V is measured, move the meter lead from the negative battery terminal to the positive terminal. If voltage is measured, disconnect the MPS. If the voltage drops, replace mixer assembly.

Master Power Relay Test

General Theory

The master power relay (MPR) provides power to the ignition coils (IC1 and IC2), oil pressure sensor (OPS), oxygen sensor (OS), and to the engine control module (ECM). The MPR coil is energized by the ECM grounding Wire 608 and power Wire 13A from the battery. The MPR connects Wire 13A to Wire 14C.



020551

Figure 10-29. Master Power Relay (MPR)

Procedure

1. Check the 15A fuse located in line with Wire 13.
 - a. If the fuse has failed, replace fuse and retest. If the fuse continues to fail proceed to [ECM Harness Fuse Testing](#).
2. Set the DMM to measure DC voltage.
3. Using the appropriate back probe, connect the red meter lead to Wire 13A at the COM terminal. Connect the black meter lead to the battery negative terminal. Record the reading. Move the red meter lead to Wire 13A at the Pin A terminal and record that reading.
 - a. If battery voltage is measured at only 1 wire, replace the wiring harness and retest.
4. Disconnect the battery.
5. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
6. Disconnect all four (4) wires from the MPR. Visually inspect all the terminals and spade connectors for looseness/damage.
 - a. If the terminals/spade connectors are found to be loose/damaged, replace the wiring harness and/or MPR and retest.

7. Set the DMM to measure resistance.
8. Connect the red meter lead to the MPR terminal for Wire 608 and connect the black lead on the terminal for Wire 13A from Terminal A at the MPR. Record the reading.
 - a. If the resistance measures greater than 165 ohms or less than 130 ohms or OL, replace the MPR and retest.
9. Connect a 12 VDC power source across the terminals for Wires 608 and 13A at Pin A on the MPR to energize the relay. Connect one meter lead to the terminal for Wire 13A at the COM terminal and the other meter lead to the terminal for Wire 14C. Record the reading.
 - a. If the resistance measures less than 5 ohms, proceed to Step 10.
 - b. If the resistance measures greater than 5 ohms or OL, replace the MPR and retest.
10. Connect the red meter lead to the disconnected Wire 13A from the COM terminal at the MPR. Using the appropriate back probe, connect the black lead to Wire 13A at the ECM. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
11. Connect the red meter lead to the disconnected Wire 13A from terminal A at the MPR. Using the appropriate back probe, connect the black meter lead to Wire 13A at the ECM. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
12. Connect the red meter lead to the disconnected Wire 608 at the MPR. Using the appropriate back probe, connect the black meter lead to Wire 608 at the ECM. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
13. Connect the red meter lead to the disconnected Wire 14C at the MPR. Using the appropriate back probe, connect the black meter lead to Wire 14C at the ECM. Record the reading.
 - a. If resistance measures less than 5 ohms, replace the ECM and retest.
 - b. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

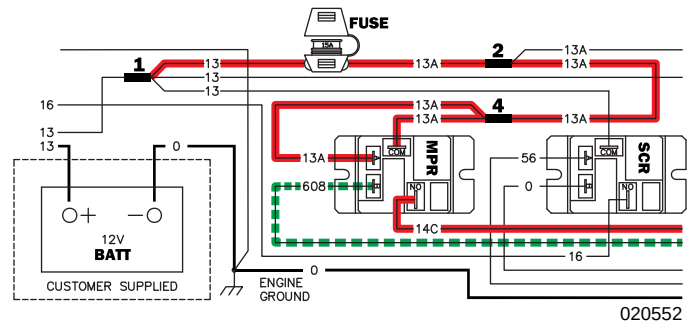


Figure 10-30. Measure Resistance on Wire 14C

Mixer Valve Power Test

General Theory

The engine control module (ECM) controls the electronic mixer control valve (EMC). The ECM uses the TTMAP sensor to verify the demand on the engine, and this will move the mixer plate position accordingly to maintain 3600 RPM.

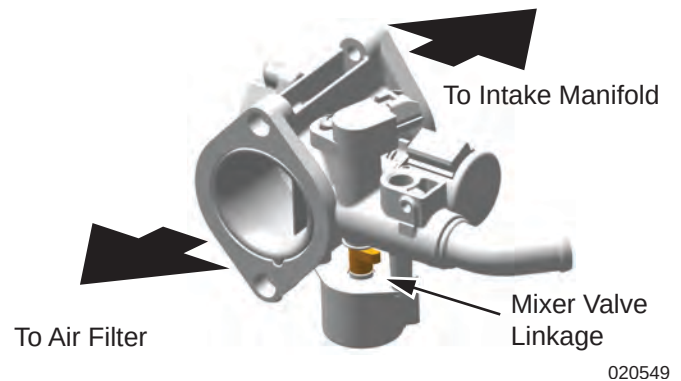


Figure 10-31. External Mixer Valve Linkage

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Verify the ETMC connector is fully seated and check the integrity of the pins.
 - a. If any pins are found to be defective, replace pins and retest.
3. Set the DMM to measure resistance.
4. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
5. Disconnect the ETMC connector. Visually inspect the ETMC harness pins for damage.

- a. If the pins are found to be damaged, replace the wiring harness and/or mixer assembly and retest.

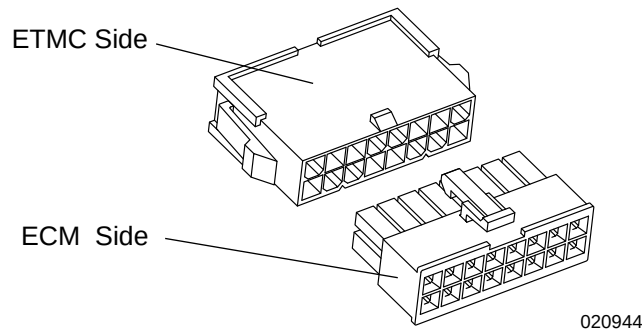


Figure 10-32. ETMC Connector Showing The ECM Connector From The Harness (Wire) Side

6. Using the appropriate back probes, connect one meter lead to Wire 605A at the ECM and the other to Wire 605A at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
7. Using the appropriate back probes, connect one meter lead to Wire 605B at the ECM and the other to Wire 605B at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
8. Using the appropriate back probes, connect one meter lead to Wire 605C at the ECM and the other to Wire 605C at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
9. Using the appropriate back probes, connect one meter lead to Wire 605D at the ECM and the other to Wire 605D at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
10. Using the appropriate back probes, connect one meter lead to Wire 605A on the mixer side of the ETMC harness and the other lead to Wire 605B on the mixer side of the ETMC harness.
 - a. If the resistance measures between 35– 45 ohms, proceed to **Step 11**.
 - b. If the resistance measures outside of the above range, replace the mixer assembly and retest.
11. Using the appropriate back probes, connect one meter lead to Wire 605C on the mixer side of the ETMC harness and the other lead to Wire 605D on the mixer side of the ETMC harness.
 - a. If the resistance measures approximately 35– 45 ohms, proceed to **Step 12**.
 - b. If the resistance measures outside of the above range, replace the mixer assembly and retest.
12. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
13. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the ECM side of the ETMC connector.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
14. Set the DMM to measure DC voltage.
15. Reconnect the ETMC harness connector.
16. Reconnect the ECM harness connector.
17. Using the appropriate back probe, connect one meter lead to Wire 766B on the mixer side of the ETMC harness. Connect the other meter lead to the negative battery terminal. With the ECM powered on, a voltage range of 0.5– 4.5 VDC should be measured.
 - a. If the voltage measured is correct, replace the ECM and retest.
 - b. If 5V is measured, disconnect the TTMAP sensor and remeasure. If voltage drops, replace mixer assembly.
 - c. If 0V is measured, move the meter lead from the negative battery terminal to the positive terminal. If voltage is measured, disconnect the TTMAP sensor. If the voltage drops, replace mixer assembly.

Oil Level Sensor Test

General Theory

The ECM monitors the engine oil level using the oil level sensor (OLS). This sensor converts a level reading to a voltage number for the ECM to monitor. This voltage is used to monitor the engine oil level to determine if the engine can be safely operated.

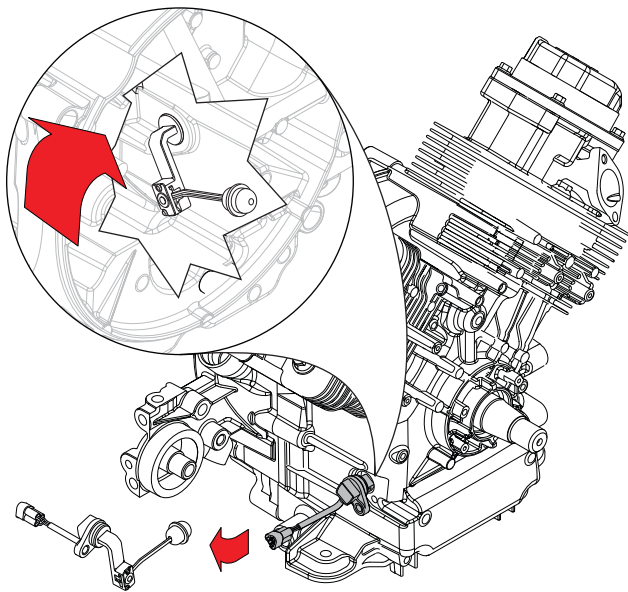
NOTE: The oil level range can be read via the Field Pro App.

If the ECM detects a voltage reading outside of the predetermined range of the OLS sensor, an E-code warning will be triggered.

NOTE: If the Power Zone 200 sees the engine oil level is low, it will set a warning.

Procedure

1. Check the engine oil level using the oil dipstick.
 - a. If the oil level is high, remove excess oil from the engine and retest.
 - b. If the oil level is low, add oil to the engine to bring it to the proper level and retest.
2. Verify the OLS connector is properly seated and connected.
 - a. If the connector is loose or not properly connected, repair connection and retest.
3. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 4**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
4. See [Figure 10-33](#). Unbolt the OLS. Slowly and carefully pull it from the engine to break the resistance of the seal holding it in place. Rotate the sensor clockwise 90 degrees and then to the left to guide it out of the block. Do not force it out.
 - a. Inspect the sensor. Verify the float is intact. If damaged, replace sensor and retest.



00080935

Figure 10-33. Oil Level Sensor Removal

NOTE: If only E-Code 2126 is present skip to Step 11.

5. Set the DMM to measure resistance.
6. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.

7. Disconnect the harness connector at the OLS. Visually inspect both the OLS and OLS harness pins for damage.
 - a. If the pins are found to be damaged, replace the OLS and/or replace the wiring harness as necessary and retest.
8. Using the appropriate back probes, connect one meter lead to Wire 899 at the ECM and the other lead to Wire 899 at the OLS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
9. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the OLS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
10. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the OLS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
11. Set the DMM to measure DC voltage.
12. Reconnect the harness connector at the OLS.
13. Reconnect the harness connector at the ECM.
14. Using the appropriate back probe, connect one meter lead to Wire 820A at the OLS. Connect the other meter lead to battery ground.
 - a. If 5V is not measured, replace the ECM and retest.
15. Using the appropriate back probe, connect one meter lead to Wire 820G at the OLS. Connect the other meter lead to battery ground.
 - a. If no voltage is measured, proceed to Step 15.
16. Check for a **short-to-power** between Wire 820G at the ECM and Wire 820G at the OLS.
 - a. If a **short-to-power** is found, replace the wiring harness and retest.
 - b. If no **short-to-power** is found, replace the ECM and retest.
17. Using the appropriate back probe, connect one meter lead to Wire 899 at the OLS. Connect the other meter lead to battery ground. With the ECM powered on, you should see a voltage range of 0.5–4.5 VDC.
 - a. If the voltage measured is outside of that range, replace the OLS and retest.
18. With the sensor connected to the harness and uninstalled from the engine block, connect to Field

Pro App and navigate to Oil Level. Slowly actuate the float arm from empty to full. If the oil level does not change from low to good, replace the sensor and retest. **Else replace the ECM and retest.**

Oil Pressure Switch Test

General Theory

The engine requires oil pressure to lubricate all the moving components in the engine. The oil pressure switch is a continuity switch that signals to the ECM if the engine has oil pressure or not. If the pressure drops below the minimum threshold, the ECM will shut down the engine to prevent damage.

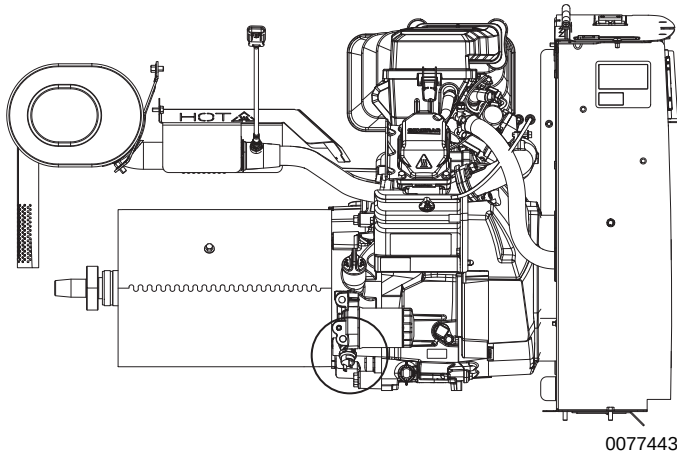


Figure 10-34. Oil Pressure Switch

Procedure

1. Check the engine oil level.
 - a. If oil level is correct, proceed to **Step 3**.
2. Add oil to bring engine to correct oil level. Start engine and monitor.
 - a. If fault is resolved, no further diagnosis is needed.
3. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 4**.
 - b. If the alarm does not return, proceed to **Step 6**.
4. Check the connectors at the OPS.
 - a. If the connectors are found to be loose or improperly connected, repair connectors and retest.
5. Disconnect the harness connector at the OPS. Visually inspect both the OPS and OPS harness pins for damage.
 - a. If the pins are found to be damaged, replace the OPS and/or replace the wiring harness as necessary and retest.
6. Remove the OPS from the engine.

7. Install a manual oil pressure gauge where the OPS was. Start the engine in MANUAL. Record the reading.
 - a. If the oil pressure is below 10 PSI (817/997cc) or 5 PSI (459cc), shut engine down and contact Generac technical support.

NOTE: Continued operation with low oil pressure will result in engine damage!
8. If the alarm did not return in Step 3 proceed to [Intermittent Alarm Test](#). If the alarm did return in Step 3 reinstall OPS on the engine.
9. Set the DMM to measure resistance.
10. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
11. Use the appropriate back probe, connect one meter lead to Wire 566 at the ECM and the other lead to Wire 566 at the OPS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the ECM harness and retest.
12. Use the appropriate back probe, connect one meter lead to Wire 14C at the ECM and the other lead to Wire 14C at the OPS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the ECM harness and retest.
13. Reconnect the ECM harness connector.
14. Set the DMM to measure continuity.
15. Connect one meter lead to each terminal on the OPS. Crank the engine and observe the meter.
 - 459cc Engine:
 - a. Once the engine passes over 5 PSI, the meter should show a loss of continuity across the sensor. If continuity is shown above 5 PSI, replace OPS and retest.
 - b. If the OPS shows a loss of continuity above 5 PSI as designed, replace the ECM and retest.
 - 817/997cc Engine:
 - a. Once the engine passes over 10 PSI, the meter should show continuity across the sensor. If no continuity is shown above 10 PSI, replace OPS and retest.
 - b. If the OPS shows continuity above 10 PSI as designed, replace ECM and retest.

Oxygen Sensor Test

General Theory

The ECM needs to adjust the air/fuel ratio to optimize engine performance, efficiency, and emissions output. The ECM requires a real-time monitor to know what adjustments to the mixer are needed. The ECM uses an oxygen sensor (OS) to make those calculations. This sensor monitors the amount of oxygen in the exhaust gas and compares it to a fresh air sample pulled from around the sensor to determine how much fuel is in the exhaust. The sensor operates in the 0–1 VDC range. The higher the voltage reading, the richer (lower oxygen amount) the exhaust gas is. The lower the voltage reading, the leaner (higher oxygen amount) the exhaust is. The ECM will then use this voltage to adjust either by adding more fuel to enrich the mixture or subtracting fuel to lean out the mixture. When the oxygen sensor is below approximately 600 °F, the voltage reading will be inaccurate. To decrease the warm-up time needed, an oxygen sensor heater circuit is used to warm-up the sensing element. If the ECM detects a voltage reading outside of the pre-determined range of the OS, an E code fault will be triggered.

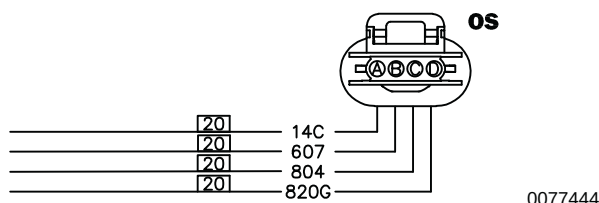


Figure 10-35. Oxygen Sensor

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#)
2. Remove the air filter and manually actuate the throttle with a finger to feel for resistance in the movement. A slight drag should be felt as the throttle moves.
 - a. If only slight drag is found, proceed to Step 3.
 - b. If there is resistance, check for obstructions causing the binding. If none are found, replace the mixer assembly and retest.
3. See [Figure 10-36](#). Check the gaskets for the air intake system for leaks or tears.

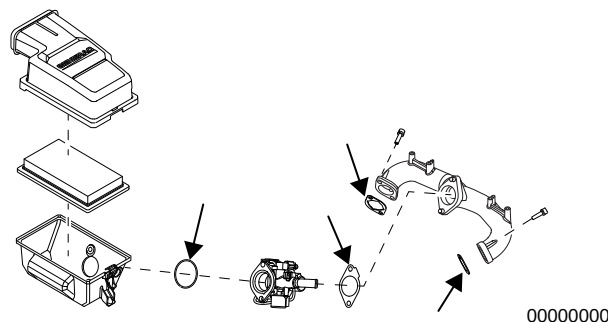


Figure 10-36. Air Intake System Gaskets

- a. If leaks or tears are found, replace damaged gaskets and retest.
4. Check the exhaust system for leaks and verify the oxygen sensor (OS) is tight.
 - a. If any exhaust leaks are found, or the OS is loose, replace damaged gaskets/tighten OS as necessary and retest.
5. Perform the [Fuel Supply and Pressure Test](#).
 - a. If out of spec, make adjustment and retest.
6. Set the DMM to measure resistance.
7. Disconnect the harness connector at the engine control module (ECM). Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
8. Disconnect the harness connector at the OS. Visually inspect both the OS and OS harness pins for damage.
 - a. If the pins are found to be damaged, replace the OS and/or replace the wiring harness as necessary and retest.
9. Using the appropriate back probes, connect one meter lead to Wire 607 at the ECM and the other lead to Wire 607 at the OS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
10. Using the appropriate back probes, connect one meter lead to Wire 804 at the ECM and the other lead to Wire 804 at the OS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
11. Using the appropriate back probes, connect one meter lead to Wire 14C at the ECM and the other lead to Wire 14C at the OS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

12. Disconnect the engine throttle/mixer connector (ETMC).
13. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the OS.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the ECM harness and retest.
14. Reconnect the ETMC connector.
15. Set the DMM to measure DC voltage.
16. Reconnect the OS harness connector.
17. Reconnect the ECM harness connector.
18. Using the appropriate back probe, connect one meter lead to Wire 14C at the OS. Connect the other meter lead to battery ground. Start the generator in MANUAL. Record the reading.
 - a. If battery voltage is measured, proceed to Step 20.
19. Turn generator OFF.
20. Disconnect the connector for Wire 14C at the master power relay (MPR). Place the red meter lead on the MPR terminal and the other meter lead to battery ground.
 - a. If battery voltage is measured, replace the wiring harness and retest.
 - b. If battery voltage is not measured, proceed to the [Master Power Relay Test](#).

NOTE: Allow the generator to run for at least 2 minutes to warm oxygen sensor to operating temperature.
21. Using the appropriate back probe, connect one meter lead to Wire 804 at the OS. Connect the other meter lead to battery ground.
 - a. If the voltage reading is outside of the operating range (0.1– 1.2V), replace the OS and retest.
 - b. If the voltage is within operating range, replace the ECM and retest.

Spark Plug Condition Test

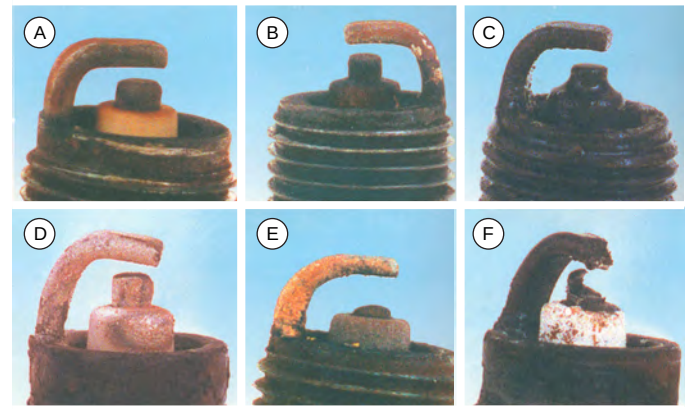
General Theory

When a no start condition is encountered and the ignition system is providing ample spark to the spark plugs, the spark plug condition needs to be checked. The spark plugs used in a gaseous powered generator are subjected to much higher ignition temperatures than those in a gasoline engine. These spark plugs are also subjected to much higher cylinder pressures due to the high compression ratios used in gaseous generator engines.

Because gaseous fuels are more difficult to ignite, the spark plug gap needs to be accurate to properly ignite the fuel in the cylinder. If the gap is incorrect or the spark plugs are worn or fouled, the engine can suffer a misfire/rough running condition.

Procedure

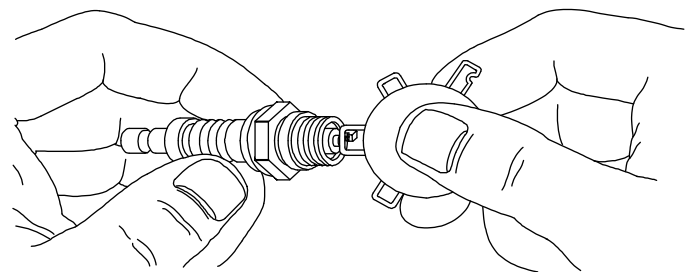
1. See [Figure 10-37](#). Remove spark plug (s) and inspect for any visible damage.
2. Replace any spark plug having burned electrodes, worn electrodes, or cracked porcelain.



0077444

Figure 10-37. Spark Plug Conditions

3. See [Figure 10-38](#). Use a wire feeler gauge to set the gap on new or existing spark plugs as per Section 3 [Generator Specifications](#).
4. Evaluate the condition of the spark plugs.
 - a. If the electrodes are worn, burned, or damaged, replace the spark plugs and retest.
 - b. If the spark plugs pass inspection, proceed to [Engine Compression Test](#) or [Cylinder Leak Down Test](#).



0077446

Figure 10-38. Checking Spark Plug Gap

Stator Battery Charger (Voltage Regulator) Test

General Theory

The engine flywheel has magnets that provide the engine stator with a way to produce current to charge the generator's 12V battery while the engine is running. This ensures the generator can maintain the battery with sufficient charge to continue operation for the duration of the outage. The nominal charging current is 10A at 13.5V at 3600 RPM operation. The stator will only provide voltage when connected to the battery with the voltage regulator (VR) unit. If the voltage on the battery is below or above the operating range of 8.5V–16.5V an E-code fault will be triggered.

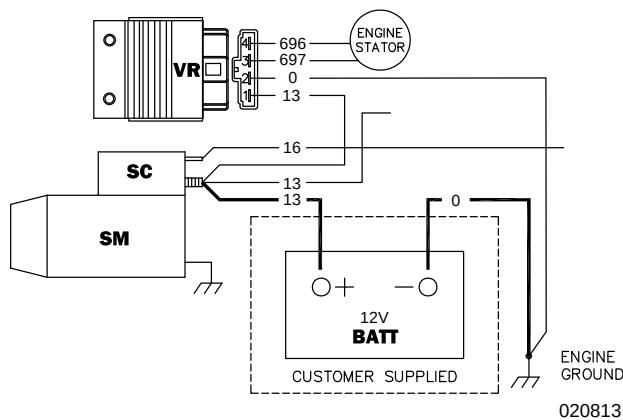


Figure 10-39. Stator Battery Charger Circuit

Procedure

1. Check the VR connector to verify it is connected correctly.
 - a. If the connector is found to be loose or improperly connected, repair the connector and retest.
2. Check the connection from the VR connector to the SC to verify it is connected correctly.
 - a. If the connector is found to be loose or improperly connected, repair the connector and retest.
3. Disconnect the battery.
4. Disconnect the J6 connector from the back of the controller.
5. Disconnect the VR connector at both the VR and the SC. Visually inspect the pins for both the VR and the VR harness for damage.
 - a. If the pins are found to be damaged, replace the wiring harness and/or VR as necessary and retest.
6. Set the DMM to measure resistance.

7. Using the appropriate back probe, insert one meter lead to Wire 13 at the VR connector and the other to the ring terminal at the SC. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
8. Using the appropriate back probe, insert one meter lead to Wire 0 at the VR connector and the other to the battery negative terminal. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, check that the ground connection for Wire 0 at the backside of the engine is clean and tight. If the connection passes inspection, replace the wiring harness and retest.
9. Using the appropriate back probes, insert the meter leads to the yellow wires at the VR connector. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the stator and retest.
10. Set the DMM to measure DC voltage.
11. Reconnect the VR connector at both the VR and the SC.
12. Reconnect the battery.
13. Place the meter leads across the battery terminals. Measure the voltage of the battery and record the reading. Leave meter leads connected here for **Step 14**.
14. Start the generator in MANUAL. Allow the generator to run for 15 seconds. Record the battery voltage reading.

NOTE: Ignore any Charger Missing AC warnings in this step.

 - a. If the battery voltage is above the reading recorded in **Step 13** and the generator does not show any faults, the Stator Battery Charger is working as designed.
 - b. If the battery voltage measures the same or less than what was recorded in **Step 13**, proceed to **Step 15**.
15. Set the DMM to measure AC voltage.
16. Disconnect the VR connector.
17. Using the appropriate back probes measure AC voltage from 696 to 697.
18. Start the generator in MANUAL. Allow the generator to run for 15 seconds. Record the voltage reading. Stop the unit.
 - a. If voltage reading is below 70 VAC replace the Engine Stator then retest.
 - b. If at least 70 VAC to discuss volts is measured, replace the VR and retest.

Starter Motor Test

Conditions Affecting Starter Motor Performance

- A binding or seizing condition in the starter motor bearings.
- A shorted, open or grounded armature.
 - Shorted armature (wire insulation worn and wires touching one another) producing low or no rpm.
 - Open armature (wire broken) producing low or no rpm and excessive current draw.
 - Grounded armature (wire insulation worn and wire touching armature lamination or shaft) producing excessive current draw or no rpm.
- Broken, damaged or weak magnets.
- Starter drive dirty or binding.

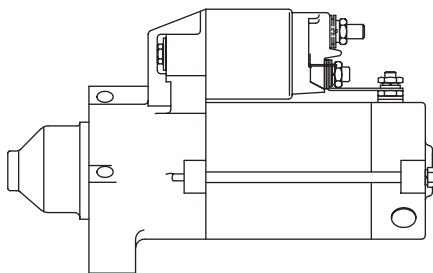


Figure 10-40. Starter Motor

General Theory

The [Battery and Cables Test](#) verified that the battery is fully charged and that the battery cables and connections are within the voltage drop specifications. The [Starter Control Relay Test](#) verified that both the circuit board is delivering DC voltage to the starter control relay (SCR) and the SCR is working as designed. The [Starter Contactor Test](#) verified the operation of the starter contactor (SC). Another possible cause of an “Engine Won’t Crank” problem is a failure of the starter motor itself.

Procedure

1. Set a DMM to measure DC voltage.
2. Connect the red meter test lead to the positive lead on the SC and connect the black meter lead to the starter motor case.
3. Set the controller to MANUAL and watch the meter. Meter should show battery voltage, the starter motor should operate, and the engine should crank.
 - a. If the meter shows battery voltage but the starter does not engage, replace the starter and retest.

- b. If battery voltage is shown and the starter motor tries to engage (pinion engages) but the engine does not crank, check for mechanical binding of the engine and/or rotor.

Starter Contactor Test

General Theory

The coil in the starter contactor (SC) must energize and its normally open contacts must close or the engine will not crank. This test will determine if the SC is not working.

Procedure

Carefully inspect the starter motor cable that runs from the battery to the starter motor. Cable connections must be clean and tight. If connections are dirty or corroded, remove the cable and clean cable terminals and terminal studs. Replace any cable that is defective or badly corroded.

See [Figure 10-41](#) for Test Points.

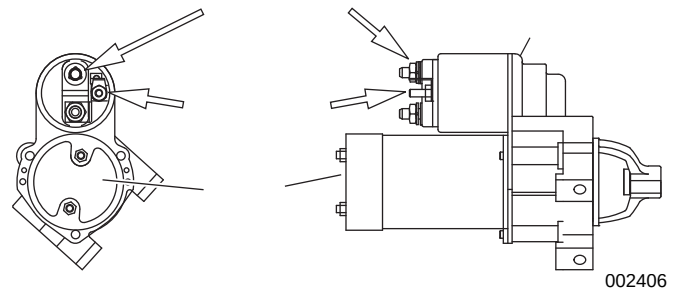


Figure 10-41. Starter Contactor

1. Set DMM to measure DC voltage.
2. Connect the positive meter lead to the positive post of the battery and connect the negative meter lead to the negative post of the battery.
 - a. If at least 12 VDC is measured, record that reading and proceed to Step 3.
 - b. If less than 12 VDC is measured, proceed to the [Battery and Cables Test](#) to check the battery condition.
3. Connect the positive meter lead to Test Point A and connect the negative meter lead to the negative post of the battery. Record the reading.
 - a. If the battery voltage measured is within 0.3 V of the measurement in Step 2, proceed to Step 4.
 - b. If the battery voltage measured is more than 0.3V different from the measurement in Step 2, proceed to the [Battery and Cables Test](#) to check for the voltage loss on the cable.
4. Connect the red meter lead to Test Point B and connect the black meter lead to the negative post

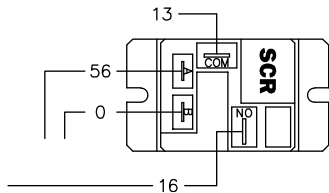
of the battery. Set the generator to MANUAL. Record the reading.

- a. If battery voltage was measured, proceed to the [Starter Motor Test](#).
 - b. If battery voltage was not measured, proceed to Step 5.
5. Set the DMM to measure resistance.
 6. Disconnect Wire 16 at both the SCR and SC. Connect the red meter lead to the SC end of Wire 16 and the black lead to the SCR end of the wire. Record the reading.
 - a. If the resistance measures less than 5 ohms, proceed to the [Starter Control Relay Test](#).
 - b. If the resistance is greater than 5 ohms or reads OL, replace the wiring harness and retest.

Starter Control Relay Test

General Theory

The starter control relay (SCR) located in the control panel must energize for cranking to occur. Once energized, the normally open contacts of the SCR will close and battery voltage will be available to Wire 16 and to the starter contactor (SC).



020554

Figure 10-42. Starter Control Relay

Procedure

1. Set the DMM to measure DC voltage.
2. Connect the positive meter lead to Wire 13 at the SCR and connect the negative meter lead to the battery negative terminal. Record the reading.
 - a. If battery voltage was NOT measured, replace the wiring harness and retest.
3. Locate and disconnect Wire 56 from the SCR. Visually inspect the terminal for Wire 56 on the SCR and the Wire 56 harness connection for damage..
 - a. If either is found to be damaged, replace the SCR and/or replace the wiring harness and retest.
4. Connect the red meter test lead to Wire 56 and the other meter test lead to the battery negative terminal. Set the controller to MANUAL. Record the reading.
 - a. If battery voltage was measured, Wire 56 is working as intended, proceed to Step 6.
5. Using the appropriate back probes, insert the red meter lead to Wire 56 at the J4 controller connector **while leaving the connector connected to the controller**. Connect the black meter lead to ground. Set the controller to MANUAL and record the reading.
 - a. If no voltage is measured, check the harness and controller pins.
 - b. If the pins are found to be damaged, replace the wiring harness and/or controller as necessary and retest.
 - c. If the harness and controller pins pass inspection, replace the controller and retest.
6. Disconnect battery.
7. Disconnect all four (4) wires from the SCR. Visually inspect all the terminals and spade connectors for damage.
 - a. If the terminals/spade connectors are found to be damaged, replace the SCR and/or replace the wiring harness as necessary and retest.
8. Disconnect the J4 harness connector from the controller.
9. Set the DMM to measure resistance.
10. Using the appropriate back probes, connect the red meter lead to Wire 56 at the J4 connector and the black lead to Wire 56 at the SCR. Record the reading.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
11. Connect the red meter lead to the Wire 13 terminal at the SC and connect the black lead to the Wire 13 terminal at the SCR. Record the reading.
 - a. If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
12. Connect the red meter lead to Wire 16 at the SCR and the black meter lead to Wire 16 at the SC. Record the reading.
 - a. If the resistance measures greater than 5 ohms or measures OL, replace the wiring harness and retest.
13. Connect the red meter lead to Wire 0 at the SCR and connect the black meter lead to the negative battery terminal. Record the reading.
 - a. If the resistance measures greater than 5 ohms or measures OL, check the ground connection at the ground block for a corroded or loose connection. If passes inspection, replace the wiring harness and retest.

14. Connect the red meter lead to the terminal for Wire 56 and the black meter lead to the terminal for Wire 0 on the SCR. Record the reading.
 - a. If the resistance measures greater than 165 or less than 130 ohms or measures OL, replace the SCR and retest.
15. Connect a 12 VDC power source across the terminals for Wires 56 and 0 on the SCR to energize the relay. Place one meter lead to the terminal for Wire 13 the other meter lead to the terminal for Wire 16 on the SCR. Record the reading.
 - a. If measured resistance is less than 150 ohms, the SCR test passes. Reinstall all connectors, reconnect battery, and proceed to the [Starter Contactor Test](#).
 - b. If continuity is not measured, replace the SCR and retest.

Throttle Position Sensor Test

General Theory

The ECM controls the electronic throttle control valve (ETC). The ECM uses the TTMAP sensor to verify the demand on the engine, and this will move the throttle plate position accordingly to maintain 3600 RPM.

Procedure

1. Clear the alarm and run the unit.
 - a. If the alarm returns proceed to **Step 2**.
 - b. If the alarm does not return, proceed to [Intermittent Alarm Test](#).
2. Verify the throttle plate can move freely when actuated manually. If there is resistance, check for obstructions causing the binding. If none are found, replace the mixer assembly and retest.
3. See [Figure 10-43](#). While visually observing the external throttle valve linkage, remove and reinstall the 15A fuse from the fuse holder in Wire 13. The throttle linkage should fully sweep then set itself for engine starting.

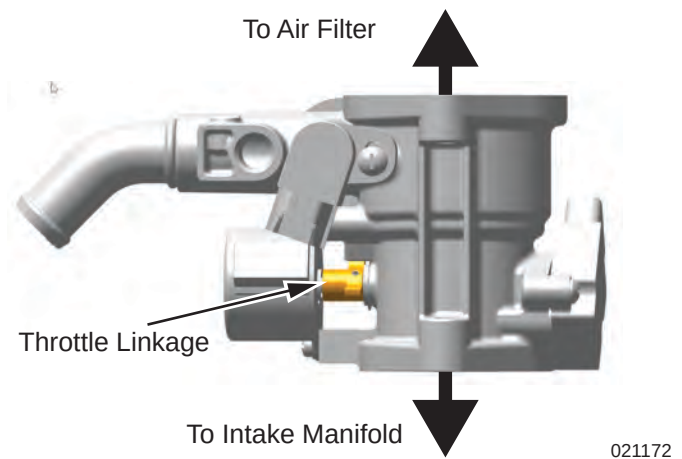


Figure 10-43. External Throttle Valve Linkage

- a. If the throttle valve linkage does not move as described, proceed to the [Electronic Throttle Control Power Test](#).
4. If testing for E-codes 3100, 3101, 3107, 3115, or 3117, skip to Step 14.
5. Set the DMM to measure resistance.
6. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace the ECM harness and/or ECM as necessary and retest.
7. Disconnect the harness connector at the TTMAP. Visually inspect both the TTMAP and TTMAP harness pins for damage.
 - a. If the pins are found to be damaged, replace the mixer assembly and retest.
8. Using the appropriate back probes, connect one meter lead to Wire 766B at the ECM and the other lead to Wire 766B at the TTMAP.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to Step 11.
9. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other lead to Wire 820A at the TTMAP.
 - a. If resistance measures greater than 5 ohms or reads OL, proceed to Step 13.
10. Using the appropriate back probes, connect one meter lead to Wire 820G at the ECM and the other lead to Wire 820G at the TTMAP.
 - a. If resistance measures less than 5 ohms, proceed to Step 14.
 - b. If resistance measures greater than 5 ohms or reads OL, proceed to Step 13.
11. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead

- to Wire 766B at the ECM and the other to Wire 766B at the ECM side of the ETMC connector.
- If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
12. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probes, connect one meter lead to Wire 820A at the ECM and the other to Wire 820A at the ECM side of the ETMC connector.
- If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
13. Disconnect the ETMC connector. Visually inspect the ETMC connector pins for damage. Using the appropriate back probe, connect one meter lead to Wire 820G at the ECM and the other to Wire 820G at the ECM side of the ETMC connector.
- If the pins are found to be damaged, replace the wiring harness and/or the mixer assembly and retest.
 - If the resistance measures less than 5 ohms, replace the mixer assembly and retest.
 - If the resistance measures greater than 5 ohms or OL, replace the wiring harness and retest.
14. Set the DMM to measure DC voltage.
15. Reconnect the ETMC harness connector.
16. Reconnect the harness connector at the TTMAP.
17. Reconnect the ECM harness connector.
18. Using the appropriate back probe, connect one meter lead to Wire 820A at the TTMAP. Connect the other meter lead to battery ground.
- If 5V is not measured, replace ECM and retest.
19. Using the appropriate back probe, connect one meter lead to Wire 820G at the TTMAP. Connect the other meter lead to battery ground.
- If no voltage is measured, proceed to Step 21.
20. Check for a **short-to-power** between Wire 820G at the ECM and Wire 820G at the mixer assembly.
- If a **short-to-power** is found, replace the wiring harness and/or mixer assembly and retest.
 - If no **short-to-power** is found, replace the ECM and retest.

21. Using the appropriate back probe, connect one meter lead to Wire 766B at the TTMAP. Connect the other meter lead to battery ground. With the ECM powered on, the meter should display a voltage range of 0.5–4.5 VDC.

- If the voltage measured falls within that range, replace ECM and retest.
- If the voltage measured is outside of that range, replace the mixer assembly and retest.

Transfer Control Wires Test

General Theory

The transfer switch is controlled by the Power Zone 200 controller in the generator. Wire 194 has 12 VDC supplied from the controller to the transfer switch. This provides battery power to the transfer relay (also powers the SACM/FCM module in some configurations). Wire 23 is switched to ground in the Power Zone 200 controller when transfer to generator is commanded by the Power Zone 200 controller. When Wire 23 is grounded, the relay is activated which causes the contactor to switch from Utility position to Standby position allowing the generator output voltage to connect to the load lugs to supply power to the customer.

If Wire 194 is shorted-to-ground in the wiring or a damaged SACM/FCM/controller output causes the voltage to drop to less than 6 VDC, the controller will turn off the output and set a Transfer Wiring Alarm. If the problem is found and corrected, the voltage will be close to battery voltage after clearing the alarm. Clearing the alarm will reset the output and return it to 12 VDC unless the overload is still present. If it is still present, the output will again turn off and set an alarm. Wire 0 is used as a ground for the 12 VDC power to the SACM/FCM.

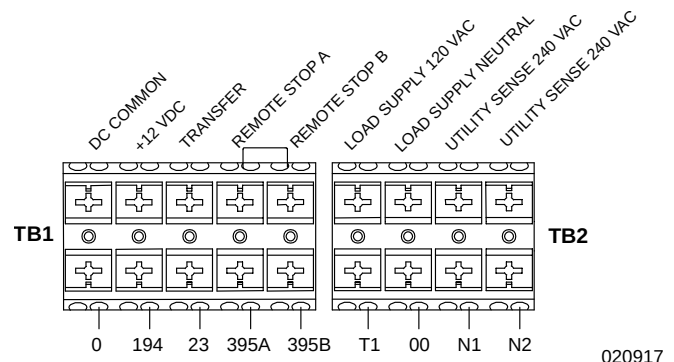


Figure 10-44. Customer Connection Block and Transfer Control Wires

Procedure

- Verify battery voltage and all fuses.
- Set the DMM to measure DC voltage.

NOTE: If troubleshooting E-code 2099 set the DMM to measure AC voltage and proceed to Step 5.

3. Locate Wires 194 and 0 at the Customer Connection Block.
4. Insert the red meter lead into the WAGO block terminal for Wire 194 and the black meter lead into the WAGO block terminal for Wire 0. Record the reading.
 - a. If the voltage measures less than 6 VDC, proceed to Testing Wire 194.
 - b. If the voltage measures 11 VDC or greater, proceed to Testing Wire 23.
5. Insert the red meter lead into the WAGO block terminal for Wire 194 and the black meter lead into the WAGO block terminal for Wire 0. Record the reading. Repeat for Wire 23 to Wire 0.
 - a. If more than 40 VAC is measured, find the mis-wired connection at TB1 & TB2 or in the transfer switch.
 - b. If 2–40 VAC is measured, install 120k Ohm resistors as directed in PIB18-06-AAC, and retest.
 - c. Power cycle the controller and retest, if still present replace the controller and retest.

NOTE: The T1 and 00 Charger input voltage is used to confirm transfer to standby has occurred. If this voltage is not seen after transfer signal has been sent, E-code 2738 (No Transfer Detected) will be set as a warning.

Testing Wire 194

NOTE: Wire 194 will not be powered if there is an active alarm for Transfer Warning. The alarm will need to be cleared at each step to perform the test correctly.

1. Set the DMM to measure DC Voltage.
2. At the transfer switch, disconnect Wire 194 at the SACM/FCM.
3. Connect the red meter lead to the disconnected Wire 194. Place the black meter lead on the terminal connection for Wire 0 on the SACM/FCM. Record the reading.
 - a. If the voltage measures less than 3 VDC, reconnect Wire 194 and proceed to Step 5.
 - b. If the voltage measures above 11 VDC, reconnect Wire 194 and proceed to Step 4.
4. Place the red meter lead on the terminal for Wire 194 on the SACM/FCM. Place the black meter lead on the terminal for Wire 0 at the SACM/FCM. Record the reading.
 - a. If the voltage measures less than 3 VDC, replace the SACM/FCM and retest.

- b. If the voltage measures above 11 VDC, and no transfer to standby occurs, refer to the Residential and Commercial Transfer Switches Diagnostic Manual for diagnosis.
5. Disconnect Wire 194 from the transfer switch side of the WAGO block in the customer connection block.
6. Place the red meter lead in the WAGO block terminal for Wire 194. Place the black meter lead in the WAGO block terminal Wire 0. Record the reading.
 - a. If the voltage measures above 11 VDC, replace Wire 194 between the customer connection block and the SACM/FCM and retest.
 - b. If the voltage measures less than 3 VDC, reconnect Wire 194 and proceed to Step 7.
7. Disconnect Wire 194 from the generator side of the WAGO block in the customer connection block.
8. Connect the red meter lead to the disconnected Wire 194. Place the black meter lead in the WAGO block terminal Wire 0. Record the reading.
 - a. If the voltage measures above 11 VDC, replace the WAGO block and retest.
 - b. If the voltage measures less than 3 VDC, reconnect Wire 194 and proceed to Step 9.
9. Disconnect the battery.
10. Remove AC utility power from the generator.
11. Disconnect the J4 connector at the controller. Visually inspect the pins for both the controller and the J4 connector for damage.
 - a. If the pins pass inspection, proceed to Step 12.
 - b. If the pins are found to be damaged, replace the wiring harness and/or controller and retest.
12. Check for a short to ground on Wire 194 between the J4 connector and the WAGO block.
 - a. If a short to ground is found, replace the wiring harness and retest.
 - b. If no short to ground is found, replace the controller and retest.

Testing Wire 23

NOTE: Confirm the MLCB is set to ON for this procedure.

1. Remove all AC and DC power from transfer switch. Verify all connections in the transfer switch are tight.
 - a. If connections fail inspection, repair as necessary and retest.
2. Set the DMM to measure DC Voltage.
3. Place the generator in AUTO and remove utility source voltage. Watch for the green LED in the

status LED indicators on the side of the controller to begin flashing.

4. Place the red meter lead on the terminal for Wire 194 at the SACM/FCM. Place the black meter lead on the terminal for Wire 23 at the SACM/FCM. Record the reading.
 - a. If the voltage measures near 12–14 VDC and no transfer to standby occurs, refer to the *Residential and Commercial Transfer Switches Diagnostic Manual* for diagnosis.
 - b. If the voltage reads close to 0 VDC even with the green LED flashing on the side of the generator, proceed to Step 5.
5. With the generator still running, remove Wire 23 from the SACM/FCM and place a jumper wire from the Wire 23 terminal to ground in the transfer switch.
 - a. If the transfer to standby occurs, reconnect Wire 23 to the SACM/FCM and proceed to Step 6.
 - b. If no transfer occurs, replace the SACM/FCM and retest.
6. With the generator still running, remove Wire 23 from the transfer switch side of the WAGO block in the customer connection block and ground the wire.
 - a. If no transfer to standby occurs, repair Wire 23 between the customer connection block and the transfer switch and retest.
7. With the generator still running, remove Wire 23 from the generator side of the WAGO block at the customer connection block. Insert a jumper wire in that terminal and ground the wire.
 - a. If transfer to standby does not occur, replace WAGO block and retest.
8. Turn generator OFF.
9. Disconnect the battery.
10. Remove AC utility power from the generator.
11. Disconnect the J4 connector at the controller. Visually inspect the pins for both the controller and the J4 connector for damage.
 - a. If the pins are found to be damaged, replace the wiring harness and/or controller and retest.
12. Set the DMM to measure continuity.
13. Using the appropriate back probe, connect the red meter lead to Wire 23 at the J4 connector. Place the black meter lead on the disconnected Wire 23 at the WAGO block. Record the reading.
 - a. If the wire does not have continuity, replace the wiring harness and retest.
 - b. If the wire has continuity, replace the controller and retest.

Wire 14 Test

General Theory

When the engine begins cranking and speed reaches 75 RPM, power is supplied to Wire 14 to open both fuel shut-off valves (SOVs). When the engine is commanded off, power to Wire 14 is removed, which in turn will de-energize both solenoids and close the SOVs.

Procedure

1. Set the DMM to measure DC voltage.
2. Locate the J4 connector on the back of the Power Zone 200 Controller.
3. Locate Wire 14 in Pin position 7 of the J4 connector.
4. Using appropriate back probes, insert the red meter lead in position 7 and the black meter lead on battery negative terminal.

NOTE: Leave the black meter lead on battery negative terminal through Step 10.
5. Press the MANUAL button on the Controller. Record the reading on the meter.
 - a. If no voltage is measured, check the connection and pins for the J4 connector. If the pins are found to be damaged, replace the wiring harness and/or controller as necessary and retest.
 - b. If the harness and controller pins pass inspection, replace the controller and retest.
6. Locate SOV1 and SOV2 in the battery compartment.

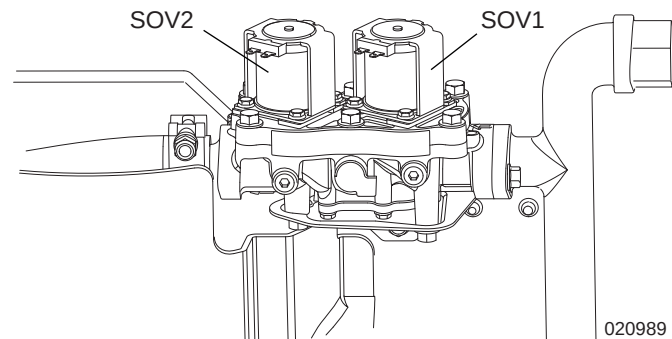


Figure 10-45. Fuel Shut-Off Valves (SOV 1 and SOV 2)

7. Using the appropriate back probe, insert the red meter lead to Wire 14 on SOV1.
8. Press the MANUAL button on the controller. Record the reading on the meter.
 - a. If no voltage is measured, replace the wiring harness and retest.

9. Using the appropriate back probe, insert the red meter lead to Wire 14 on SOV2.
10. Press the MANUAL button on the controller. Record the reading on the meter.
 - a. If no voltage is measured, replace the wiring harness and retest.
11. Set the DMM to measure resistance.
12. Disconnect Wire 0 on SOV 1. Connect the red meter lead to Wire 0 and the other lead to the ground bar. Record the reading.
 - a. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.
13. Disconnect the wires to both SOV1 and SOV2. Connect the meter leads across the terminals of each solenoid. Record the reading.
 - a. If resistance measured between 5–60 ohms, proceed to Step 14.
 - b. If the resistance measured less than 5 ohms, more than 60 ohms, or OL, replace the fuel shut off valve assembly and retest.
14. Disconnect the harness connector at the ECM. Visually inspect the pins for both the ECM and the ECM harness for damage.
 - a. If the pins are found to be damaged, replace either the wiring harness or ECM as needed and retest.
15. Connect the red meter lead to Wire 610 at SOV2. Using the appropriate back probe, insert the black meter lead in the ECM side of Wire 610 Pin location 3L. Record the reading on the meter.
 - a. If resistance measures less than 5 ohms, replace the ECM and retest.
 - b. If resistance measures greater than 5 ohms or reads OL, replace the wiring harness and retest.

Intermittent Alarm Test

General Theory

The condition that triggered the alarm on the generator is no longer present. Intermittent alarms may be caused by improper battery voltage, loose connections, faulty parts, old firmware, external sources of electrical noise, etc.

Procedure

1. Check GenService for any appropriate bulletins.
2. Check each electrical ground on the unit for required torque specifications (**found in the beginning of this book**). There are several in the control box, one on each engine foot, and one on the back side of the engine block near the rear engine foot.
3. Perform the [Battery And Cables Test](#).
4. Verify each device is fully updated.
5. Run the generator at full load (if possible).
6. While running the generator, perform harness wiggle testing. Using the known fault code, gently wiggle the connector to each sensor (if applicable) and to the ECM connector and to all ground points if able.
7. If not able to recreate the fault condition contact **Technical support**.

Power Zone 200 Fuse Testing

NOTE: This test requires having several replacement 7.5 Amp fuses on hand.

Replace the 7.5 Amp fuse. If the fuse immediately fails proceed to Procedure A. Run the unit. If the fuse fails after a crank attempt proceed to Procedure B.

Procedure A

1. Check all harness grounds for tightness. Refer to the **torque tables at the beginning of the book** for torque values.
 - **Check 1 harness ground on each engine foot.**
 - **Check 1 harness ground on the back of the engine block near the rear engine foot.**
 - Check all harness grounds in the control box.
2. Disconnect the battery.
3. Disconnect J6 connector from the controller.
4. Set the DMM to measure resistance. Using the appropriate back probes, connect the red meter lead to Wire 877 at J3-9 and the black lead to the negative battery cable. Record the results.
 - a. If more than 4 kohms is measured proceed to **Step 7**.
5. Disconnect the ECM connector and J3 connector and measure again.
 - a. If OL is not measured replace the harness and retest.
6. Plug the ECM connector back in and remeasure.
 - a. If less than 8 kohms is measured replace the ECM and retest.
7. Connect J3 connector and disconnect the ECM and measure again.
 - a. If less than 9 kohms is measured replace the controller and retest.

8. Reconnect the ECM. Set the DMM to measure resistance. Using the appropriate back probes, connect the red meter lead to Wire 13 at J4-1 and the black lead to the negative battery cable. Record the results.
 - a. If greater than 100 kohms is measured proceed to Step 10.
 9. Disconnect J4 connector and measure again.
 - a. If less than 100 ohms is measured, replace the harness and retest.
 10. Disconnect J4 if not disconnected already. Set the DMM to measure resistance. Place a meter lead on pin J4-1 on the Power Zone 200 controller and the other meter lead to the negative battery cable.
 - a. If 120k to 140 kohms is not measured, replace the controller and retest.
 11. Disconnect J4 connector. With the DMM set to measure resistance check the resistance. At the controller connect a meter lead to the pin for Wire 13 at J4-1 and the other meter lead to the pin for Wire 13 at J4-2. Record the results.
 - a. If more than 5 ohms, replace the controller and retest.
 12. Plug each connector back in and reconnect the battery. Install a new fuse.
 - a. If the fuse still fails replace the controller and retest.
 13. If the fuse fails again replace the harness and retest.
- to the pin for Wire 0 at J4-10 and the other meter lead to the pin for Wire 0 at J4-11. Record the results.
- a. If greater than 5 ohms, replace the controller and retest.
6. With the DMM set to measure resistance, use the appropriate back probes to connect a meter lead to Wire 14 at J4-7 and the other meter lead to the negative battery cable. Record the results.
 - a. If greater than 5 ohms is measured proceed to **Step 9**.
 7. Disconnect each fuel solenoid. Set the DMM to measure resistance. Using the appropriate back probes connect a meter lead to Wire 14 at J4-7 and the other meter lead to the negative battery cable. Record the results.
 - a. If less than 5 ohms is measured, replace the harness and retest.
 8. Unplug each fuel solenoid. With the DMM measuring resistance. Connect a meter lead to one of the fuel solenoid pins and the other meter lead to the other fuel solenoid pin. Record the results. Perform this on both fuel solenoids.
 - a. If less than 5 ohms is measured, replace the fuel solenoids and retest.
 9. Set the DMM to measure resistance. Connect a meter lead to one of the Wire 14 connectors for the fuel solenoids and using the appropriate back probes connect the other meter lead to Wire 14 at J4-7. Record the results. Perform this for each fuel solenoid connector Wire 14.
 - a. If greater than 5 ohms is measured replace the wiring harness and retest.
 10. Set the DMM to measure resistance. On the controller connect a meter lead to the pin for Wire 14 at J4-7 and the other meter lead to the pin for wire 0 at J4-10. Record the results.
 - a. If less than 100 kohms is measured, replace the controller and retest.
 11. Unplug J4 connector. Set the DMM to measure resistance. Using the appropriate back probes connect a meter lead to Wire 56 on the harness side at J4-12 and the other meter lead to battery negative. Record the results.
 - a. If greater than 130 ohms and less than 165 ohms is measured, proceed to **Step 15**.
 12. Disconnect Wire 56 on the SCR. Set the DMM to measure resistance. Using the appropriate back probes connect a meter lead to Wire 56 at J4-12 and the other meter lead to the connector at the SCR for Wire 56. Record the results.
 - a. If greater than 5 ohms is measured, replace the harness and retest.

Procedure B

1. Check all harness grounds for tightness. Refer to the **torque tables at the beginning of the book** for torque values.
 - **Check 1 harness ground on each engine foot.**
 - **Check 1 harness ground on the back of the engine block near the rear engine foot.**
 - Check all harness grounds in the control box.
2. Disconnect the battery. Disconnect J6 connector.
3. With the DMM set to measure resistance, check the resistance. With the appropriate back probes connect the red meter lead to Wire 4 at J7 and the black meter lead to Wire 0A at J7. Record the results.
 - a. If 4–15 ohms is measured, proceed to **Step 5**.
4. Proceed to [Internal Diode Test](#) and test the Brushes and the Rotor then return here.
5. Disconnect J4 connector. Set the DMM to measure resistance. At the controller connect a meter lead

to the pin for Wire 0 at J4-10 and the other meter lead to the pin for Wire 0 at J4-11. Record the results.

- a. If greater than 5 ohms, replace the controller and retest.

6. With the DMM set to measure resistance, use the appropriate back probes to connect a meter lead to Wire 14 at J4-7 and the other meter lead to the negative battery cable. Record the results.

- a. If greater than 5 ohms is measured proceed to **Step 9**.

7. Disconnect each fuel solenoid. Set the DMM to measure resistance. Using the appropriate back probes connect a meter lead to Wire 14 at J4-7 and the other meter lead to the negative battery cable. Record the results.

- a. If less than 5 ohms is measured, replace the harness and retest.

8. Unplug each fuel solenoid. With the DMM measuring resistance. Connect a meter lead to one of the fuel solenoid pins and the other meter lead to the other fuel solenoid pin. Record the results. Perform this on both fuel solenoids.

- a. If less than 5 ohms is measured, replace the fuel solenoids and retest.

9. Set the DMM to measure resistance. Connect a meter lead to one of the Wire 14 connectors for the fuel solenoids and using the appropriate back probes connect the other meter lead to Wire 14 at J4-7. Record the results. Perform this for each fuel solenoid connector Wire 14.

- a. If greater than 5 ohms is measured replace the wiring harness and retest.

10. Set the DMM to measure resistance. On the controller connect a meter lead to the pin for Wire 14 at J4-7 and the other meter lead to the pin for wire 0 at J4-10. Record the results.

- a. If less than 100 kohms is measured, replace the controller and retest.

11. Unplug J4 connector. Set the DMM to measure resistance. Using the appropriate back probes connect a meter lead to Wire 56 on the harness side at J4-12 and the other meter lead to battery negative. Record the results.

- a. If greater than 130 ohms and less than 165 ohms is measured, proceed to **Step 15**.

12. Disconnect Wire 56 on the SCR. Set the DMM to measure resistance. Using the appropriate back probes connect a meter lead to Wire 56 at J4-12 and the other meter lead to the connector at the SCR for Wire 56. Record the results.

- a. If greater than 5 ohms is measured, replace the harness and retest.

13. Leave J4 connector and the SCR disconnected. Set the DMM to measure resistance. Use the appropriate back probes to connect a meter lead to Wire 56 at J4-12 and the other meter lead to battery negative. Record the results.
 - a. If less than 5 ohms is measured, replace the harness and retest.
 14. Disconnect Wire 0 at the SCR. Set the DMM to measure resistance. Measure the resistance across the two bare pins on the SCR. Record the results.
 - a. If less than 130 ohms or greater than 165 ohms is measured, replace the SCR and retest.
 15. Reconnect each connector and reconnect the battery. Disconnect J6 connector. Install a new fuse. If the fuse still fails replace the controller and retest.
 16. If the fuse fails again replace the harness and retest. If it fails again, replace the controller and retest.
- a. If less than 5 ohms is measured replace the OPS and retest.
 5. Connect the MPR and the ECM, replace the fuse if needed, and reconnect the battery.
 - a. If the fuse fails, replace the ECM and retest.
 6. Disconnect the fuse. Connect the ETMC. Reconnect the fuse.
 - a. If the fuse fails, replace the mixer assembly and retest.
 7. Disconnect the fuse. Connect the OS. Reconnect the fuse. Press manual and off in quick succession. Clear the alarm if one presents.
 - a. If the fuse fails, replace the OS and retest.
 8. Disconnect the fuse. Connect the OPS. Reconnect the fuse. Press manual and off in quick succession. Clear the alarm if one presents.
 - a. If the fuse fails, replace the OPS and retest.
 9. Disconnect the fuse. Connect IC1. Reconnect the fuse. (Perform this twice if twin cylinder). Press manual and off in quick succession. Clear the alarm if one presents.
 - a. If the fuse fails, replace the IC1 and retest.
 10. Reconnect any other remaining connector. Run the unit.
 - a. If the fuse fails, replace the ECM and retest.
 11. Replace each IC and retest.

ECM Harness Fuse Testing

NOTE: This test requires having several replacement 15 Amp fuses on hand.

Procedure

Replace the fuse and run the unit. If the fuse fails proceed to testing below.

1. Disconnect the battery and J6. Disconnect the ECM. Unplug Wire 13A from the MPR. Set a DMM to measure resistance. Connect a meter lead to pin for Wire 13A at the MPR and the other meter lead to the negative cable. Record the results.
 - a. If less than 5 ohms is measured, replace the harness and retest.
2. Unplug Wire 13A and Wire 608 to expose the pins for the MPR coil. Set a DMM to measure resistance. Connect a meter lead to pin for Wire 13A at the MPR and the other meter lead to the pin for Wire 608 at the MPR. Record the results.
 - a. If less than 130 ohms or greater than 165 ohms is measured, replace the MPR and retest.
3. Unplug IC1 (and IC2 if present), the ETMC, the OPS Wire 14C, and the OS. Set a DMM to measure resistance. Connect a meter lead to pin for Wire 14C at the MPR and the other meter lead to the negative battery cable. Record the results.
 - a. If less than 5 ohms is measured, replace the harness and retest.
4. Set a DMM to measure resistance. Connect a meter lead to a pin on the OPS, connect the other meter lead to the negative battery cable. Record the results. Do this twice for each OPS pin.

Engine Cranks But Does Not Run, Lacks Power, Runs Rough, Hunts or Backfires

The engine needs spark, compression, fuel, and air to be able to start and run. If any of these elements are missing, the engine cannot run. Various conditions may cause the engine to not stabilize. When the ECM sees the RPM is low, it opens the throttle and the mixer—so the engine speed recovers but it overshoots. When the ECM sees that RPM is high, it closes the throttle and the mixer—so engine speed recovers, but it undershoots. This cyclical back and forth in engine speed corresponds to an engine hunting scenario. This could be caused by various things including but not limited to old firmware, low engine compression, faulty parts, improper fuel pressure, incorrect fuel setting, bad spark, air blockage, and more.

Procedure

1. Verify the Power Zone 200 controller and the ECM are fully updated and the correct fuel type is chosen.
2. Verify the air filter is not clogged or wet, and the air intake and the throttle body is not obstructed.

- a. If obstructions are found, correct the condition and retest.
3. Check oil level.
 - a. If low, add oil until full and retest.
4. Perform [Battery and Cables Test](#).
 - a. Perform repairs as needed and retest.
5. Perform [Engine Speed Test](#).
 - a. Perform repairs as needed and retest.
6. Start the unit. Is the engine able to come off the starter at all?
 - a. If not, proceed to [Wire 14 Test](#).
7. Perform the [Fuel Supply and Pressure Test](#).
 - a. Replace any parts if needed and retest.
8. Verify installed fuel pipe meets sizing requirements in installation manual.
9. Perform the [Ignition Coil Test](#).
 - a. Replace any parts if needed and retest.
10. Perform the [Engine Compression Test](#).
 - a. If low, check rocker stud torques. Use the torque sheet at the beginning of the book for the torque values.
11. Perform the [Cylinder Leak Down Test](#).
 - a. If low, check rocker stud torques. Use the torque sheet at the beginning of the book for the torque values. Check for excessive wear on the guide plates.
12. Look for visible signs of damage. Listen for noises, see if audible sounds come from the engine or from the alternator.
 - a. If holes or dents or grinding or scraping noises come from the engine, replace the engine.
 - b. If holes or dents or grinding or scraping noises come from the alternator, replace the alternator.
13. Contact **Technical support**.

Section 11: *Electrical Data*

Introduction

Introduction Go to www.generac.com (<http://www.generac.com/service-support/product-support-lookup>) for the most current wiring diagrams and electrical schematics. Use model or serial number.

A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459

GROUP WD

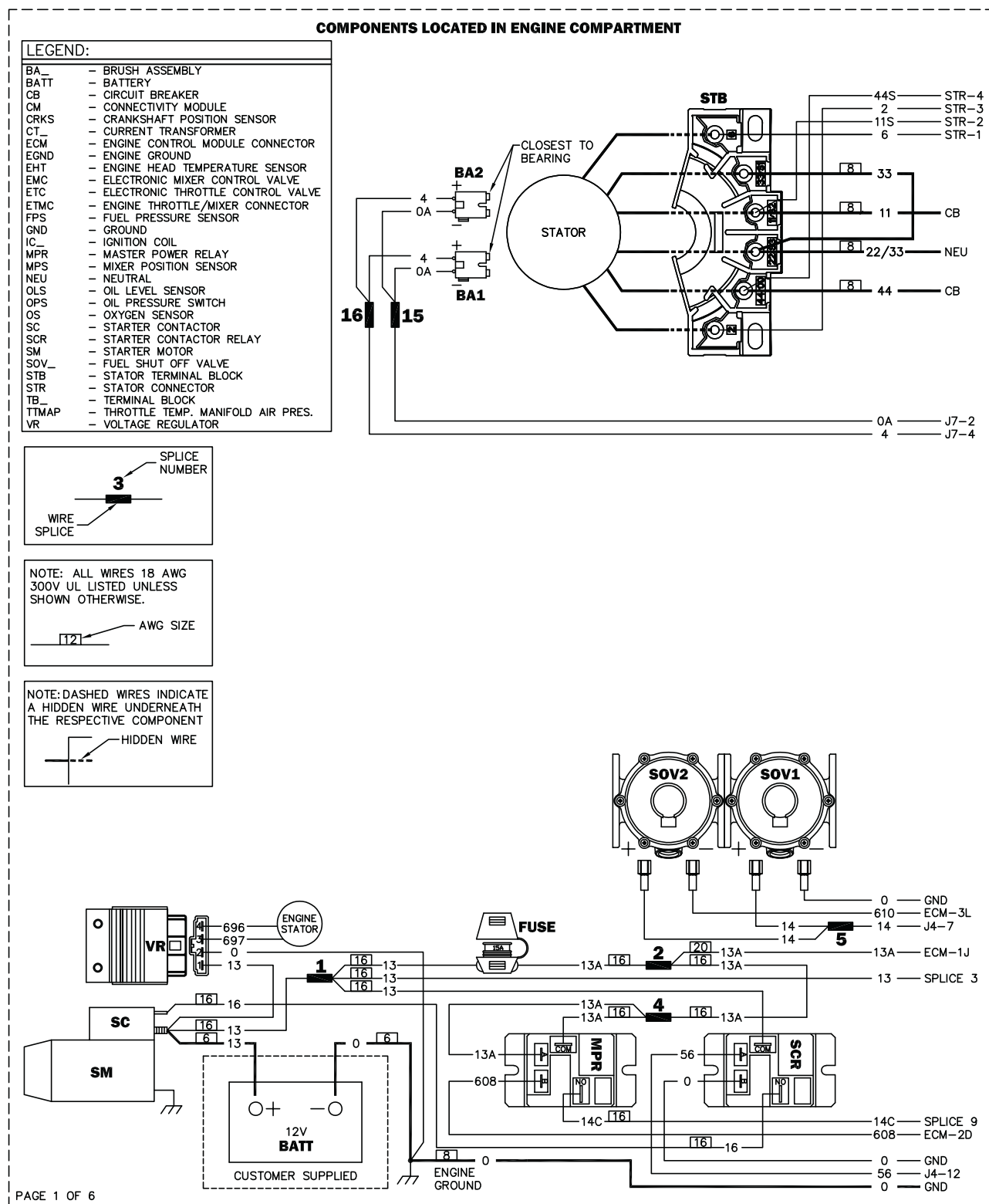
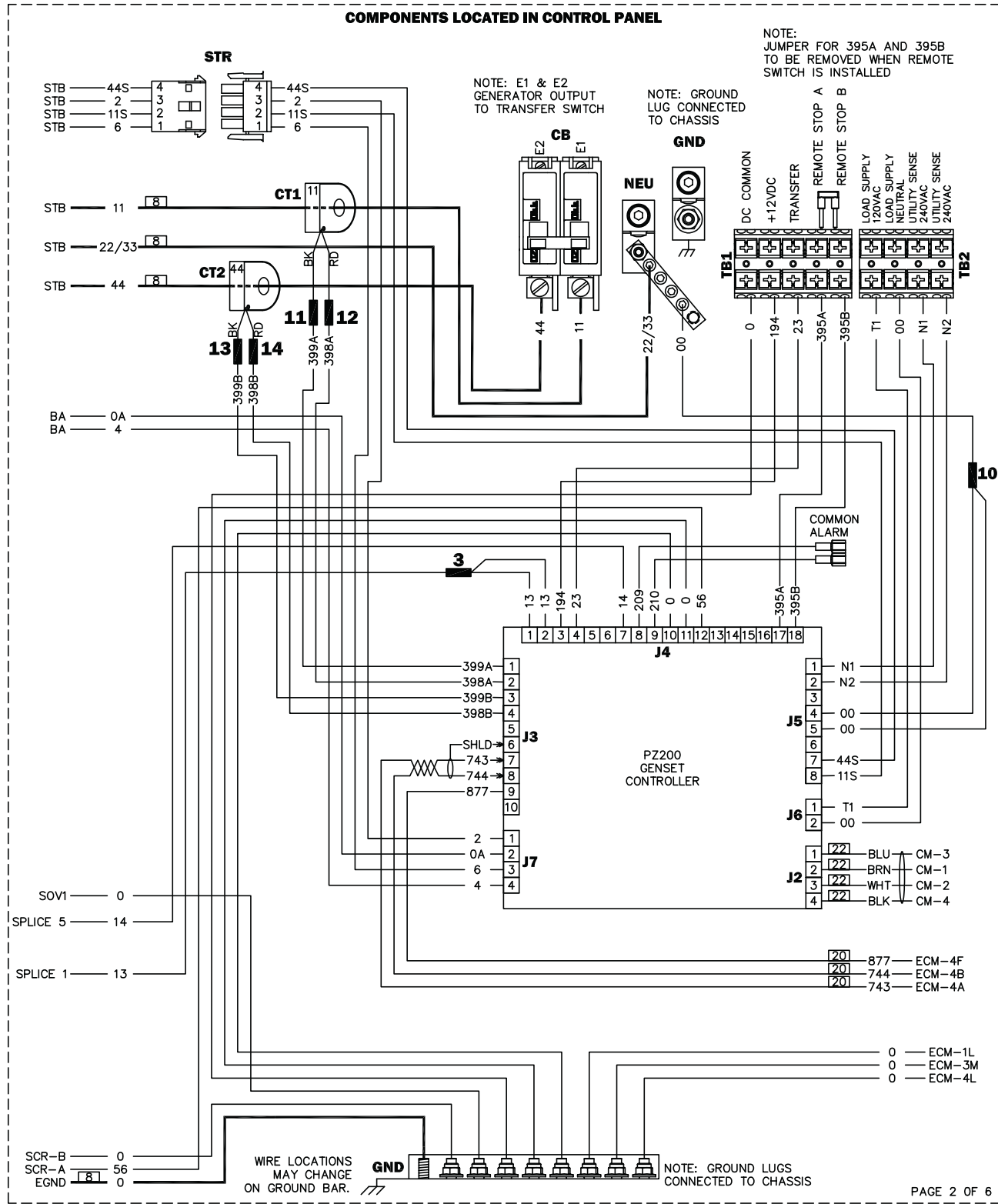


Figure 11-1. A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459 (page 1 of 6)

GROUP WD



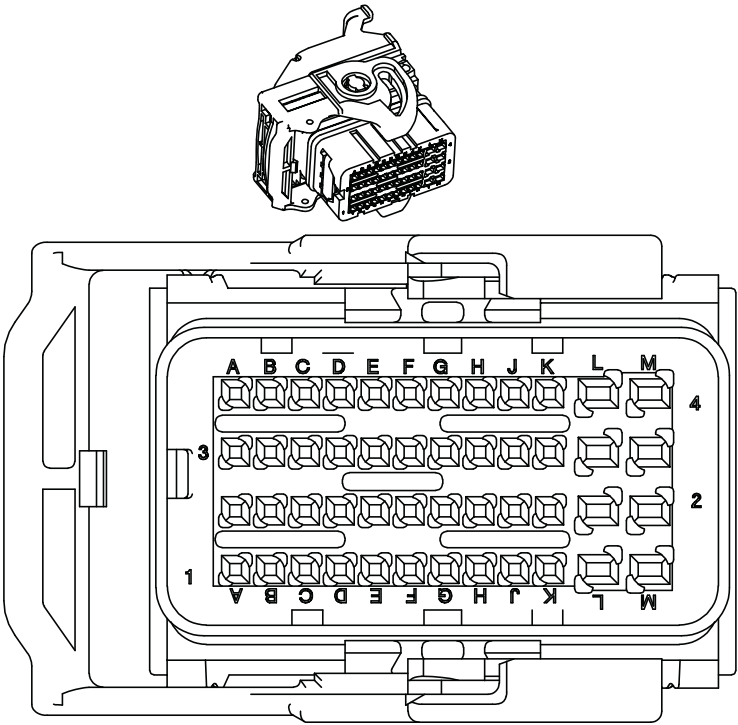
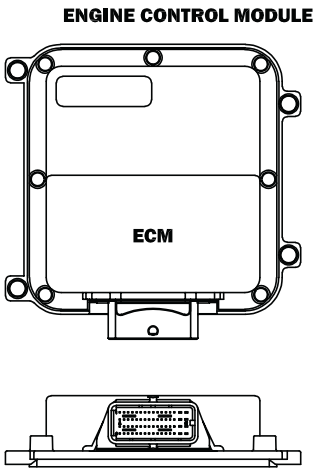
REVISION: CN-0100171-B
DATE: 05/02/2025

PAGE 2 OF 6

WIRING - DIAGRAM
PZ200 SSI 60HZ 459
DRAWING #: A0005639540

Figure 11-2. A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459 (page 2 of 6)

COMPONENTS LOCATED IN CONTROL PANEL



PIN LIST

PIN	WIRE	TO
1A	605A	ETMC-13
1B	605B	ETMC-14
1C	79A	CRKS-A
1D	79B	CRKS-B
1E	770A	ETMC-9
1F	771A	ETMC-11
1G	770B	ETMC-10
1H	771B	ETMC-12
1J	13A	SPLICE 2
1K	14C	SPLICE 6
1L	0	GND
1M	-	-

PIN LIST

PIN	WIRE	TO
2A	605C	ETMC-15
2B	606	ETMC-7
2C	766A	ETMC-6
2D	608	MPR-B
2E	-	-
2F	607	OS-B
2G	754	ETMC-3
2H	-	-
2J	899	OLS-C
2K	714	EHT-1
2L	-	-
2M	-	-

PIN LIST

PIN	WIRE	TO
3A	605D	ETMC-16
3B	766B	ETMC-4
3C	797	ETMC-1
3D	-	-
3E	-	-
3F	-	-
3G	-	-
3H	-	-
3J	-	-
3K	820A	SPLICE 7
3L	610	SOV2
3M	0	GND

PIN LIST

PIN	WIRE	TO
4A	743	J3-7
4B	744	J3-8
4C	566	OPS
4D	-	-
4E	-	-
4F	877	J3-9
4G	-	-
4H	-	-
4J	804	OS-C
4K	820G	SPLICE 8
4L	0	GND
4M	451	IC1-C

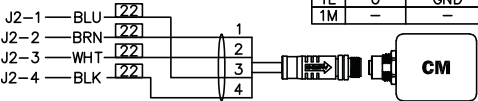


Figure 11-3. A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459 (page 3 of 6)

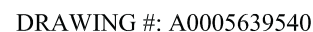
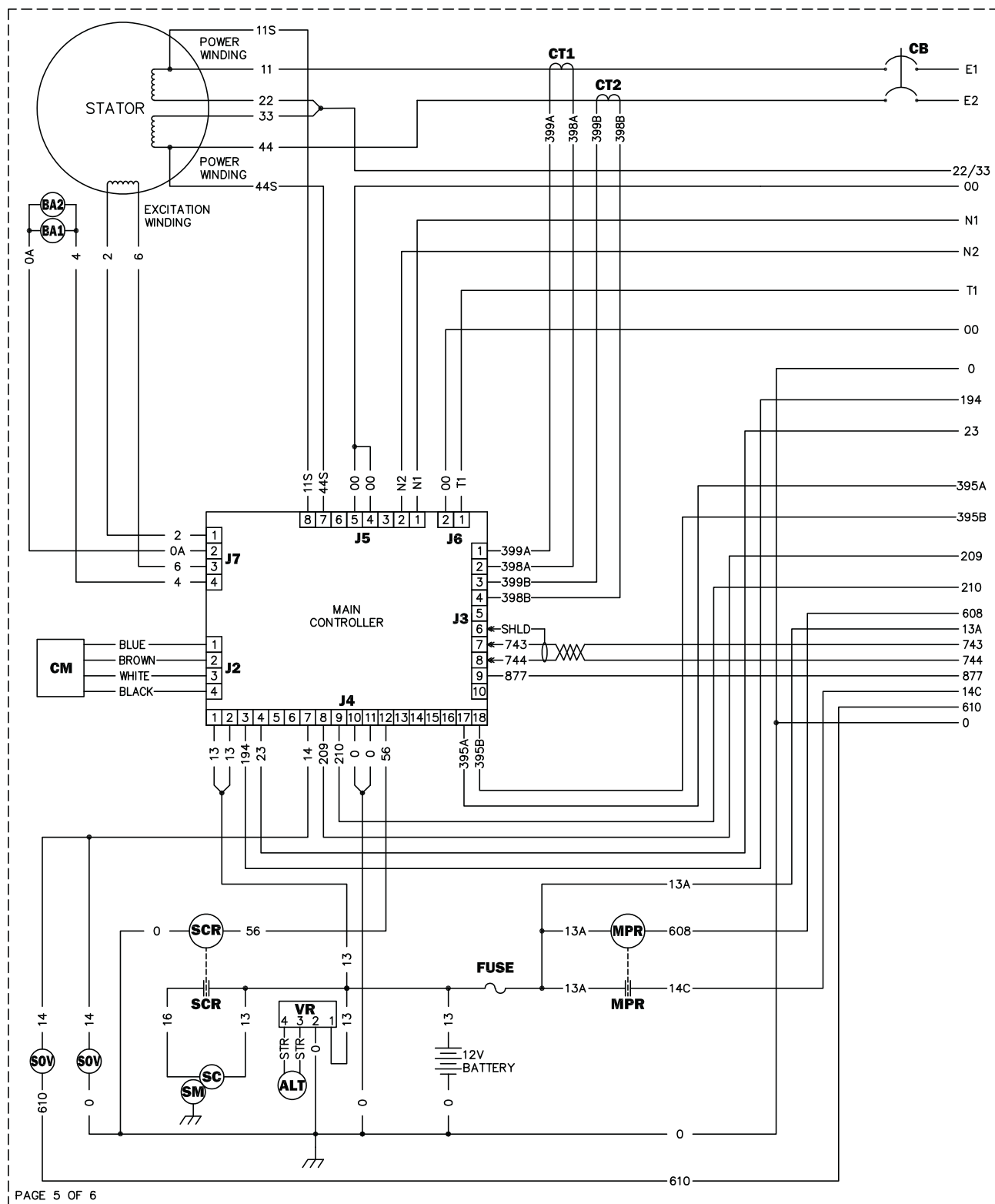


Figure 11-4. A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459 (page 4 of 6)

GROUP SD



SCHEMATIC - DIAGRAM

REVISION: CN-0100171-B

PZ200 SSI 60HZ 459

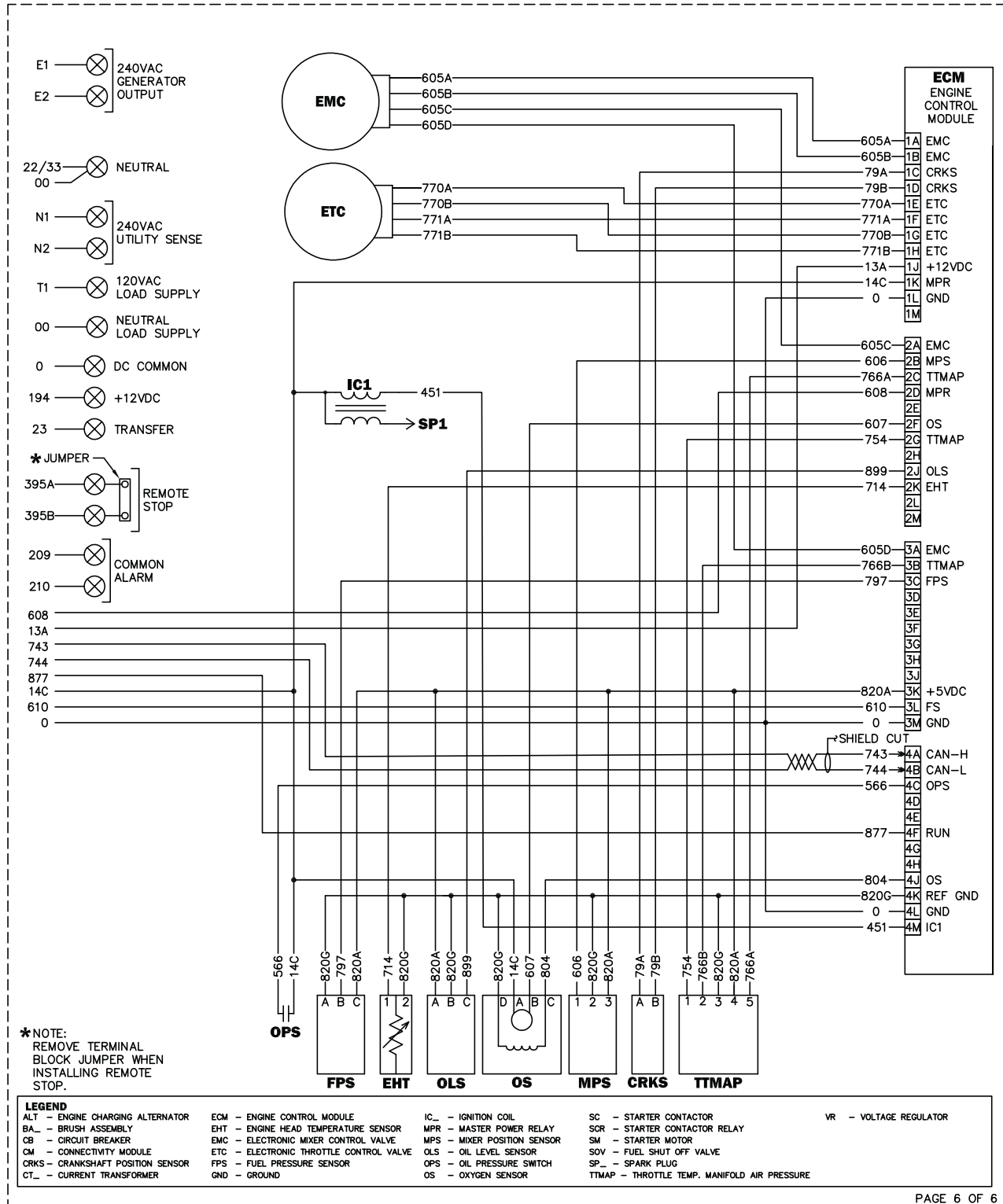
DATE: 05/02/2025

PAGE 5 OF 6

DRAWING #: A0005639540

Figure 11-5. A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459 (page 5 of 6)

GROUP SD



PAGE 6 OF 6

SCHEMATIC - DIAGRAM

REVISION: CN-0100171-B

PZ200 SSI 60HZ 459

DATE: 05/02/2025

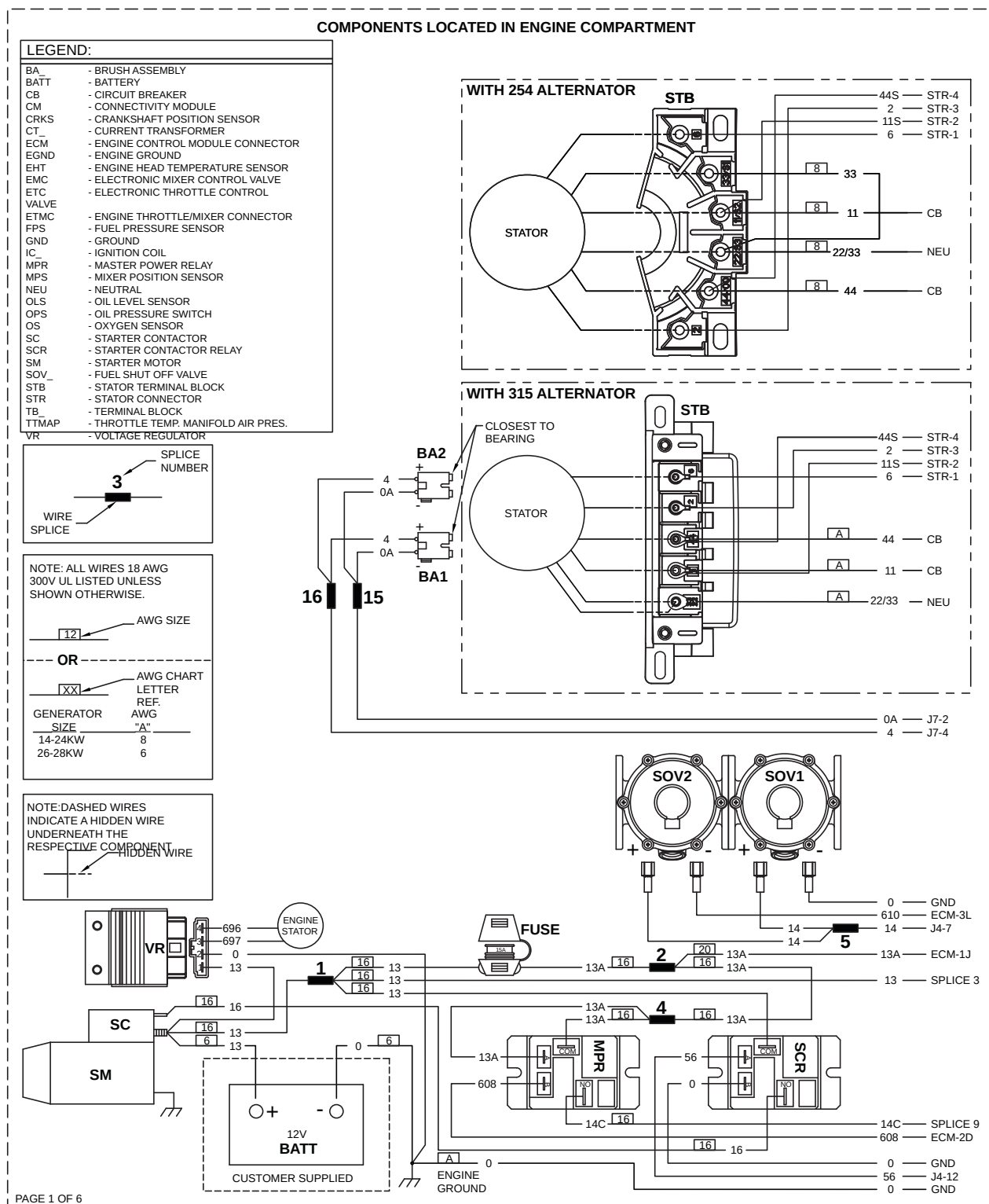
PAGE 6 OF 6

DRAWING #: A0005639540

Figure 11-6. A0005639540-B WD/SD HSB PZ200 SSI 60HZ 459 (page 6 of 6)

A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997

GROUP WD



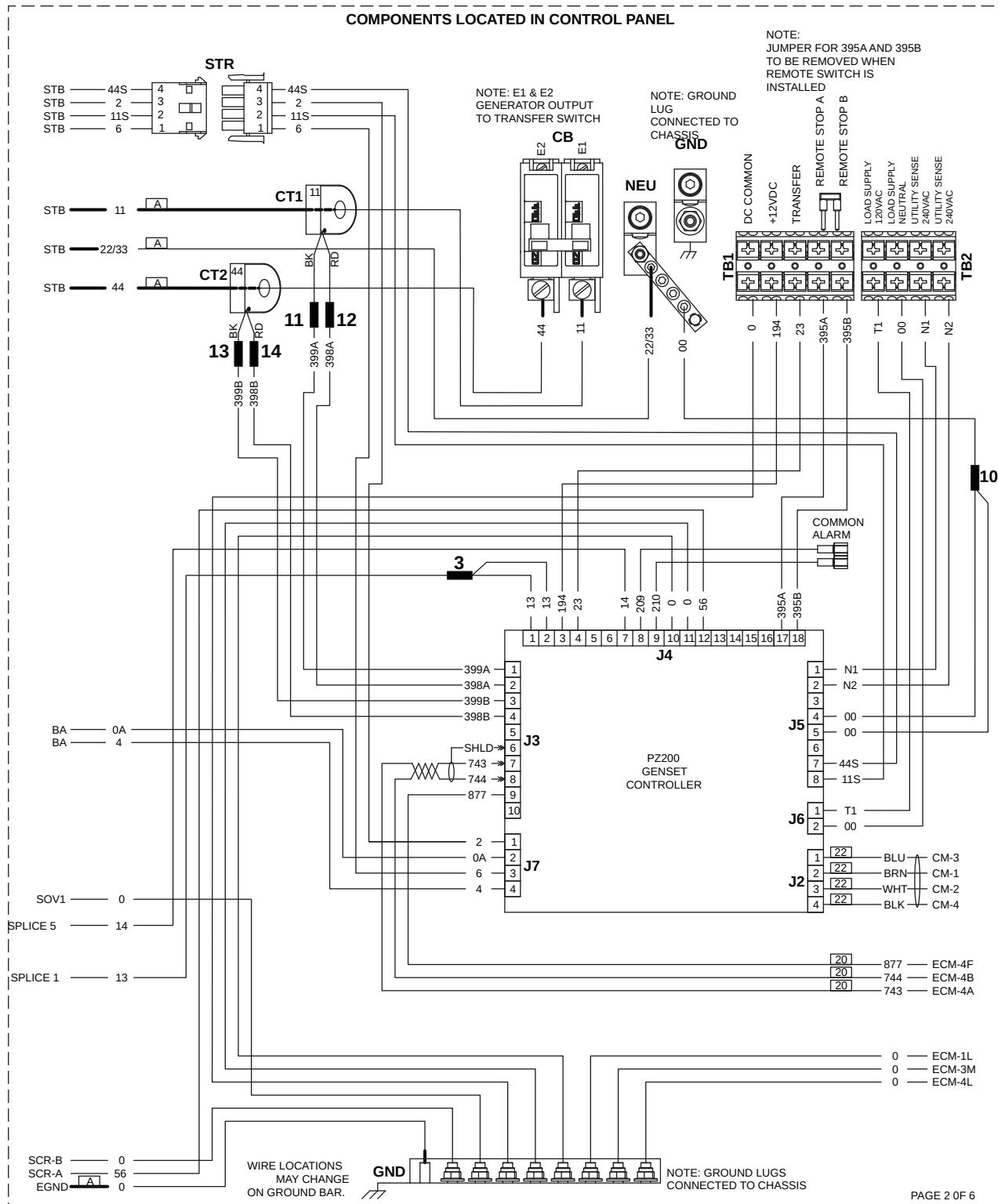
REVISION: CN-0100171-B
DATE: 05/02/2025

PAGE 1 OF 6

WIRING - DIAGRAM
PZ200 SSI 60HZ 817/997
DRAWING #: A0005639538

Figure 11-7. A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997 (1 of 6)

GROUP WD



REVISION: CN-0100171-B

DATE: 05/02/2025

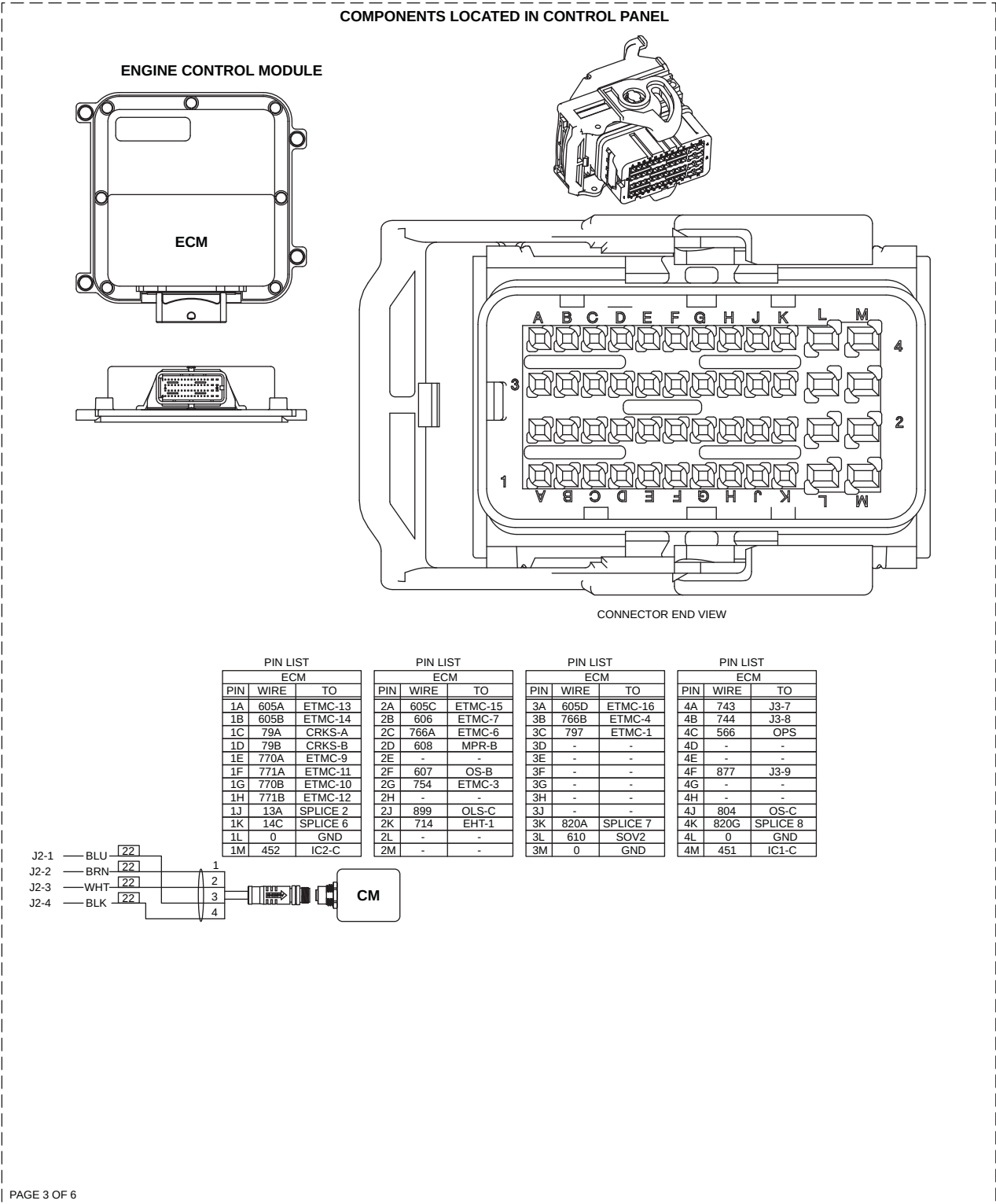
PAGE 2 OF 6

WIRING - DIAGRAM

PZ200 SSI 60HZ 817/997

DRAWING #: A0005639538

Figure 11-8. A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997 (page 2 of 6)



PAGE 3 OF 6

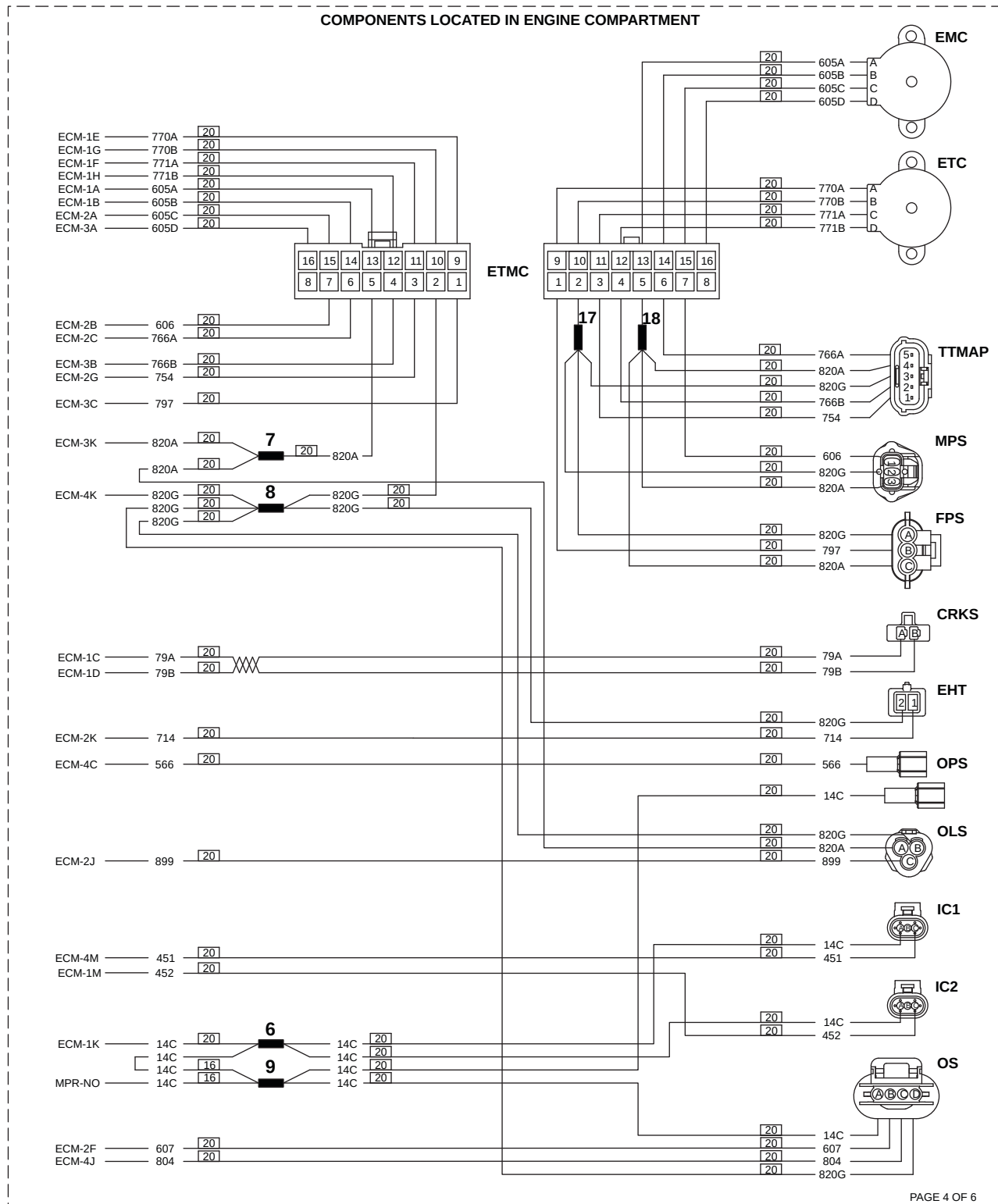
REVISION: CN-0100171-B
DATE: 05/02/2025

PAGE 3 OF 6

WIRING - DIAGRAM
PZ200 SSI 60HZ 817/997
DRAWING #: A0005639538

Figure 11-9. A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997 (page 3 of 6)

GROUP WD



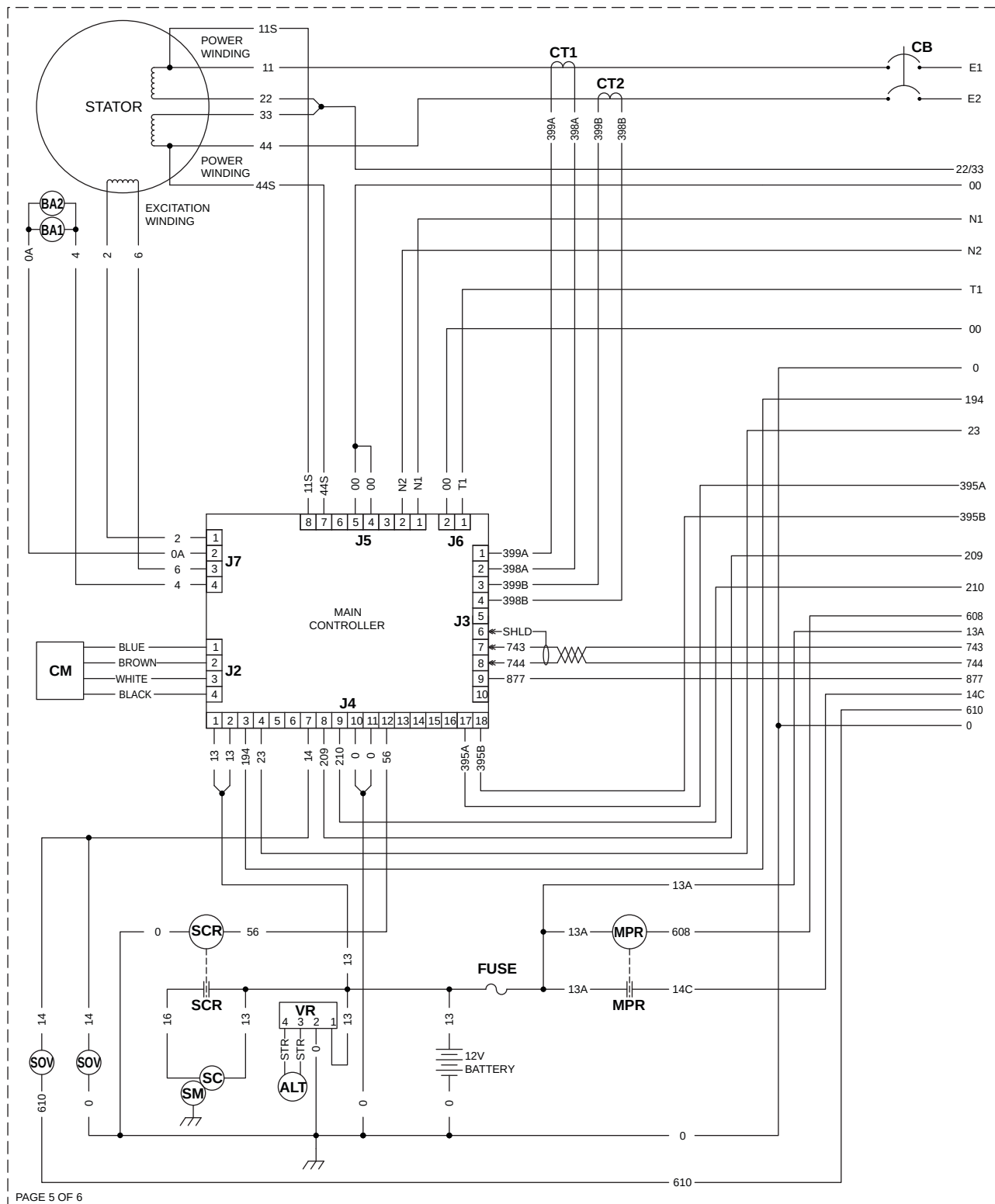
REVISION: CN-0100171-B
DATE: 05/02/2025

PAGE 4 OF 6

WIRING - DIAGRAM
PZ200 SSI 60HZ 817/997
DRAWING #: A0005639538

Figure 11-10. A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997 (page 4 of 6)

GROUP SD



PAGE 5 OF 6

SCHEMATIC - DIAGRAM

REVISION: CN-0100171-B

PZ200 SSI 60HZ 817/997

DATE: 05/02/2025

PAGE 5 OF 6

DRAWING #: A0005639538

Figure 11-11. A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997 (page 5 of 6)

Figure 11-12. A0005639538-B WD/SD HSB PZ200 SSI 60HZ 817/997 (page 6 of 6)

A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997

GROUP WD

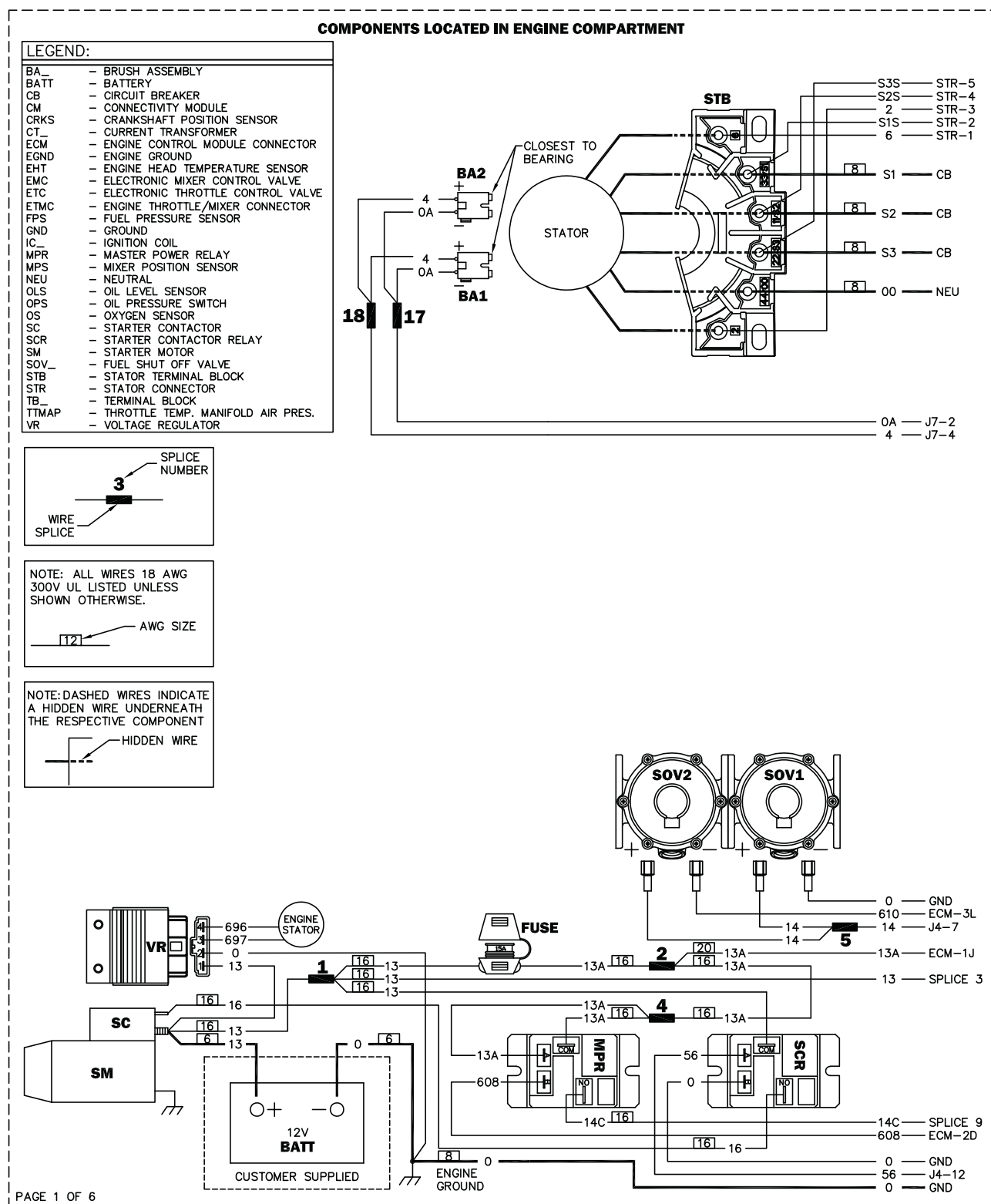
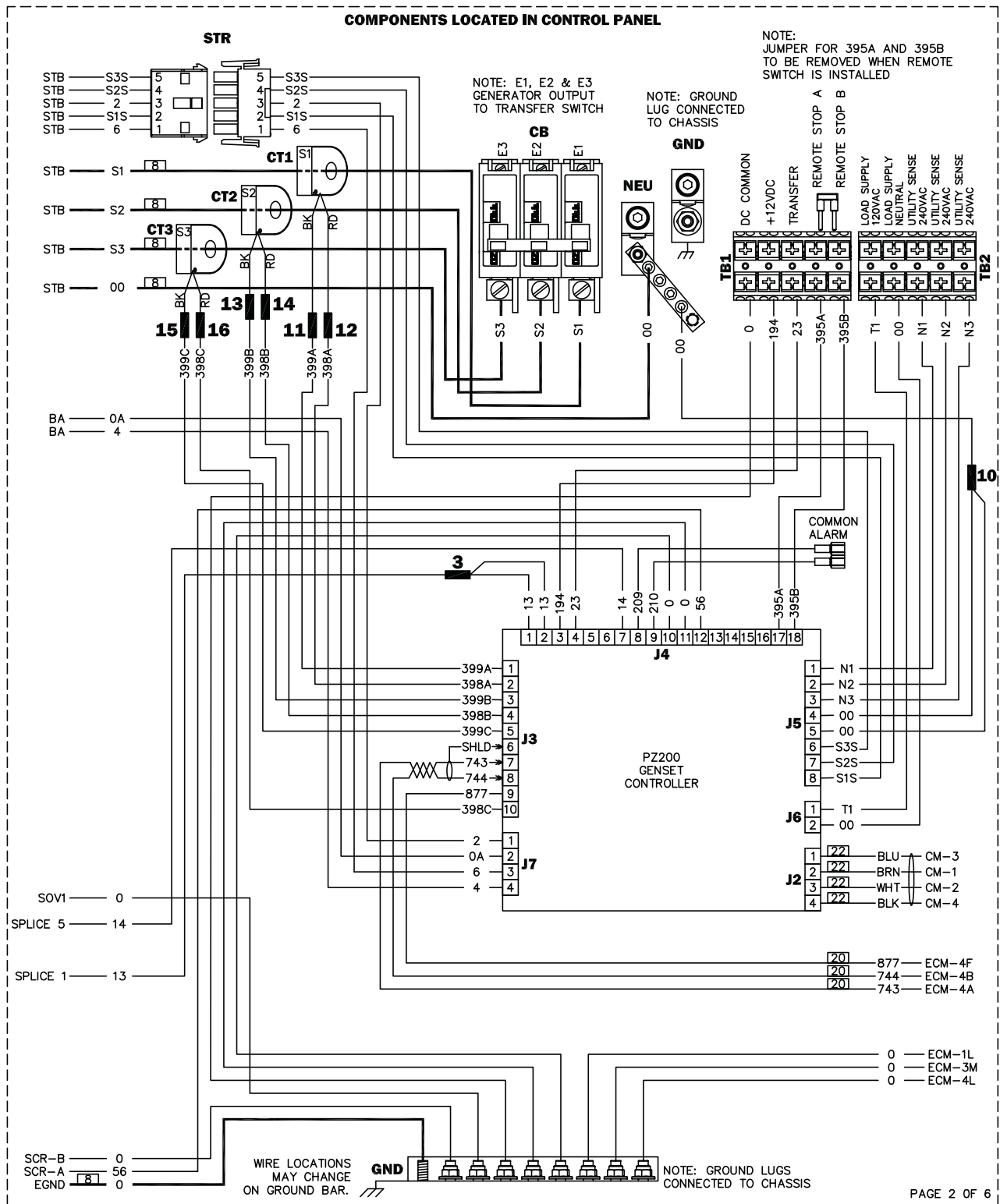


Figure 11-13. A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997 (page 1 of 6)

GROUP WD



REVISION: CN-0100171-B

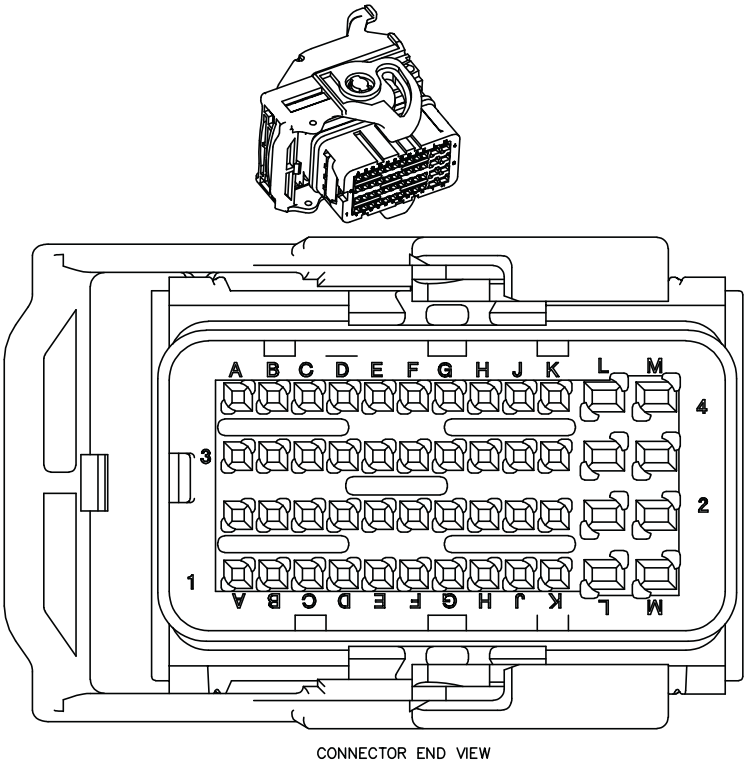
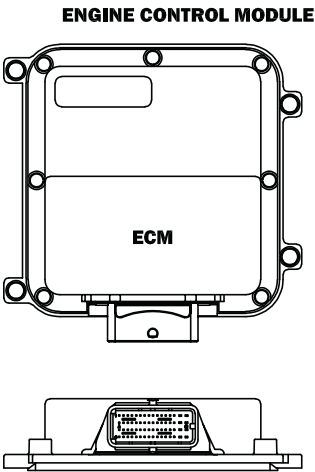
DATE: 05/02/2025

PAGE 2 OF 6

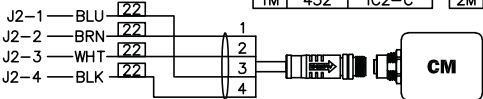
WIRING - DIAGRAM
PZ200 SSI 60Hz 3PH 817/997
DRAWING #: A0005639541

Figure 11-14. A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997 (page 2 of 6)

COMPONENTS LOCATED IN CONTROL PANEL



PIN LIST			PIN LIST			PIN LIST			PIN LIST		
PIN	WIRE	TO	PIN	WIRE	TO	PIN	WIRE	TO	PIN	WIRE	TO
1A	605A	ETMC-13	2A	605C	ETMC-15	3A	605D	ETMC-16	4A	743	J3-7
1B	605B	ETMC-14	2B	606	ETMC-7	3B	766B	ETMC-4	4B	744	J3-8
1C	79A	CRKS-A	2C	766A	ETMC-6	3C	797	ETMC-1	4C	566	OPS
1D	79B	CRKS-B	2D	608	MPR-B	3D	-	-	4D	-	-
1E	770A	ETMC-9	2E	-	-	3E	-	-	4E	-	-
1F	771A	ETMC-11	2F	607	OS-B	3F	-	-	4F	877	J3-9
1G	770B	ETMC-10	2G	754	ETMC-3	3G	-	-	4G	-	-
1H	771B	ETMC-12	2H	-	-	3H	-	-	4H	-	-
1J	13A	SPLICE 2	2J	899	OLS-C	3J	-	-	4J	804	OS-C
1K	14C	SPLICE 6	2K	714	EHT-1	3K	820A	SPLICE 7	4K	820G	SPLICE 8
1L	0	GND	2L	-	-	3L	610	SOV2	4L	0	GND
1M	452	IC2-C	2M	-	-	3M	0	GND	4M	451	IC1-C



PAGE 3 OF 6

WIRING - DIAGRAM

PZ200 SSI 60HZ 3PH 817/997

DRAWING #: A0005639541

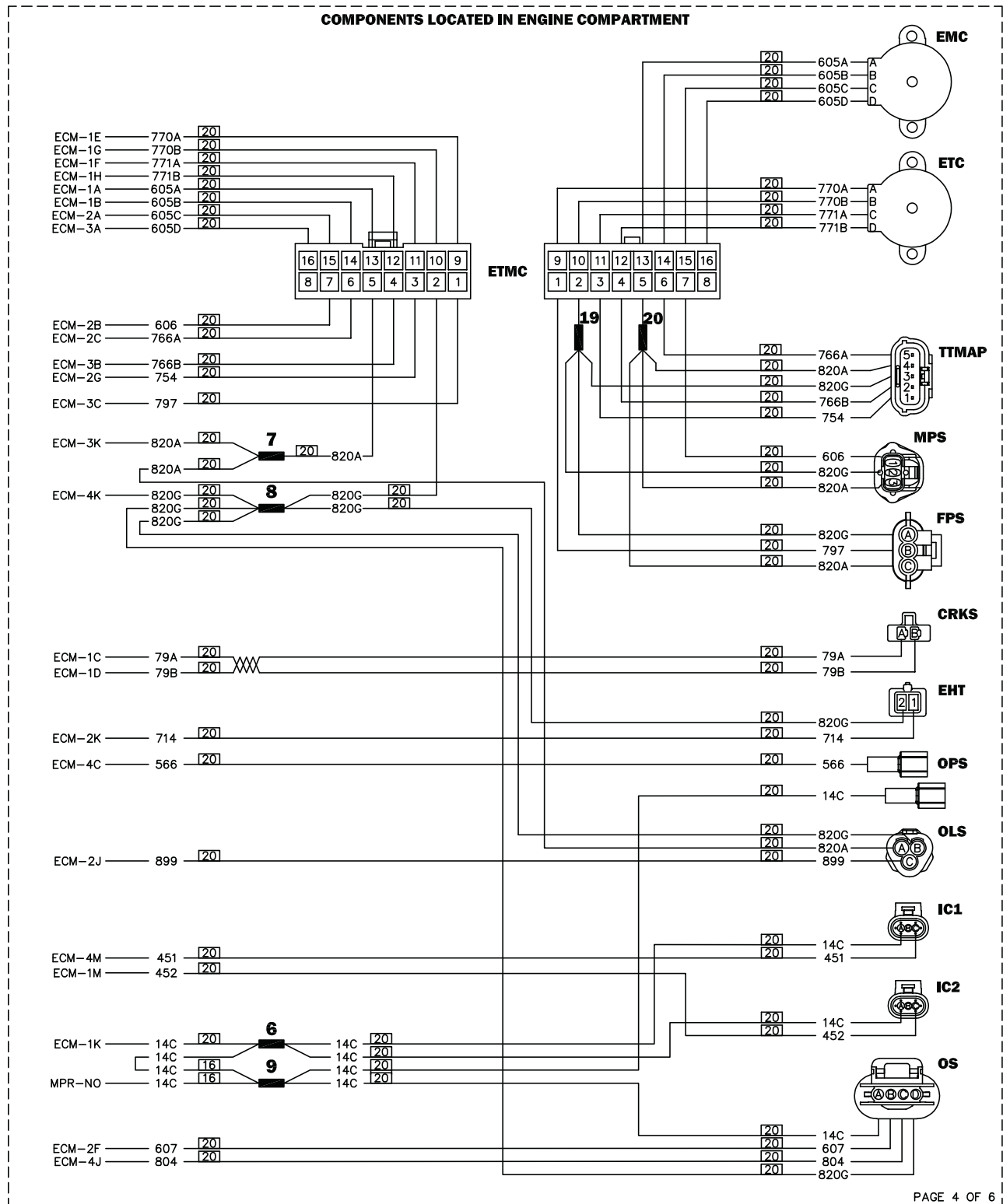
REVISION: CN-0100171-B

DATE: 05/02/2025

PAGE 3 OF 6

Figure 11-15. A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997 (page 3 of 6)

GROUP WD



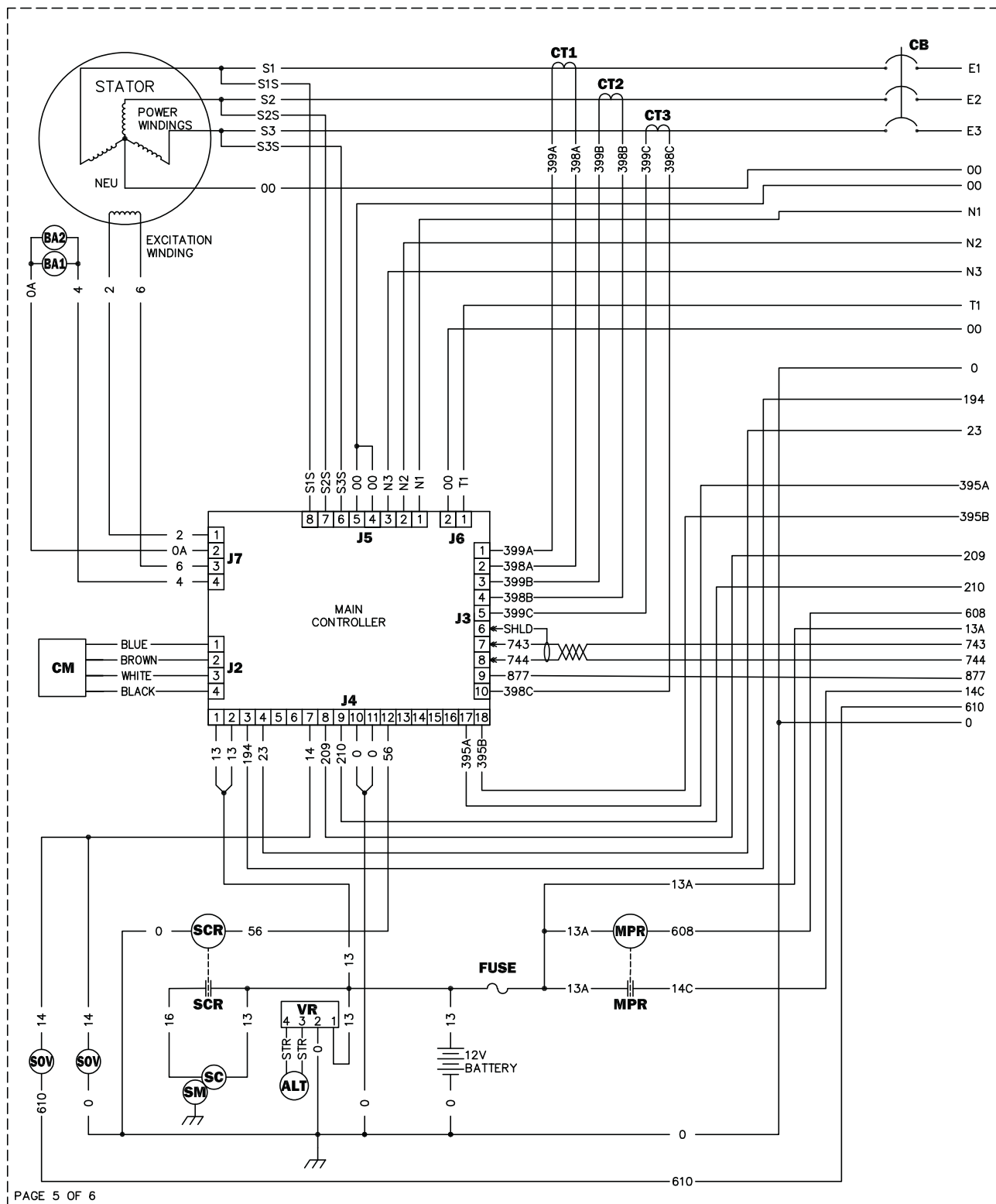
REVISION: CN-0100171-B
DATE: 05/02/2025

PAGE 4 OF 6

WIRING - DIAGRAM
PZ200 SSI 60Hz 3PH 817/997
DRAWING #: A0005639541

Figure 11-16. A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997 (page 4 of 6)

GROUP SD



SCHEMATIC - DIAGRAM

PZ200 SSI 60HZ 3PH 817/997

DRAWING #: A0005639541

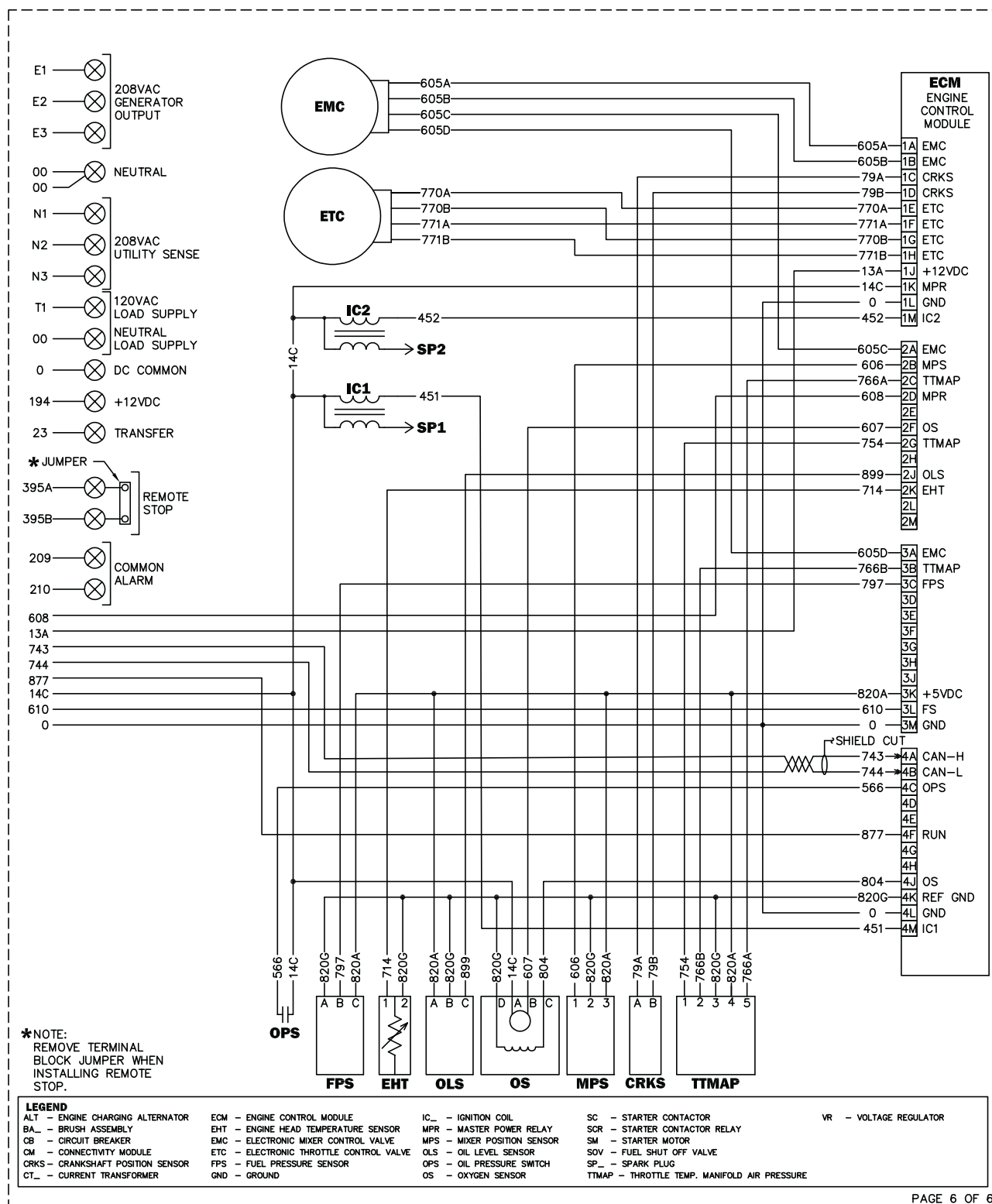
REVISION: CN-0100171-B

DATE: 05/02/2025

PAGE 5 OF 6

Figure 11-17. A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997 (page 5 of 6)

GROUP SD



PAGE 6 OF 6

SCHEMATIC - DIAGRAM

PZ200 SSI 60Hz 3PH 817/997

DRAWING #: A0005639541

REVISION: CN-0100171-B

DATE: 05/02/2025

PAGE 6 OF 6

Figure 11-18. A0005639541-B WD/SD HSB PZ200 SSI 60Hz 3P 817/997 (page 6 of 6)

This page intentionally left blank.

Section 12: Major Disassembly

Major Disassembly

Safety



Automatic start-up. Disconnect utility power and render unit inoperable before working on unit. Failure to do so will result in death or serious injury.

(D000191)

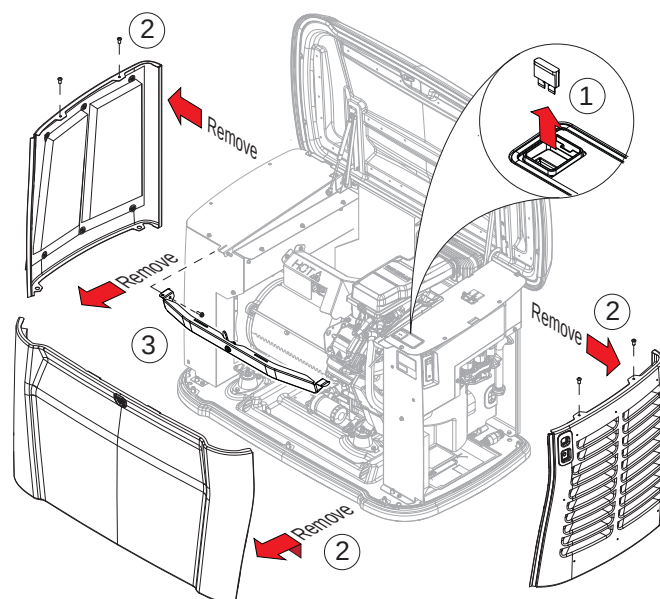
Required Tools

- Stator holding adapters (PN 0K8824)
- Rotor protector sheet (PN 0K8210)
- Vibration dampener puller
- 3 inch M12x1.75 Bolt
- Standard mechanics tool set
- Rubber mallet or dead-blow hammer
- Torque wrenches (Inch lbs and Foot lbs)
- 3 or 4 small 2"x4" blocks of wood

Preparation for Disassembly

1. Turn off fuel supply to generator.
2. Remove utility power source from the generator.
3. Unlock the roof.
4. Lift the roof.
5. Set the generator controller to OFF.

Disassembly



00077426

Figure 12-1. Remove Front and Side Panels

1. Remove the 7.5 fuse from the controller.
2. Remove the front panel and side panels.
 - a. Lift and remove front panel.
 - b. Remove two screws from each side panel and lift panels.
3. Disconnect negative battery terminal.
4. Use a 10 mm socket to remove two (2) bolts from the center support.

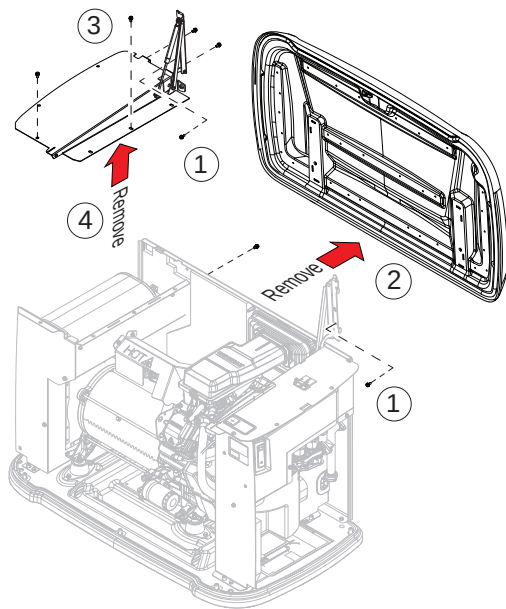


Figure 12-2. Remove Roof

1. Remove one bolt from each bracket on the underside of the roof.
2. Remove roof from brackets by sliding up and lifting away.

WARNING

Personal injury. Excessive weight. Two person lift. Use only appropriate techniques when lifting equipment. Improper lifting techniques could result in equipment damage, death or serious injury.

(W000822)

3. Remove 4 screws from top exhaust cover.
4. Remove top exhaust cover.

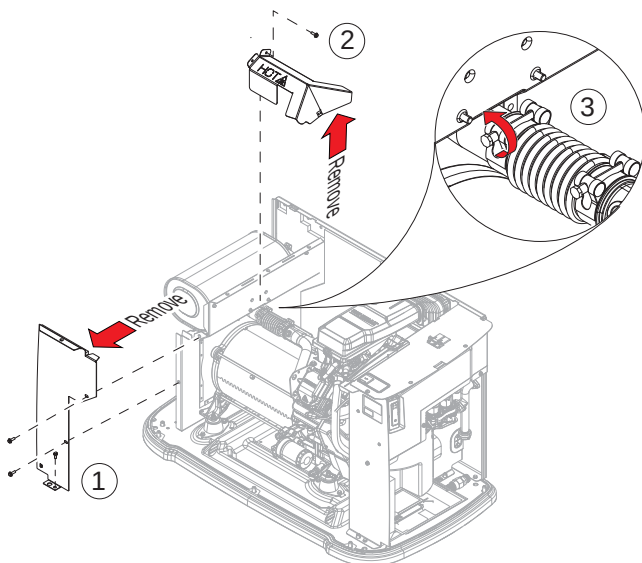


Figure 12-3. Remove Exhaust Covers

1. Remove bolt from bottom of front exhaust cover and two screws from the front of the cover.
2. Use a 10 mm socket to remove two bolts from the exhaust flex pipe cover. Remove the exhaust flex pipe cover.
3. Loosen the bolt on the rear exhaust flex pipe clamp.

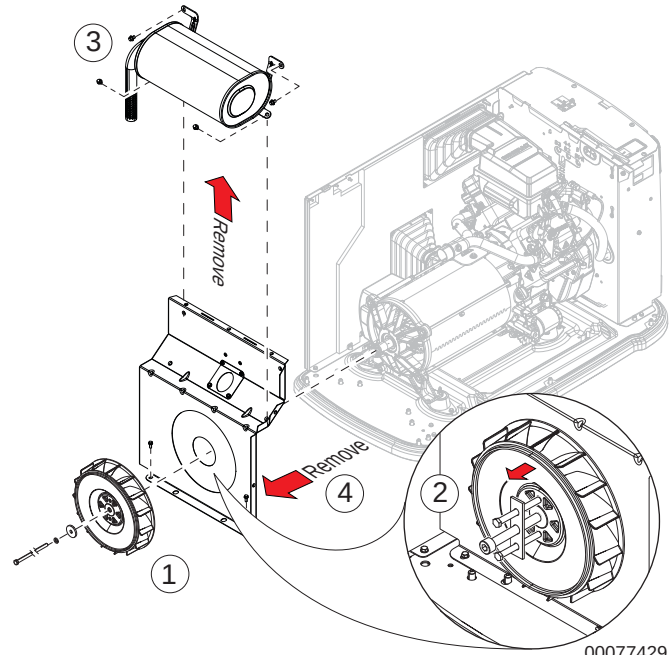


Figure 12-4. Remove Fan From Rotor

1. Use a 9/16" socket to remove rotor bolt.
2. Thread a 3 inch M12 x 1.75 bolt into rotor end, leaving about 1/2" of thread exposed in preparation for next Step.
 - a. Attach a vibration dampener puller or suitable alternative puller to the fan using two M8 x 1.25 bolts.

NOTE: Thread the puller bolts at least 12" into the fan.

 - b. Tighten the central puller shaft with a wrench to loosen the fan from the rotor shaft.
 - c. Remove the fan from the rotor.

NOTE: Use a wrench to prevent the fan from rotating while using the vibration dampener puller.
3. Remove bolts and muffler from rear exhaust divider panel.
4. Remove rear exhaust divider panel.

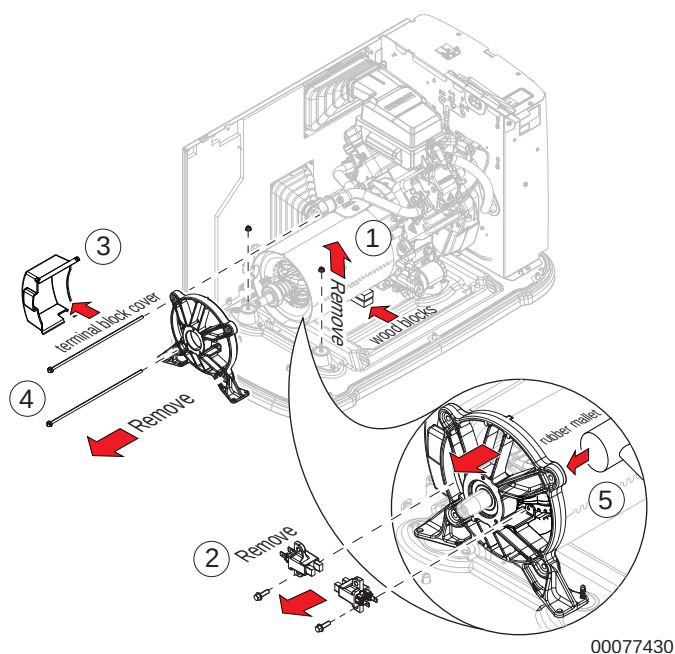


Figure 12-5. Remove Rear Bearing Carrier

1. Lift the back end of the alternator up and place a 2" x 4" piece of wood under the stator. Use additional blocks of wood if needed.
2. Disconnect wires from both brush assemblies and remove brush assemblies.
3. Remove the stator terminal block (STB) cover
 - a. Cut the cable ties holding the stator terminal block (STB) cover.
4. Use a 13 mm socket to remove the four (4) stator bolts.
5. Use a small rubber mallet or suitable dead-blow hammer to remove the rear bearing carrier.

NOTE: Use short deliberate blows and alternate tapping between the upper lobes of the bearing carrier until it is free from the alternator can and rotor shaft bearing.

Modify Existing Rotor Bolt for Rotor Removal

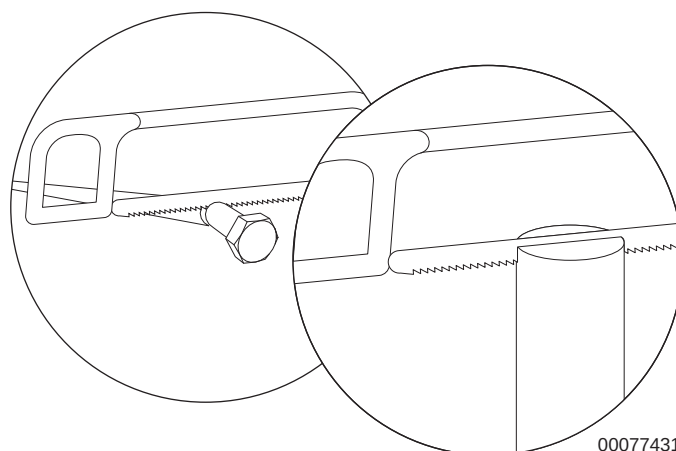


Figure 12-6. Modify Rotor Bolt

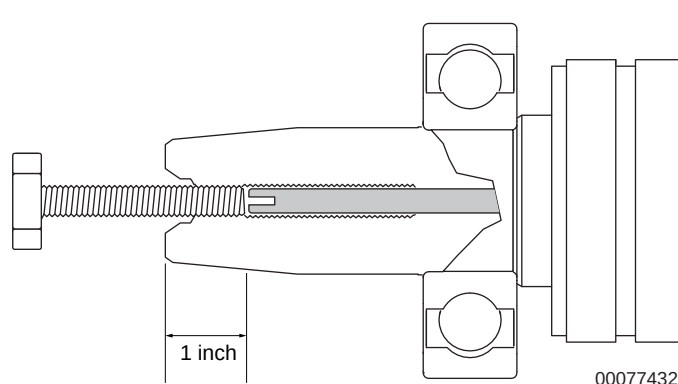
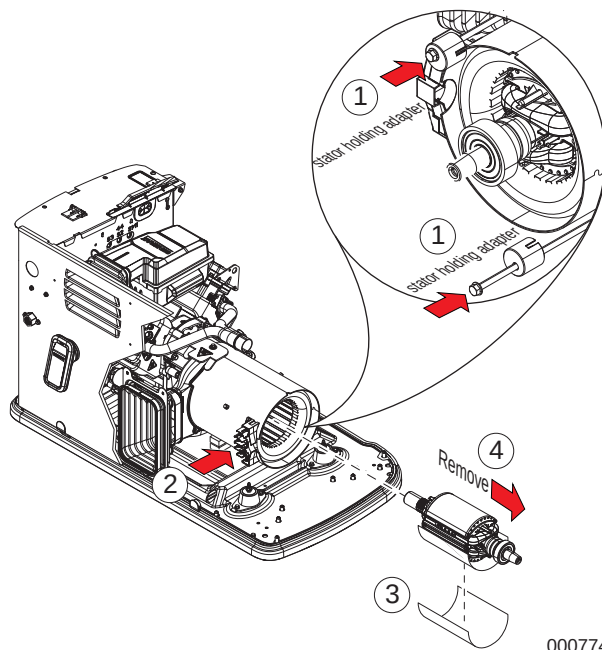


Figure 12-7. Insert Modified Rotor Bolt

1. Remove rotor bolt and cut 1.25 in (38.1 mm) from head of bolt, leaving bolt 16.25 in (41.28 cm) length.
2. Use a hacksaw or suitable cutting wheel to cut a slot in the cut end of the rotor bolt.
3. Using a flat blade screwdriver, thread the cut rotor bolt into the crankshaft, allowing about an inch of threads exposed for the following steps.
4. Screw a 3" M12 x 1.75 bolt into rotor end.
5. Apply torque to the 3" M12 x 1.75 bolt until rotor breaks free from the tapered engine crankshaft.



00077433

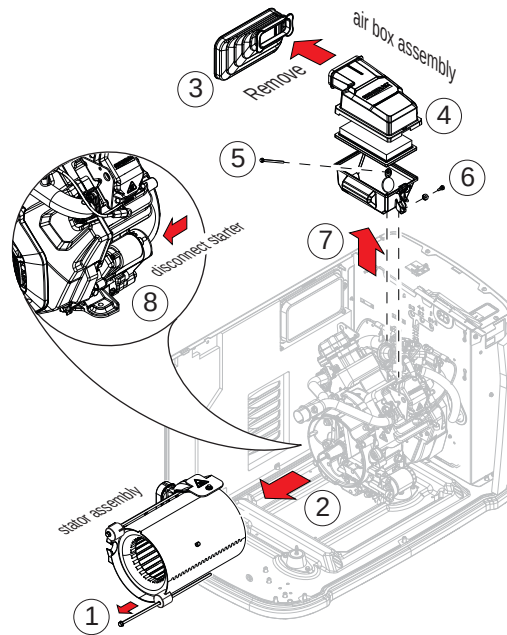
Figure 12-8. Instal Stator Holding Adaptors

1. Insert two previously removed stator bolts into the two stator holding adaptors (P/N 0K8824).
2. Remove stator wires from stator terminal block (STB).
3. Insert the rotor protector sheet (A) (PN 0K8210) between the rotor and stator assembly.
4. Carefully slide rotor (on protective sheet) out of the stator can. Place the rotor in a safe area to prevent any damage.
5. Remove the modified rotor bolt from the crankshaft.

WARNING

Personal injury. Excessive weight. Lift using appropriate lifting eyes and use appropriate lifting equipment to lift unit. Incorrect lifting techniques could result in death, serious injury, or equipment damage.

(W000224)



00077434

Figure 12-9. Disassemble Air Box

1. Remove two stator bolts and stator holding adaptors.
2. Remove stator.

WARNING

Personal injury. Excessive weight. Lift using appropriate lifting eyes and use appropriate lifting equipment to lift unit. Incorrect lifting techniques could result in death, serious injury, or equipment damage.

(W000224)

3. Remove air box cover boot.
4. Remove air box cover.
5. Remove air filter.
6. Disconnect starter wires.

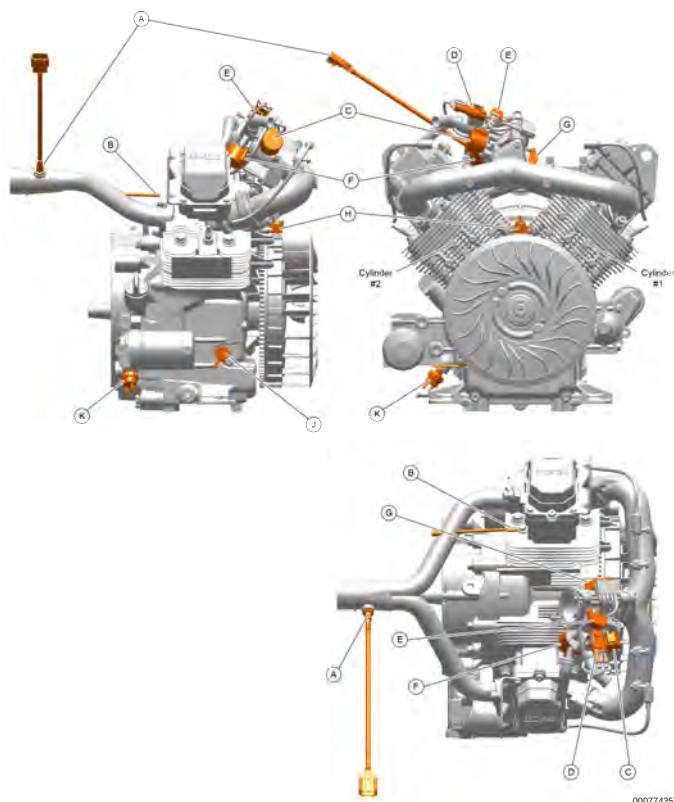
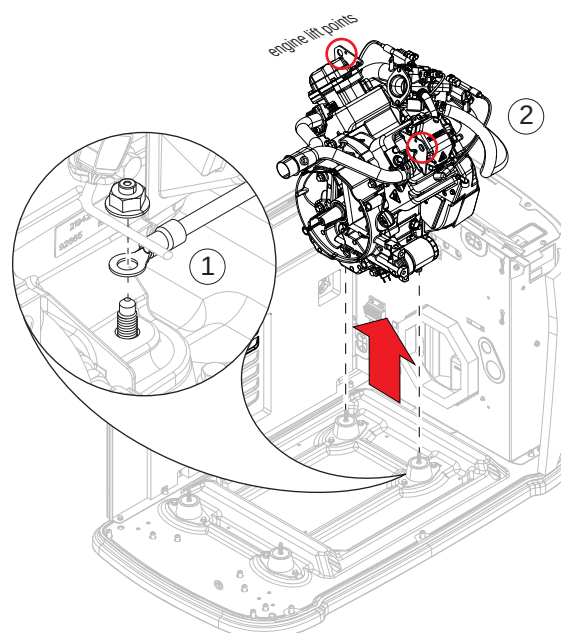


Figure 12-10. Engine Control Location

See diagrams in [Engine Control Module](#). Disconnect harness connectors from all engine mounted controls and sensors.



00077436

Figure 12-11.

1. Remove nuts from two engine mounts and disconnect engine ground wire.
2. Use appropriate equipment to lift engine from base.

WARNING

Personal injury. Excessive weight. Lift using appropriate lifting eyes and use appropriate lifting equipment to lift unit. Incorrect lifting techniques could result in death, serious injury, or equipment damage.

(W000224)

Assembly

1. Follow disassembly steps in reverse.

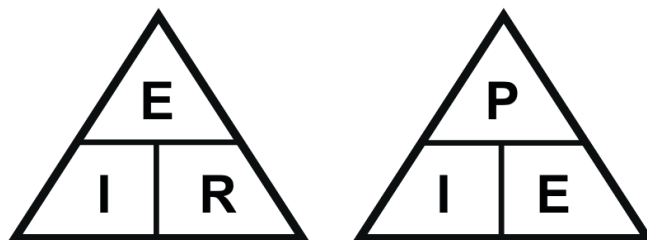
Electrical Formulas

To Find	Known Values	1-phase	3-phase
Kilowatts (kW)	Volts, Current, Power Factor	$(E \times I) / (1000)$	$(E \times I \times 1.73 \times PF) / 1000$
KVA	Volts, Current	$(E \times I) / (1000)$	$(E \times I \times 1.73) / 1000$
Amperes	kW, Volts, Power Factor	$(KW \times 1000) / (E)$	$(KW \times 1000) / (E \times PF \times 1.73)$
Watts	Volts, Amps, Power Factor	Volts x Amps	$E \times I \times 1.73 \times PF$
No. of Rotor Poles	Frequency, RPM	$(2 \times 60 \times \text{Frequency}) / (\text{RPM})$	$(2 \times 60 \times \text{Frequency}) / (\text{RPM})$
Frequency	RPM, No. of Rotor Poles	$(\text{RPM} \times \text{Pole}) / (2 \times 60)$	$(\text{RPM} \times \text{Pole}) / (2 \times 60)$
RPM	Frequency, No. of Rotor Poles	$(2 \times 60 \times \text{Frequency}) / (\text{Rotor pole})$	$(2 \times 60 \times \text{Frequency}) / (\text{Rotor pole})$
kW (required for Motor)	Motor Horsepower, Efficiency	$(HP \times 0.746) / (\text{Efficiency})$	$(HP \times 0.746) / (\text{Efficiency})$
Resistance	Volts, Amperes	E / I	E / I

To Find	Known Values	1-phase	3-phase
Volts	Ohm, Amperes	$I \times R$	$I \times R$
Amperes	Ohms, Volts	E / R	E / R

E = Volts**I = Amperes****R = Resistance (Ohms)****PF = Power Factor**

Term	Symbol	Measurement	Constant	Shift		Result	
Current	I	Amps	Voltage E	Resistance Increase	↑	Current Decrease	↓
Wattage	P	Watts	Voltage E	Resistance Decrease	↓	Current Increase	↑
Voltage	E	Volts	Resistance R	Voltage Decrease	↓	Current Decrease	↓
Resistance	R	Ohms	Resistance R	Voltage Increase	↑	Current Increase	↑
			Current I	Resistance Decrease	↓	Voltage Decrease	↓
			Current I	Resistance Increase	↑	Voltage Increase	↑
			Power P	Voltage Increase	↑	Power Increase	↑
			Power P	Voltage Decrease	↓	Power Decrease	↓
			Power P	Current Increase	↑	Power Increase	↑
			Power P	Current Decrease	↓	Power Decrease	↓

**Figure 12-12.**

This page intentionally left blank.

Part No. A0004542981 Rev. E 08/21/2026
© 2026 Generac Power Systems, Inc.
All rights reserved.
Specifications are subject to change without notice.
No reproduction allowed in any form without prior written
consent from Generac Power Systems, Inc.



Generac Power Systems, Inc.
S45 W29290 Hwy. 59
Waukesha, WI 53189
1-888-GENERAC (1-888-436-3722)
www.generac.com